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Public vs. private equity: The widest equity return gap in two decades

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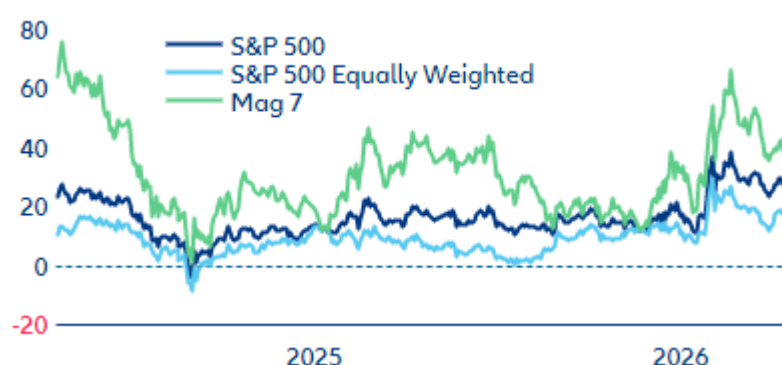
In Summary

- **Public equities have repriced the future while private equity is still monetizing the past.** Since the October 2022 trough, the S&P 500 has compounded at more than 20% a year for three straight years, while Private Equity buyout portfolios assembled at the 2020-2021 peak, and now carrying more expensive debt into a thin exit market, have lagged on every benchmark and horizon. At this point in time, the gap between liquid and illiquid equity returns seems to be the widest in two decades.
- **The rally that opened the gap is narrow and earnings-led, not a liquidity melt up.** The Magnificent Seven alone delivered more than half the S&P 500's three-year total return, strip them out and the index is an ordinary performer, and the equal weighted version has compounded at about half the pace, a gap unseen since the late 1990s. But the move rests on delivered earnings and a real AI capex super cycle rather than multiple expansion, so the premium looks more durable than the dot-com peak.
- **Yet private equity's long-run premium is real, this cycle has just interrupted it.** On a like for like basis, matched to public indices for geography, size, sector and leverage and net of fees and carry, buyout has beaten public markets over three decades, with MSCI estimating pooled direct alpha of roughly 400bps a year since 1994. The exception is the 2021-2023 vintages, which currently show a negative direct alpha of about ~800bps against the MSCI ACWI, the first consecutive run to trail public markets in the series. Consequently, the data points to rough PE performance ahead even allowing for the fact that these vintages are still young and J-curve distorted.
- **Private equity can no longer count on rising valuations and must now drive returns by growing the underlying business.** For over a decade, buyout returns came mostly from rising valuations, selling companies for a higher multiple than was paid. That is mostly gone as borrowing costs have roughly doubled and exit valuations have stopped climbing, so returns now come from growing the business. The contrast on the financing side is also large, with the largest listed companies still holding more cash than debt, so benefiting from higher rates, while buyout owned companies sit on floating rate debt due to be refinanced in 2026-2028, where higher rates only add to their interest bill.
- **As capital drains from private equity, private debt is catching the outflows.** The same jump in interest rates that made buyout deals more expensive made lending the more attractive trade. Big long term investors continue to like senior private credit because it pays a steady, contractual cash yield, its value barely moves and losses have been low, none of which buyout has delivered lately. So new money is rotating out of private equity and into private debt.
- **2026 will decide whether the public-private gap was a passing extreme or a lasting regime change.** Our central case assumes an economic mix that keeps returns positive but well below the post-Covid peaks with public equity settling in the low teens, while buyout follows in the mid-teens through 2027-2028. All in all, the way forward for private equity seems to be less about how much to own and more about how to own it. This means favoring managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed and resisting the temptation to buy the label without the underlying skill.

Public markets have sprinted ahead

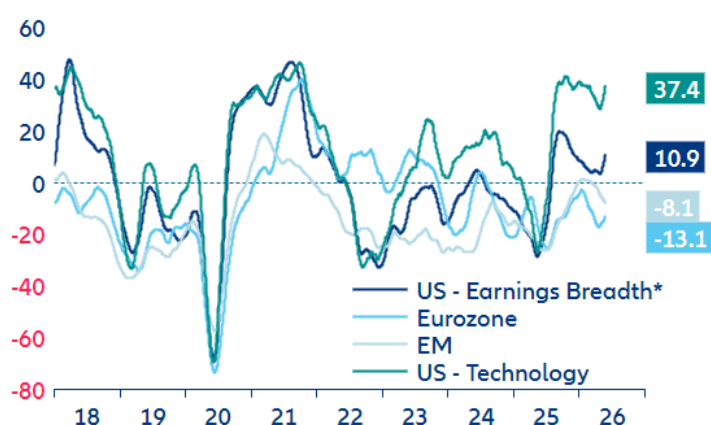
Since the October 2022 trough, public equities have staged one of the sharpest re-ratings in modern memory. The S&P 500 has compounded at more than 20% a year for three straight years, with other large-cap benchmarks close on its heels. But the headline flatters the market. The rally has been extraordinarily narrow: The Magnificent Seven (*Mag 7* - Apple, Microsoft, Alphabet (Google), Amazon, Nvidia, Meta (Facebook) and Tesla) alone delivered more than half of the S&P 500's three-year total return. Stripped of those seven names, the index is a thoroughly ordinary performer. The dispersion shows up in valuation, too. The top ten names now trade at a roughly 50% forward price/earnings premium to the rest of the index, wide by any historical yardstick. But that premium rests on an earnings path well above the dot-com peak of the 2000s, a sign that the fundamentals are more resilient now than they were then (Figure 1). Earnings breadth is mending as well (Figure 2), if grudgingly: Barely a third of S&P 500 members beat the index in 2025, up from a record low the year before. At the same time, small caps have scarcely shown up at all. The equal weighted cut of the same indices tells a more sober story still, with the S&P 500 equal weight compounding at about half the pace of its capitalization weighted parent, a gap unseen since the late 1990s. All in all, what reads at the index level as a broad bull market is, mechanically, the re-rating of a very small number of balance sheets.

Figure 1: Equity performance y/y%



Sources: LSEG Datastream, Allianz Research

Figure 2: Earnings breadth (3mma)



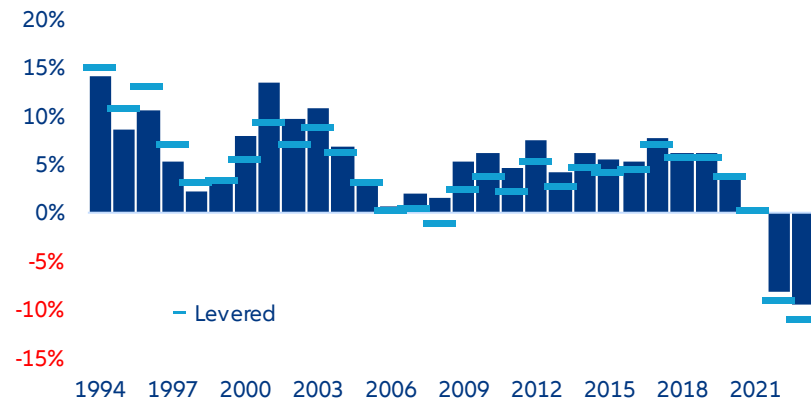
Sources: IBES, LSEG Datastream, Allianz Research

Note: Earnings breadth measures the share of index constituents receiving upward earnings estimate revisions.

The scale of the recent gap is easier to appreciate against private equity's long-run record. On a like-for-like basis, comparing each buyout fund with a public index matched for geography, size, sector and leverage, and measuring excess return net of fees and carry, private equity has, in fact, beaten public markets over the past three decades. MSCI estimates pooled buyout direct alpha of roughly 4% a year since 1994. That long-run premium is precisely what the current cycle has interrupted. Broken out by vintage, buyout funds delivered positive alpha against global equities in almost every cohort since the mid-1990s. The exception is the 2021-2023 vintages, which currently show

direct alpha of roughly -5% to -12% against the MSCI ACWI, the first run of consecutive vintages to trail public markets in the series. Of course, those readings come with an important caveat since the funds are young, and early fee drag (the J-curve) mechanically depresses returns that typically recover as portfolios mature, so those returns are provisional rather than final. But even allowing for that, the message is consistent: The vintages bought into the cheap-money peak are, so far, the worst relative performers in a generation and may still lead to some struggling years ahead (Figure 3).

Figure 3: Buyout direct alpha vs. public equity by vintage (in %)

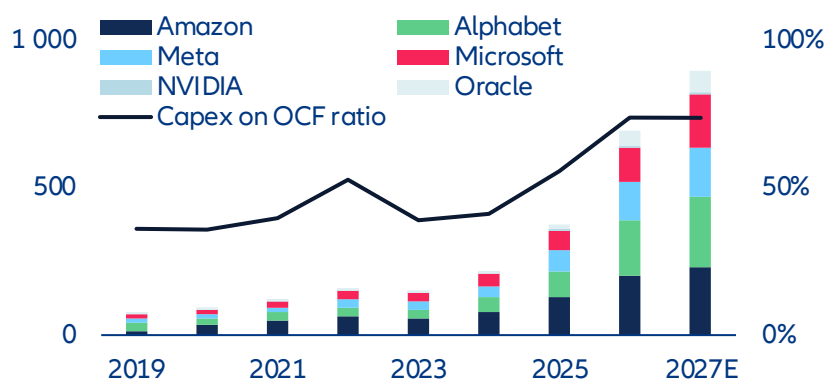


Sources: MSCI Private Capital Universe; Lester, O'Shea & Warren (2025), Allianz Research

Note: Direct alpha is the annualized excess return of buyout funds over a leverage- and sector-matched public index, net of fees and carry. 2021-2023 vintages are shaded to indicate they remain J-curve-distorted and are not yet representative of since-inception performance; 2024-2025 are omitted as too young to have generated representative exit data.

The character of the move matters as much as the magnitude. This has not been a liquidity-driven melt up of the 2020-2021 variety. It has been an earnings- and margin-led re-rating built on top of a genuine capex super cycle (Figure 4). Along these lines, forward earnings for the S&P 500 are up by roughly a third since the 2022 trough, and the bulk of the index's price return is explained by delivered earnings rather than multiple expansion. That distinction matters for how we read the gap to private markets performance (Figure 5). Public equity has not simply re-priced risk, it has also re-priced a specific, identifiable growth thesis around AI infrastructure, hyperscale compute and the operating leverage of asset-light platforms. Private equity, holding a portfolio assembled for a different regime, has had no comparable thesis to ride, and software, one of its biggest post-Covid allocations, has suffered from major potential disruptions. The second feature worth flagging is what the rally has done to the investable opportunity set itself. Free float in the largest names has shrunk as buybacks have accelerated and insider holdings have stayed concentrated, even allowing for the supply that the latest mega-IPOs have brought back. The effective supply of mega-cap equity available to incremental buyers is therefore tighter than headline market capitalization suggests (Figure 6). At the same time, the US now has roughly half as many listed companies as it did at its late 1990s peak, and the firms that do go public are older and larger than they used to be. So when a private equity sponsor comes to sell a company today, the public market it can exit into is smaller, more concentrated in a few large names and pickier about what it will absorb than the market sponsors were counting on when they bought in over 2018–2021. Consequently, the gap between public and private valuations in that light does not look like a temporary mispricing that will simply close, but a lasting result of a deeper shift in where growth capital is raised and how it is priced.

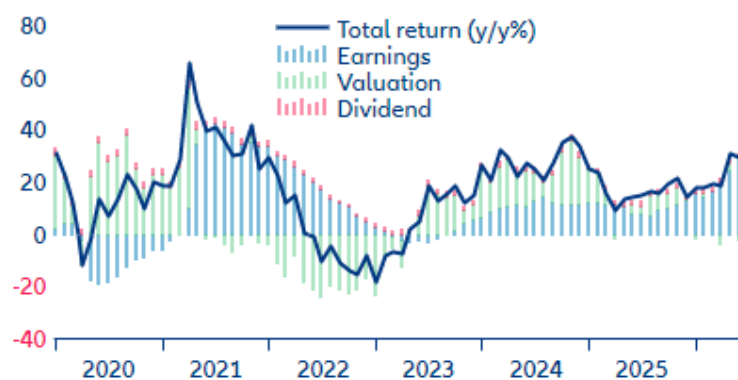
Figure 4: AI capital expenditure



Sources: Bloomberg, Allianz Research

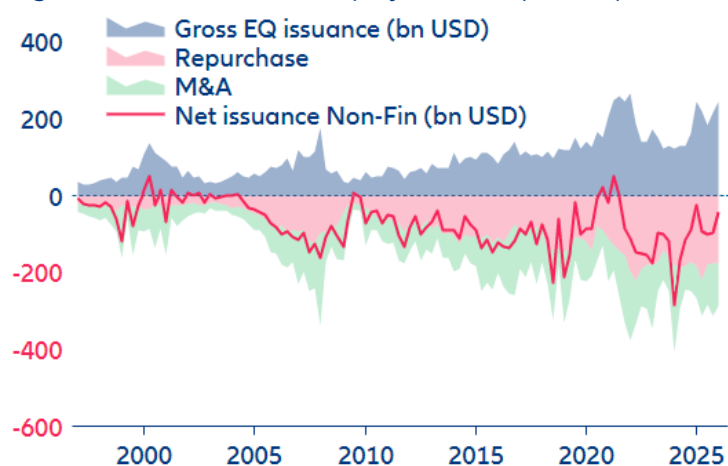
Note: OCF – Operating Cash Flow

Figure 5: S&P 500 return breakdown



Sources: LSEG Datastream, Allianz Research

Figure 6: US non-financial equity issuance (bn USD)

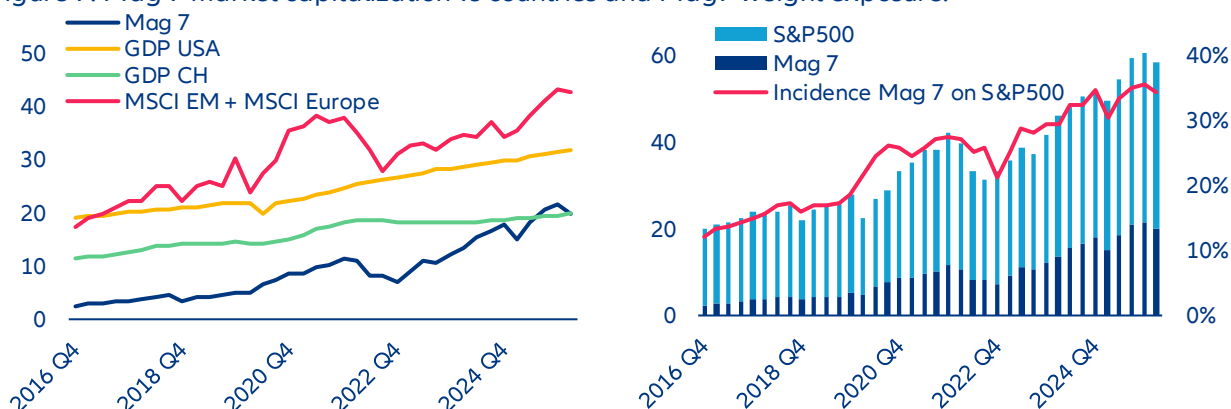


Sources: LSEG Datastream, Allianz Research

Public markets have absorbed private equity's growth engine

The narrowness of the public equity rally deserves closer inspection, because it is both the source of the divergence with private equity and a risk factor in its own right. The Mag 7 now represent roughly a third of S&P 500 market capitalization, up from around an eighth a decade ago, and their combined market value exceeds the GDP of every country except the US and China (Figure 7). Of course, the concentration is also self-reinforcing, because as these companies grow, passive and benchmark tracking flows mechanically direct more capital toward them, deepening their index weight independently of any fresh view on fundamentals.

Figure 7: Mag 7 market capitalization vs countries and Mag7 weight exposure.



Sources: Bloomberg, Allianz Research

Earnings concentration is also at extremes, and it now runs in parallel with the concentration in price. The same handful of mega-caps that dominate index weight also generate a disproportionate and rising share of S&P 500 earnings, so the benchmark's profits, not just its market value, increasingly depend on a single cohort. On valuation, the index trades well above its long-run average forward multiple, but the premium is almost entirely a mega-cap phenomenon; strip out the largest names and the rest of the market sits much closer to its historical norms, leaving a two-tier index in which the most expensive part is also the largest part (Figure 8). Because so much of the index now sits in long duration growth names, the multiple is also unusually sensitive to moves in rates and growth expectations; a small shift in either can swing the level materially. As an anchor for that sensitivity, a reversion of the cluster's valuation premium to its long-run average alone, before any change to underlying earnings, would cost the S&P 500 close to 10-20% of its value.

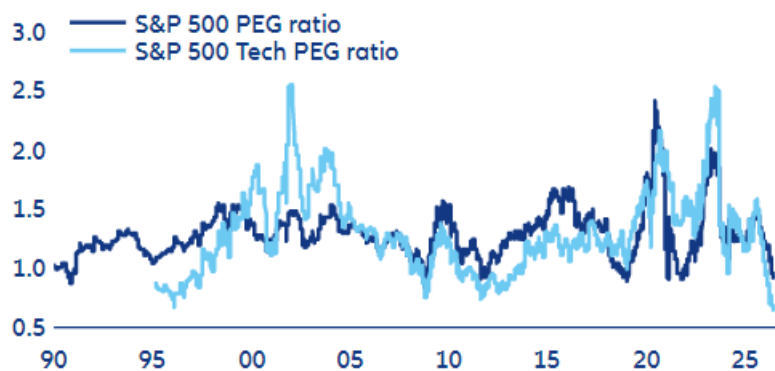
Figure 8: S&P 500 12m forward PE ratio



Sources: IBES, LSEG Datastream, Allianz Research

The same concentration invites two sharply opposing readings, and the contest between them is itself a market risk. The bullish case argues that the Mag 7 are nothing like the 2000 dot-com leaders as they generate hundreds of billions of dollars of free cash flow a year between them, sit on net cash balance sheets and dominate the decade's most powerful secular trend: AI compute, cloud infrastructure and advertising-data networks. Consensus forward earnings growth comfortably exceeds the broader market's, and on a PEG (Price/Earnings-to-Growth ratio) basis the cluster trades at parity or a discount. Nonetheless, the bearish case argues that the AI thesis rests on hyperscale capex sustaining at current levels, that enterprise AI revenue has yet to scale into the returns that capex implies and that the concentration itself is a source of fragility. An earnings disappointment from one or two names could de-rate the whole index in days. Both views can be right at once but which one prevails comes down to whichever is dominating the narrative at the time (Figure 9).

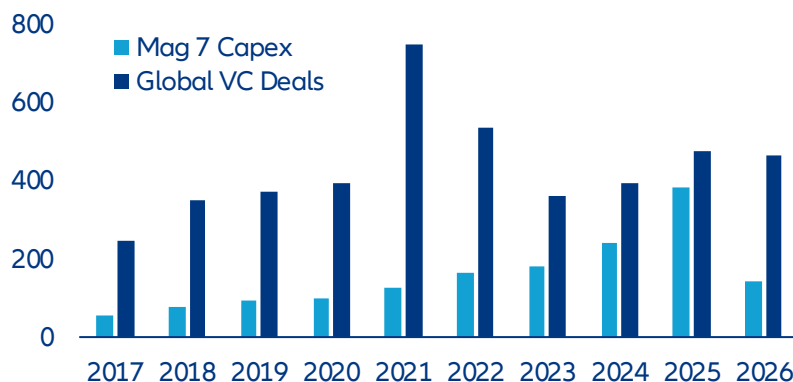
Figure 9: S&P 500 PE to growth ratio



Sources: IBES, LSEG Datastream, Allianz Research

Why the concentration matters for the public-private comparison is subtler than it first appears. The Mag 7 are themselves "long duration" growth assets, capex heavy and R&D intensive, with revenue streams that compound over a decade rather than monetize in a five-year LBO window. In effect, US public markets have absorbed the kind of growth equity exposure that used to migrate to private markets. Combined hyperscale capex now approaches the entire global venture capital (VC) deployment. Private investors who, in earlier cycles, would have funded the next generation of compute or AI infrastructure through late-stage venture rounds can now access that exposure through publicly listed mega caps, at any size and at low friction (Figure 10).

Figure 10: Mag 7 Capex vs VC deployment (USD Bn)



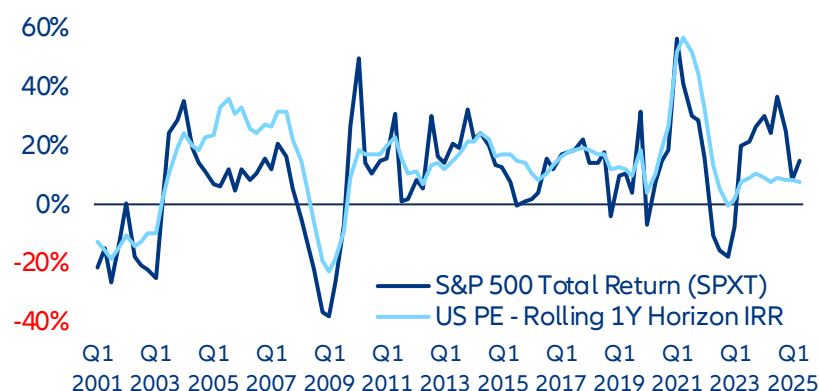
Sources: Bloomberg, PitchBook, Allianz Research

The corollary is uncomfortable for private markets. The universe of high-growth, scaled, profitable technology companies available exclusively to private investors has shrunk. At the time of writing, some of the private mega caps remain private holdings (though not for long, as SpaceX, Anthropic and OpenAI have IPO'd or filed to do so),

but the depth of comparable opportunity in private markets has narrowed precisely as fundraising and deployment has slowed.

Private equity, by contrast, is still digesting the previous cycle. The asset class entered the 2022 rate shock with peak fundraising, peak entry multiples, peak leverage and aggressive growth assumptions. The 2020-2021 vintages, in particular, were priced for a world of cheap money and continued multiple expansion, a world that ended within 18 months. Even top-quartile buyout funds returned single-digit IRRs in 2025, well behind both the S&P 500 and MSCI World, and the average vintage of the past decade barely cleared mid-single-digits (Figure 11).

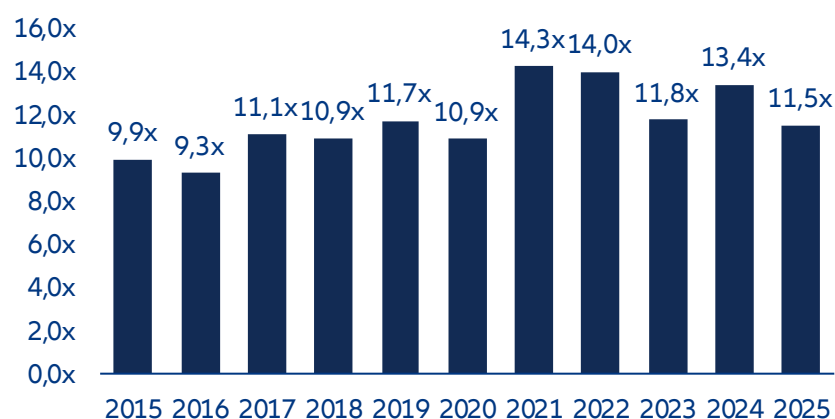
Figure 11: US Private Equity 1y IRR vs S&P 500



Sources: Bloomberg, PitchBook, Allianz Research

The lag has many causes, but the most mechanical is a reset in the cost of capital. A typical deal a decade ago cleared at around 10 times EBITDA, with half its capital in single-digit yield debt; today's clears closer to 13 times, with materially less leverage at far higher financing costs. Sponsors paid those record prices expecting to exit into further multiple expansion. That expansion has not arrived, and because thinner leverage cushions the entry, the drag on equity returns runs deeper than the change in multiples alone would suggest (Figure 12).

Figure 12: North America buyout entry multiples

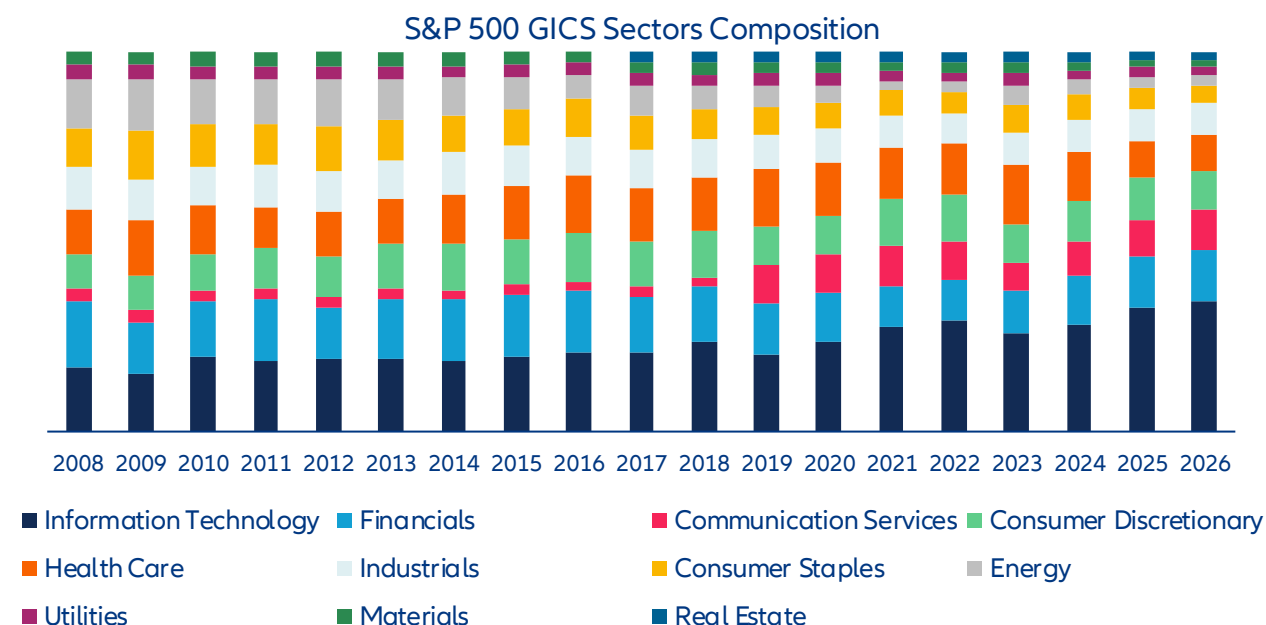


Sources: Pitchbook, Allianz Research

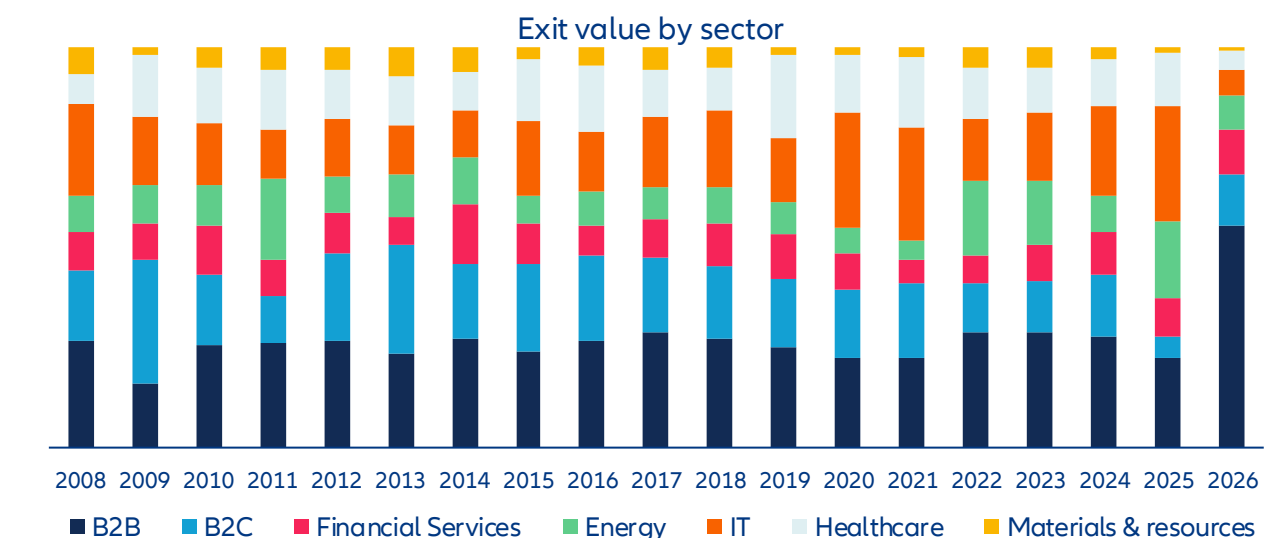
The mismatch is as much about sector mix as about timing. Public benchmarks are now dominated by mega-cap technology, semiconductors and communication services, the very cohort the AI capex cycle has re-rated, while financials, healthcare and industrials make up the balance. Private equity's deployment skews the other way, with software and technology being the single largest buyout sector for most of the past decade, peaking at roughly a third of deal value in the 2020-2021 vintages, with healthcare, business services and industrials filling out most of the remainder.

The catch is that the software the public market rewards and the software private equity owns are not the same exposure. Listed technology leadership is concentrated in hyperscale compute, AI infrastructure and platform businesses with net-cash balance sheets; the typical Private Equity-owned software asset is a mature, leverage-financed vertical or horizontal SaaS business bought on a high revenue multiple, precisely the cohort most exposed to slowing seat growth, AI-driven displacement risk and a higher discount rate. So even where the two universes appear to overlap on the word "software," they sit on opposite sides of the AI trade: public markets long the enabler, private portfolios long the potentially disrupted incumbent (Figure 13-15).

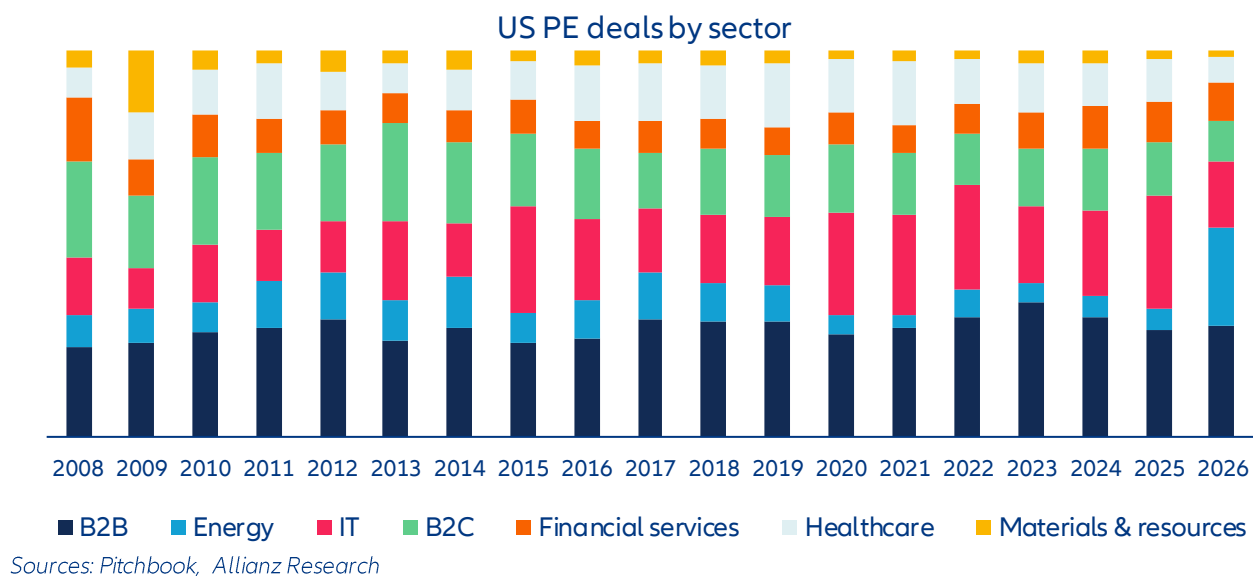
Figure 13-15: US Public vs Private EQ sector allocation (in %)



Sources: Bloomberg, Allianz Research



Sources: Pitchbook, Allianz Research



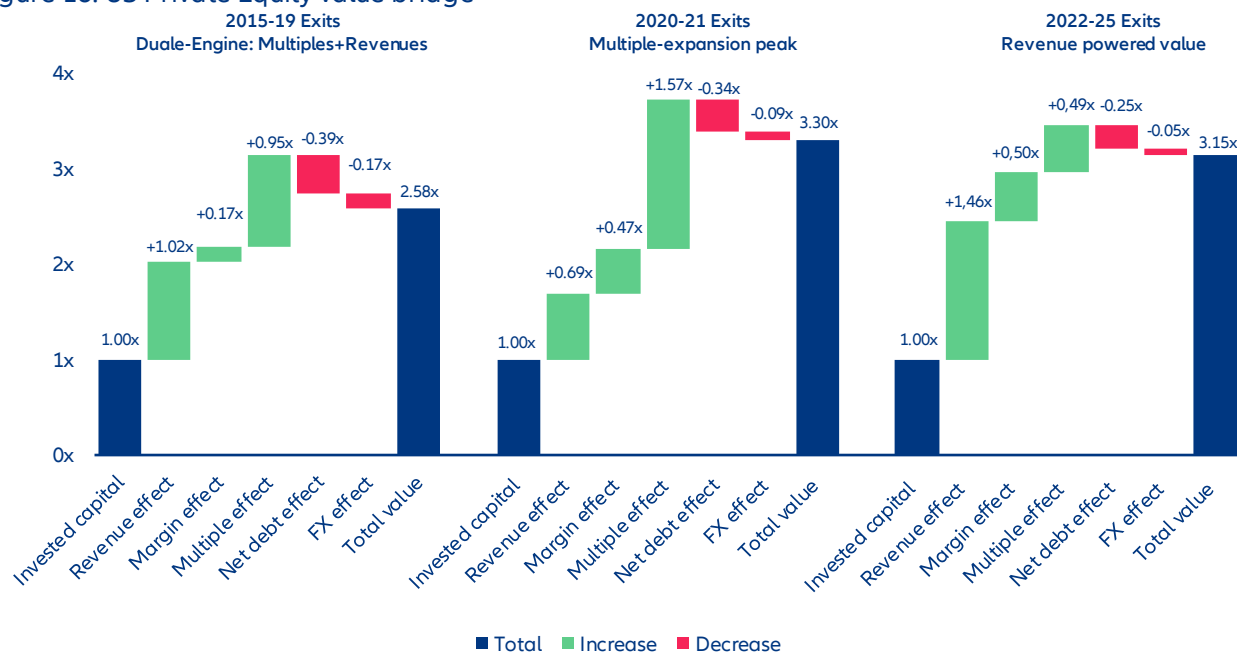
The Private Equity engine has changed, not just stalled

What ultimately separates public and private equity today is not how much they have returned, but what is driving those returns. Strip a buyout down to its parts and the value created comes from three places: the business growing its earnings, the multiple on those earnings rising between purchase and sale and the debt used to finance the deal. For most of the past decade, one of those did the heavy lifting. It was not, as is often assumed, leverage; across successive cohorts of exited deals, debt has been close to neutral and frequently a quiet drag once its rising cost is counted. The real engine was the multiple. Sponsors bought well, watched valuations climb and sold into a market that reliably paid more than they had, with growth in the underlying business playing only a supporting role.

That engine has stalled. Purchase multiples sit at or near record highs, so there is little room left for further re-rating, and exit multiples are no longer drifting up to reward patient owners. With cheap leverage gone and the multiple tapped out, only one source of value is left standing: the business itself. Revenue growth, once the junior partner, is now the dominant contributor to private equity returns, and the deals that work are the ones where the company genuinely grows rather than the ones where the multiple obligingly expands.

This is exactly where the two universes part ways. Public market returns are still, overwhelmingly, a re-rating story, but a narrow one, concentrated in a handful of net-cash mega caps that the AI trade has repriced upward. Private markets have been denied that same rerating, and the businesses sponsors own carry debt rather than cash, so their returns now have to be earned the hard way, one point of revenue growth at a time. Put simply, public equity is still being lifted by the market; private equity increasingly has to build its returns from the ground up. The bar has risen accordingly: only operationally well-versed managers in genuinely growing sectors can clear it (Figure 16).

Figure 16: US Private Equity value bridge



Data as of Q2 2025. Pooled value bridge across a subset of exited buyout holdings, weighted by invested capital.
Source: MSCI Private Capital Universe, MSCI Private Asset and Deal Metrics data

The exit market is where the Private Equity cycle's hangover shows most clearly. Buyout funds hold tens of thousands of unsold portfolio companies worth trillions of dollars, and median holding periods have stretched markedly. The starkest symptom is liquidity, the cash investors actually get back. Unlike a public equity, which can be sold on any trading day, a private holding returns capital only when the fund exits a deal, and that channel has narrowed sharply. Distributions to LPs (limited partners, the investors who fund the buyout) stayed below the 20% mark for a fourth straight year, the longest such stretch on record, and a depth last seen during the 2008-09 crisis. After three years of investors paying in more than they took out, net cash flow edged only modestly above breakeven in 2025, a change in direction rather than a true recovery.

The other exit routes tell the same uneven story. Deal and exit values rose sharply from a low base in 2025, but the rebound was narrow, driven by a handful of mega-deals while the mid-market stayed constrained. The IPO window has cracked open, but aftermarket pricing suggests public investors are still cautious on sponsor owned businesses, with Private Equity-backed listings tending to trade less buoyantly than venture backed ones. The secondary market has absorbed part of the gap, transacting a record volume in 2025 at a blended 87% of NAV (net asset value, the carrying value of the fund's holdings), with continuation vehicles taking a meaningful share, but it clears only a small slice of the unsold inventory each year. The plumbing is working again; the backlog behind it will probably take years to drain (Figure 17).

Figure 17: TTM US Private Equity buyout fund distributions as a share of beginning NAV



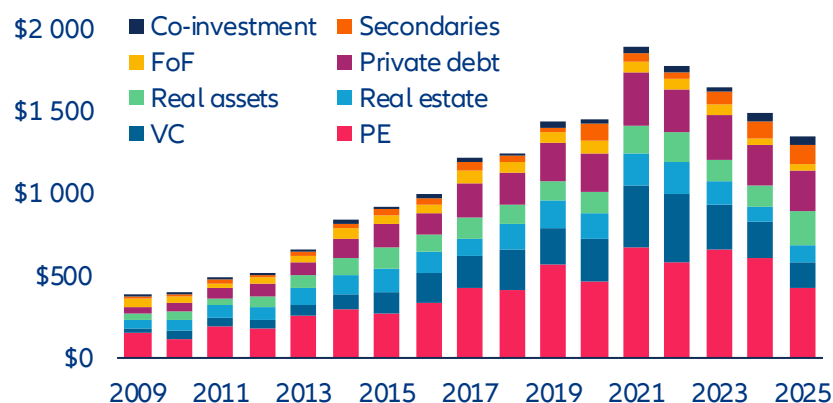
Source: Pitchbook, Allianz Research (Data as of Q3 2025)

Note: TTM = trailing twelve months. The line shows cash distributions paid to investors over the prior twelve months as a share of the fund's beginning net asset value (NAV); lower bars mean less cash is being returned.

Private credit has absorbed what private equity gave up

Private credit deserves its own section because it has been the structural beneficiary of the very forces that have pressured private equity. The same rate reset that raised LBO yields and made buyout math harder turned lending into the more attractive side of the trade, and as banks retreated from leveraged lending under regulatory pressure and sponsors still needed acquisition finance, direct lending stepped into the gap at unprecedented scale. Global direct-lending AUM has more than quadrupled since 2018 and is on track to keep growing through 2028. All-in yields on senior unitranche loans to mid-market, sponsor-backed borrowers remain in the low double digits, equity-like returns earned from a senior, secured position in the capital structure rather than the bottom of it. That combination is exactly what private equity has failed to deliver over the past three years: For long-duration insurers and pension funds, private credit has provided contractual cash yield, low mark-to-market volatility and a credit-loss experience that has stayed below long-run averages despite higher rates (Figure 18).

Figure 18: Private capital raised (USD bn) by strategy

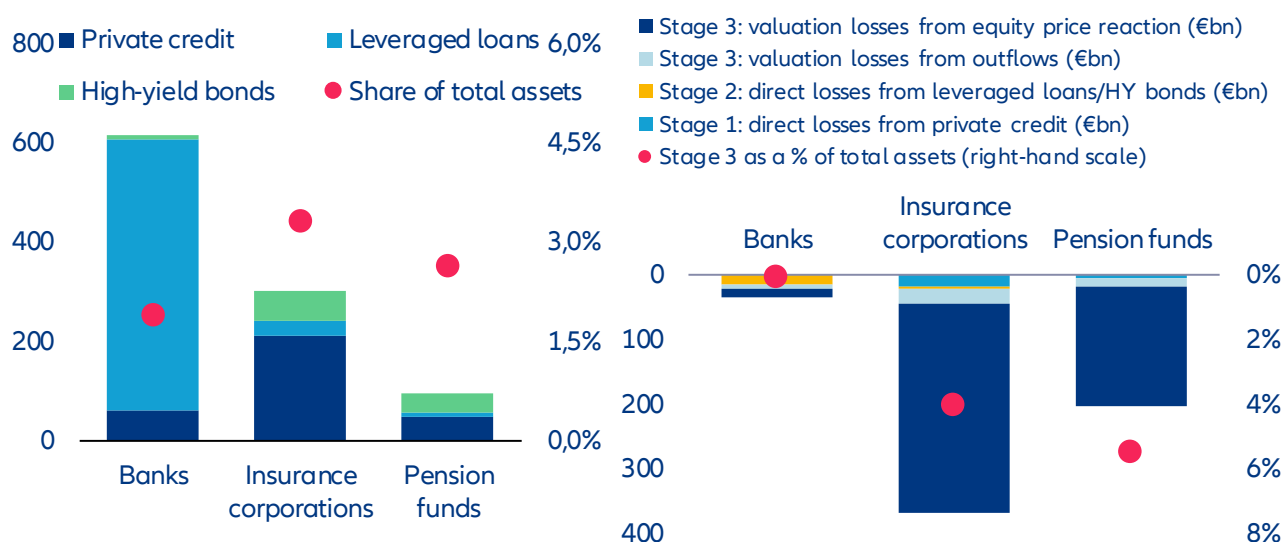


Sources: Pitchbook, Allianz Research

Two cautions are worth registering. First, private credit's default experience has been benign, with realized losses well below long-run averages, but the asset class has not yet been tested through a full credit cycle at current scale. The borrower base is also more levered than at any prior measurement point, and payment-in-kind (PIK) structures, in which coupons accrue rather than pay in cash, have grown sharply, suggesting some borrower-level stress that is being deferred rather than recognized. Second, private credit is increasingly intertwined with the insurance

balance sheets that own much of the asset class. The IMF and the ECB have flagged this as a systemic concern; when liquidity pressures emerge, insurance regulators may impose forced de-risking, creating sudden price-discovery events in a market built on stable, marked-to-model valuations. Neither concern is a near-term catalyst. Both argue for portfolio construction that favors senior secured exposure, granular underwriting and clear pricing discipline over chasing the highest-yielding mezzanine and second-lien strategies. (Figure 19-20)

Figure 19-20: Direct exposures of Eurozone entities to private credit and leveraged debt markets (left); Simulated losses for Eurozone financial entities from broad defaults in private credit and a severe shock in the software sector (right)



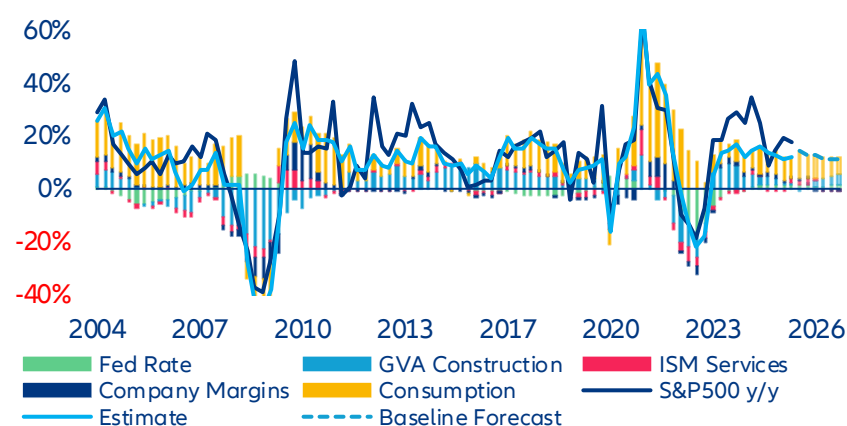
Sources: ECB, Allianz Research

The broader implication for private credit is that the asset class should be treated as a core, structural holding rather than a tactical reach for yield. The case does not depend on rates staying high. Even as financing costs normalize, the bank retreat from leveraged lending looks structural, the borrower base is largely captive and the asset's contractual cash yield and low mark to market volatility answer precisely the liquidity problem that has made private equity so hard to hold. For insurers and pension funds matching long-dated liabilities, that makes private credit a natural anchor allocation rather than a trade. But "core" tends to mean senior, not adventurous. The allocation belongs at the top of the capital structure, in senior secured exposure run by managers who can underwrite through a full cycle. Mezzanine, second-lien and opportunistic strategies still earn their place as return enhancers, but they are best treated more tactically, sized around that senior core rather than forming it. The suggested posture, in short, is a larger and more permanent allocation held more conservatively.

Outlook 2026: three drivers, three scenarios

The next 12 months will decide whether the 2022-2025 divergence between public and private market returns was a cyclical extreme or the start of a longer regime change. The central case assumes a prolonged but contained Middle East conflict, the Federal Reserve maintaining high rates and the AI capex cycle sustaining without a major guide down, a configuration that keeps returns positive across asset classes but well below the peaks of the post-Covid recovery. Under those conditions, public equity returns would be moderate into the low teens. Consumption and construction sentiment, the two biggest supports for the S&P 500 since 2010, hold up, while higher for longer rates shift from neutral to a modest drag. Private equity follows with a lag, with buyout returns settling in the mid-teens through 2027-2028 as financing conditions stabilize and the income flows that drive most of the variation in buyout returns recover from their 2023-2024 trough (Figure 21).

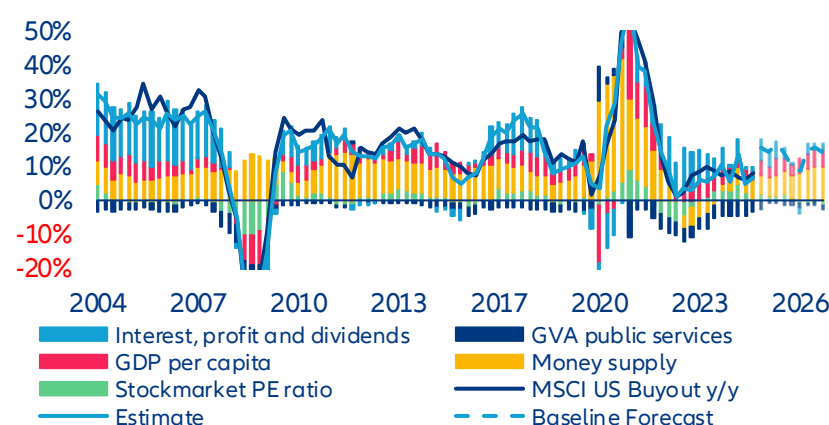
Figure 21: S&P 500 return decomposition and forecast (y/y%)



Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Figure 22: MSCI US Private Equity buyout return decomposition and forecast (y/y%)

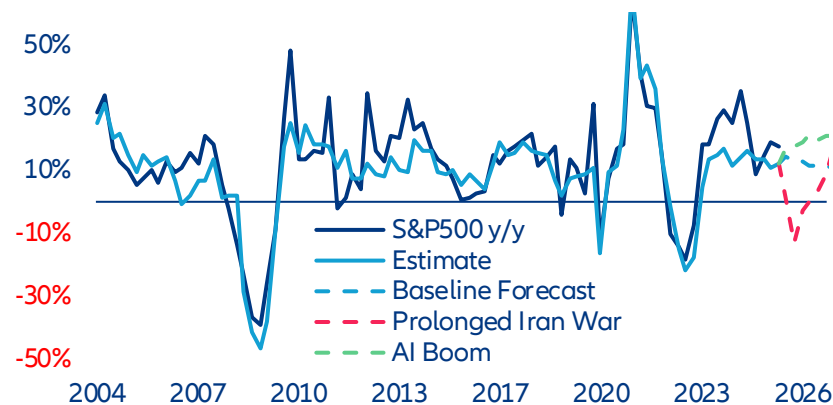


Sources: LSEG Datastream, Oxford Economics, MSCI, Allianz Research

Three drivers will decide which scenario plays out, each operating on a different horizon and through a distinct channel. In the near term, the resilience of the Middle East deal is the dominant variable. If oil and gas trade continue to normalize through the negotiated settlement, the path for both asset classes would look very different than if disruption to the Strait of Hormuz reignites beyond September. In that second case, a forced Fed tightening would turn rates from a modest drag into an active headwind for public equity. In this regard, consumption which is itself sensitive to rates through mortgage costs and credit availability, and the US consumer would enter any oil driven tightening from an already fragile position. The shock would also hit the real economy directly, with construction activity stalling as input cost visibility would collapse and commodity pass through would erode real

incomes. Under that path, public equity would turn negative in 2026, with buyouts following one to two quarters later as valuations reset with their characteristic lag (Figure 23).

Figure 23: S&P 500 scenario analysis (y/y%)

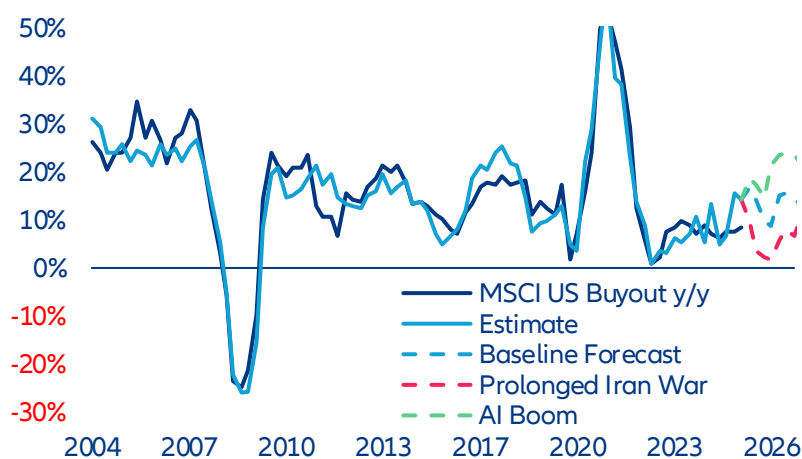


Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Over the medium term, the path of rates and the sustainability of AI capex are likely to become the binding constraints, and their interaction will determine whether the baseline recovery holds or extends. A higher for longer rate environment would delay the recovery of buyout profits, holding the downside path well below the cost of illiquidity in 2026. If AI capex were to guide down, whether through enterprise revenue disappointment, regulatory friction or balance sheet discipline, large cap tech margins would be hit first, and operating leverage means the damage would reach returns faster than the analyst revision cycle would suggest. From there it would propagate into buyout exit valuations. At that point a 10 to 15% public market de-rating is the single factor most capable of turning a baseline buyout year into the 4% downside outcome (Figure 24).

Figure 24: US Private Equity buyout scenario analysis (y/y%)



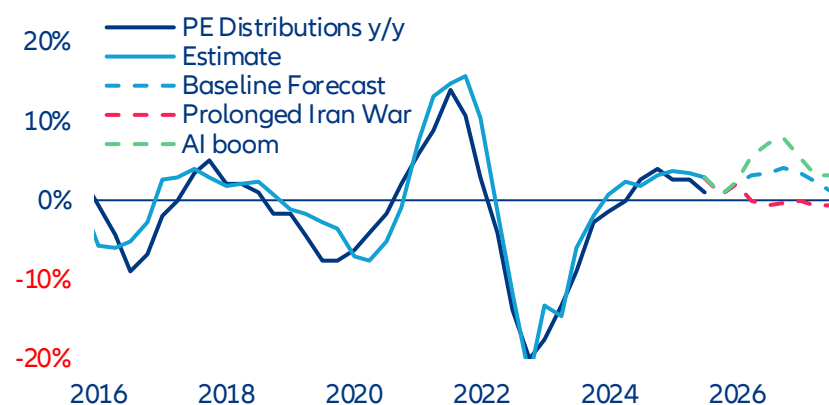
Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

The upside case requires both the near- and medium-term constraints to ease together, and when they do, the dynamics are mutually reinforcing in a way that produces outsized returns. In this scenario, with the Middle East conflict de-escalating after the June 2026 agreement, the Fed gains leeway and AI investment accelerates toward the 1990s capex cycle, with digital technology spending and business investment rising sharply. In that configuration the outcomes are assertive. Public equity peaks above 20% in 2027 as the consumption, margin and Fed rate drivers align at once, historically the setup that produces the sharpest single year public equity returns.

Buyout surpasses 20% shortly after as multiple expansion feeds through into exit valuations and money supply conditions ease, with the equity market PE ratio and the interest and profit rate drivers all contributing at the same time. Europe participates, but structurally less so. With the ECB's more gradual easing delivering less impulse than the Fed, Eurozone growth stays subdued and the AI preparedness gap, lower adoption capacity and smaller hyperscaler exposure mean the productivity gains accrue disproportionately to the US, compounding the transatlantic divergence of the post-2020 period.

The central message of the downside is not the magnitude of the return shock but its asymmetric transmission across asset classes. In the downside scenario, public equity would reprice immediately and sharply as margins and consumption deteriorate together under the stagflationary mix of input cost pressure and slowing growth, then would recover gradually as base effects and eventual policy normalization rebuild the factor stack. Private equity NAVs would adjust with a two to three quarter lag, technically positive in 2026 but close to the cost of illiquidity. The losses would be more contained in return space yet materially worse in liquidity terms, and nowhere is that gap more visible than in distributions. Historically, borrowing spreads and the CAPE ratio drive distribution activity, with rates the most persistent drag since 2022 and equity issuance, a proxy for IPO activity, the cyclical amplifier that has been conspicuously absent through the drought. The mega-IPO pipeline is where that factor becomes concrete. Successful mega-IPOs at premium multiples would lift the equity issuance driver, validate private AI marks across the buyout book and reopen the exit channel for sponsors holding inventory, converting years of paper gains into LP cash on a scale that dwarfs any prior exit event. Nonetheless, a collapse of the June agreement and renewed Iran escalation would break the sequence entirely, with spreads widening, the CAPE ratio deteriorating, rates staying higher and the IPO window closing before the queue clears (Figure 25).

Figure 25: US Private Equity distributions y/y abs change



Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

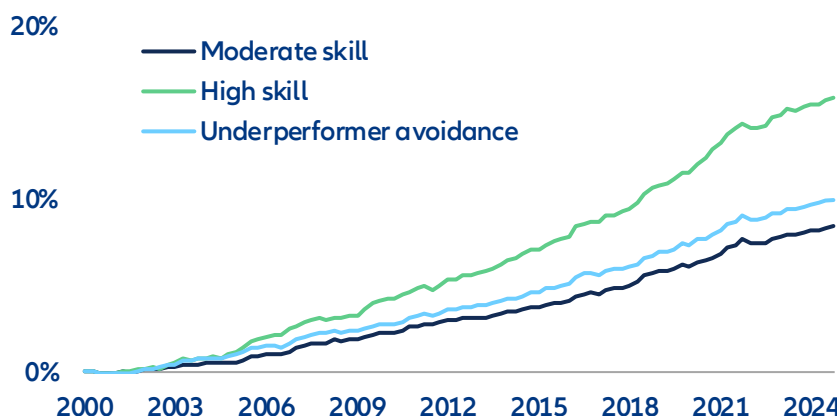
Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

From beta to selection on both sides of the divide

All in all, the story is, at heart, a story about the fading power of beta. For most of the last two decades, simply being in the room was enough, public markets rode a long bull run and private equity compounded on cheap leverage and steadily expanding multiples. That era has closed on both sides of the divide. Public benchmarks now carry their returns on a narrow cohort of AI-exposed mega caps, while private equity is still working through vintages bought at the top of the cycle. What used to be a rising-tide question, how much to allocate to private markets, has become a question of which managers, which vintages, and which exposures. The center of gravity has shifted from beta to selection.

That shift is not a matter of conviction but of arithmetic. The dispersion between the best and worst private equity managers has widened to the point where the median fund tells you very little, and the returns that justify the asset class increasingly sit in its upper reaches. Skilled selection, on the evidence, is now worth more than the allocation decision itself, but it is a genuine skill, not a default setting, and an allocator without a durable edge is as likely to give that premium back as to capture it. Beta has become a commodity, the scarce input is judgement (Figure 26).

Figure 26: Average Private Equity buyout cumulative total portfolio alpha



Sources: Pitchbook, Allianz Research

For investors looking forward, that reframes the central question. The debate worth having is no longer "public or private," but "differentiated or undifferentiated" within each, because passive index beta and generic, broad-exposure buyout have quietly become the same crowded trade, priced for a tailwind that is no longer blowing. The democratization of private markets, through evergreen and semi-liquid wrappers, only sharpens the point as it widens access to an asset class precisely as access stops being the thing that pays, and the funds that actually generate the premium remain the hardest to reach. The way forward is less about how much to own and more about how to own it. Investors should favor managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed, and resisting the temptation to buy the label without the underlying skill. The next market correction will be the first real test of which private market structures were built to endure and which were built to sell, and the distinction between robust and fragile will become visible quickly. The alpha, on both sides of the liquidity divide, now belongs to those with the conviction and the capability to select it.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends, (v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures, and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein, save for any information required to be disclosed by law.

Allianz Research | 17 June 2026

Southern Europe won the last decade. The next one is less obvious

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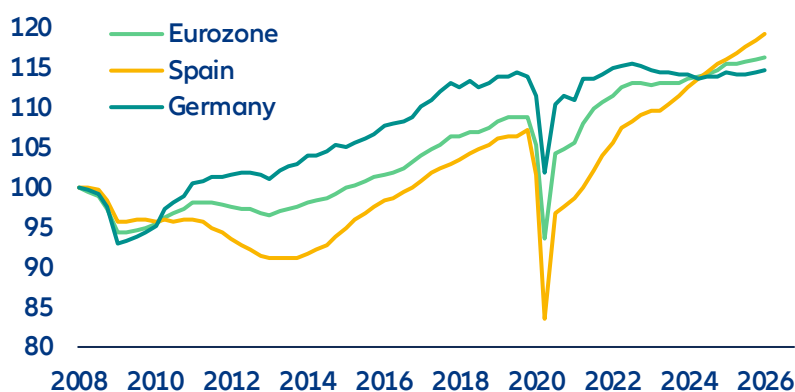
In Summary

- **Southern Europe has delivered an impressive turnaround this past decade, broader and more durable than many expected.** Spain, Portugal and Greece are running ~11% above pre-pandemic output, while Germany has stagnated for four consecutive years. Sovereign spreads that peaked above 250bps during the debt crisis have compressed to around 70. Southern banking systems, once the epicenter of Eurozone fragmentation risk, have largely completed their NPL clean-up (it is German and French banks that are now carrying rising asset quality pressures). This is not a cyclical blip but reflects an earned shift through labor market reforms, fiscal consolidation and the NGEU investment impulse – the largest coordinated public investment program the Eurozone has ever deployed. Spain's renewable buildout – now supplying more than half of its electricity – has added a further dimension: an industrial energy cost advantage that did not exist before 2021 and that has no equivalent in the North. The reversal is real but increasingly reflected in market pricing.
- **The growth model has delivered output convergence, but productivity – the engine of durable income catch-up – remains the unfinished business.** Employment-led recovery has run its course as the primary driver; Greece remains below its 2008 per capita income level. Sustaining the narrative through the next phase requires a more difficult set of structural changes: judicial efficiency, capital market deepening, R&D intensity and a completed Banking Union. The NGEU disbursements ending in 2026 sharpen that test: what phases out is not just a fiscal impulse but a reform discipline mechanism, and where implementation has been compliance-driven rather than institutionally embedded, the risk of slippage rises. The fiscal impulse is fading, with no clear replacement, and private cross-border capital flows remain too limited and episodic to fill that gap organically – the handover from public investment anchor to market finance is assumed but not yet visible in the data. These are not reasons to dismiss the story – they are the conditions that would extend it.
- **The political calendar and spread geometry call for selectivity.** France, Italy, Spain and Greece all face national elections in 2027, creating fiscal pressure points at the precise moment external conditionality fades and the ECB continues quantitative tightening. At around 70bps, BTP and Bonos spreads leave limited room for positive surprises and some room for temporary widening if credibility wavers. This is not a crisis configuration – sovereign balance sheets are in materially better shape than in 2011 and the ECB's Transmission Protection Instrument (TPI) provides a credible backstop – but it is a configuration in which episodic volatility is more likely than further compression. The region's reform record since 2012 is real and the underlying story has not exhausted itself. What comes next is more differentiated: returns will accrue to investors who discriminate by country, sector and reform trajectory rather than treating Southern Europe as a monolithic allocation.

The convergence scorecard: the wreckage of the 2010s and the post-pandemic catch-up

The Eurozone's defining fault line of the 2010s – the North–South divide – has materially narrowed over the past five years. The 2008 financial crisis and the subsequent Eurozone sovereign debt crisis produced a split of historic proportions. Between 2008 and 2013, Spanish GDP fell by roughly -9%, unemployment peaked at 28% and youth unemployment exceeded 55%. Greece suffered the most severe peacetime output contraction in a developed economy in the post-war era. Italy and Portugal followed trajectories nearly as damaging, albeit with somewhat better access to external support. In the years that followed, Mediterranean economies experienced only a sluggish recovery, while the core bloc (Germany, France, the Netherlands and Belgium) continued to expand, led by Germany's export competitiveness and manufacturing rebound, powered by surging global demand for automobiles and machinery in emerging markets.

Figure 1: GDP 2008=100



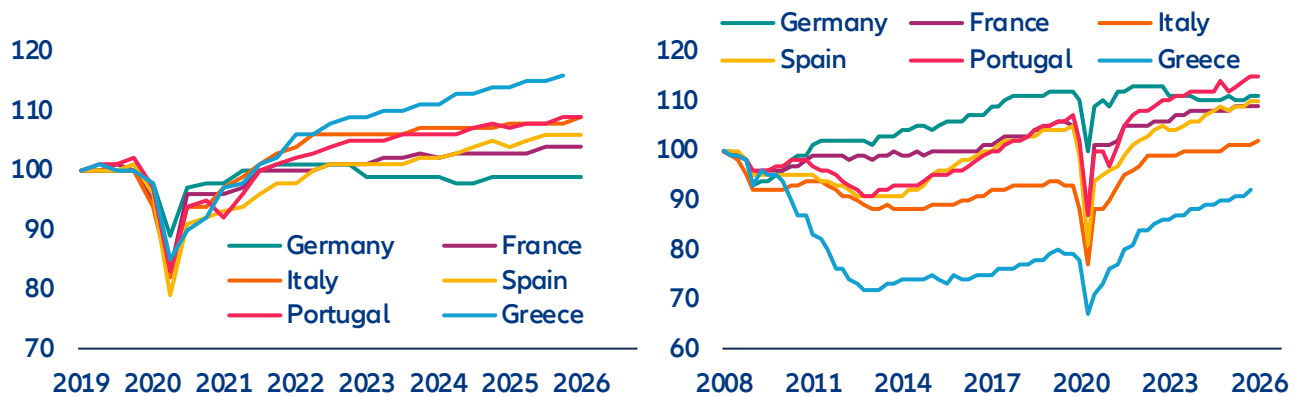
Sources: LSEG Datastream, Allianz Research

The post-2021 recovery marked a clear shift in the geography of Eurozone growth, with Southern economies moving from chronic underperformance to cyclical leadership. This shift reflects three reinforcing mechanisms. First, reforms implemented in the aftermath of the sovereign debt crisis – often under EU and IMF conditionality – improved competitiveness, labor market flexibility and external adjustment, laying the foundations for stronger employment and export performance. Second, the recovery has been strongly sector-biased, with contact-intensive services – particularly tourism – benefiting disproportionately. In Southern economies, tourism typically contributes between 10% (in Italy) and 20% of GDP (in Greece and Portugal) once indirect and induced effects are included. Third, the deployment of Next Generation EU (NGEU) funds supported public investment and credit conditions, amplifying domestic demand and accelerating structural upgrading, including through infrastructure and digitalization.

At the same time, Germany's growth model, built on cheap energy inputs and external demand, has been structurally disrupted. The loss of Russian gas, combined with weaker demand from China, has exposed vulnerabilities linked to underinvestment and slower technological adaptation, contributing to prolonged stagnation between 2022 and 2025. The resulting convergence combines catch-up and compression – a critical distinction. The narrowing of income gaps reflects not only structural improvement in the periphery, but also a rebalancing of relative growth across the Eurozone.

However, this convergence remains incomplete when assessed through a structural lens. While output growth has converged, convergence in GDP per capita – the more relevant measure of welfare – has been significantly more limited. Despite strong post-pandemic performance, longer-run comparisons point to persistent scarring effects from the sovereign debt crisis. Greece remained below 2008 per capita income levels in 2025, Spain has broadly closed the gap, while Portugal has exceeded it, reflecting sustained reform implementation and a strong tourism-led rebound.

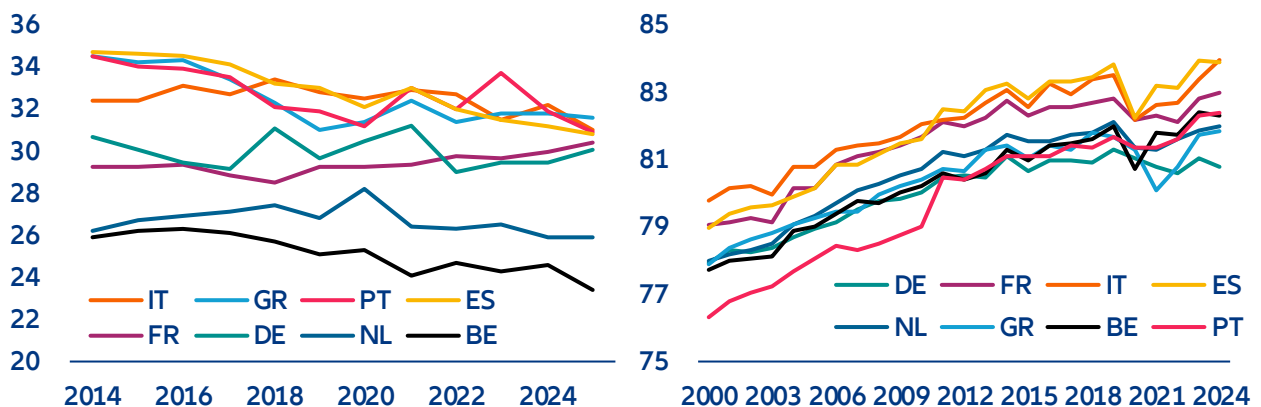
Figure 2: GDP per capita, 2019=100 and 2008=100



Sources: LSEG Datastream, Allianz Research

This divergence reflects the nature of the current growth model. Since 2021, growth has been predominantly employment-driven rather than productivity-driven. This has delivered rapid gains in output, but only partial convergence in income levels. The human dimension of the crisis decade remains visible. Southern Europe leads the EU in life expectancy yet carries persistently higher income inequality than its northern peers (Gini coefficients of 32–34 versus 25–30 in core economies), with recent gains that have not reached the bottom of the distribution. The current phase therefore risks resembling “convergence by slowdown” in the core as much as genuine structural catch-up in the periphery.

Figure 3: Gini coefficient and life expectancy



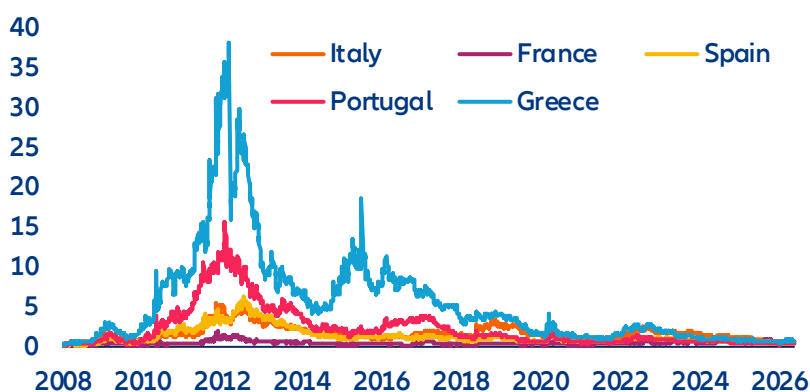
Sources: LSEG Datastream, Allianz Research

The capital market scorecard

Financial markets have broadly confirmed the convergence narrative. Italian 10-year spreads compressed from a peak of 250bps in 2022 to an average around 70bps in early 2026; Greek government bond spreads consistently trade tighter than Italy. EGB spreads are nowadays broadly back to the levels of 2008. NGEU has also been a key driver of this compression, with its combined impact on growth, fiscal balances and policy credibility accounting for a significant share of the observed convergence¹. Equity markets echo this: Italian, Spanish and Greek indices have all outpaced the Stoxx Europe 600 since 2022, with gains driven by value rotation and banking rehabilitation rather than a fundamental re-rating. Measured from the sovereign crisis period (2008=100), Southern European markets experienced a prolonged phase of underperformance followed by a gradual recovery.

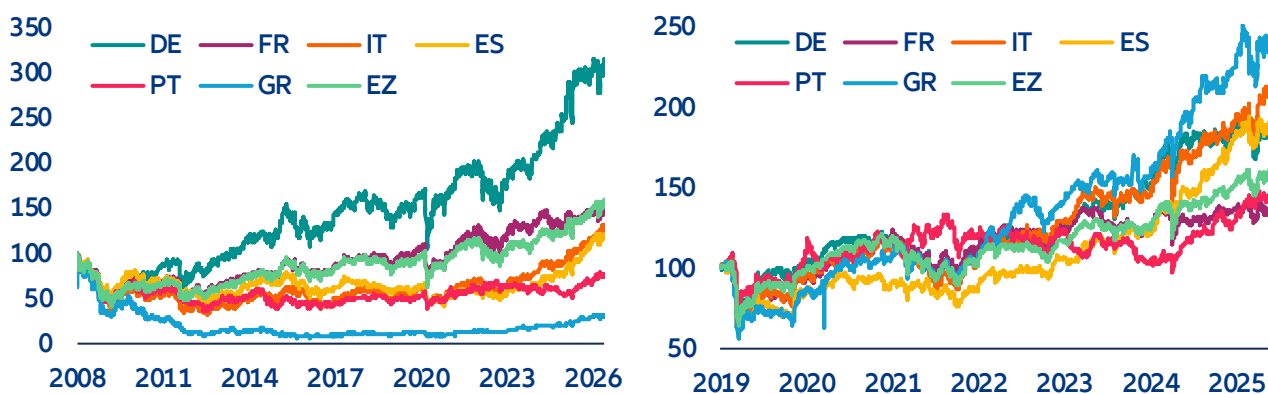
¹ [What to watch | November 14, 2025 | Allianz](#)

Figure 4: 10y government bond spreads over Bund



Sources: LSEG Datastream, Allianz Research

Figure 5: Equity index (2008=100, 31/12/2019=100)

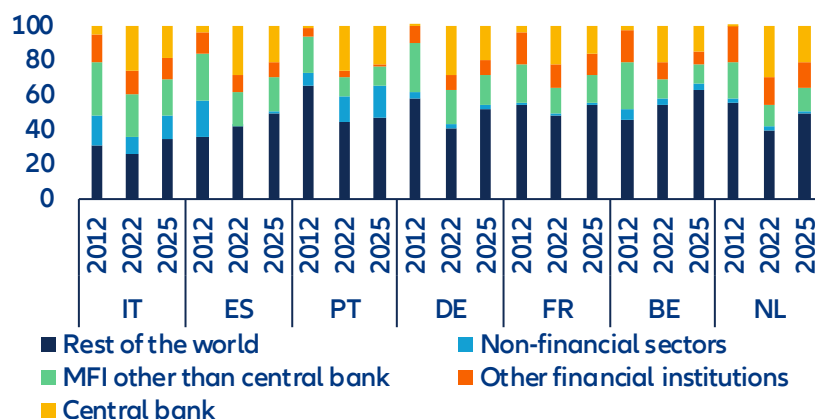


Sources: LSEG Datastream, Allianz Research

A substantial share of this re-rating reflects a catch-up from deeply discounted levels rather than a transition to premium valuation. The initial phase of financial convergence – the compression of crisis-era risk premia – has largely played out. Whether it becomes durable will depend on the transition to real convergence through productivity gains. Future performance will depend less on multiple expansion and more on the underlying drivers of earnings growth.

One important structural caveat: the ECB's gradual withdrawal as price-setter via QT means marginal sovereign funding increasingly depends on hedge funds and active managers with shorter horizons, making spreads more reactive to fiscal signals than during the QE era. For Italy this matters most – retail absorption capacity is approaching its limits, and marginal funding depends increasingly on investors with firmer fiscal conditions and shorter time horizons. The fact that spreads have compressed despite this shift is a testament to the improvement in underlying fundamentals elsewhere in the South; for Italy, it is a risk that remains latent.

Figure 6: Government debt holdings by investor type



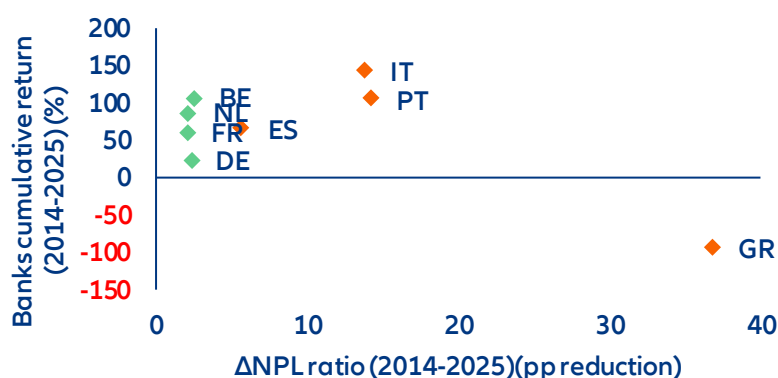
Sources: LSEG Datastream, Allianz Research

Banking sector: the structural inversion markets have not fully priced in

Banking sector repair is arguably the most underappreciated development in European finance over the past decade. A decade ago, Southern banks were the epicenter of Eurozone fragmentation risk: non-performing loan (NPL) ratios stood at 17% in Italy, 18% in Portugal and close to 50% in Greece. The subsequent clean-up – driven by ECB supervisory pressure under the Single Supervisory Mechanism, stricter provisioning and active NPL disposals – was slow and painful, but is now largely complete. By Q3 2025, all four Southern banking systems continue to reduce NPL volumes: Spain currently stands at 2.4%, Italy -2.7%, Portugal 2.6% and Greece 2.8%.

What makes this development particularly relevant is the direction of travel elsewhere. Germany and France are now experiencing rising NPL volumes, driven by commercial real estate and corporate sector stress. A decade after the sovereign debt crisis, it is core European banks – not Southern ones – that are carrying the asset quality problem. This inversion is structural, not cyclical, and is not yet fully reflected in prevailing narratives about fragmentation risk in the Eurozone.

Figure 7: NPL ratio change (%) vs Banking Equity Index Cumulative Return (2014 Q3-2025Q4)



Sources: LSEG Datastream, Allianz Research

Improved asset quality also has direct macro-financial implications. The sharp reduction in NPL stocks freed up capital, reduced provisioning needs and strengthened profitability, increasing banks' capacity to extend credit to the private sector. In that sense, banking sector repair is not just a legacy issue but a precondition for the broader economic convergence, as it restored the transmission of monetary policy and supported investment and growth.

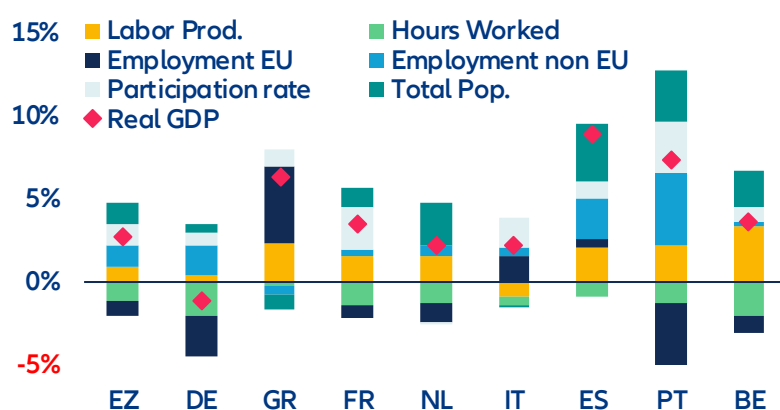
What did the South do right?

Labor market dynamics: the core driver of Southern resilience

Labor market performance is central to explaining the strength and persistence of Southern Europe's recovery. Since 2021, employment has not lagged output but has acted as the primary engine of expansion. Labor input – through both rising participation and higher employment rates – has accounted for the bulk of GDP growth, pointing to a distinctly labor-intensive recovery phase.

This dynamic reflects both cyclical demand and structural adjustment. Participation has increased across demographic groups, while migration has played a critical stabilizing role. Foreign workers accounted for more than half of labor force growth in recent years, easing labor shortages at a time when supply constraints were binding, particularly during the 2022 energy and supply shock. At the same time, earlier labor market reforms – especially in Southern Europe – have improved hiring dynamics, reduced rigidities and increased resilience to shocks.

Figure 8: Supply-side GDP decomposition, 2022-2025 cumulative growth

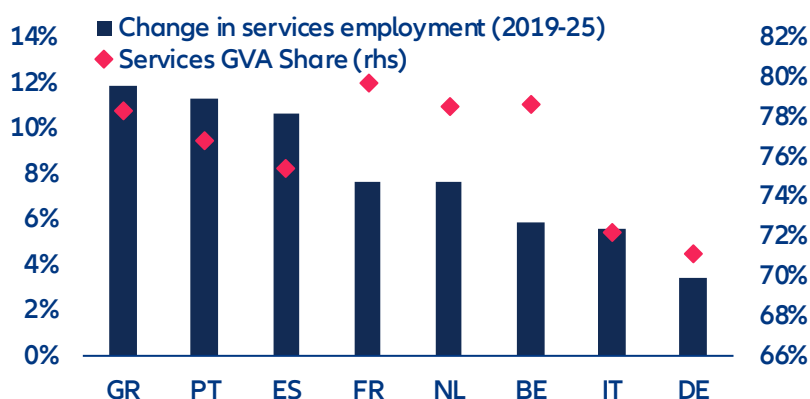


Sources: LSEG Datastream, Allianz Research

Spain provides the clearest example of a structural break. The 2021 labor market reform directly targeted the long-standing duality between temporary and permanent contracts. The impact is visible in the contractual structure: temporary employment declined by more than 10pps between Q4 2021 and Q4 2025, while hiring shifted decisively towards open-ended contracts. The share of fixed-term contracts – previously close to 90% of new hires – fell to below 60%, and youth temporary employment declined sharply, converging towards the EU average.

On the demand side, the sectoral composition of the recovery has amplified these effects. Strong services demand – particularly tourism – translated into high employment elasticity, allowing output growth to feed rapidly into job creation. Compared to the pre-crisis period, employment has responded more quickly and more proportionately to economic activity. Importantly, employment gains have been predominantly private-sector driven, pointing to robust underlying demand dynamics rather than administrative expansion.

Figure 9: Employment in services vs services GVA/ total GVA



Sources: LSEG Datastream, Allianz Research

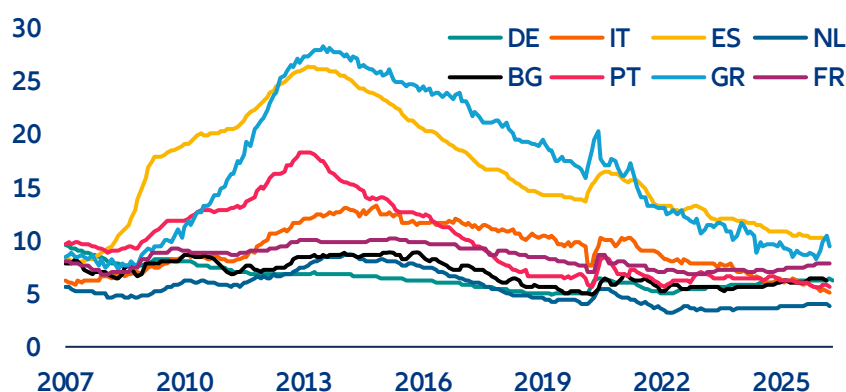
These shifts are not merely statistical. Greater contract stability alters firm incentives, making investment in training and skills more profitable, improving match quality and supporting a gradual shift towards higher-quality employment. At the same time, increased labor availability – particularly in high-demand sectors such as construction, hospitality and ICT – has allowed employment to expand without triggering proportional increases in unemployment or strong second-round inflation effects. Migration composition also matters: Non-EU workers are disproportionately represented in shortage sectors and often exhibit high participation rates.

At the aggregate Eurozone level, compositional changes have contributed to the decline in unemployment.

A larger share of older and more educated workers – groups with structurally lower unemployment rates – has mechanically compressed aggregate unemployment, while migration has expanded labor supply without raising joblessness.

These gains come with clear trade-offs. The recovery has been driven predominantly by the extensive margin, while average hours worked remain below pre-pandemic levels. This reflects both cyclical and structural factors, including the expansion of part-time employment, the shift towards services with structurally lower working hours and precautionary labor hoarding amid persistent shortages. As a result, labor input has increased mainly through employment gains rather than higher utilization per worker, dampening productivity growth and weakening the pass-through from output growth to aggregate income. Strong employment performance therefore partly masks an incomplete normalization of labor utilization. The implication is that output convergence has translated only partially into income convergence, as employment gains have not been matched by commensurate productivity growth. The key medium-term question is whether improvements in labor market structures – contract stability, participation and labor supply – will translate into a sustained reacceleration of productivity.

Figure 10: Unemployment rate (%)



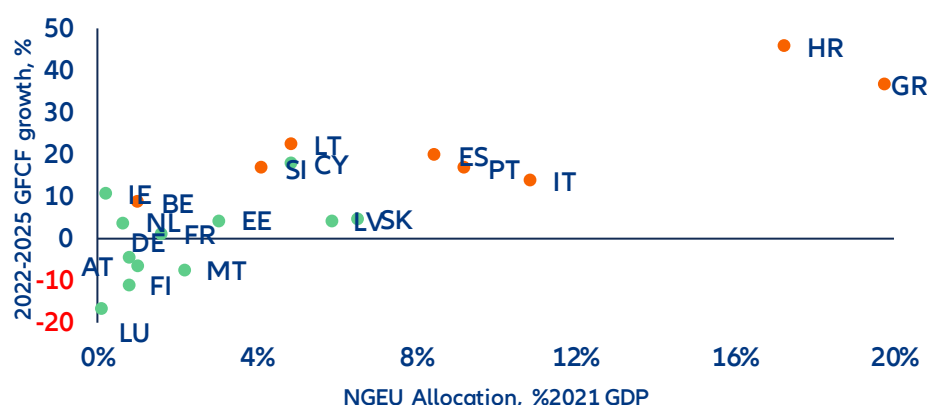
Sources: LSEG Datastream, Allianz Research

NGEU: front-loaded investment impulse with reform conditionality

The scale of the NGEU allocation to Southern Europe is easily underestimated. Greece received funds equivalent to nearly 20% of GDP, Italy close to 11%, Portugal 9.3% and Spain 8.5% – allocations that dwarfed those of core economies, with Germany receiving less than 1% of GDP and France just 1.6%. For economies where national fiscal space remained severely constrained, this represented a public investment impulse with no post-war precedent outside of crisis interventions.

The macroeconomic transmission operated through three distinct channels. First, the direct investment effect: Disbursements concentrated from 2022 onwards drove a visible recovery in public investment as a share of GDP across the South, reversing a decade of austerity-driven compression. Second, the fiscal accounting effect: Grants do not count as borrowing – they flow through the expenditure side without adding to national debt. For high-debt economies running primary surpluses, this was analytically significant: it allowed investment to expand while headline debt ratios continued to fall, contributing directly to the snowball dynamic and to the spread compression that followed. Third, the conditionality effect: By linking disbursements to predefined milestones, the NGEU transformed reform commitments into binding obligations, lowering the political cost of reform by externalizing part of the decision process. Spain's labor market reform, Portugal's digitalization agenda and Greece's administrative overhaul were all advanced under this framework.

Figure 11: NGEU allocation vs GFCF cumulative growth



Sources: Eurostat, Allianz Research. Orange dots represent country which have requested loans on top of grants.

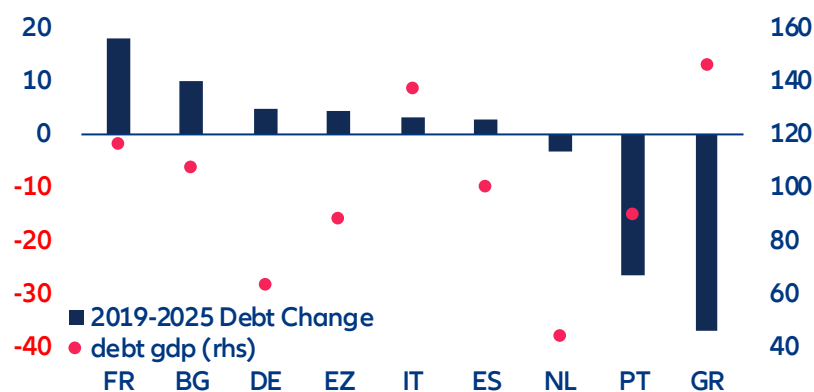
NGEU did not operate in isolation from fiscal consolidation – it amplified it. By flowing through the expenditure side without adding to national debt, grants improved headline balances and contributed directly to the debt dynamics that followed. The two pillars are complementary: NGEU created the fiscal room; Southern governments, for the most part, used it well.

Fiscal consolidation: the pillar that is cracking

Following the sovereign debt crisis, Southern economies implemented prolonged and politically costly consolidation strategies – cutting deficits, rebuilding primary balances and restoring institutional credibility at significant social cost. After the pandemic, they made the most of a favorable window: strong nominal growth and still-low effective interest rates combined to generate a powerful snowball effect, pushing debt ratios down materially. Portugal cut its debt ratio by over 40pps to below 90% of GDP since 2020. Spain delivered primary surpluses from 2022 onwards. Greece regained investment-grade status by 2025. Monetary policy set the lower bound for spreads across the board, but fiscal consolidation determined the extent of the re-rating.

Italy's picture is more nuanced. The return to a primary surplus in 2024 was a genuine improvement and should not be dismissed. However, the surplus has not been large enough to stabilize the debt ratio against a backdrop of weakening growth and past generous policy (Superbonus tax credit) – debt has continued to drift upward, leaving the fiscal position rather vulnerable. Italy's spread compression reflects ECB backstop credibility and a period of relative political stability as much as fundamentals, and it is the case that most warrants watching as financing conditions normalize and the ECB's role as marginal buyer fades.

Figure 12: Change in debt-to-GDP ratio



Sources: Eurostat, Allianz Research

Energy transition: from structural vulnerability to competitive advantage

The 2022 energy crisis triggered a structural shift in the distribution of energy exposure across the Eurozone – one that cuts against the traditional North–South framing. Southern economies – particularly Spain and Portugal – accelerated investment in renewable capacity, materially reducing their dependence on imported gas. Spain now sources more than 50% of its electricity from renewables, lowering its exposure to gas price volatility. On electricity, where the renewable build-out has been fastest, Southern Europe has moved from structurally higher-cost to lower-cost relative to the German industrial benchmark.

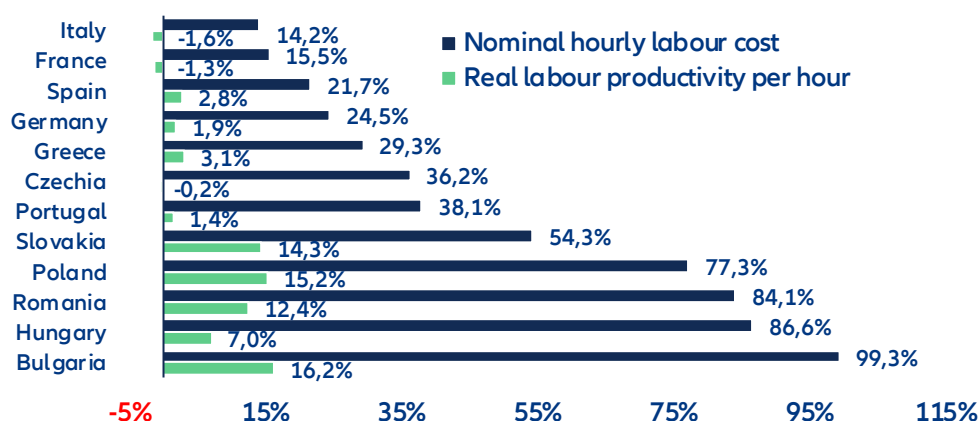
The implications extend beyond near-term growth. Energy has become a binding constraint for capital allocation in energy-intensive sectors, including digital infrastructure and AI. As a result, energy-intensive industries – chemicals, steel, aluminum and cement – are increasingly locating in energy-advantaged jurisdictions. Southern Europe, with its expanding renewable base and Atlantic LNG access, is becoming more competitive in attracting this reallocation.

However, the adjustment remains incomplete. Three residual exposures deserve explicit quantification. First, oil dependency: Southern economies remain net importers of crude, with transport fuels accounting for roughly 35–40% of final energy consumption. Second, EV adoption is uneven – the EV share of new car sales reached ~23% in Germany versus ~14% in Spain and ~8% in Italy in 2024 – leaving Southern transport sectors structurally more exposed to oil price volatility for at least the next decade. Third, energy poverty remains a distributional vulnerability: Approximately 13% of EU households in Southern member states report an inability to adequately heat their homes (Eurostat 2024), limiting the political durability of energy transition policies.

External tailwinds: the CEE rotation

Beyond the South gaining on the North, a broader reordering of comparative advantage is unfolding across the European economy. Some of the factors that previously powered Central and Eastern Europe's convergence are increasingly becoming sources of pressure, exactly as they are turning into relative advantages for Southern Europe. Since end-2019, nominal hourly labor costs rose by +99% in Bulgaria and +87% in Hungary, against productivity gains of just +16% and +7% respectively, while productivity in Czechia remained broadly flat despite strong wage growth (Figure x). The drivers are structural rather than cyclical. Labor markets in CEE have tightened to record levels, with unemployment at 3.1% in Poland and 2.8% in Czechia, while working-age populations have shrunk sharply since 2015 (Bulgaria –13.5%, Poland –11.3%, Romania –8.8%). The low-wage manufacturing model that anchored the region's integration into European supply chains is therefore becoming structurally harder to sustain as labor scarcity and rising costs erode part of the competitiveness advantage that originally underpinned it. At the same time, Southern Europe is benefiting from almost the opposite configuration: more moderate labor-cost pressures, stronger labor-market normalization and a more services-oriented growth structure supported by tourism and renewable investment.

Figure 13: Nominal labor cost and real productivity, cumulative % change since 2019



Sources: LSEG Datastream, Allianz Research

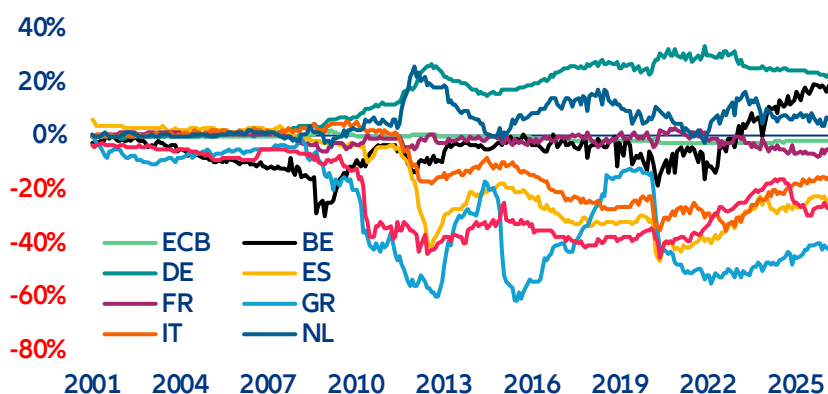
This divergence is reinforced by the changing geography of industrial competitiveness. Southern Europe's renewable build-out, together with Germany's loss of cheap Russian gas, has materially altered relative energy costs across Europe. In H2 2025, ex-tax electricity prices for medium-sized industrial users stood at EUR10.6 cents/kWh in Spain and EUR10.7 in Portugal, compared with EUR17.1 in Hungary, EUR16.0 in Slovakia, EUR15.6 in Romania and EUR14.6 in Czechia – a divergence of roughly 5–6 cents/kWh that did not exist before 2021. At the same time, CEE economies remain heavily concentrated in sectors tied to the weakening German industrial cycle, particularly autos. Motor vehicles account for 24.7% of manufacturing value added in Slovakia, 21.3% in Czechia, 15.8% in Hungary and 14.6% in Romania, all above Germany's own 19.5% and several multiples of Spain's 9.1% or Italy's 5.9%. The region is therefore simultaneously among the most exposed to high industrial energy costs and the most dependent on a German automotive model facing intensifying Chinese competition and weak external demand. Southern Europe, by contrast, carries neither exposure to the same extent. The result is a broader reordering of comparative advantage across Europe: Germany remains constrained by the structural weaknesses exposed after the energy shock, Southern Europe is increasingly gaining competitiveness in energy, services and investment attraction, while Central and Eastern Europe is confronting the limits of the very model that originally drove its convergence success.

The post 2026 cliff

2026 is the decisive test – political and institutional – as the external anchor fades. The NGEU is explicitly time-bound, with disbursements scheduled to end in 2026. Italy, Spain, Portugal and Greece together account for almost 60% of the total EU envelope, making the post-2026 transition a system-level test. The key question is whether convergence gains are becoming structurally embedded – in administrative practices, legislation and firm behavior – or remain largely compliance-driven. Where implementation remains compliance-driven, the risk of policy slippage increases sharply once the external anchor weakens.

This creates a dual risk. First, the withdrawal of policy support may expose underlying growth dynamics that remain heavily reliant on labor rather than productivity. Second, the loss of a binding reform framework could make policy slippage more likely, particularly in high-debt economies facing renewed fiscal constraints as financing conditions normalize. Convergence is currently policy-supported, not yet fully self-sustaining: Private cross-border capital flows remain limited and episodic. Capital flight has stopped, but it has not clearly reversed – Target2 imbalances have stabilized but have not structurally declined.

Figure 14: TARGET2 balances (% of nominal GDP)



Sources: LSEG Datastream, Allianz Research

The institutional gap: A completed Banking Union is the critical missing piece for the transition from public to private-led convergence. Without it, private cross-border flows remain too shallow to replace the NGEU impulse post-2026. Despite a common monetary policy, financial integration remains incomplete: Cross-border bank lending and portfolio allocation within the Eurozone still exhibit strong home bias, with domestic sovereign exposures dominating bank balance sheets. The absence of a European Deposit Insurance Scheme means risk-sharing remains incomplete: ECB backstops continue to substitute for missing institutional integration rather than operating within a fully unified financial system. Initiatives such as the Savings and Investments Union aim to deepen capital markets and reduce home bias, enabling more efficient cross-border allocation of savings, but the scale of private risk-sharing remains limited relative to the size of the Eurozone economy, and the transition from public to private investment as the primary driver of adjustment remains incomplete.

Political risk. The political sustainability of reform is not guaranteed. France, Italy, Spain and Greece all face national elections in 2027, putting fiscal discipline and reform commitment to a simultaneous political test across Eurozone economies. Populist movements in Italy and Greece remain a latent risk to reform continuity. Spain's political fragmentation – governing with a minority coalition dependent on regional nationalist parties – adds procedural uncertainty even where the policy direction is broadly reform-positive. Reform gains have not come without a social price: fiscal consolidation in the early phase involved material cuts to healthcare, education and social transfers that compressed living standards for the bottom half of the income distribution, and that political legacy has not fully faded.

From policy supported convergence to structural test. The first phase of Southern rebalancing – compressing crisis-era risk premia, rehabilitating banking systems, restoring fiscal credibility – has largely run its course, driven by a combination of domestic policy effort and EU-level architecture. From a macro perspective, the catch-up is real but unfinished: employment-led growth has delivered output convergence without closing the income gap, and the post-2026 withdrawal of the NGEU anchor will test how much of the improvement is structurally embedded. For markets, spreads at decade lows and equity valuations normalized leave little room for further re-rating on recovery alone: Outperformance now requires a differentiated view on which economies can convert institutional gains into productivity-led growth.

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Allianz Research | 16 June 2026

US-Iran Deal: Markets Price Peace, Economies Still Pay for War

In Summary

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- **Markets have celebrated the end of the Strait of Hormuz conflict. The economy has not yet earned that relief - and for most households, corporates, and governments, things may get worse before they get better. The physical reopening is a multi-month process, not a switch.** The US-Iran MoU and 60-day ceasefire mark a meaningful turning point, but they are a de-escalation framework, not a final settlement. Normalization will be slow, frictional, and reversible, and markets have frontloaded the good news before the transmission lags have even peaked. Mine-clearing alone takes 30 to 50 days. After that, commercial traffic returns only as fast as shipowner confidence allows - the 1988 Iran-Iraq precedent shows normal conditions took more than three months to restore even with US naval escorts. Our base case restores 65% of the disrupted 4-5 mb/d within three months and 80% within four, with full normalization by year-end. The corridor reopens under managed, not free, navigation. Any stumble in nuclear negotiations or ceasefire adherence on Lebanon restarts the clock.

- **Energy prices ease, but inflation peaks later than markets think.** Brent stabilizes around USD 80/bbl in Q3 before easing to USD 75/bbl in Q4 and USD 67/bbl by end-2027. Past energy increases are still passing through supply chains, utility bills, and rents. US headline CPI averages 3.3% in 2026 and core CPI peaks at 3.1% in Q4; Eurozone inflation peaks around 3.4% in Q4 before averaging 3.1% in 2026. Real wage growth turns positive only in Q1 2027. Markets pricing disinflation from today are running at least two quarters ahead of the data.

- **The shock hits every balance sheet - but Europe bears the deepest scar.** The US absorbs the blow with structural advantages: a net energy exporter, it captures a terms-of-trade windfall through higher mining revenues and fiscal receipts, while tax rebates cushion households and AI-driven investment sustains corporate capex. Europe has no such offset. Governments across the Eurozone have deployed just EUR12bn YTD - 0.1% of GDP - with Germany's defense and infrastructure stimulus the sole demand anchor. Households face a sharper and more persistent squeeze: energy bills represent a larger share of disposable income than in the US, floating-rate mortgage exposure is higher across the UK and Netherlands, consumer confidence has not recovered to pre-war levels, and real purchasing power will remain compressed through most of 2026. For corporates, the energy cost relief, most visible in transport and petrochemicals where fuel runs 25-40% of operating costs, runs directly into this demand void. Pricing power is the dividing line: airlines and branded pharma can defend margins through surcharges; automotive OEMs and generic drugmakers cannot. Input costs are easing but wage bills remain elevated into 2027, and listed-company cash has already fallen 3% to EUR 36.7trn. Any ceasefire relapse would recouple all three pressures simultaneously - reigniting input costs while households are still digesting the first wave and fiscal space is exhausted - with Europe more exposed throughout given higher energy intensity, greater trade openness, and no room left for transfers.

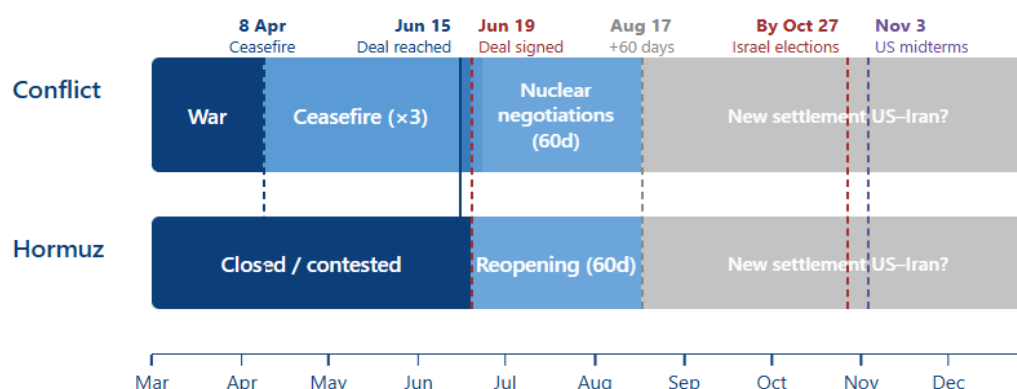
- **For central banks, the risk of a policy mistake is the highest since 2022.** Both the Fed and ECB are expected to deliver one further hike in H2 - the Fed likely in September as core CPI ticks above 3%, the ECB before year-end - before disinflation opens the door to cuts in H2 2027. Both face a time consistency problem as inflation pressure will fade quickly while their data-dependency on backward-looking measures continues to justify tightening well into Q4. Overtightening into an already-fragile household and corporate backdrop remains a plausible - and underpriced - scenario for both banks. But the US retains more growth momentum to digest further tightening, while the Eurozone's growth is already fragile.

- **For capital markets, the signal is rotation and carry, not re-rating.** Equities never priced the conflict as a systemic shock - MSCI US up 8%, MSCI Europe +5% YTD - so there is little upside to unlock on good news but significant room to gap lower on bad. Valuations sit in the 85th percentile with earnings in the 92nd percentile (MSCI World, 20-year history); a 5-10% pullback through summer is plausible. A tilt toward Europe, cyclicals, and energy-sensitive sectors is warranted over a broad melt-up. On credit, EUR IG at 77bps and US IG at 72bps are near all-time tights; the deal protects the carry trade rather than opening new compression. In rates, the short end and belly of European curves offer the most compelling outperformance (35bps front, 20bps belly), while long ends remain vulnerable to fiscal supply - bull steepening, not a parallel rally. In private markets, gains show up in activity - firmer exits, smoother refinancings - not in marks.
- **Relief is the call. It is not the all-clear.** Markets have frontloaded the peace dividend; the economic data has not confirmed it. Inflation will get worse before it gets better. Household and fiscal stress will persist well into 2026. And the deal remains conditional on nuclear negotiations and a multi-front ceasefire with a poor track record. The asymmetry is uncomfortable: upside on good news is modest; downside on bad is fast, disorderly, and correlated. The cost of being slow to react would far exceed the cost of staying cautious.

How does the deal impact our economic outlook?

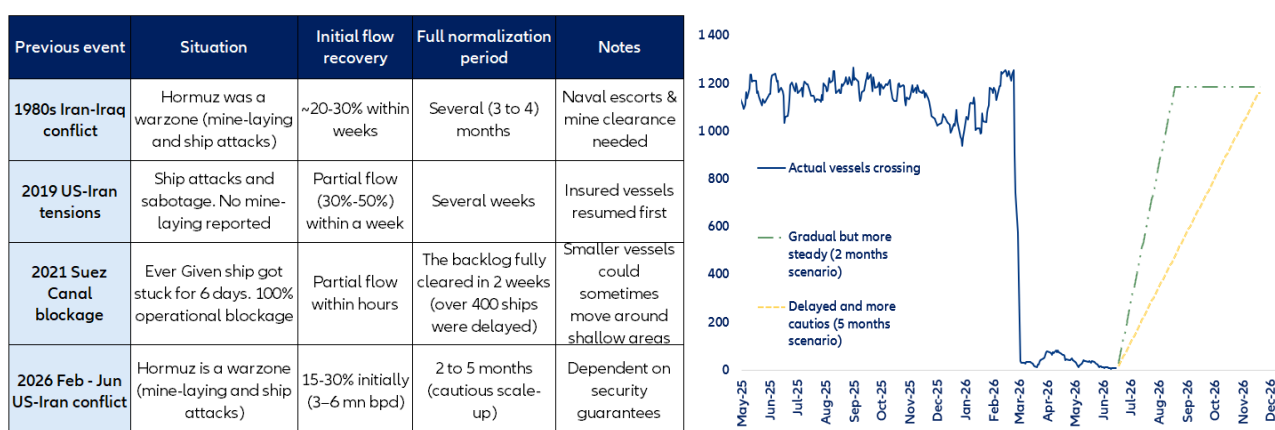
The US and Iran have reached an interim agreement, due to be signed on 19 June, establishing a 60-day ceasefire and paving the way for the reopening of the Strait of Hormuz. After four months of disruption that rattled energy markets and prompted central banks to resume monetary tightening, the announcement marks a significant turning point, reducing upside risks to energy prices and inflation. The agreement is best understood as a sequenced de-escalation framework rather than a final settlement. It establishes a 60-day ceasefire across multiple regional fronts, creating a window for negotiations on Iran's nuclear program and broader geopolitical arrangements. A key provision is the granting of a temporary waiver allowing Iran to sell oil freely during this period, implying a potentially rapid increase in exports relative to current constrained levels, even in the absence of a full sanctions rollback. The deal also envisages a phased approach to sanctions relief, conditional on compliance and progress in negotiations. While the exact sequencing remains unclear, the US position appears to tie any permanent lifting of sanctions to verifiable commitments regarding Iran's nuclear activities. This conditionality introduces a significant implementation risk, as delays or setbacks in negotiations could quickly derail the process. At the same time, the agreement formally incorporates de-escalation across all active fronts, including Lebanon, although participants themselves have identified this dimension as one of the most fragile elements of the arrangement. From an energy-market perspective, the agreement contains several important provisions. The Strait of Hormuz is expected to reopen from 19 June following mine-clearance operations, while in the first hours of the deal announcement President Trump indicated traffic would return "free-of-charge", Iran later signaled it intended to charge "fees for services" for vessels transiting the waterway leaving Tehran with some influence still on the Strait. It is yet unclear what form will these fees take but environmental charges have been floated.

Figure 1: Potential timeline of events until year-end



What needs to be watched for the deal to be sustainable? The sustainability of the agreement will depend on its implementation during the first 60 days. Four indicators will be particularly important to monitor. First, progress on maritime security, notably the pace of mine-clearing operations, which could take from 30 to 50 days to be completed before shipping lanes are deemed fully operational. Second, the speed at which commercial traffic resumes through the Strait of Hormuz (ships in and out), which ultimately depends on the level of confidence in the sector and the security guarantees provided to shipowners. Third, the evolution of war-risk insurance premiums and freight rates, which will provide an early indication of whether shipowners and charterers view the security environment as sufficiently stable to resume regular operations. Finally, progress in negotiations over Iran's nuclear program and the maintenance of ceasefires across other regional fronts, including Lebanon, as these remain the key political conditions underpinning the agreement. Historical experience suggests that confidence is restored more slowly than physical access. Following the 1988 Iran-Iraq ceasefire, commercial traffic required several weeks to start recovering despite the deployment of US naval escorts, while a return to normal operating conditions took more than three months (see Figure 2A). This implies that even under a favorable scenario, the Strait may not return to pre-crisis throughput before Q4 (Figure 2B). Periods of relative calm are therefore likely to be punctuated by short-lived kinetic incidents, with the key test being whether such episodes remain contained. Ultimately, the durability of the agreement will be measured not by the absence of security incidents, but by the ability of the parties to prevent isolated disruptions from escalating into a renewed closure of a maritime corridor that handles roughly one-fifth of global oil demand.

Figure 2A and 2B: Historical benchmarks for maritime traffic recovery in the Middle East; and scenarios for vessel traffic* recovery through the Strait of Hormuz



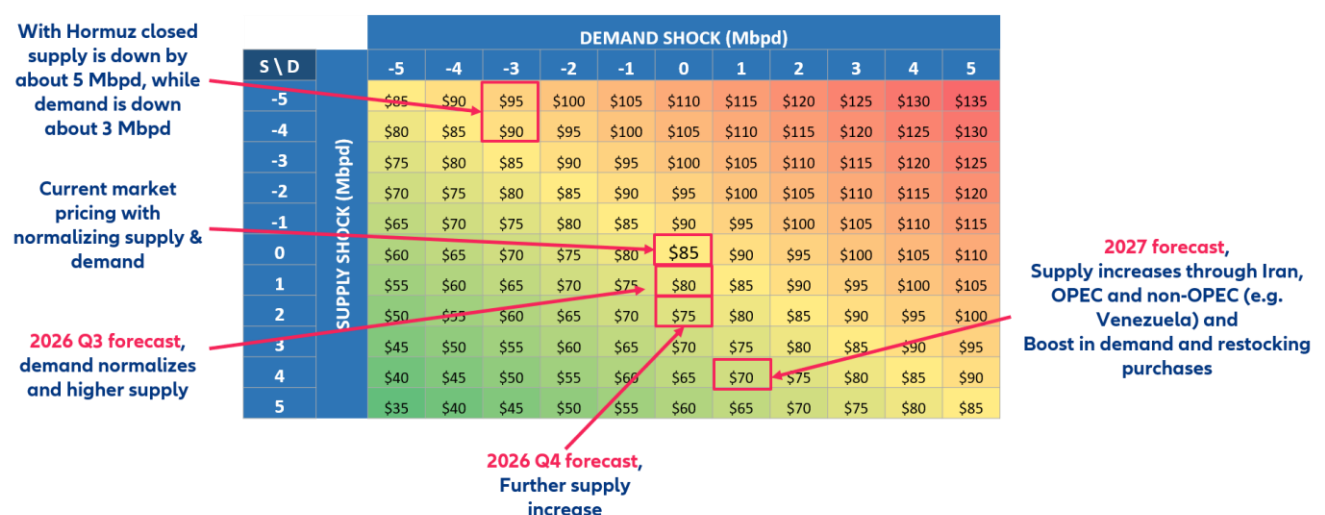
Source: Bloomberg, Allianz Research. *Traffic: number of vessels crossing the Hormuz Strait (7 day rolling total)

Going forward we see a progressive normalization of oil and gas trade flows through year-end rather than an immediate return to pre-conflict conditions. This reflects both existing inventory constraints and the time required to fully restart Gulf production and shipping logistics. In the meantime, partial rerouting of flows, higher exports from the US, Canada, Russia and Latin America, together with strategic reserve releases and moderate demand destruction, would continue to offset a large share of the supply disruption. Most of the effective supply gap would close within three to four months, with roughly 65% of missing supply restored within three months and close to 80% within four months. Crucially, the absence of direct destruction to oil and gas infrastructure prevents the disruption from evolving into a prolonged structural supply shock. We now expect Brent to average 80 USD/bbl in Q3 2026 as the initial supply shock partially unwinds. Currently, the market is absorbing a net disruption of roughly 4-5 mb/d on the supply side while demand destruction of approximately 2-3 mb/d persists, particularly across Asian economies that were hardest hit by the supply shortage and elevated prices. These two forces offset one another in Q3: recovering supply is bearish, but the concurrent demand recovery limits the downside. The net effect is prices stabilizing around 80 USD/bbl as market recalibrate to a world in which the risk of a prolonged Hormuz closure has been removed – for now. By Q4 2026, oil prices should soften further to 75 USD/bbl as supply normalization overshoots the demand recovery. Persian Gulf producers, eager to recapture market share and revenue after

months of shut-ins, are assumed to restore output at a pace slightly ahead of consumption's rebound, tilting the quarterly balance into mild surplus. Natural gas market conditions are also expected to ease for European and Asian buyers and we expect prices to average 43 EUR/MWh in 2026. We also revise our 2027 annual average oil price forecast to 70 USD/bbl - reflecting a structurally more supplied market. Approximately 4 mbpd of incremental production could hit the market : a blend of Iranian output returning to near high levels, OPEC unwinding residual voluntary cuts as the supply emergency passes, and non-OPEC growth from the US and Venezuela. On the demand side, 1 mb/d of incremental consumption offsets part of the supply surge: global activity recovers, Asian industrial demand rebounds, and, critically, large-scale inventory restocking by refiners and strategic reserve managers provides durable support to physical demand. The result is a well-supplied but not a "super glut", anchoring Brent in the high-60s to mid-70s range, with the key risk being larger-than-expected supply increases.

By the time the transportation in the strait of Hormuz (in and out) and resumption of the oil and gas production is full (year-end), mitigating factors will continue to be at play, i.e. demand destruction helping to slow the depletion of global reserves, while fiscal support measures would aim to cushion the inflationary shock. Demand destruction is progressively helping rebalance markets. At present, oil demand destruction is running at around 2.5mbpd, though this largely reflects the immediate disruption effects following the escalation in the Middle East conflict. Assuming a gradual reopening starting in June and normalization of trade flows in the next four months, average demand destruction would moderate to around 0.4mbpd for 2026 as a whole, remaining broadly consistent with a global slowdown rather than a recession scenario.

Figure 3: Brent price forecast based demand and supply shocks



Source: Allianz Research

The good news is that the transmission of the energy shock would not be full which contains the stagflationary flavor of this crisis. The fastest pass-through typically occurs through manufacturing and services input prices as well as energy-related producer prices, with effects peaking within roughly two to six months after the initial shock. Over the following six to twelve months, inflationary pressures progressively broaden as firms pass higher energy and transport costs through supply chains, pushing up intermediate and consumer-goods producer prices as well as industrial and retail selling prices. The most persistent effects are likely to be avoided, i.e. rise of core inflation, consumer-goods prices, inflation expectations and negotiated wages.

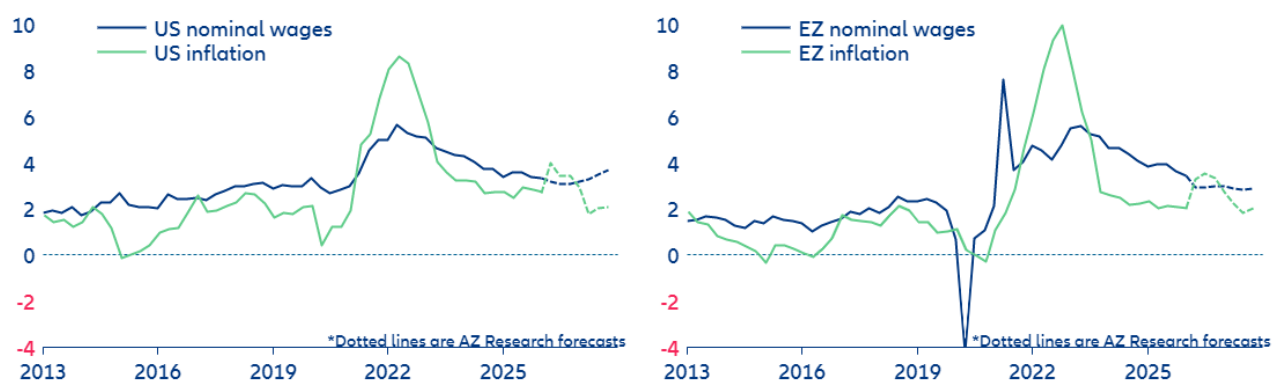
In the US, we continue to expect GDP growth to hold up (+2.0%) thanks to supportive fiscal policy, strong AI-related investment and positive wealth effects. The US being a net energy exporter, its economy benefits from a positive terms of trade shock supporting aggregate income, with the income gains in the mining sector accruing to some households and the government through higher receipts. Nevertheless, the rise in inflation is a net negative for US household, whose aggregate real disposable income has declined in Q2, but fiscal policy is supportive, with tax rebates in H1 2026 providing some partial offset. Furthermore, strong equity valuations encourage households to save less to smoothen their spending on non-energy items. Meanwhile, on the corporate side, capex intentions

remain solid amid strong demand for AI-related business equipment and software. The AI buildout and manufacturing output are relatively immune to high global energy prices thanks to abundant gas supply and low industrial electricity prices, though supply-chain bottlenecks in the global electronics sector is a concern. Recent business surveys indicate that the expansion of the US manufacturing sector is likely to continue in coming months. On the inflation front, we expect headline CPI at +3.3% in 2026. Energy inflation is expected to decline from June. However, higher energy prices relative to their pre-Middle East crisis level, and only a gradual reduction of global supply chains strains should push up core prices. We expect core CPI inflation to tick up above +3% by October, from +2.8% in May. In this environment of increasing core inflation, we expect the Fed to re-affirm its anti-inflation credentials by increasing interest rates by 25bps, in September.

Europe has borne the brunt of the Hormuz shock, with weaker growth and rising inflation as the price to pay for being a net energy importer. Unlike the US and Asia, the Eurozone cannot offset the drag through an AI-driven investment boom - the region simply lacks the large-cap tech champions that are currently powering capital expenditure elsewhere. The energy shock has hit European industry disproportionately hard, feeding into input costs and squeezing margins, as confirmed by the further deterioration in the Eurozone Composite PMI to 47.5 in May, with the services sector - more exposed to confidence effects from the Iran conflict - contracting sharply. Price pressures are building on both input and output sides, reinforcing the case for a more restrictive monetary stance even as growth falters. Germany is a partial exception: fiscal stimulus channeled into defense and infrastructure is providing some near-term support, with Q1 GDP expanding +0.3% q/q (excluding Ireland) on the back of strong exports and higher government spending, and the German Composite PMI holding broadly stable. France, by contrast, saw its Composite PMI collapse to 43.5 in May, its lowest since November 2020.

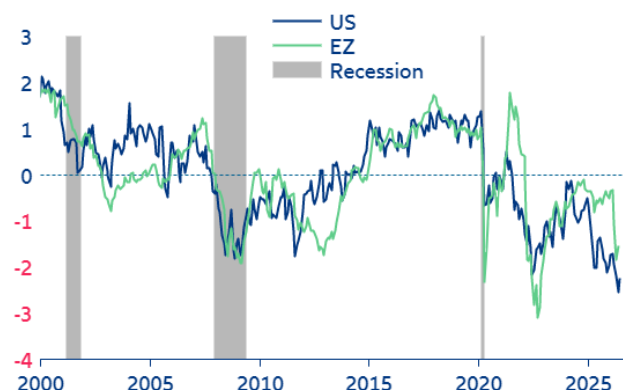
In China, we continue to expect a resilient economy, driven primarily by exports and high-tech manufacturing while domestic demand remains weak amid persistent property-sector stress and subdued consumer confidence. We forecast growth at 4.7% in 2026 and 4.4% in 2027 and maintain our inflation forecast at 1% in 2026 and 1.2% in 2027. As the US-Iran deal will provide a gradual return of oil supply to Asian markets, near-term headwinds to our inflation forecast appear to have eased. However, as conditions normalize, we continue to expect modest inflationary pressure to persist in the near term. We expect the People's Bank of China to remain on hold through 2026 as inflation remain below the 2% policy threshold. While the US-Iran deal will ease inflationary pressures on input prices, the inflationary impact of the return of positive Chinese PPI growth will continue to materialize over the coming quarters. The typical transmission lag between producer and export prices means some modest pass-through remains likely. As a result, we continue to expect Chinese exporters' price competitiveness to slightly erode, while destination markets could see some contained inflationary spillovers. As a leading exporter of manufactured goods, China continues to benefit from strong global demand for clean-energy equipment and AI hardware. However, trade headwinds remain a source of uncertainty. The US has signaled a potential 12.5% tariff under Section 301 targeting forced labor practices to be implemented late July while several additional investigations remain open. Combined with continued RMB appreciation and rising geopolitical tensions, tariff uncertainty could challenge Chinese exporters momentum.

Figure 4: Inflation has surpassed wages leading to a temporary loss in purchasing power



Source: LSEG Datastream, Allianz Research

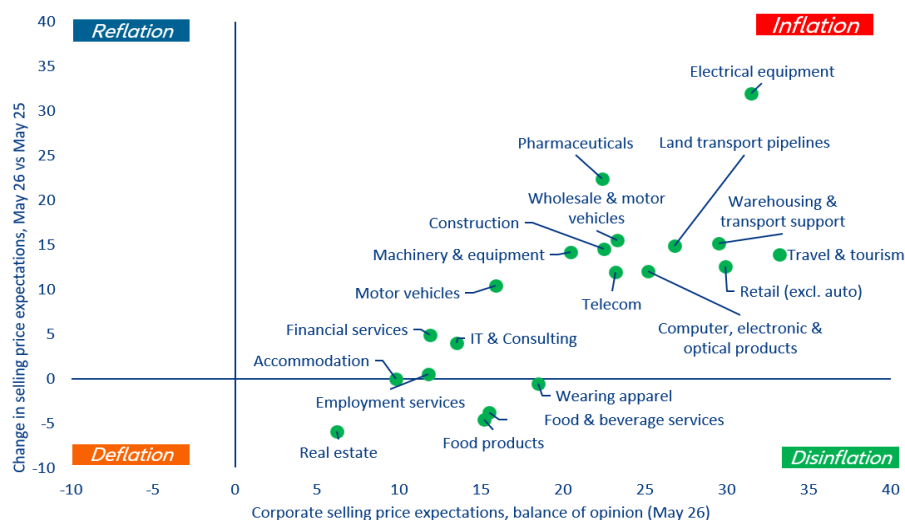
Figure 5: Consumer sentiment has dropped significantly in Europe and the US



Source: LSEG Datastream, Allianz Research

At the corporate level, a gradual reopening is a genuine relief, but pricing power remains the dividing line that a deal break would expose again. The easing in energy prices loosens the squeeze that had reached balance sheets through four channels, namely higher energy costs, logistics disruption, input-cost inflation and weaker demand, with the sectors hit first, transport and petrochemicals where fuel runs 25-40% of operating expenses, the ones that benefit most as crude normalizes. The relief is uneven, set by pass-through capacity: airlines (fuel surcharges and ancillary fees), fertilizer producers and branded pharma can hold price, and semis are cushioned by AI demand, whereas automotive OEMs, computer hardware and generic-drug makers have effectively no pricing power and gain the least margin headroom from cheaper energy. Cash buffers remain solid in aggregate, with listed-company cash falling -3% to EUR36.7trn in 2025 and non-financials more resilient at -1%, though liquidity is concentrating in AI-infrastructure and defense/energy-security names while energy, healthcare and real estate erode. The corporate relief on costs, however, runs into some headwinds on the revenue side: household real purchasing power could remain compressed through most of 2026, and that compression is itself a direct drag on top-line growth and thus on the margin expansion that cheaper energy would otherwise permit. CPI in both the euro area and the US expected to peak around Q4 2026 as energy pass-through is still working its way through utility bills and supply chains. Households are still in a defensive posture: savings rates have edged up, credit-card delinquencies are rising at the margin in the US and eurozone consumer confidence has not recovered to pre-war levels. The practical consequence for corporates is a squeeze on volumes even where pricing power nominally exists. The margin arithmetic therefore looks less clean than the energy line alone implies: input costs are easing but wage bills could remain elevated into 2027, meaning the Q4 2026 peak in CPI marks an inflection point rather than an immediate relief of the operating-cost structure. Again the European exposure is the sharper one: household energy bills carry a larger share of disposable income than in the US, mortgage re-pricing is more immediate in markets such as the UK and the Netherlands where floating-rate structures dominate, and fiscal space to cushion the squeeze through transfers is more constrained after the 2022-23 spending. The caution, consistent with the rest of this note, is that the relief is reversible: should the ceasefire fail and the closure run beyond six months, input-cost inflation that is now only spreading into cyclicals such as pulp and paper and auto-parts would re-intensify, pricing power itself would begin to erode, turning a margin squeeze back into revenue contraction, and Europe would be more exposed than the US throughout given higher energy intensity, greater trade openness and less fiscal room.

Figure 6: Selling price expectations for corporates in the Eurozone, May vs Feb 2026



Sources: Eurostat, Allianz Research

Figure 7 - Short-term pricing power assessment of sectors

Sector	United States	Eurozone
Transport		
Airlines	✓	✓
Shipping	✓	✓
Road	✓	✓
Transport Equipment		
Aviation & Space	✓	✓
Trucks	✓	✓
Shipbuilding	✓	✓
Chemicals		
Petrochemicals	✗	✗
Basic inorganics (fertilizers)	✓	✓
Polymers (plastics)	✓	✓
Specialty chemicals	✓	✓
Pharmaceuticals		
Big Pharma (branded)	✓	✓
Generics	✗	✗
Pulp & Paper		
Graphic paper	✗	✗
Paper packaging & board	✓	✓
Tissue & Hygiene	✓	✓
Machinery & Equipment		
Industrial & robotics	✓	✓
Construction machinery	✓	✓
Agricultural machinery	✓	✓
Construction		
Infrastructure / Civil Engineering	✓	✓
Residential	✗	✗
Automotive		
Vehicles	✗	✗
Parts & accessories	✓	✗
Tires	✗	✗
Electronics & telecom hardware		
Semiconductor	✓	✓
Computer: High/Low-end machine	✓	✗
Smartphone: High/Low-end machine	✓	✗
Household equipment		
Household appliances	✓	✓
Furniture	✗	✗
Textile		
Apparel - luxury/non-luxury	✓	✓
Footwear	✓	✓
Accessories	✓	✓
Retail		
Food	✓	✓
Non-food	✓	✓

✗	No pricing power
✓	Mild pricing power
✓	Strong pricing power

Source: Allianz Research

What is the impact for capital markets?

Middle East de-escalation creates room for outperformance in rates. The short-end and the belly of the curve stand out as the most compelling areas, with the case for steepening and duration strongest there. The repricing of rates was driven by different forces on either side of the Atlantic, which shapes where the most compelling opportunities lie.

In Europe, the repricing was inflation-driven, meaning that a de-escalation scenario and lower energy prices will hit the root cause directly. After last week's ECB hike market still prices one more move into year-end. A durable peace deal plus a lower oil price could give the ECB confidence to stay on hold instead. For now our base case stays one more hike, broadly in line with the market. But the upside is real with ~35bp at the front and up to 20bp vs forwards at the curve belly over the next one to two years

The US offers a more mixed picture. The repricing is only partially inflation-driven, it also reflects renewed US strength and the return of US exceptionalism, which a durable deal could amplify rather than unwind. Front-end holds some outperformance potential of ~25bps as de-escalation may keep the Fed on hold against one hike priced over 12m by markets, especially if AI-driven productivity disinflation starts to show through. Our base case still pencils in one hike in 2026, but risk is tilted to pricing it out again. Belly outperformance on the USD curve is less clear than for the EUR curve, since the growth and exceptionalism legs won't reverse mechanically on lower energy. A significant near-term uncertainty is the Fed's upcoming meeting under incoming Chair Warsh, which could overshadow or offset any peace-deal reaction. Both the ECB and the Fed remain explicitly data-dependent, and inflationary repercussions of the energy price shock are already visible — including early evidence that reflation is broadening beyond energy. That context means neither central bank is likely to move ahead of the data, whatever the geopolitical backdrop. The risk bias deviates however, with the ECB risking a lagged error of commission (hiking too much on stale data), while the Fed risks a deliberate policy shift under new leadership regardless of geopolitics. The fault line is the long end: even on a deal, structural US inflation and fiscal supply leave long-end yields vulnerable, so a curve steepening rather than a parallel rally is the more likely path and the part that could disappoint. Nevertheless, US TIPS look more attractive than EUR linkers, and caution on long-end rates globally remains warranted — even in a scenario of normalizing energy flows — given structural inflation dynamics and elevated upside risks in US and global rates.

In the case of equities de-escalation is helpful, but it is unlikely to create a large new rally because at less than -10% maximum drawdown, equities never priced the conflict as a systemic shock. The June 14 US-Iran deal reduces one of the obvious downside tails, but this is still a partial and fragile geopolitical reversal not a full peace settlement. That distinction matters because the market starting point is already fairly strong (as of June 12 MSCI US was up around 8% since the start of the Iran war, MSCI EM is up around 6.5%, and the MSCI Europe, despite being the relative laggard, is still +5%ytd). In other words, equities have looked through the conflict because earnings, AI leadership and risk appetite have mattered more than geopolitics. The benefit here is relief and not re-rating. Our oil baseline of 80 USD/bbl in Q3, declining to 75 USD/bbl in Q4 and 70 USD/bbl on average in 2027, reflects a gradual supply normalization that removes the inflation and rates scare scenario rather than creating a positive impulse. This should support Europe, cyclicals, transport, airlines and energy-intensive sectors that were carrying a modest war discount, and it lowers the probability of a policy accident. However, at this stage the market is not reversing a war discount, it is simply taking out part of a tail risk that had mostly remained in commodities. On a confirmed deal we would expect consolidation rather than a broad melt-up, with the relief accruing to Europe, cyclicals and the energy-sensitive sectors that carried the discount rather than to the index as a whole. The fault line cuts the other way: because the conflict was never priced as a systemic shock, there is little discount to unwind on good news but plenty of room to gap lower on bad news, leaving the tape asymmetric and headline-driven if the ceasefire frays. This comes against the backdrop of already ambitious market pricing, with valuations in the 85th percentile and earnings growth in the 92nd percentile (MSCI World, 20-year history), leaving little room for upside surprises. In this situation holding exposure, adding strategic risk only very gradually or opportunistically and focusing on rolling hedges at moderate cost offers continues to offer the best risk-reward. From a positioning perspective, the key risk for broader risk assets (including credit) is not just a potential breakdown of the deal, but the second-round effects of earlier shocks on growth, inflation expectations, and rates. Notable downwards revisions could turn earnings revision momentum and pull equities by 5-10% easily through summer. With AI-led

enthusiasm driving equities to new highs, the focus shifts to delivery: any disappointment on monetisation or the burden of sustained capex could trigger renewed volatility. Last weeks' 30% swing in Oracle offered a high-velocity glimpse of how markets can behave on such news — albeit for the market overall it would be far more muted form, as we might see toward end-2025.

For credit the news protects the carry trade, rather than opening a major new spread tightening phase. Credit has been even clearer than equities in saying that this was not priced as a funding shock. EUR IG remains around 77 bps and US IG around 72 bps, both already close to all time tights. Similarly, Euro HY (BB/B) is around 213 bps and has tightened roughly 19 bps since the start of the war. That is not the footprint of a market worried about a major geopolitical credit event. It is the footprint of strong demand for carry, resilient earnings and still open refinancing markets. De-escalation is therefore constructive because it reduces the risk of an oil spike, renewed rate volatility and a confidence shock. But at these spread levels, the most realistic impact is spread support rather than strong compression. On a final resolution we would therefore expect only limited further tightening from already-tight levels, with spreads holding firm rather than compressing meaningfully. The fault line is precisely those tight levels: with spreads close to all-time tights there is almost no cushion, so the risk is heavily skewed to widening if macro or rate concerns return, and HY would feel it first. Inside High Yield, discrimination between more B- and BB-ratings remains at decade highs (2.1x ratio) and indicates some caution in run for carry. IG should remain well bid as macro volatility falls and HY should continue to benefit more from risk appetite and higher rates, but will remain vulnerable as the margin for error if the ceasefire fails or oil risk reappears is low. In this situation we remain cognizant of downside risks but gradually and also selectively embrace opportunities for carry where meeting portfolios have been awaiting more clarity for implementation needs. Carry remains preferred to price appreciation assets in this fragile and again time-consuming outlook. For credit specifically, in a demand led market, the focus is less on fundamentals—which remain broadly average—and more on technicals. Strong inflows continue to anchor spreads at low levels but also increase the risk of a more abrupt reversal than would typically be implied by fundamentals in a normal cycle. Areas we watch are any further divergences within the ratings spectrum, with B/BB segments reaching new highs (<2.5x), as well as emerging signs of fatigue in technology issuance, where bid-cover ratios falling below 3x, new-issue concessions below 20bps and attrition rising above 1/3 could indicate normalization to start.

Lastly, for private markets the interim agreement is likely to improve liquidity and transaction confidence, but it is not a valuation reset. The relevant anchor is that public equities and liquid credit recovered most war-related losses, so they do not have a large visible war discount to reverse, the move is therefore one of mechanics rather than marks. The impact is instead practical and routes through three channels. First, firmer and re-rated public comparables, particularly in energy infrastructure and grid/electrical contracting, where listed names trade at historic EV/EBITDA premiums and a consolidation wave is still building, give Private Equity a clearer exit and valuation reference points, easing exit and IPO discussions that had stalled while sponsors sit on a record backlog of unsold companies and high dry powder. Second, resilient credit markets and the receding oil and rate volatility tail support refinancings, letting sponsors clear debt raises and recaps that were on hold. Third, lower energy tail risk improves the margin outlook for industrials, logistics, consumer services and transport-linked assets, where fuel had run 25-40% of operating costs at the peak. Along these lines, Private Credit will also benefit from a lower default tail risk but lender spreads, as is the case for liquid credit, are already compressed, with median direct lending spreads having fallen from roughly 700 bps in early 2023 to around 500 bps, so the return impact is likely to be modest. Infrastructure and real estate are likely to get a mixed but generally stabilizing impulse with lower energy volatility helping utilities, tenants and transport assets, and a lower oil inflation path reopens room for the rate cuts that rate-sensitive real estate and residential construction depend on, while some of the energy-security and defense-adjacent premia that re-rated during the conflict may soften at the margin, even as the structural, data center- and electrification-driven grid buildout continues regardless. In practice we would expect the move to show up in activity rather than marks, with deal flow, exits and refinancings picking up gradually rather than valuations re-rating higher.

Pressure points to watch. Every constructive call above is conditional, and the conditionality is the message, this is a half-baked reopening, not a settled peace, and the relief rests on a peace deal, that remains anchored on unresolved negotiations over Iran's nuclear program and a ceasefire on all fronts, including Lebanon, and the track

record of failed de-escalation attempts suggests it can unravel quickly. Because markets never fully priced the war, the move on good news is modest while the room to reverse on bad news is large, so the risk profile is asymmetric and the reaction to any setback would likely be fast and disorderly rather than gradual. In this context, a failure to adhere to the deal, a renewed closure of the Strait, a loss of central bank look-through or a tightening in dollar funding would not arrive in isolation, but would recouple oil, rates, credit and equities at once and turn an orderly normalization into a sharp, correlated drawdown. We therefore treat the base case as constructive but provisional as it warrants close and continuous monitoring of the early warning signals around oil, volatility, credit spreads and long end rates. In short, relief is the call, but it is not the all clear, and the cost of being slow to react would be far greater than the cost of staying cautious.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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From kickoff to cash flow – Economic spillovers from the Football World Championship 2026

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In Summary

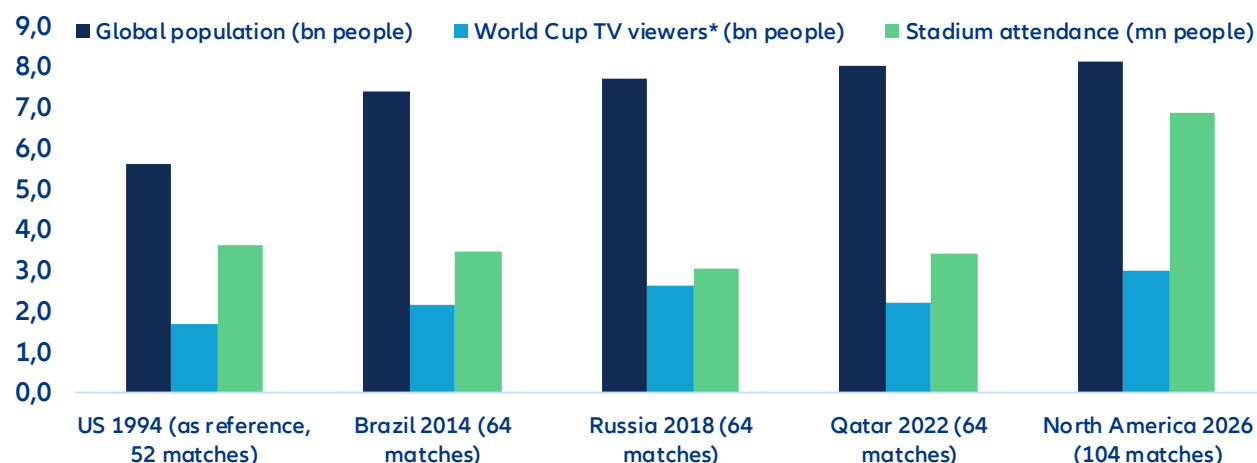
- **The Football World Cup 2026 will be the largest tournament in football history and a structural departure from previous editions, expanding from 32 to 48 teams and from 64 to 104 matches across 16 host cities in the US, Canada and Mexico.** It will test whether a World Cup can successfully evolve from a single-country event into a distributed continental experience without sacrificing atmosphere, attendance or commercial performance. Over a six-week period, it is expected to mobilize approximately 6.5mn attendees, including 2.6mn international visitors, generating an estimated USD9bn in GDP across North America during June–July 2026. For comparison, Taylor Swift's Eras Tour and Beyoncé's Renaissance World Tour, with 149 and 56 shows respectively, generated approximately USD2.1bn and USD579mn in revenue. FIFA projects record commercial revenues of USD13bn for the 2023–2026 cycle but the macroeconomic impact remains more concentrated than transformative, with tourism-related spending accounting for the dominant transmission channel.
- **From a demand perspective, total tourism-related expenditure is expected to reach around USD8bn, split between USD6.8bn in foreign tourism exports and USD1.2 bn in domestic consumption, after accounting for crowding-out effects.** Despite the stricter US visa barrier (refusal rate of 33% on average for non-European qualified nations, 74% for Senegal, 61% for Iran), the US captures the largest share, with an estimated USD5.4bn boost, followed by Mexico (USD1.4bn) and Canada (USD1.2bn). This reflects around 40% international visitors and 60% domestic attendees, each staying an average of 6–10 days and spending between USD180 and USD350 per day depending on the host country. Air travel adds a further USD1.0bn in incremental airline revenues, reinforcing the importance of mobility-linked sectors in overall value creation. Security operations spending will add USD1bn to the economic boost. Most of this spending is government consumption.
- **The distribution of gains will be highly uneven across sectors and geographies.** Lodging and airlines emerge as the clearest winners, supported by peak hotel occupancy rates of 90–95%, with room prices rising by up to +15–20% in selected host cities following the draw phases. Airlines benefit from structurally constrained capacity growth ranging between +0.4% and +2.1% in Q2 2026, supporting strong pricing power on key domestic and international routes. Meanwhile, food & beverage, retail and entertainment industries also stand to gain meaningfully from elevated match-day consumption, particularly in Mexico where football-related social spending is deeply embedded in consumer behavior.
- **However, the macroeconomic impact remains modest relative to the size of host economies, translating into a GDP uplift of approximately USD6.1bn in the US (+0.1pp quarterly growth), USD1.7bn in Mexico (+0.3pp) and USD1.3bn in Canada (+0.2pp).** The event is therefore best characterized as a high-intensity, short-duration demand shock rather than a structural growth driver, with benefits concentrated in tourism-sensitive sectors and constrained by substitution effects, capacity bottlenecks and regulatory frictions. Ultimately, the 2026 World Cup will deliver clear sectoral winners – hotels, airlines and urban tourism ecosystems – while reinforcing the importance of execution, mobility infrastructure and cross-border coordination in shaping the final economic outcome.

Bigger and better? The tournament that outgrew its borders

The Football World Cup 2026 is set to become the largest football tournament ever organized and arguably the most ambitious sporting event in modern history. But will this ultimately lead to stronger economic gains? For the first time, FIFA has expanded the competition from 32 to 48 national teams, increasing the number of matches from 64 to 104 and extending the tournament across three host countries: the US, Canada and Mexico. The competition, which begins in June 2026 and concludes in July, represents a structural break from previous World Cups. Rather than concentrating activity in a single host nation, FIFA has effectively created a continental-scale event stretching across North America, spanning 16 host cities, thousands of kilometers of travel corridors and multiple regulatory jurisdictions. From an economic perspective, the tournament should be viewed less as a sporting event and more as a six-week tourism boost, and a consumption and media platform capable of mobilizing 6.5mn attendees (2.6mn international visitors) and generating USD9.1bn of GDP in June-July across the three nations.

The scale of the tournament is unprecedented. Previous World Cups typically attracted between 3.0-3.4mn stadium spectators. With 104 matches and significantly larger aggregate venue capacity (67,000 people per stadium on average), FIFA 2026 could become the first tournament in history to potentially surpass 7mn cumulative attendees (if all tickets get sold), effectively doubling the scale of fan participation observed in recent editions (Figure 1). The FIFA itself published a 2023–2026 total potential revenue of USD13bn (vs USD6.4bn for Qatar), an increase of USD6.6bn compared to the previous 2019-2022 cycle, driven mainly by higher income from TV broadcasting rights (USD4.3bn total, +USD964mn vs Qatar), ticket sales and hospitality rights (USD3.1bn total, +USD2.6bn) and marketing rights (USD2.7bn total, +USD927mn), with additional smaller gains from licensing and other revenue sources, reflecting strong commercial momentum ahead of the expanded tournament.

Figure 1: FIFA World Cup reach in previous tournaments, vs expectations for 2026



Sources: Reuters LSEG, UN, FIFA, Allianz Research. Note: People watching at least 20 minutes of a match on TV

However, the decentralized nature of this World Cup—spanning multiple countries—creates both opportunities and challenges, with clear winners and losers. The distribution of economic gains will be driven not only by where travelers choose to go, but also by each country's preparedness, infrastructure and capacity to welcome them effectively. One of the most distinctive features of FIFA 2026 is that none of the three host nations needed to build a new generation of World Cup stadiums from scratch. Unlike recent tournaments that required massive upfront infrastructure investment, North America is largely leveraging existing assets. Qatar, in contrast, invested roughly USD220bn ahead of the previous World Cup, largely in long-term infrastructure projects, with USD7–10bn dedicated to build stadiums. This time, all 16 venues were operational before the tournament and require mostly renovations, capacity expansions and FIFA-compliance operational upgrades, with approximately USD2–2.5bn spent in this regard.

This makes FIFA 2026 one of the most capital-efficient World Cups in modern history. However, the primary challenge is not stadium readiness but rather transportation and hospitality logistics. The key execution risk is not stadium readiness but the ability of airports, hotels, transportation networks and border authorities to accommodate millions of visitors moving across 16 cities. Cities such as New York, Los Angeles, Dallas, San Francisco and Miami possess mature tourism ecosystems, extensive hotel inventories and world-class airport infrastructure.

However, congestion risks remain significant. Airports in several host cities already operate near capacity during peak travel periods and the World Cup is expected to create temporary spikes in passenger volumes that will test the resilience of transportation networks.

Canada's preparation efforts have focused more heavily on accommodation and urban mobility. Toronto and Vancouver are highly developed metropolitan areas with strong public transportation systems and excellent reputations for event management. However, both cities are likely to face accommodation pressures. As shown in Table 1, Vancouver (hosting seven matches) has the lowest hotel capacity among all host cities, while Toronto will host six matches with nearly three times the room inventory. This imbalance suggests that Vancouver may experience tighter lodging conditions and greater upward pressure on room rates during the event. Indeed, current hotel pricing already reflects Vancouver's constrained lodging supply. Among all cities, Vancouver has the highest accommodation cost of the tournament, with prices double the amount observed in half of the hosting cities of the list. Toronto, by contrast, benefits from a significantly larger hotel inventory and more extensive transportation infrastructure, which should provide greater capacity to absorb visitor demand, although meaningful pricing pressures are still expected.

Table 1: Comparing hospitality capacity across World Cup 2026 venues

Hosting cities	Hotel rooms capacity	Stadium capacity (# people)	Avg hotel price (USD)
Atlanta	95 000	71 000	\$331
Boston	65 000	65 878	\$611
Dallas	150 000	80 000	\$395
Houston	100 000	72 220	\$236
Kansas City	30 000	76 416	\$344
Los Angeles	103 000	70 240	\$384
Miami	65 000	65 326	\$378
NYC (final match)	140 000	82 500	\$624
Philadelphia	80 000	69 596	\$376
Seattle	40 000	68 740	\$446
San Francisco	150 000	68 500	\$283
Toronto	85 000	45 500	\$496
Vancouver	28 000	54 500	\$785
Guadalajara	65 000	46 232	\$335
Mexico City (open. match)	80 000	87 523	\$509
Monterrey	55 000	53 500	\$538

Sources: FIFA, Kick off adventures, Allianz Research

Mexico's preparation strategy reflects a different set of priorities. Significant investments have been directed toward airport modernization, urban transportation improvements and stadium renovations. Public and private spending associated with the tournament is estimated at roughly USD1.5-2.0bn, including upgrades to the three host venues – Estadio Azteca in Mexico City, Estadio BBVA in Monterrey and Estadio Akron in Guadalajara – as well as investments in airports, roads, public transport systems and surrounding urban infrastructure. While infrastructure quality remains less homogeneous than in the US, Mexico enters the tournament with an important competitive advantage: affordability to international visitors. For many football fans, attending matches in Mexico is materially less expensive than attending equivalent matches in major American cities. This affordability extends beyond accommodation costs to include food, transportation and entertainment expenses, potentially positioning Mexico as the most attractive destination for international supporters operating under budget constraints.

However, lower costs alone do not guarantee proportionally larger economic gains. Part of the spending generated by the tournament is likely to be concentrated in a limited number of host cities and sectors, while import leakages, profit repatriation by international operators and substitution effects may reduce the net domestic impact. Furthermore, accommodation bottlenecks, security concerns and transportation congestion could constrain visitor spending and limit the extent to which tourism demand translates into broader economic activity. As observed in previous host countries, headline visitor numbers and spending estimates often overstate the final contribution to economic growth once these offsetting factors are taken into account.

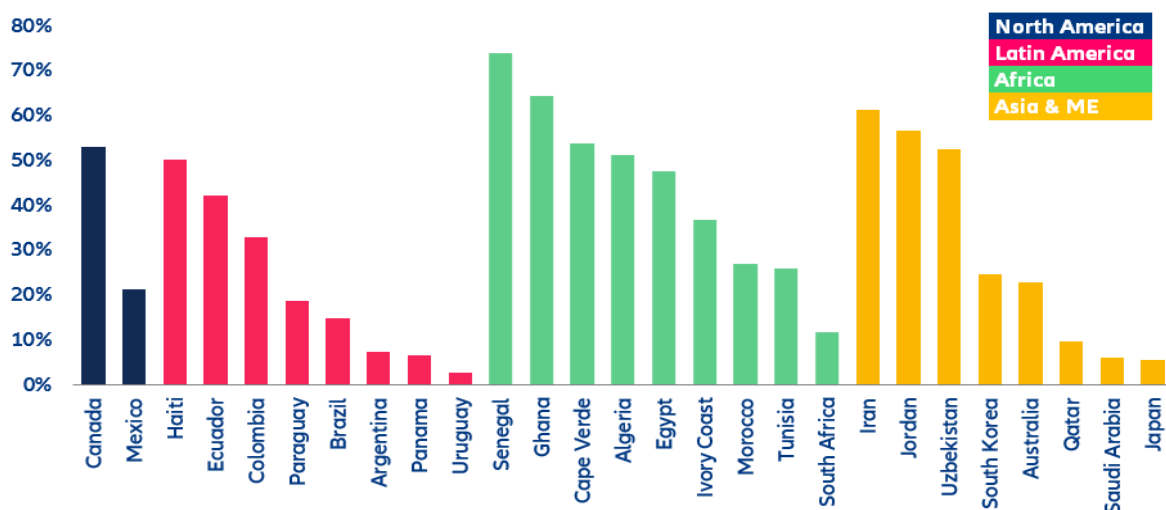
Breaking borders on the pitch, reinforcing them at the gate

While the US is a major global tourism hub – ranking third worldwide in international arrivals with nearly 70mn visitors in 2025 – and will host the largest share of matches in this World Cup, the surge in inbound tourism is not likely to be as strong as one might expect this year. Three factors help explain this: (1) The White House's growing travel ban, (2) differing football cultures across the three host nations and (3) exchange rate dynamics between countries.

One of the most underappreciated risks facing the tournament relates to immigration and visa policy. Eleven of the 16 host cities are located in the US, while Canada and Mexico will host in two and three cities, respectively. The US will stage 78 of the tournament's 104 matches (including the final in NYC), ensuring that the majority of economic activity will naturally accrue to American cities. Yet, the geographic footprint also introduces complexities not observed in previous tournaments. Fans following their national teams across multiple venues may be required to undertake several domestic flights, travel across international borders and navigate significantly different immigration systems. The operational challenge facing FIFA is therefore considerably greater than in Qatar 2022, where all matches were located within a relatively compact geographic area. Success will depend not only on stadium operations but also on the seamless functioning of airports and border control agencies.

Visa-related barriers vary significantly across the three host countries. First, the tourism visa cost difference is noticeable. The US presents the highest friction, with a standard B1/B2 visa fee of USD185 for most non-visa-waiver travelers, alongside stricter screening requirements. Canada offers a lower-cost and more streamlined process, with most visitors paying around CAD100 (USD72) for a tourist visa or only a minimal fee under the eTA system for eligible nationalities. Mexico, by contrast, has the lowest barrier, with tourist entry typically free or included in airfare for air travelers. Beyond the price, entry procedures also differ significantly. Unlike previous World Cups, where entry procedures were often simplified or partially suspended for ticket holders, the 2026 tournament will occur during a period of heightened scrutiny surrounding international mobility. Canada and Mexico generally maintain relatively accessible tourist visa systems for many nationalities. The US, however, presents a more complicated picture. Visa waiting times remain lengthy in several countries, approval rates vary significantly by nationality and border procedures have become increasingly strict under the new Trump Administration. As shown in Figure 2, US State Department data indicates that applicants from 11 of the 48 countries qualified for the World Cup face visa refusal rates exceeding 40%. While these factors may appear secondary from a sporting perspective, they are highly relevant from an economic standpoint because international attendance remains one of the primary drivers of incremental tourism spending.

Figure 2: US visa refusal rates for tourist and business travelers (B-visas only) by nationality in 2025



Sources: US Department of State, Allianz Research. Note: Excluding the 16 European qualified nations

The issue becomes even more important when viewed through a geopolitical lens. Football's fastest-growing fan bases are increasingly located in the Middle East, Africa, Central Asia and certain regions of Latin America but visitors from these regions may perceive travel to the US as administratively burdensome or politically uncertain. In

some cases, specific nationalities face significant travel restrictions or heightened scrutiny during visa processing. Although the Trump administration recently announced that soccer fans from 50 countries subject to an initial USD15,000 visa bond requirement (a deposit) would be exempt – provided they can demonstrate travel to the US for the World Cup with valid match tickets – nationals of Iran and Haiti remain subject to entry restrictions. In addition, travelers from Côte d'Ivoire and Senegal are subject to partial restrictions under the expanded scope of the travel ban. The result may not be a complete absence of international visitors, but rather a redistribution of demand toward Canada and Mexico.

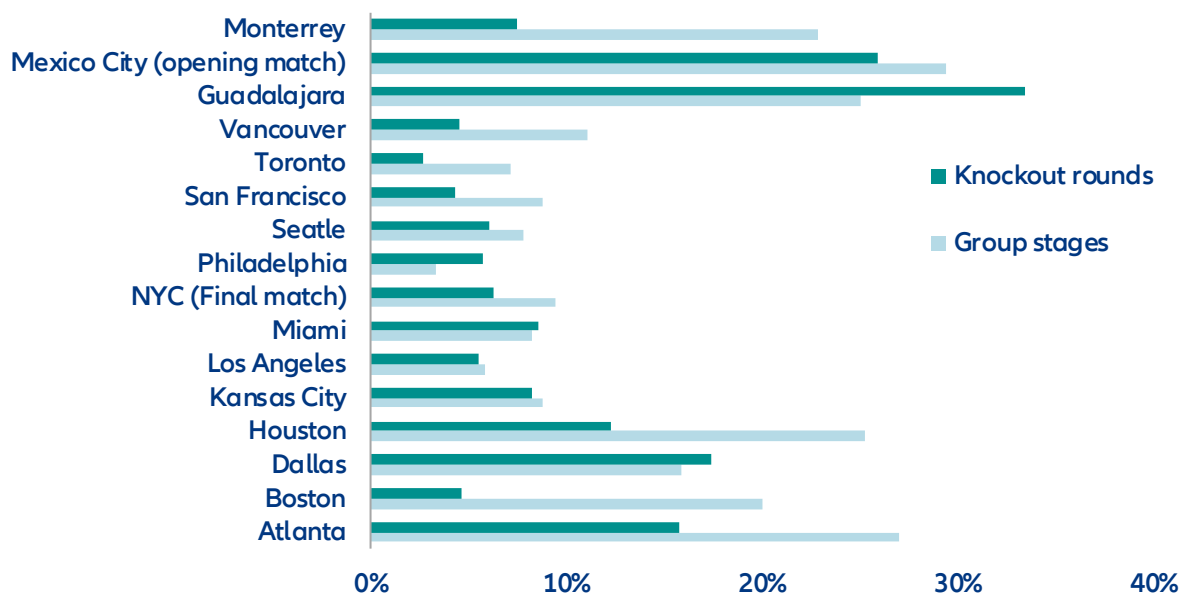
Soccer culture, and how the sport is experienced and consumed, also differs considerably across hosting countries. A defining characteristic of FIFA 2026 is the stark contrast among its hosts. Mexico remains the most football-centric nation, where the sport is deeply embedded in national identity and cultural life. Local demand for matches is expected to be exceptionally strong, and the stadium atmospheres in Mexico City, Guadalajara and Monterrey are likely to rank among the most vibrant of the tournament. The US presents a different profile. Although soccer has experienced significant growth over the past two decades, it remains secondary to the NFL, NBA and college sports in terms of cultural relevance and media attention. Canada occupies an intermediate position, where soccer has benefited from increasing popularity among immigrant communities and younger demographics, but still lacks the historical depth observed in Latin America and Europe. Available inventory tickets suggest that stadium occupancy is unlikely to reach 100% across all matches in the US. Remaining ticket inventory appears to be concentrated in selected group-stage matches, particularly in the US and Canada, while demand for Mexico-hosted matches remains comparatively stronger.

Three countries, three currencies. Exchange rates clearly also have a major influence on travelers' preferences. Exchange rate dynamics are likely to play a meaningful role in shaping travel demand, visitor flows and spending patterns. In recent years, the US dollar has remained relatively strong against most major and emerging-market currencies, increasing the overall cost of travel to US host cities and potentially discouraging price-sensitive international visitors. By contrast, the Canadian dollar has traded at a discount to the USD, enhancing Canada's value proposition for international travelers seeking a similar North American experience at a lower effective cost. Meanwhile, the Mexican peso, despite periods of notable strength driven by robust economic fundamentals and nearshoring investment flows, continues to offer a significantly lower price point for accommodation, food, transportation and entertainment compared with many northern destinations.

Follow the money: The net winners

The lodging sector is positioned to be among the clearest beneficiaries of the tournament, although outcomes will vary considerably by city. New York, Miami, Los Angeles, Dallas and San Francisco possess extensive hotel inventories and significant experience hosting large-scale international events. These cities are likely to absorb demand relatively efficiently, albeit at elevated room rates. In contrast, cities such as Seattle and Vancouver face more meaningful capacity constraints. Limited room inventories is generating substantial price increases during peak periods, benefiting hotel operators but potentially discouraging some visitors. As shown in Figure 3, hotel room prices increased by approximately 15% following the group stage draw, with the three Mexican host cities recording the most pronounced gains (all exceeding +20%). Guadalajara in particular experienced especially strong demand, reflecting its historical position as the most affordable accommodation market among the three Mexican cities. As the final stages of the tournament shift away from Mexico, with Dallas and Atlanta set to host the semifinals, hotel room prices rose by an average of +11% following the knockout round draw.

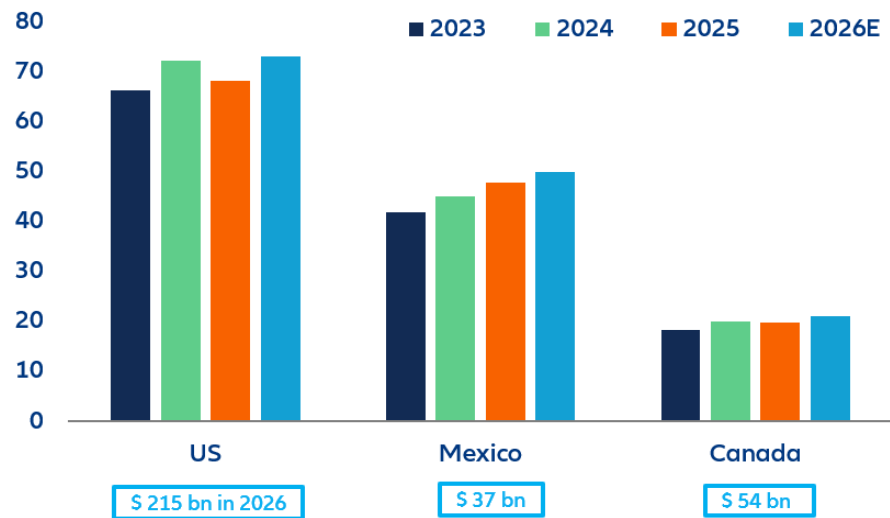
Figure 3: Observed hotel room price change after the draw of



Sources: Lighthouse, Allianz Research.

Overall, the event is expected to provide a meaningful boost to the broader tourism sector in 2026. However, part of the y/y increase – particularly in the US – also reflects a low base in 2025, when traveler confidence was weakened amid uncertainty around Trump-era travel ban policies. Indeed, while international arrivals increased by +6.2% y/y in 2025 in Mexico, they declined by -5.5% in the US (see Figure 4) while were flat in Canada. Against this backdrop, the event could provide some relief to the US tourism sector amid the recent and still ongoing slowdown, with international visitor volumes again down -1.7% y/y in the first quarter of this year. The World Cup could therefore lift 2026’s US international arrivals to approximately 72–74mn visitors, implying a net uplift of +3% to +6% versus a normal baseline year, leading the country to collect around USD215bn of tourism receipts for the entire year.

Figure 4: Tourism evolution by country: number of international visitors (million people) and expected tourism receipts for 2026FY in USD (blue boxes)



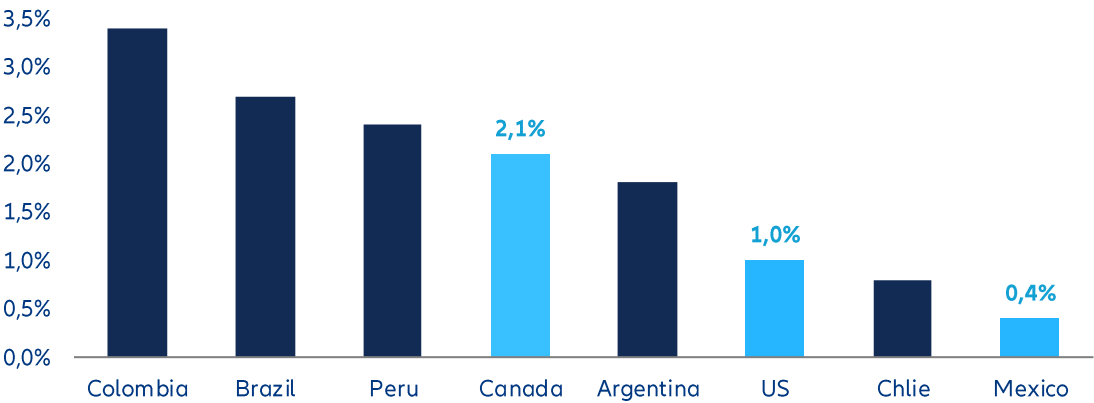
Sources: UN Tourism, Allianz Research.

The food and beverage industry should experience a meaningful boost in parallel, but media and broadcasting companies remain among the largest beneficiaries overall. Stadium consumption, fan festivals, restaurants and bars typically generate strong sales growth during major sporting events. The effect may be particularly

pronounced in Mexico, where football consumption patterns are closely associated with social gatherings and beverage purchases. Nevertheless, the World Cup's greatest economic value has always derived from its global audience rather than in-person attendance. Advertising revenues, sponsorship agreements, streaming subscriptions and broadcasting rights collectively account for a significant share of FIFA's commercial ecosystem. Construction firms, by contrast, are likely to experience only modest incremental gains because most infrastructure spending has already occurred during the preparation phase.

Beyond ground-based tourism activities, airlines also stand out as one of the most direct beneficiaries, turning the period into an opportunity to improve margins after four months of abnormally high jet-fuel prices. The tournament's geographic footprint virtually guarantees elevated demand for both international and domestic flights, particularly as supporters have to move between host cities that are kilometers away. Clearly airlines – alongside hotels – are positioned as the two sectors with the strongest pricing power during the tournament, driven by limited substitution options and surging demand. This is particularly evident on long-haul domestic routes such as Seattle–Miami, Los Angeles–New York, and international ones such as Mexico City–Vancouver, where flight times exceed five hours and alternative modes of transport are effectively non-viable. Since the onset of the Strait of Hormuz conflict, jet fuel prices have averaged roughly twice their 2025 average level. Therefore the World Cup is likely to provide airlines with a temporary opportunity to enhance yields and margins through dynamic pricing, capacity optimization and higher ancillary revenues. Yet, while the event should drive strong demand on selected routes, its impact is expected to be tactical rather than transformational for the industry's overall profitability. Besides the oligopoly nature of the business, the sector's capacity may not be at ideal levels. According to IATA, scheduled passenger seat capacity in Q2 2026 is expected to grow by just +1.0% in the US and +0.4% in Mexico (Figure 5), suggesting that airlines will enter the World Cup with relatively constrained supply. As a result, the tournament is likely to be characterized by strong pricing power rather than significant capacity expansion, supporting higher fares and improved yields. While Canada's capacity growth is projected at a higher 2.1%, its smaller and more concentrated aviation market may still face bottlenecks during peak tournament periods. Overall, the data point to a demand-driven environment where airlines are better positioned to maximize profitability through revenue management and network optimization than through volume growth.

Figure 5: Americas air passenger seat capacity scheduled for Q2 2026 by country, y/y growth rate



Sources: IATA, Allianz Research

A positive but modest GDP boost of USD9.1bn across North America

The World Cup represents a positive yet modest demand shock across North America during the June-July 2026 period, with foreign tourism spending being the most important channel. From a macro-economic perspective, sporting events entail four types of spending which impact GDP growth of the domestic economy: foreign tourism spending, including air travel (counted as services exports in the national accounts); domestic tourism spending (counted as household services consumption); security/public administration spending (mostly government consumption) and infrastructure spending (investment). The economic literature on sporting events points to crowding-out effects in the tourism industry partially offsetting the positive spending impulse: some regular tourists typically avoid host cities, while some locals prefer to stay at home. This substitution effect is typically large for domestic tourism spending. Previous World Cup tend to confirm this: South Africa's only identified about 310,000 foreign tourists who came specifically for the World Cup (vs an annual baseline of approximately 8mn visitors). Russia 2018 tells a similar story: Total tourist volume was similar to the prior year. The tournament shifted the composition of visitors from neighboring countries toward more distant origins; it did not expand the total. "Operations" spending (venue management, broadcasting support, hospitality services, temporary workers etc.) is not a separate GDP component: it ultimately flows in visitors spending (therefore included into household consumption and services exports), security and public services (part of government consumption) and broadcasting services (services exports). We do not count infrastructure spending which has already occurred and was accounted into GDP mostly in prior quarters and years, as our focus is on the June-July period¹. Marketing/broadcasting rights and tickets purchased accruing to Switzerland-based FIFA would not affect GDP of the host countries, but the transfer of income towards Switzerland would be negative for the current account of host countries and create a wedge between GDP (total spending in the domestic economy) and GNI (Gross National Income, i.e. GDP adjusted by the flows of income with foreign countries).

Tourism spending itself is expected to increase by USD8bn in North America, split between USD5.4bn in the US, USD1.4bn in Mexico and USD1.2bn in Canada, largely dominated by foreign spending. Tourism spending is the product of the number of visitors, their average stay and their daily expenditures (on food, local transportation, accommodation, retail purchases etc). The unprecedented scale of the 2026 World Cup should increase the number of visitors and/or the length of the average stay. Table 2A and 2B depicts our forecast of foreign and domestic tourism spending across the three countries, laying out our assumptions on the number of visitors, the average stay, daily expenditure, air travel expenses (accruing to local airlines) and displaced tourism. On average, we expect foreign visitors to spend USD350 per day in the US, USD280 in Canada, and USD180 in Mexico, with daily spending by local visitors smaller. The average stay is expected to be 8-10 days for international visitors and 3-4 for domestic tourists. Air travel accruing to domestic haulers (counted as services exports) is estimated at USD0.8bn for US airlines and USD0.2bn for Mexican and Canadian airlines. Latest estimates suggest that the six-week tournament will attract around 6.5mn visitors over the course of the competition (including 3.8mn in the US), split between 4mn local attendees and 2.6mn international visitors. In all, the aggregate tourism spending (international + domestic) would amount to USD8bn across North America, split between USD6.8bn in tourism exports (foreign tourism spending) and USD1.2bn in household consumption (domestic tourism spending). These numbers are net of displaced tourism spending (crowding out), which are sizeable for domestic tourism. In the US alone, the net total boost of spending is expected to be USD5.4bn (USD4.6bn for foreign tourism and USD0.8bn for domestic tourism), versus USD1.4bn in Mexico and USD1.2bn in Canada.

¹ In contrast, services spending such as flight ticket purchase is recorded into GDP when the flight takes place, not when the ticket is purchased. The impact of infrastructure spending on GDP is, moreover, harder to estimate because of potential positive spillovers effects magnifying the initial impact (fiscal multiplier above 1).

Table 2A: Foreign tourism spending and domestic tourism spending forecasts

Country	Foreign visitors (mn)	Average stay (days)	Daily expenditure (\$)	International air transport travel (\$bn)	Gross foreign spending (\$bn)	Displaced tourism (\$bn)	Net tourism export gain (\$bn)
United States	1.5	10	350	0.8	6.0	-1.4	4.6
Mexico	0.7	9	180	0.2	1.3	-0.1	1.2
Canada	0.4	8	280	0.2	1.1	-0.1	1.0
Total	2.6	—	—		8.4	-1.6	6.8

Table 2B: Domestic tourism spending and domestic tourism spending forecasts

Country	Domestic visitors (mn)	Average stay (days)	Daily expenditure (\$)	Gross domestic spending (\$bn)	Substitution / leakage (\$bn)	Net GDP effect (\$bn)
United States	2.3	4	250	2.3	-1.5	0.8
Mexico	1.0	3	120	0.4	-0.2	0.2
Canada	0.7	3	180	0.4	-0.2	0.2
Total	4.0	—	—	3.1	-1.9	1.2

Sources: FIFA, UN Tourism, various, Allianz Research.

Security operations spending will add to the economic boost. Governments will increase spending on police overtime, border controls, intelligence activities, cybersecurity, emergency medical services and venue security. Most of this spending is government consumption. In total, ballpark estimates put these to an aggregate amount of USD0.7bn in the US, USD0.3bn in Mexico and USD0.1bn in Canada – only accounting for the spending during the June-July period (earlier preparation spending could increase these figures by 20-30%).

In total, the FIFA World Cup is expected to boost GDP over June-July by USD6.1bn in the US (+0.1pp quarterly growth), USD1.7bn in Mexico (+0.3pp) and USD1.3bn in Canada (+0.2pp). Table 3 summarizes the channels of impacts and the total GDP effect across the three host countries for the June-July period. The foreign tourism channel is by far the largest, at USD6.8bn, with domestic tourism (USD1.2bn) and security spending (USD1.1bn) being close to each other. In total, US GDP is expected to receive a boost of USD6.1bn over June-July, Mexico USD1.7bn and Canada USD1.3bn. In terms of quarterly non-annualized GDP growth (mostly falling on Q3), this represents a boost of +0.1pp (+0.4pp annualized) above a counterfactual without FIFA World Cup in the US, +0.3pp (+1.2pp annualized) in Mexico and +0.2pp (+0.8pp annualized) in Canada.

Table 3: Summary of channels of impacts of FIFA World Cup on June-July 2026 GDP

Country	Net foreign tourism (\$bn)	Net domestic tourism (\$bn)	Security spending (\$bn)	Total tournament GDP impulse June-July (\$bn)	Quarterly GDP growth impact (p.p.)
United States	4.6	0.8	0.7	6.1	0.1
Mexico	1.2	0.2	0.3	1.7	0.3
Canada	1.0	0.2	0.1	1.3	0.2
Total	6.8	1.2	1.1	9.1	

Sources: FIFA, various, Allianz Research.

The US is expected to capture the majority of economic benefits simply because it hosts the majority of matches, though relative to its economy the effect will be small. The largest gains will likely be concentrated in transportation, hospitality, food services, entertainment and retail. Cities such as New York, Los Angeles, Dallas and Miami are particularly well positioned due to their combination of international connectivity, large hotel inventories and established tourism ecosystems. Corporate hospitality is also expected to play a major role, particularly for knockout-stage matches where premium seating and sponsorship packages command substantial prices.

Mexico's aggregate economic impact will be smaller in absolute terms but more meaningful relative to the size of its economy and of its tourism sector. In 2025, international overnight visitors nearly reached 48mn, i.e. +6% compared to 2024. Early 2026 indicators also point to continued strength, with international arrivals during the first quarter rising by a similar pace year-on-year and air passenger traffic from the US and Canada remaining on an upward trend. Given the close proximity of the three host countries, the tournament could also encourage significant "nearcation" travel, with fans opting for shorter trips within North America instead of long-haul destinations due to ongoing geopolitical tensions, disruptions to key international transport routes, higher transportation costs and security concerns affecting some long-distance destinations. From a macroeconomic perspective, the tournament is unlikely to have a transformative effect on growth given the size of the Mexican economy and the temporary nature of the event. However, it could contribute around 0.1-0.3pp to GDP growth in 2026 (i.e. between USD2bn and USD6bn) through higher tourism revenues, stronger private consumption and event-related investment.

Canada's gains will likely fall between those of the US and Mexico on a per-visitor basis but remain small in aggregate due to its limited number of host cities. With one of the smallest stadiums (BMO Field in Toronto), only 13 matches out of 104 will take place in Canada, with Toronto and Vancouver representing the country's sole host cities for the tournament. The country's strength lies in attracting relatively affluent travelers willing to pay premium prices for accommodation, transportation and hospitality experiences. Consequently, although Canada hosts only 13% of matches, spending per visitor may compare favorably with other markets.

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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Emerging markets in a fragmented world: From geography to resilience - the 4Rs framework

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In Summary

- **The Iran war marked the first major oil shock that did not trigger a broad emerging markets sell-off.** Markets repriced countries based on strengths and weaknesses rather than the traditional EM–DM divide. This supports a resilience-based investment framework built around the “4Rs” – Resource Position, Reserve Strength, Rate Credibility and Refinancing Structure – which increasingly explains cross-country differentiation more effectively. Although institutional mandates, benchmarks and trading-desk structures will continue to rely on the EM–DM distinction for the foreseeable future, portfolio construction frameworks that lean primarily on this historical classification risk becoming progressively less relevant.
- **Resource Position, not the EM–DM label, defines the fault line of the Iran shock.** Economies with large import dependencies such as Egypt, Romania, South Korea, Greece and the UK, have faced the strongest repricing pressures, while commodity exporters have benefited from improved terms of trade. Even in a downside scenario with oil prices above USD180/bbl, the pain would be concentrated within the energy-importing cohort.
- **Reserve Strength increasingly separates resilient sovereigns from vulnerable “triple-deficit” economies, irrespective of EM or DM classification.** Since the 2013 taper tantrum, many EMs have rebuilt fiscal discipline, strengthened current-account positions and stabilized debt trajectories, entering the Iran shock with roughly 1pp of GDP more fiscal headroom than at the onset of Covid-19. EM economies now account for roughly 60% of global GDP in PPP terms, up from around 40% in 2000. FX reserve buffers have continued to strengthen across the Middle East, Central Asia and Emerging Europe, while several advanced economies remain mired in persistent fiscal deficits and deteriorating external balances.
- **Rate Credibility has structurally improved across EMs and increasingly resembles DM-style monetary frameworks.** Inflation targeting is now the norm across most major EMs, and EM central banks tightened by an average of 780bps during the post-pandemic cycle versus around 400bps in the DM, frontloading hikes despite weaker growth. The Iran war has confirmed this convergence: no major EM central bank has been forced into emergency hikes, capital controls or disorderly stabilization. EM FX volatility has fallen materially, with several G10 currencies experiencing higher volatility during 2024–25. Outliers, notably Türkiye, Argentina and Nigeria, remain, but the broader convergence trend is intact.
- **Refinancing Structure has fundamentally changed the transmission of external shocks.** Foreign-currency debt shares have fallen by roughly 20–40pps across major countries including Brazil, Mexico, India, Indonesia and several CEE. Deeper domestic institutional investor bases and larger local-currency bond markets have reduced vulnerability to USD and Fed, turning FX depreciation into a macroeconomic adjustment mechanism rather than a solvency trigger. Currently, EM currencies adjusted in an orderly manner with no widespread defaults or emergency IMF interventions; even in a downside scenario of further Fed and ECB tightening, the likely outcome is slower convergence rather than a return to a crisis like the 1990s or early 2010s.
- **The compression of EM risk premia increasingly reflects structural convergence rather than cyclical overvaluation.** The excess spread of EM hard-currency debt over comparable DM credit has largely disappeared on a rating-adjusted basis. Even in a downside Iran escalation scenario, EM hard-currency spreads would likely widen from around 178bps to approximately 235bps and peak near 280bps, materially below the roughly 700bps reached during the Covid-19 shock. A compelling long-term case increasingly lies in EM local-currency debt, where structurally higher real yields continue to generate superior long-term risk-adjusted returns relative to DM fixed income.

The line between EMs and DMs is blurring

Emerging vs. developed is not a label – it is an investment hypothesis about institutions, policy credibility and the distribution of tail risks that is being tested by the fragmentation of the global economy. The traditional EM–DM split has been a shorthand for differences in the strength of institutions, the predictability of policymaking and the degree to which investors can trust rules to be stable over time. These institutional features matter because they shape inflation outcomes, fiscal discipline, market depth and, ultimately, the risk premium investors demand. Importantly, recent market experience has also highlighted that DM does not automatically mean risk-free: volatility in some developed government bond markets has risen materially since 2022, and fiscal uncertainty has become a more visible driver of term premia and curve behavior. In parallel, a structural fragmentation of the global economy, the US-China decoupling, the partial US retreat from its role as anchor of the rules-based order, friendshoring, supply-chain regionalization and growing regulatory divergence across blocs, has begun to redistribute risk away from the EM-DM axis and toward country-specific positioning within a contested international system. The two forces compound each other: DM credibility is eroding from within just as the global architecture that historically rewarded DM positioning is being rewritten from the outside.

Introducing the 4R framework: a resilience-based alternative to EM–DM

Throughout this paper we organize the analysis around four structural dimensions that explain cross-country market segmentation better than the EM–DM taxonomy:

- Resource Position: energy dependence, commodity exposure, vulnerability to terms-of-trade shocks such as the Iran war
- Reserve Strength: fiscal discipline, external balances, FX reserves, exposure to twin or triple deficits
- Rate Credibility: central bank independence, inflation-targeting frameworks, convergence of monetary policy reaction functions
- Refinancing Structure: local-currency funding shares, depth of domestic investor bases, resilience to USD and Fed cycles

The subsequent analysis maps the evidence onto these four dimensions of the “4Rs” framework.

A large and growing share of EM debt trades on fundamentals that increasingly resemble DM-style underwriting like credible central banks, improving external buffers and solid policy frameworks while EM risk dynamics improve. Parts of the DM universe display EM-like stress points through political constraints and fiscal slippage. The fragmenting global economic system reinforces this EM-DM convergence, as several EMs are positioned to benefit from ongoing supply-chain and trade realignments while parts of the DMs see their competitive positions deteriorating. In parallel, dispersion across countries remains high, meaning “EM beta” is less informative than identifying where fundamentals warrant investment-grade-like pricing and where high-yield-like tail risks still dominate.

A conceptual difference appears in EM sovereign debt. EM hard currency government bonds are, by construction, a spread asset because default/restructuring risk is non-zero and must be priced. DM government bonds, in contrast, are – just like EM local currency sovereign bonds – typically treated as the benchmark “risk-free” curve in their own currency. A significant change in a country’s economic and political outlook is first and foremost digested on FX markets. Investors should recognize that term premia can still move meaningfully with inflation uncertainty, fiscal outlook and market functioning, even absent classical default risk. For portfolio construction, the practical takeaway is that investors should separate spread risk from rate, term-premium and FX risk, and to avoid relying on labels when the underlying risk drivers have shifted.

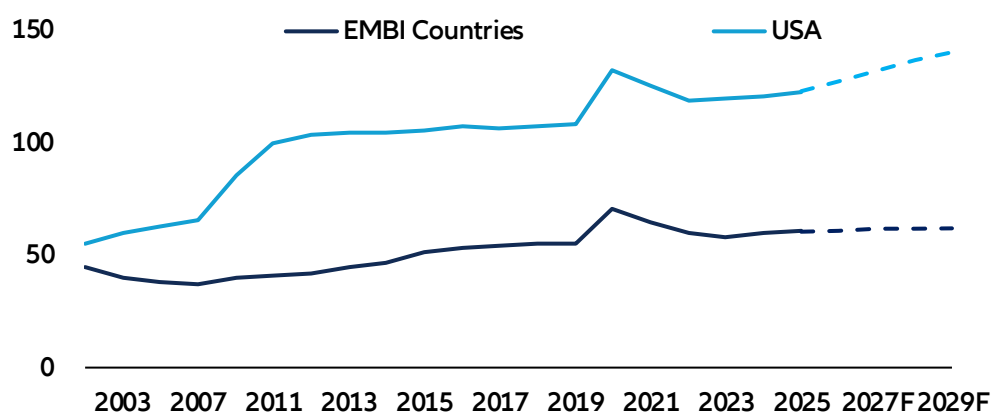
Improved fundamentals: fiscal discipline, external position

This section examines the first two Rs — Resource Position and Reserve Strength. The current shock is the latest data point in a deeper structural shift: the distinction between developed and emerging markets is blurring. Political instability, rising populism and unsustainable public finances were traditionally associated with EM

countries, but they have become descriptions for some of the world's largest and most 'developed' economies in recent years. While each country's political backstory is unique, the common theme is a pronounced deterioration of macroeconomic fundamentals – a dynamic that global fragmentation increasingly reinforces as weakening institutions and fracturing trade relationships feed on each other in a self-reinforcing spiral. Today, glaring fiscal imbalances in the US and other 'advanced' economies point to an inversion of the traditional world order, with little prospect of a meaningful reversal.

In strong contrast to their DM peers, EMs have seen strengthening fundamentals over the past decade. Spurred by the upheaval of the 2013 taper tantrum, which exposed underlying economic imbalances, many EMs have undergone significant rebalancing, strengthening their resilience. Even after the Covid-19 pandemic took hold, many EM policymakers remained fiscally prudent, resulting in primary fiscal balances returning to surplus within just a few years and debt-to-GDP stabilizing at modest levels, improved external positions across EM sovereigns and stronger growth outlooks for EMs compared to DM peers.

Figure 1: General government gross debt, EMBI countries and US, 2003–2029F (% of GDP)

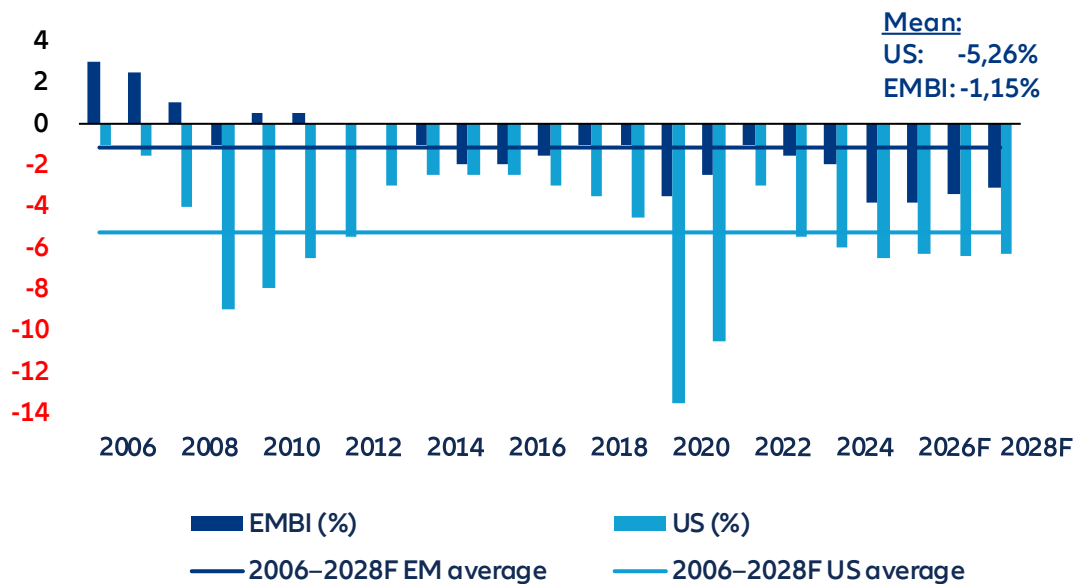


Note: Dashed lines indicate IMF/Bloomberg forecasts.

Sources: IMF, Bloomberg (forecasts), Allianz Research

The Iran war is a live test of Reserve Strength. The early evidence suggests EMs have entered the shock with more headroom (1pp of GDP) than the Covid-19 shock overall. However, this aggregate masks a wide dispersion in fiscal positions, from those that experienced forced reforms in recent years due to shocks and cyclical beneficiaries to those deteriorating. The successive shocks of Covid-19, the Ukraine war and the 2022–23 inflation surge functioned as a stress test that catalyzed reform in two distinct cohorts of improvers. Pakistan, Ghana and Romania were forced into adjustment by near-default conditions and IMF programs, with Argentina also in this group given Milei's dramatic primary surplus delivery in 2024–25. Colombia, Brazil and to a more ambiguous extent Mexico are in the second cohort, having improved through commodity tailwinds and limited tariff exposure. On the deteriorating side, Indonesia, Thailand, Bangladesh, Hungary and Nigeria have each seen pronounced erosion since 2020, driven respectively by subsidy expansion, fiscal stimulus, political transition costs, pre-election spending and oil-revenue volatility.

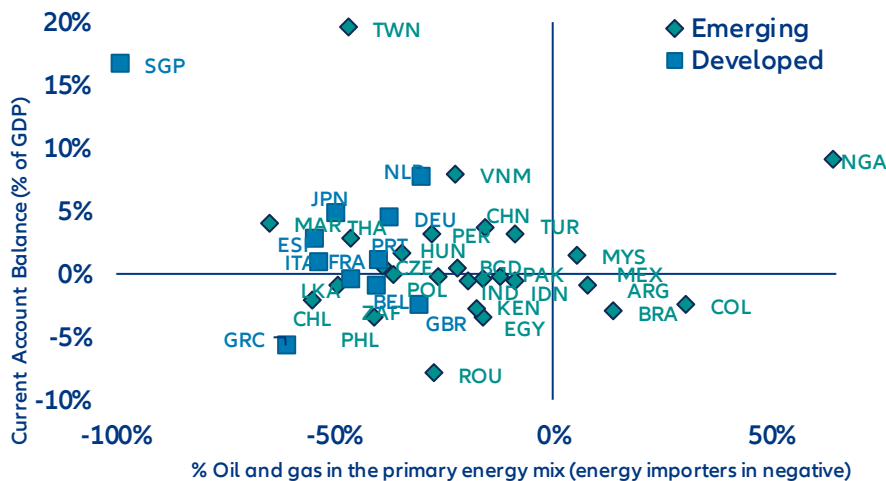
Figure 2: General government fiscal balance, EMBI countries and US, 2006–2028F (% of GDP)



Note: Horizontal lines show period averages (2006–2028F).
Sources: IMF, Bloomberg (forecasts), Allianz Research

The Iran war shock has revived the severity of global imbalances, but the differentiation it has produced is **not** between EM and DM – it runs along Resource Position, separating energy importers from exporters. In prior decades, an external shock of this magnitude would have triggered a near-mechanical sell-off of EM assets against developed-market safe havens, with the EM-DM dividing line acting as the primary axis of risk repricing. The 2026 episode has yet to produce such pattern. Those countries more exposed to the ongoing war are those EMs and DMs with current account deficits and large imports of hydrocarbons, which include countries such as Romania, Egypt, as well as the UK or Greece. If the conflict escalates further, this distinction would sharpen rather than dissolve: in a downside scenario of sustained supply shortages and oil prices spiking above USD180 per barrel, averaging close to USD100 for 2026, the most acute repricing would fall on this energy-importing, current-account-deficit cohort rather than on EMs as a whole. The remaining group would still experience a sell-off, but a materially reduced one, consistent with the broader resilience indicators discussed throughout this section. Even in a severe escalation, in other words, the dividing line would run through energy exposure, not through the conventional EM-DM taxonomy.

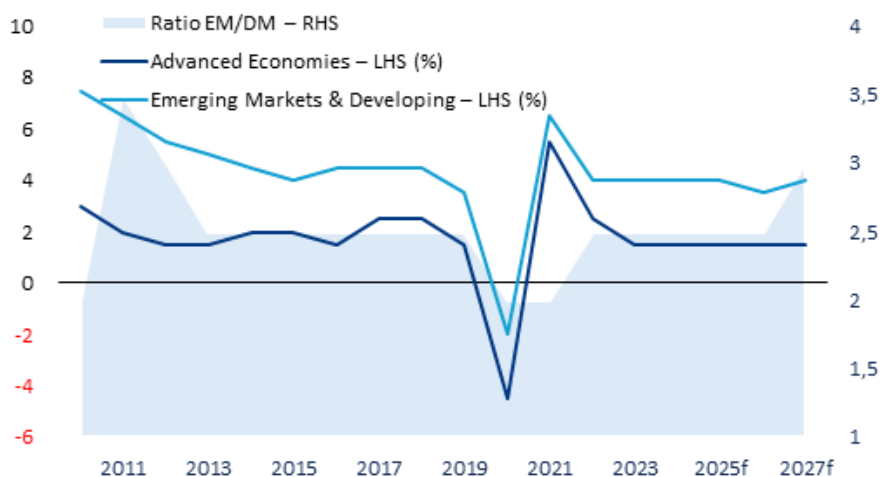
Figure 3: Heterogenous current account imbalances, but EMs better positioned with regards to energy independence than DMs



Sources: LSEG, Allianz Research

EM growth continues to outperform that of DMs, and the relative growth differentials are expected to widen further, continuing to reweight EMs at the center of the global economy. Emerging and developing economies now account for roughly 60% of global GDP in purchasing-power-parity terms, up from approximately 40% in 2000, a 20pps reallocation of global economic weight. The growth trajectory indicates a widening distance between EMs and DMs, causing a further acceleration of the EM catch up towards DM purchasing power levels. In addition, the Iran war shock has not disrupted this trajectory. Hence, the center of gravity of the global economy will continue to move towards the EM bloc in the years to come.

Figure 4: GDP growth, advanced economies and emerging markets & developing markets, 2011–2027F (% y/y)



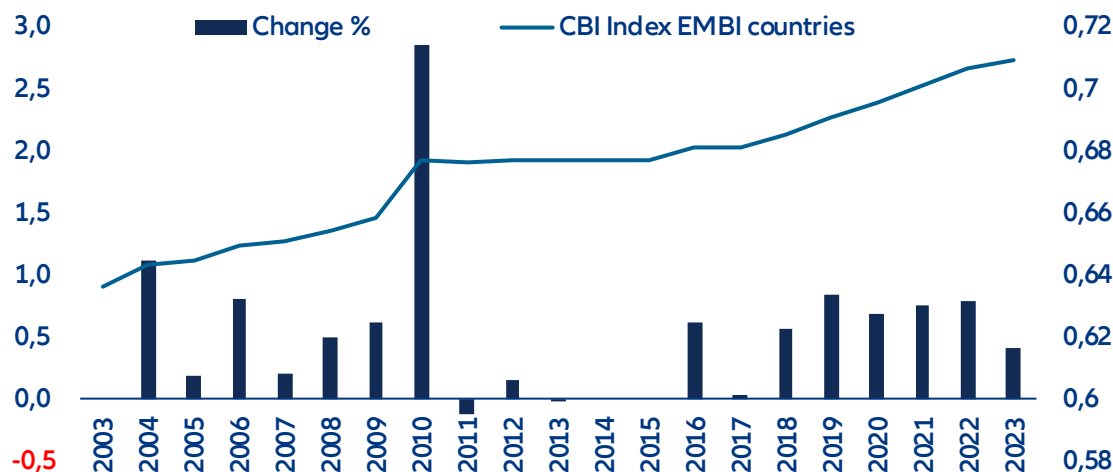
Note: . Shaded area shows the EM/DM growth ratio (RHS).

Sources: IMF, Bloomberg (forecasts), Allianz Research

Monetary policy convergence: evidence and implications

The third R – Rate Credibility – captures three decades of EM progress on central bank independence and policy frameworks, accelerated by the GFC and pandemic shocks, although regions and income levels diverge. As shown in Figure 5, the most pronounced increase in central bank independence among EMBI countries occurred in the aftermath of the GFC, and while the index plateaued temporarily in the subsequent years, it resumed a continuous upward trajectory from 2016 onward. The upper-middle-income group (which captures most of the "core" emerging markets: Brazil, Mexico, South Africa, Malaysia, Thailand etc.) shows the most consistent upward trend, narrowing the gap with high-income countries throughout the 1990s–2010s. While Latin America remains ahead and the closest to DM convergence, MENA (with the exception of Israel) remains the laggard region. Eastern and Central Europe experienced a dramatic catch-up wave tied to EU accession. While low-income and low-middle income countries have experienced the quickest convergence towards DM standards in the last 15 years, the reforms undertaken by West Africa Economic and Monetary Union (WAEMU) countries sharply improved Africa's convergence, though it remains uneven. These reforms include central bank mandates covering everything from board composition, governor appointment rules and limits on lending to the government to financial and budgetary independence, reflecting a structural rather in how EM central banks are designed to operate. Overall convergence is being driven by regional integration (experienced in the EU and WAEMU), external inducements especially tied to IMF-induced reforms, given the conditionality of IMF programs on central bank independence, as well as crisis and shocks that trigger reforms.

Figure 5: Central Bank Independence (CBI) Index for Emerging Market Bond Index (EMBI) countries



Note: The line depicts the Central Bank Independence (CBI) Index, a weighted composite of six sub-dimensions (board independence, policy autonomy, policy objectives, lending restrictions, financial independence, and reporting requirements), scaled from 0 (no independence) to 1 (full independence). Bars show the year-on-year change in percentage points.

Source: CBI Database

Inflation targeting frameworks are the new normal. Twenty years ago, most EM central banks managed exchange rates or monetary aggregates. Today, South Africa, Brazil, Mexico, Chile, Colombia, India, Indonesia, Philippines, Thailand, Türkiye (nominally), Egypt (post-2024 reform), Ghana, Kenya and others run explicit inflation-targeting regimes with published targets, forecasts and forward guidance. The South Africa Reserve Bank is the latest to shift from a 3–6% range to a 3% point target, in itself a convergence move towards DM-style inflation anchoring. EM central banks tightened by an average of 780bps during the post-pandemic cycle, versus around 400bps in advanced economies, a decisive demonstration of improved policy credibility.

The Iran war also serves as a live test of increased EM central bank credibility and EM-DM monetary policy convergence. So far, no major EM central bank has been forced into emergency hikes, capital controls or disorderly stabilization measures, a significant break from the historical pattern of oil shocks triggering balance-of-payments crises across EM economies. The differentiation once again runs through fundamentals rather than the EM-DM label. In a downside scenario of further escalation, this resilience would be tested but not overturned: sustained energy-price pressure would likely prompt the Fed to deliver two further 25bp hikes and the ECB three, tightening global liquidity and raising refinancing costs for sovereigns with larger hard-currency debt loads. Crucially, even this tightening would be milder than the cumulative policy response seen during the Covid-19 and Ukraine-war inflation episodes, and its impact would be more contained. Hard-currency debt now represents a materially smaller share of total EM sovereign debt than in previous shock episodes, limiting the mechanical transmission of tighter global liquidity into EM balance sheets. In other words, the downside scenario would slow the convergence but not reverse it.

Increased orthodox monetary policy has reduced the inflation risk premium embedded in currencies, reducing volatility self-reinforcing the investor base. As inflation-targeting regimes have matured and central-bank independence has been entrenched, the inflation risk premium embedded in EM exchange rates has fallen. Lower pass-through reflects anchored inflation expectations, which in turn reduce the volatility and tail-risk character that historically defined EMFX as an asset class. During 2024/25, EMFX volatility was overtaken by G10 FX moves due to several parallel moves: the yen's performance, Trump's tariffs announcements and EM strength. The 2026 Iran war episode has provided new evidence: EM currencies have adjusted to the oil shock through orderly depreciation rather than crisis-style breaks, EMBI+ spreads have remained contained and no EM central bank has been forced into emergency hikes or capital controls, a stark contrast to the chaotic EM responses of 1973 or 1997.

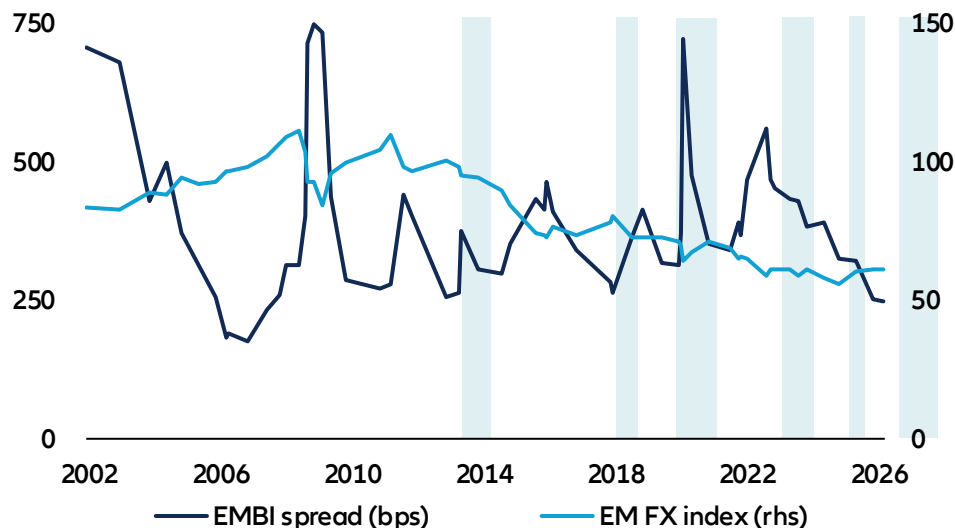
While the convergence has been taking place, a gap remains between EMs and DMs due to more government interference in key countries, market perception, as well as several outliers undermining the average. The convergence thesis is supported by the median EM central bank, but relevant central banks of the EM asset class, such as China or India, operate with higher government influence or within distinct frameworks. Meanwhile, a

meaningful tail of outliers continues to undermine the unweighted average, such as Türkiye (the 2021–23 unorthodox episode under political pressure, with credibility now being rebuilt), Argentina (whose monetary regime debate goes beyond central bank framework questions), and Nigeria (where multiple FX windows and "stealth tightening" remain in place despite Governor Cardoso's reform agenda). Egypt and Pakistan are mid-transition cases where IMF programs have catalyzed reform but track records are short. Notably, Brazil, often cited as a test case given the government pressures on the central bank, has emerged as a convergence success as political pressure did not prevent the September 2024 reversal back into hikes.

Shock absorption and resilience in EMs

This section turns to the fourth R – Refinancing Structure – alongside the broader shock-absorption toolkit. The structural fragility that has historically defined EMs, including exposure to external shocks, dollar funding and commodity volatility, has paradoxically become the foundation of their resilience. Repeated stress events over the past two decades forced EMs to adopt institutional and structural reforms and the evidence is visible in market behavior itself. EM hard-currency (HC) spread widenings firmly recovered after each crisis and show a level of resilience since the global rate hiking cycle triggered a sell-off in 2021–2022, which would have seemed unthinkable just a few years earlier (Figure 6). Even more astonishing is the stability in EM currencies. Previous crises like the GFC in 2008–2009 or the EM sell-off in 2015 triggered major losses in EM currencies, but since then EM currencies show a remarkable stability, as moderate FX losses typically fall short of the EM excess carry. The 2018 EM sell-off, the 2020 Covid-19 shock, the 2022 Ukraine-driven inflation surge and the 2026 Iran war have each tested and reinforced the asset class. Fundamental strengthening as well as larger and deeper shock absorbers like FX reserves levels, higher FX flexibility to absorb and lower level of HC debt are decreasing external vulnerabilities linked to the USD and the Fed.

Figure 6: EM hard-currency spreads and FX as stress indicators and recovery dynamics across major crisis episodes, 2002–2026



Note: The time series approximate the development of EM hard-currency spreads (JP Morgan's flagship EMBI index) and EM FX (JP Morgan's EM currency basket EMCI) around major crisis affecting EM debt, namely the Global Financial Crisis 2008-2009, taper tantrum 2013, the EM sell-offs in 2015 and 2018, Covid-19 in 2020 as well as the global rate hiking cycle in 2021-2022. Intervals between major crisis events are smoothed for readability.

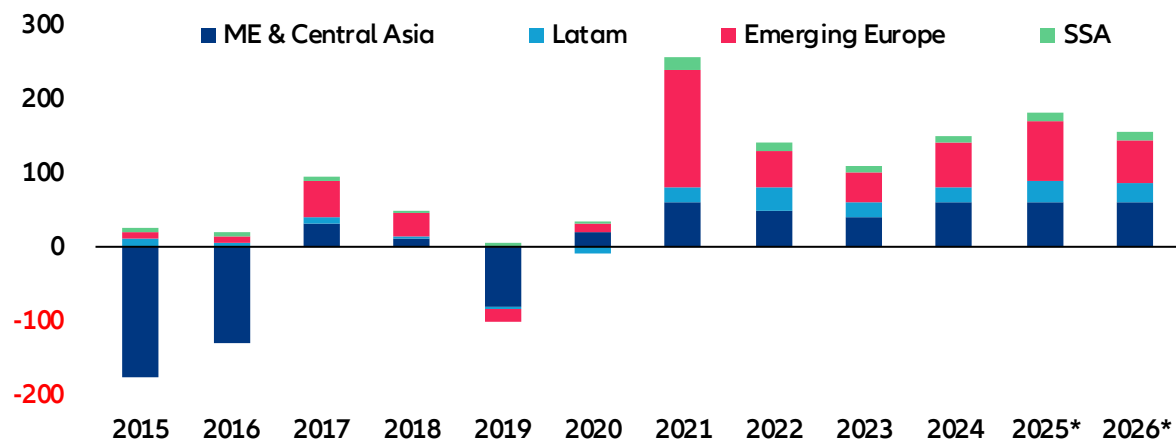
Sources: JPMorgan, Allianz Research

EM FX reserves have proven remarkably resilient over the long term, despite repeated bouts of external stress.

As shown in Figure 7, aggregate reserve accumulation across EM regions turned sharply positive again after the pandemic-driven drawdown in 2019 and 2020, with the Middle East, Central Asia and Emerging Europe contributing to the bulk of inflows since 2021. While the pace of accumulation has moderated from its post-

pandemic peak, the structural trend remains constructive, with forecasts for 2025 and 2026 pointing to continued reserve building across most regions. Sub-Saharan Africa and Latam remain the most vulnerable in terms of reserve adequacy, but even here the direction of travel has improved relative to the acute stress periods of 2015 to 2016 and 2019. Taken together, the data suggests that EM reserve buffers are broadly stronger today than a decade ago, providing a meaningful cushion against external shocks, including the kind of energy-price and financing-condition volatility associated with the current geopolitical environment.

Figure 7: EM FX reserves by region, 2015–2026F (USD bn)

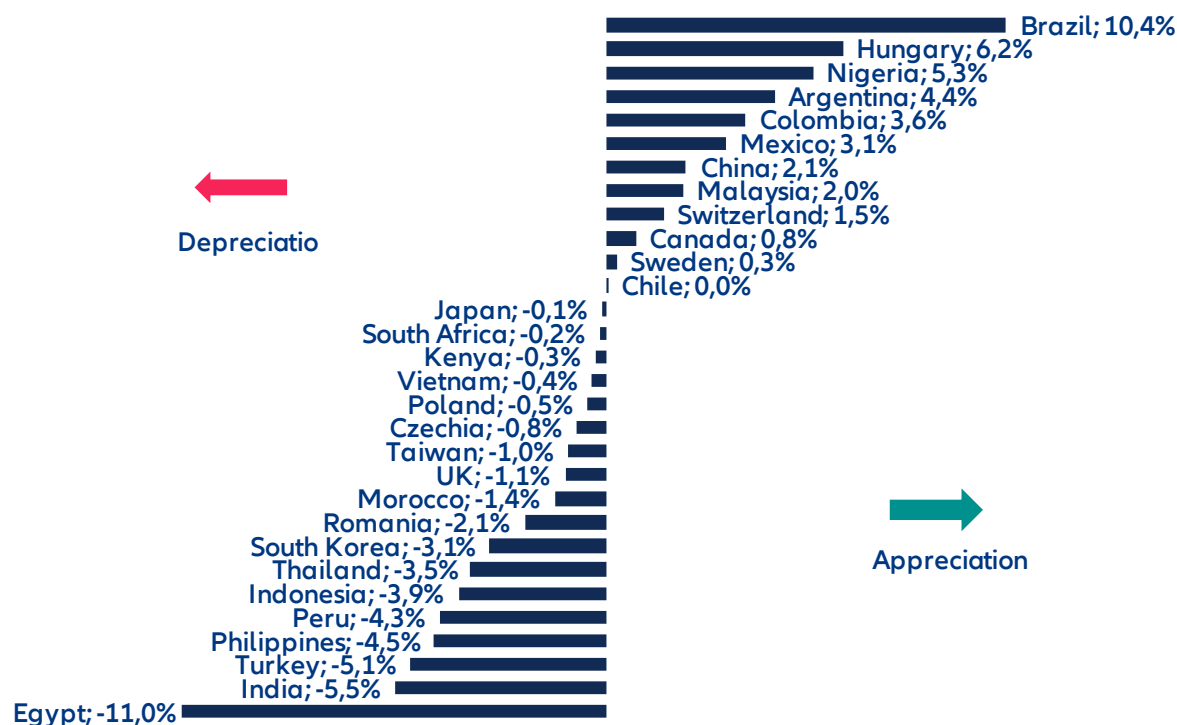


Note: as of April 2025, *indicates forecasts.

Source IMF, Bloomberg (forecasts), Allianz Research

Beyond reserve buffers, the broader shift toward flexible exchange rate regimes has been the second pillar of EM shock absorption, allowing currencies to function as adjustment mechanisms rather than crisis triggers as they often did in prior decades. The ongoing energy-linked shock is a great example. The dividing line between energy importers vs exporters, and exposure to Middle Eastern hydrocarbons, holds true in the currency space as well. The most exposed Asian countries have led the depreciation cycle since the start of the war, given their heavy exposure to Gulf energy imports, among them the South Korean won, together with other energy-importing countries such as Romania, Morocco or Türkiye. On the opposite side, energy exporters, many in the LATAM region, have benefited the most from the shock. Two notable outliers complete the picture: Hungary, where the post-election political transition has overridden the energy-trade signal, and China, whose low oil-import intensity and continued access to discounted Russian energy have largely insulated it from the shock. The Egyptian pound stands out as one of the worst-performing LCU, where pre-existing external imbalances have been compounded by geographic proximity to the conflict. Egypt is, however, also illustrative of a broader structural change: it is one of a cluster of historically crisis-prone EMs – alongside Ghana, Nigeria, Pakistan, Ethiopia and Sri Lanka – that have shifted from managed pegs and multiple exchange-rate windows toward market-determined floats since 2022, in most cases under IMF programs. This matters for the downside scenario. In previous shock episodes, these economies typically defended overvalued pegs until reserves were exhausted, at which point the currency broke and a balance-of-payments crisis followed. With more flexible regimes now in place, a further escalation of the energy shock would still produce a sharp depreciation in this cohort, but as a gradual adjustment rather than a sudden-stop rupture, with the currency absorbing the shock incrementally instead of triggering it. The contrast with the 1997 Asian crisis or Egypt's own pre-2024 experience is the clearest illustration of how regime choice has changed the character of EM shock transmission.

Figure 8: Currency performance 2026 YTD



Sources: LSEG, Allianz Research

Refinancing Structure has been transformed: EM sovereigns have shifted their funding mix from hard currency to local currency over the past two decades, with foreign-currency debt shares falling by 20–40pps across leading EMs. Brazil, Mexico, India, Indonesia, Thailand and the major CEE sovereigns now fund themselves predominantly in domestic currency, supported by deeper domestic institutional investor bases and growing foreign participation in local currency bond markets. The implication is structural: a stronger dollar no longer inflates EM debt burdens to the same extent as it did in 1997, 2013, or still in 2018. Local currency depreciation has become an adjustment mechanism rather than a solvency event which is the single most important reason the Iran war shock has so far been absorbed without disorderly defaults, capital controls or IMF emergency programs.

However, the USD continues to be a strong external influence on EM moves, causing central banks to be ahead of the Fed. Given that the USD anticipates Fed expectations, EM central banks across the board track the Fed. During the 2021–22 inflation surge, EM central banks in Brazil, Mexico, Chile, Czech Republic, Hungary and Poland tightened ahead of the Fed. That is a complete reversal of the historical pattern where EMs reacted defensively to Fed moves. In the subsequent easing cycle, they started to ease only after the Fed's easing cycle began in 2024. During the latest Iran war response, the monetary authorities of South Africa, Egypt and Israel all framed their pauses to monetary easing identically, referring to "inflation expectations anchoring" and "data dependence". No EM central bank has hiked aggressively nor imposed capital controls, in contrast with the 1973 or 1979 oil shocks, when EM responses were disordered and followed devaluations and defaults. This demonstrates the transition towards a converged reaction function: EM central banks now follow their own script anticipating the Fed rather than just acting in reaction to its moves.

What does this mean for EM debt investors?

The 4Rs framework has clear implications for portfolio construction, but the EM–DM taxonomy will not disappear from the investment universe overnight. Benchmark providers, institutional mandates and asset-manager desk structures (EM versus DM credit, EM versus DM rates) remain organized around the historical classification, and these institutional features will continue to shape how capital is allocated for the foreseeable future. With that caveat in mind, improving monetary and fiscal credibility together with an increasing capacity to absorb shocks – captured across all four Rs – reduces the downside of EM debt investments and changes their role in institutional portfolios. These structural improvements carry direct implications for portfolio construction. For risk-

constrained investors – pension funds, insurers, sovereign wealth funds – the historically cautious stance toward EM debt was grounded in real volatility: the "Fragile Five" of 2013, the "Troubled Ten" that followed, serial default cycles and the chronic vulnerability to Fed tightening. That era reflected genuine fragility. But as fiscal credibility improves, monetary frameworks mature and shock-absorption capacity deepens, the risk premium embedded in EM debt increasingly overstates actual portfolio risk. The practical consequence is meaningful: lower volatility and reduced tail risk justify higher EM debt allocations within existing risk budgets, without requiring investors to take on more aggregate portfolio risk. For liability-driven investors in particular, this is not a marginal shift – it represents a structural re-rating of an entire asset class, and portfolios that fail to reflect this new reality may be leaving risk-adjusted return on the table.

Despite the long-term improvement in EM fundamentals and the convergence with DMs, investors in EM hard-currency debt were historically paid a significant premium for "EM risk" over "DM risk". On a like-for-like basis across rating classes, EM hard-currency debt traded at a meaningful spread over comparable developed market bonds. This excess reflected genuine incremental risks, including thinner liquidity, greater sensitivity to dollar strength and commodity cycles, political instability, currency mismatch on sovereign balance sheets and a structurally narrower investor base that demanded compensation for volatility and sudden-stop risk. History vindicated some of that caution. The Tequila, Asian and Russian crises of the 1990s, Argentina's repeated defaults and the 2013 taper tantrum all demonstrated that EM spreads could widen sharply in ways investment-grade DM bonds rarely did. More recently, as convergence has progressed and EM fundamentals have matured, this structural risk premium has compressed (Figure 9). The excess spread of EM over DM credit has now largely disappeared, and relative valuations are broadly fair, with little incremental compensation remaining for investors crossing the EM/DM boundary. The compression of the premium does not, however, imply the disappearance of cyclical risk. In a downside scenario of further escalation in the Iran conflict, we assess that the average EM hard-currency spread would widen from its pre-crisis level of 178bps to around 235bps over 2026, with a peak of close to 280bps during the year. The scale of this move would be modest by historical standards. During the Covid-19 shock, the EMBI Global Diversified spread widened from roughly 300bps to a peak of around 700bps in March 2020, more than doubling in the space of weeks, before retracing over the following months. The downside repricing we envisage for the Iran scenario, by contrast, would represent a widening of around 100bps at its peak. This more contained response suggests that convergence has altered the nature of EM spread risk, rendering it more cyclical and less crisis-prone, without eliminating it entirely.

Figure 9: EM sovereign over US credit pick-up on a rating-adjusted basis



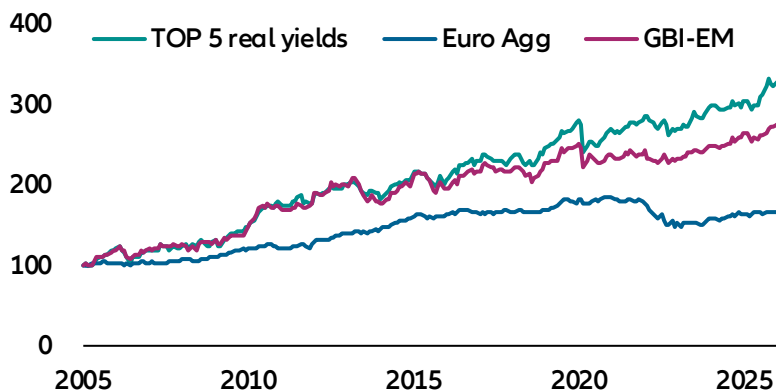
Notes: To calculate the pick-up Emerging Markets hard currency debt investors get for "EM risk" over "DM risk", we compare the spread over US Treasuries for EM Sovereign bonds (JP Morgan EMBIG) with the option-adjusted spread for US Credit (Bloomberg) per rating bucket (AA, A, BBB, BB, B). The pick-up reflects the weighted average of the spread difference based on rating bucket weights in the EMBIG.

Sources: JP Morgan, Allianz Research

An EM risk premium can still be earned when we move our focus from credit to rates. EM local currency bonds offer much higher nominal yields compared to DM bonds. From a DM-based investor's point of view, we have to take FX risk into account. There is a long history of EM currency depreciation against major DM currencies that exceeds 2% p.a. in a long-term average. Nevertheless, exposure to EM local currencies pays off as local excess carry exceeds FX losses in the long-term. The root cause of higher EM performance is the real yield pick-up. After accounting for the inflation embedded in higher nominal EM rates, EM local debt has historically offered real yields

that are substantially higher than what investors can earn in DM markets, where demographics, productivity slowdowns and central bank intervention have compressed real rates. This matters because real yields are not merely an academic construct: empirically, they are among the strongest predictors of long-term fixed income returns, capturing the true compensation investors receive for bearing interest rate risk. For a DM investor, that persistent real yield advantage is the genuine investment case for EM local currency debt. Not only did EM local currency bonds significantly outperform a Euro Aggregate investment over the past 20 years for a EUR-based investors; selecting the highest real yielders only further boosted EM local currency outperformance (Figure 10).

Figure 10: EM local currency debt outperforms DM fixed income for EUR-based investors, higher real rates are the major driver of the outperformance

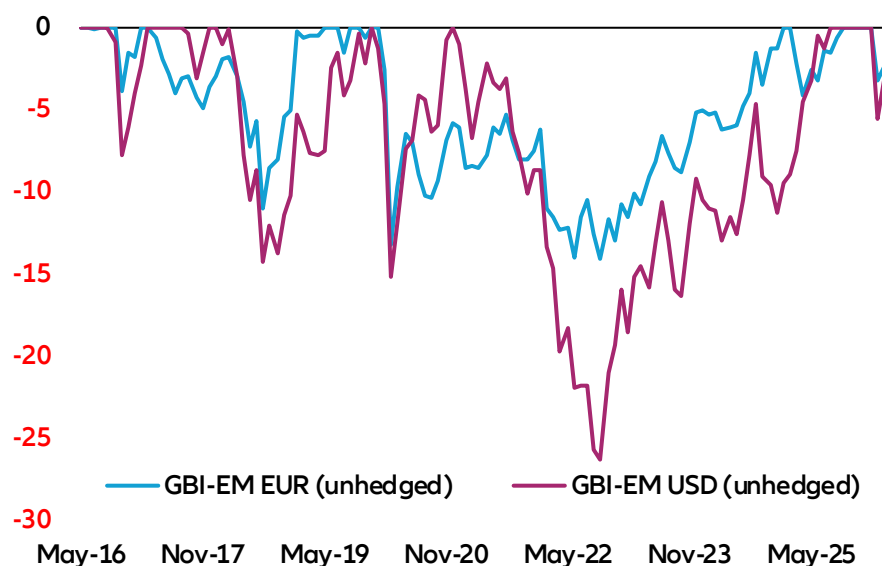


Notes: To analyze the importance of the real yield, we have tested a trading strategy that invests equal-weighted into five issuers offering the highest expected real yields of the GBI-EM investment universe, JP Morgans flagship EM local currency index. At each point in time, the expected real yield is derived by subtracting the inflation forecasts over the next 5 years from the 5yr nominal bond yield.

Sources: LSEG Workspace, Oxford Economics, Allianz Investment Management

While EM local currency FX clearly adds volatility to DM investors, EUR-based investors benefit from mitigation by their home currency. The carry of EM local currency cash is a strong performance contributor for DM-based investors, while FX is without doubt the strongest source of volatility. For a EUR-based investor, FX volatility looks meaningfully different and in some ways more favorable than it does through a USD lens. Because the USD functions as the world's reserve and safe-haven currency, other currencies tend to weaken against it in unison when global risk appetite deteriorates and investors seek shelter. The EUR is more closely tied to risk sentiment than to safe-haven dynamics, giving it a degree of procyclicality that it shares with many EM currencies. The practical implication is that a EUR-based investor holding EM local currency debt experiences meaningfully lower cross-currency volatility than a USD-based peer investor. When EM currencies weaken, the euro tends to weaken too, partially cushioning the FX impact. Figure 11 shows that the EUR's absence of safe-haven status becomes a structural advantage when accessing EM local debt: The investor's base currency aligns more naturally with the asset class, reducing FX volatility and drawdowns that make the same trade more costly for USD-based investors.

Figure 11: GBI-EM drawdown from peak measure in EUR vs. USD (%)



Notes: Drawdown measured as percentage decline from prior peak on a monthly total return basis. EUR unhedged reflects local currency returns converted to EUR without currency hedge. USD unhedged reflects the same index from a USD investor perspective.

Sources: JP Morgan Markets, Allianz Research

The strong progress in EM monetary policy credibility increases the diversification benefits of investing into EM local currency debt for DM investors. EM central banks have moved decisively away from the exchange rate stabilization bias that historically characterized their policy frameworks, adding on average 780bps to policy rates during the post-pandemic tightening cycle compared to just 400bps for advanced economies (IMF, 2024). Crucially, this tightening was front-loaded and aggressive even at the cost of short-term economic pain, signaling a genuine commitment to domestic inflation targeting over growth stabilization. For a globally diversified, DM-based fixed-income investor, a decent EM local currency allocation provided significant diversification benefit during the painful global rates repricing following the Covid-19 and Russia-Ukraine shocks: A sharp EM sell-off driven by rates and FX has been followed by a much faster recovery compared to home currency fixed-income investments.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Matryoshka market: Nested convictions in equities

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Summary

- **Geopolitical resolution is the master assumption.** Equity markets are pricing a framework agreement by September, consistent with the prediction market consensus and the pronounced kink in VIX term structure at Q3. That bet has logic: the global (political) cost of ongoing conflict and mid-term elections make a deal valuable to the US administration by autumn. But a genuine agreement requires Iran to provide credible security guarantees for the Strait of Hormuz. Tail and downside risks are visible in options but not in equity valuations. In our downside scenario, equities could correct by 25% and 20%, respectively, in Europe and the US. Indeed, if resolution fails and oil remains structurally elevated, every subsequent layer inherits a more adverse starting point: margins compress, central banks are constrained and the stagflation risk premium returns not as a tail scenario but as a working assumption.
- **The earnings story is concentrated exposure to some sector tailwinds, not broad-based resilience.** The S&P 500 has delivered double-digit EPS growth for six consecutive quarters, and European corporates surprised with Q1 growth of +11.5% y/y. Record margins should be read as limited remaining buffer. Beyond the headline figure, approximately 17pps of S&P 500 EPS growth in Q2 2026 originate from the IT sector alone – growing at an estimated +66% y/y and representing roughly 31% of total index earnings. Excluding technology, the rest of the S&P 500 expands at around +12%, in line with a 6% nominal GDP world. Europe tells a parallel story: exclude energy, the direct beneficiary of elevated crude, and STOXX 600 EPS growth falls from +11.5% to approximately +7%. Furthermore, a 20% equity correction would additionally impair US consumption by an estimated 1-1.5pp of GDP through the wealth-effect channel – a feedback loop markets are pricing as exogenous.
- **Non-cyclical index composition concentrates rate sensitivity rather than reducing it and the pending mega-IPO wave deepens the crowding.** The shift in major equity indices away from cyclicals toward technology, healthcare, utilities and high-dividend defensives is read as structural insulation. Non-cyclical sectors divide into bond-proxy profiles (utilities, healthcare) and long-duration growth profiles (large-cap technology): both are materially sensitive to real interest rates, through yield competition and discount rate mechanics respectively. The index has not removed rate risk, it has hidden it in less visible form. The pending mega-IPO wave deepens this structural vulnerability. SpaceX (USD1.75tn), OpenAI (USD850bn) and Anthropic (USD850bn) – combined USD3.45tn – would represent approximately 4-5% of S&P 500 market capitalization at index inclusion, making their combined entry the largest supply event in US equity market history. All three would land in or adjacent to technology, further crowding the most rate-sensitive index segment.
- **The real equity risk premium is a breakeven, not a buffer.** The negative nominal equity risk premium (ERP) is rationalized two ways: equities offer inflation protection that nominal bonds do not, and dominant technology constituents carry a structural growth premium that justifies a negative nominal spread, by analogy with long-duration bonds whose value sits in distant cash flows. On the real ERP measure (i.e. earnings yield minus real risk-free rate) the picture is thinner still, sitting at approximately 2.5%. That is not a buffer, it is a

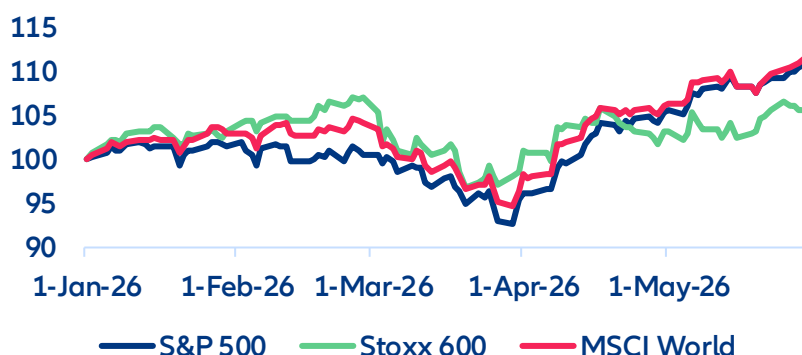
breakeven. A rise in real rates of 50–75bps eliminates it entirely. At that level, the reallocation from equities into investment-grade credit becomes a mechanical trade for liability-matched institutional investors. The multiple stays compressed until either real rates fall or earnings growth widens the premium. In a scenario where geopolitics and growth have simultaneously deteriorated, neither condition arrives quickly.

- **Technical positioning amplifies fundamental moves non-linearly while the mega-IPO pipeline adds further risk.** The 2025–2026 rally has been amplified by structural technical tailwinds: CTA trend-following models near maximum long, option dealers suppressing intraday volatility and FINRA margin debt near cycle highs. Each is mechanically reversible, and the reversal is self-reinforcing rather than gradual. A 5–7% fundamental-driven decline can cascade into a 15–20% technical drawdown before positional overhang clears. The mega-IPO pipeline introduces another amplifier poorly captured by conventional risk models. Passive vehicles must rebalance to new index weights on a defined schedule, selling existing holdings irrespective of valuation or momentum. These rebalancing flow would represent some of the largest single-session institutional selling pressure seen outside of crisis conditions, enough to breach the technical thresholds that convert a managed correction into a self-sustaining cascade without any prior fundamental deterioration. A lockup expiry cliff at approximately 180 days post-listing adds a second, later-dated supply wave.
- **In the adverse scenario, all three policy levers are simultaneously constrained.** Since 2010, equity pricing has reflected not only fundamental value but policy optionality – the assumption that a geopolitical deal, fiscal stimulus or central bank rate cuts are available in any deterioration. The underpriced risk is not the absence of any single lever but the correlation of constraints across all three. The geopolitical channel has been partially activated - ceasefire negotiations are themselves a policy response. The fiscal channel has structural limits: Europe’s substantial post-2022 commitments to energy security and industrial policy did not fully offset structural competitiveness damage from sustained energy repricing. The central bank channel requires either sovereign market dysfunction or a banking stress event to activate – high-bar triggers in an inflationary environment.

A geopolitical resolution is the master assumption of equity markets

As of May 2026, the S&P 500 and Euro Stoxx 600 sit within a few percent of their all-time highs. Implied volatility has spiked on event days but mean-reverted quickly. It seems that equity markets are looking through the war in Iran – one of the most geopolitically significant events in the Middle East in a generation (see Figure 1). However, the market is not ignoring the war. It is pricing a specific outcome: resolution over the summer. That outcome is also holding a set of convictions on other key issues from interest rates to earnings to economic policies. In that sense equity markets can be seen as a Matryoshka doll: it is not a deceptive object, but the outer vivid layer hides how many layers remain and what sits at the center.

Figure 1: Equity performance (Jan-26 = 100)



Sources: LSEG Workspace, Allianz Research

The assumption of a resolution makes sense but it comes with risks. The options market's implied volatility term structure shows a pronounced kink around Q3 2026: elevated near-term volatility fading sharply into year-end, consistent with pricing a binary resolution event. Confirming this conviction, the prediction market for an Iran ceasefire or framework agreement by September sits above 60%. Implicitly, what market participants and analysts are expecting is that the political cost of the ongoing crisis will lead to a resolution – one way or the other. Surging inflation, fuel shortages, the potential social and economic risks coupled with the perspective of US mid-term elections make a deal by autumn “inevitable”. This is not an irrational bet. But it is a specific, time-bound, politically-conditioned bet. If it was to break down, it would have more consequences down the line.

The critical issue is the concession structure required for a genuine agreement. A durable deal requires Iran to move simultaneously on two fronts: a verifiable freeze or rollback of its nuclear program, and some form of security guarantees for the strait of Hormuz. A partial framework agreement is plausible and may be sufficient to sustain equity markets temporarily. But a nominal settlement that unravels, a spoiler event – proxy escalation, third-party strike – or a US administration that claims resolution without genuine Iranian compliance, would leave the problem intact. The probability-weighted payoff from being long equities in a world where a September resolution fails is underrepresented in current levels. If the conflict persists and oil remains structurally elevated, the stagflation risk premium returns. In our downside scenario, equities could correct by 25% and 20%, respectively, in Europe and the US. Brent reprices higher, energy input cost pass-through hits margins and inflation persistence that follows constrains central bank flexibility. Every subsequent layer of analysis would inherit a more adverse starting point.

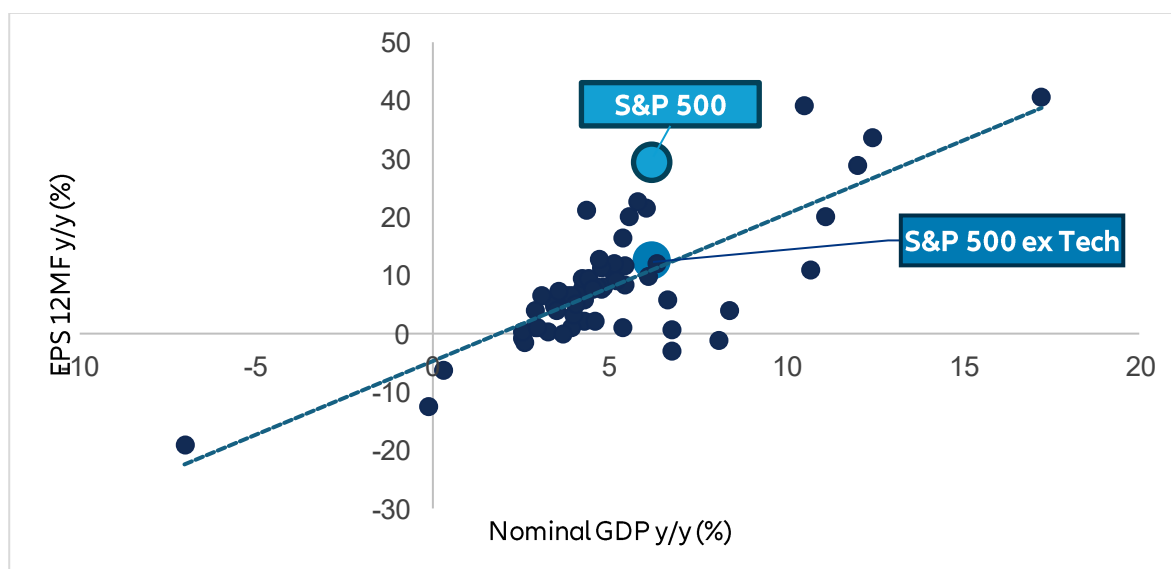
Record margins and earnings signal asymmetric downside

The earnings picture entering 2026 has been much better than feared. The S&P 500 index has been recording double-digit earnings growth for over six consecutive quarters now, and the trend is expected to continue throughout 2026. On the other side of the Atlantic, the STOXX Europe 600 surprised with Q1 earnings jumping by +11.5% y/y (Figure 3). Europe, often dismissed as structurally lagging, is increasingly demonstrating that earnings resilience is becoming a cyclical force of its own. Revisions have been net positive for two consecutive quarters. Growth forecasts have held – US GDP is tracking above 2%, European growth has recovered modestly from the 2024-25 soft patch and inflation has remained below the 3.5-4% threshold that historically triggers meaningful demand destruction. Corporate margins are at or near record highs. The market reads this as durable fundamental support: a solid earnings floor. Inflation is treated as a manageable pass-through phenomenon that is fueling turnover and not hurting volume significantly. The implicit assumption is that margin resilience reflects a relative pricing power.

Record margins are a sign of limited remaining buffer. When margins are at historical peaks, the distribution of outcomes is skewed: the scope for positive earnings surprise is narrow, while the scope for compression if costs rise or volumes soften is substantial. Markets are treating elevated margins as a baseline from which to project forward earnings. They should instead be read as a vulnerability indicator – the starting point from which any adverse shock produces a larger-than-average revision cycle.

Behind the headline: one sector is doing the heavy lifting. Aggregate earnings growth can provide valuable insights, though it is important to consider the nuances to make a fully informed assessment. Figure 2 breaks down the S&P 500's current 12-month forward EPS trajectory, comparing it to the historical relationship between corporate earnings and nominal GDP growth. This relationship has shown remarkable stability across cycles from 2012 to 2025. The conclusion is clear: if we exclude the technology sector, the remaining 69% of S&P earnings align with the regression line, with an estimated growth rate of around 12% year-on-year, in line with the 6% nominal GDP growth rate. This indicates a stable and predictable performance, consistent with market cycles. The macro is performing its intended function. The addition of technology has led to a 29% increase in EPS growth, a level that would typically require an 8-10% nominal GDP to maintain. The index is not being priced according to the economic conditions of the market it is intended to represent. It is being priced for one sector's ability to defy the economy it inhabits.

Figure 2: EPS 12-month forward vs nominal GDP (Q1 2012 – Q2 2026)



Sources: LSEG, Allianz Research

The arithmetic behind this discrepancy matters. It is estimated that approximately 17pps of headline S&P EPS growth in Q2 2026 originate from the IT sector, accounting for roughly 31% of total index earnings. This sector is projected to expand at an estimated rate of +66% year-on-year. This pace is not indicative of the broader earnings cycle, but rather a capex monetization thesis, concentrated in a select group of names, that is meaningfully ahead of any macro anchor. The remaining components of the index are expanding at a rate that aligns with the broader economic environment. However, this implies limited potential for further growth and limited capacity to offset deficiencies in other components of the index.

This reframing of the asymmetry argument provides greater precision. The concern is not that margins will compress uniformly across the index; it is that a single sector will decelerate. A moderation from +66% to a still-exceptional +30% IT earnings growth – the kind of deceleration that would accompany any credible concern about AI capital spending returns, hyperscaler spending fatigue or monetization delays – would have a materially negative effect on the headline S&P EPS number, with limited offset from the rest of the market. The index's earnings recovery, which is currently supported by a broad range of stakeholders, is, in essence, a highly concentrated exposure to a single factor: the continued compounding of AI capex monetization at its current rate. The "record margin" story is, arithmetically, a single-sector story – strip out IT and the index is operating in line with a 6% nominal GDP world.

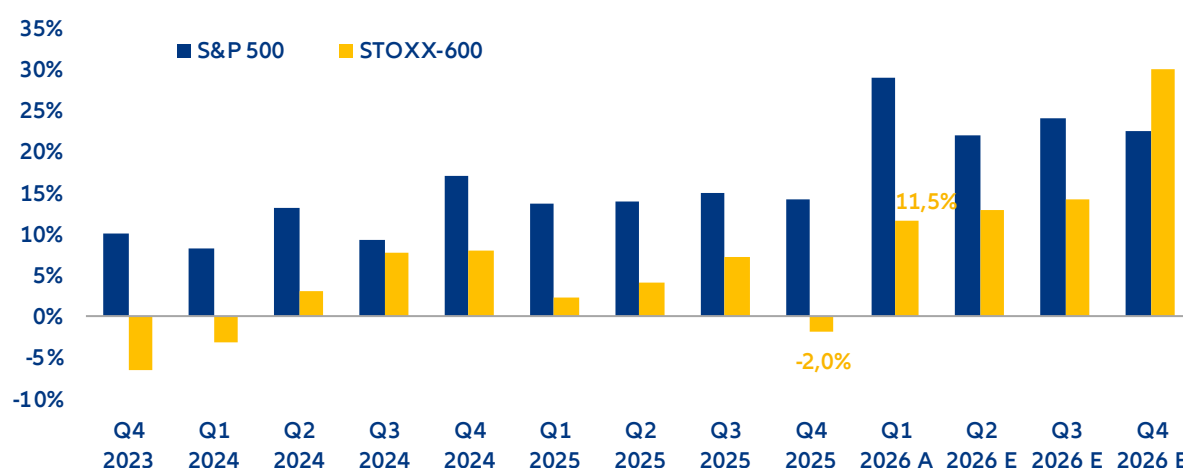
One underappreciated risk is the circular reference in the current macro configuration. US equity markets and US consumption are not independent variables. Household wealth effects from equity performance materially influence consumer spending, which in turn supports corporate revenues and earnings. A 20% equity market correction – the order of magnitude in our downside scenario – would impair US consumption by an estimated 1-1.5pp of GDP. That growth impairment feeds back into earnings revisions, which would extend the equity correction. Markets are pricing equities as though consumption is an exogenous growth driver but it is partly endogenous to equity performance itself.

Sector composition of index hides heterogenous trends and concentrate risks

Q1 2026 corporate results and earnings revisions paints a two-speed market. While both regions show improving forward EPS trajectories into 2026, sector dynamics are playing a huge role in the diverging performance across the Atlantic. While the continued earnings expansion of the S&P 500 remains largely concentrated within the technology sector (EPS: +55% growth in Q1) – primarily driven by AI-related capex acceleration and hyperscaler monetization dynamics – the STOXX Europe 600 (historically weaker and more volatile) has recently shown a meaningful improvement in market sentiment and earnings expectations, despite Europe's historically more fragile

and cyclical macro backdrop. After delivering the worst earnings growth in seven quarters in Q4 2025 (-2.0% y/y), European corporates are heading to the best quarter in three years, recording a double digit growth of +11.5% y/y not seen since Q1 2023 (Figure 3). Yet, this noteworthy growth is fundamentally explained by one single sector: energy, registering an EPS growth of around +50% in Q1. Excluding this sector, the STOXX-600's average EPS growth rate is around +7%. The escalation of geopolitical tensions in the Middle East has sustained elevated crude prices and widened upstream profitability, allowing major oil & gas companies to deliver a substantial expansion in operating margins and cash-flow generation, ultimately translating into significantly stronger P&L performance across the sector. Furthermore, consensus earnings expectations for the upcoming quarters of 2026 have materially improved as well, supported by the strong momentum in the European energy complex. By the end of 2025, EPS estimates for Q2 and Q3 growth pointed to +3.1% and +6.6%, respectively, while current estimates point to double-digit figures of +12.8% and +14.2%. This upward revision reflects not only a sustained period of elevated crude prices following more than three months of geopolitical escalation in the Middle East, but also the increasing likelihood that heightened energy market volatility will persist well into 2026, supporting structurally stronger upstream earnings and reinforcing the positive revisions momentum across the sector.

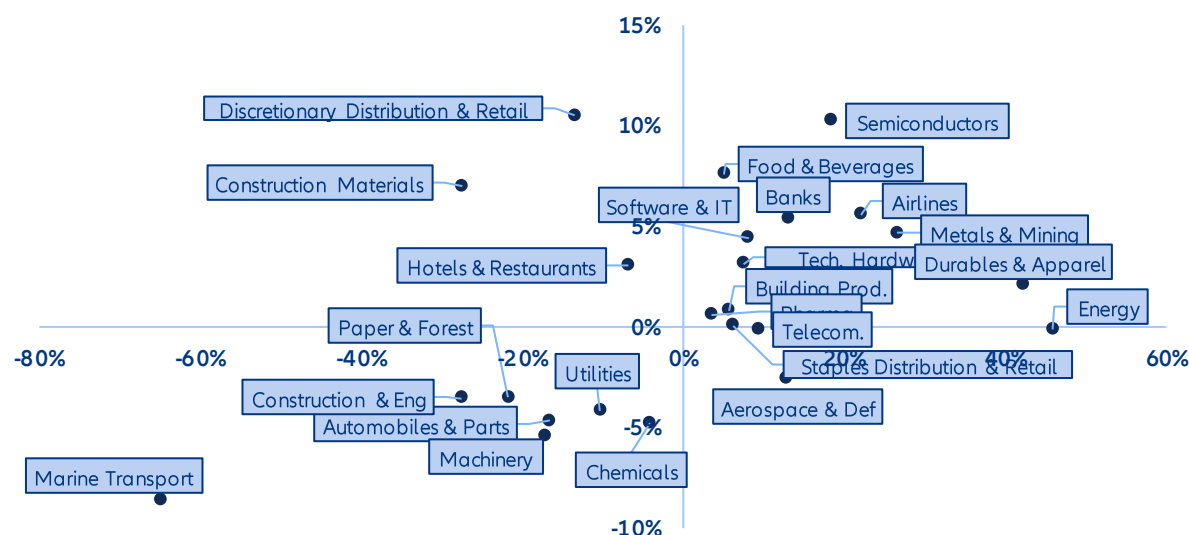
Figure 3: EPS y/y growth rates for companies in the S&P 500 vs STOXX 600 indices



Sources: LSEG I/B/E/S, Allianz Research

European Q1 2026 earnings dynamics point to a market that is resilient on the surface but highly differentiated at the sector level. The entire universe of European corporates have reported aggregated EPS growth of +10.3% y/y, achieved despite an exceptionally subdued top-line environment, with revenues expanding only +0.3%, underscoring the extent to which European corporates have leaned on pricing discipline, cost-out programs and operating leverage to preserve margins in a still-fragile macro and sentiment backdrop. Sector dispersion has been a key feature: Energy stands out as the primary earnings engine, benefitting from the sustained geopolitical tensions and the elevated crude dynamics, while marine transport delivered outsized profitability amid high bunker oil prices and limited pricing power. At the same time, several traditionally cyclical, industrial-heavy segments – including chemicals, machinery, paper & forest products and automobiles (Figure 4) – emerged as notable underperformers, reflecting a combination of structurally elevated fixed-cost bases (particularly energy-linked input costs in chemicals and paper) and higher demand elasticity, which together amplified margin pressure over the period.

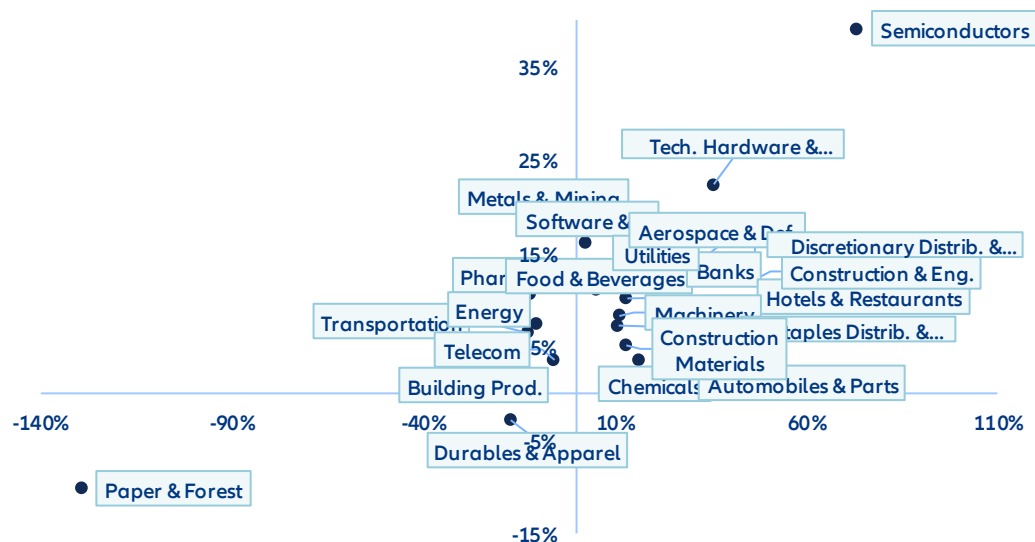
Figure 4: Revenue (Y axis) and EPS (X axis) y/y growth rates for European corporates in Q1 2026, by sector



Sources: Refinitiv Datastream (as of 27 May 2026), Allianz Research

Unlike the US market where earnings expansion remains heavily concentrated in a narrow set of tech-driven growth engines, Europe’s earnings resilience appears materially broader across the real economy. This reduces concentration risk at the index level, but simultaneously leaves the region more exposed to cyclical and commodity-sensitive dynamics rather than to a single dominant secular growth theme. The divergence is particularly evident in the composition of earnings leadership. In Europe, upgrade momentum has been driven predominantly by energy and other macro-sensitive sectors, reflecting the region’s higher operating leverage to commodity prices and geopolitical developments. By comparison, the US earnings profile remains structurally more durable and less dependent on commodity-linked upside, with growth continuing to be anchored by technology-intensive industries. As shown in Figure 5, semiconductors (+39.1% y/y) and technology hardware (+22.4%) recorded the strongest revenue growth rates of any US sector in Q1, reinforcing the extent to which AI-related capital expenditure and digital infrastructure investment continue to dominate the US corporate earnings cycle. Paper & forest products stood out as the clearest earnings casualty of the quarter in the US, caught between deteriorating pricing dynamics, structurally elevated energy and freight costs, and weakening end-market demand, a combination that effectively erased operating leverage across the industry.

Figure 5: Revenue (Y axis) and EPS (X axis) y/y growth rates for US corporates in Q1 2026, by sector



Sources: Refinitiv Datastream (as of 29 May 2026), Allianz Research

Non-cyclical composition concentrates rate sensitivity. The S&P 500 and Euro Stoxx 600 carry a significantly higher weight in non-cyclical sectors than in prior cycles. Non-cyclical sectors can be divided into two groups: bond-proxy profiles (utilities, healthcare, high-dividend energy stocks) and long-duration growth profiles (large-cap Technology). Both are materially sensitive to real interest rates – the former through yield competition and leverage cost, the latter through discount rate effects on terminal value. The index shift away from cyclicals has not reduced rate sensitivity but it has concentrated it in a less visible form. In a rising real rate environment, the "defensive" parts of the index are among the most exposed and the earnings quality argument for holding them dissolves precisely when it is most needed.

The anticipated listings of OpenAI and Anthropic, alongside the already-discussed SpaceX entry, represent a structurally distinct event from the ordinary IPO cycle. Together, these three companies carry private-market valuations of USD1.75trn (SpaceX) and USD850bn (for each OpenAI and Anthropic) – a combined USD3.45trn, roughly equivalent to the entire current weight of the financial sector in the S&P 500. This trend signals continued risk appetite and is expected to deepen index liquidity, attract fresh capital inflows and sustain the momentum dynamic that has supported equity returns in early 2026. Their combined entry into capitalization-weighted indices would be the largest supply event in US equity market history, and their sectoral concentration compounds the risk. All three would land in, or adjacent to, technology – the index segment already identified as carrying the most concentrated rate sensitivity. The simultaneous addition of three megacap AI-and-infrastructure names does not diversify the index; it deepens a crowding that already exists.

The mechanics are straightforward but their scale is not. Passive and semi-passive vehicles – which now account for over 55% of US equity AUM – are rules-bound buyers at index inclusion. They have no discretion on timing or price: they must acquire the weight the index assigns, funded by proportional sales of every other existing constituent. For additions of this size, the required reallocation is not a rounding error. It is a structural flow event, one that persists for the adjustment period and leaves the index systematically lighter in every other sector. At current index weights, a combined entry at this scale would represent 4-5% of S&P 500 market capitalization on day one of full inclusion – forcing capital redistribution that would suppress every incumbent constituent's relative performance during the rebalancing window – regardless of fundamentals. The AI premium that has inflated Technology valuations through 2025 and into 2026 would, paradoxically, be the mechanism through which the rest of the index is diluted. Anticipation of these listings partially prices in the demand side, the supply-side redistribution impact on incumbents remains largely absent from current consensus.

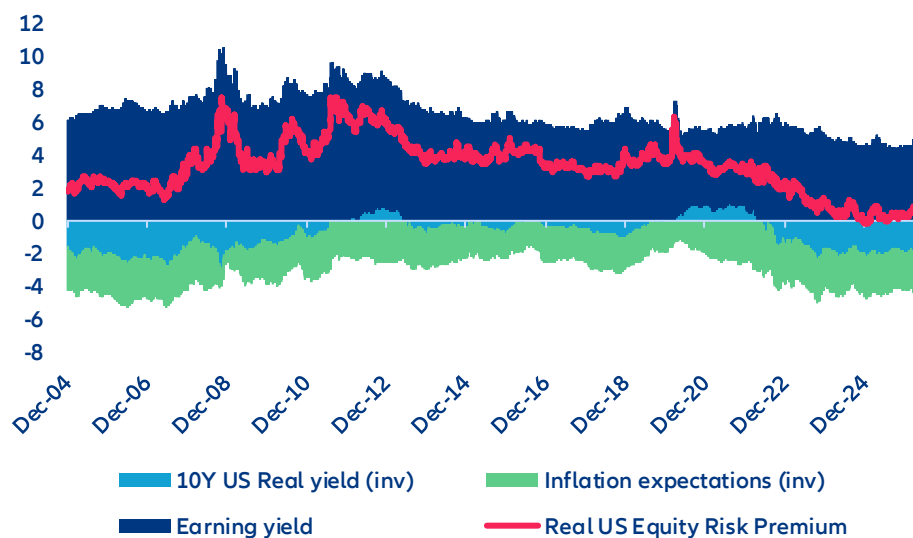
Valuation and rates

The real equity risk premium is a breakeven, not a buffer. The negative nominal equity risk premium is rationalized two ways. Equities are real assets whose cash flows index to inflation, while nominal bonds do not – in an inflationary regime, the inflation-protection argument compresses the ERP markets are willing to accept. The structural growth premium of dominant technology constituents is treated as deserving of a negative nominal premium, by analogy with long-duration bonds where value sits in distant cash flows. On the real ERP measure – earnings yield minus real (inflation-adjusted) risk-free rate – the picture looks more supportive, sitting at approximately 2.5%. This is thin by historical standards but positive, and the market treats it as sufficient justification for current institutional equity weights.

The inflation-hedge argument is valid, but only while real rates remain contained. It breaks down precisely when it is most needed: in a regime where inflation is persistent and central banks respond by keeping real rates elevated. In that environment, the inflation-hedge and the rate-headwind operate on the same asset simultaneously, and the rate headwind dominates for non-cyclical, long-duration equity profiles – which is most of the index. The competitive pressure from credit is more immediate than equity multiples suggest. Investment-grade credit at BBB spreads of approximately 120bps, or high-yield at approximately 180bps, delivers real returns comparable to equities, with materially lower volatility, senior claim in the capital structure, and defined maturity.

The 2.5% real ERP is not a buffer; it is a breakeven. A rise in real rates of 50-75bps eliminates it. At that point, the reallocation from equities into investment-grade credit is a mechanical, structurally-driven trade – not a sentiment event, and not reversible until real rates fall or earnings growth widens the premium again. The de-rating that follows is structural rather than cyclical, driven by the rate level and not by risk appetite. The multiple stays compressed until the underlying arithmetic turns; in a scenario where geopolitics and growth have also deteriorated, neither condition arrives quickly.

Figure 6: S&P500 earnings yields and risk premium (%)

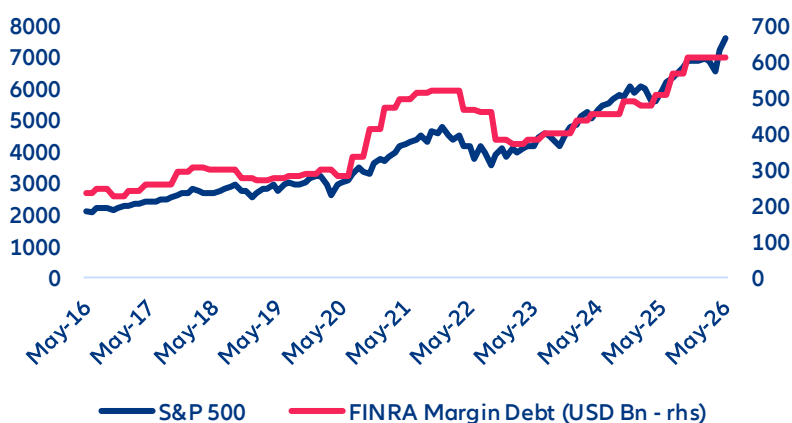


Sources: LSEG Workspace, Allianz Research

The same technical mechanics that amplified the rally can worsen a correction.

Technical positioning amplifies fundamental moves non-linearly. The 2025-2026 rally has been supported by structural technical tailwinds as well as by fundamentals. Commodity Trading Advisors (CTA) trend-following models are near maximum long exposure, reflecting the sustained upward trend in major indices. Options dealer positioning – driven by concentrated retail options activity at round-number strikes – reinforces trend directionality. FINRA margin debt (i.e. funds that investors have borrowed from brokers) has recovered from its 2022 lows to levels near cycle highs (see Figure 7). Markets read these conditions as healthy indicators of broad participation and sustainable momentum.

Figure 7: S&P 500 and FINRA margin debt



Sources: LSEG Workspace, Allianz Research

Each of these tailwinds is mechanically reversible, and the reversal dynamic is self-reinforcing rather than gradual. CTA momentum models do not taper when trend signals reverse, they flip. The transition from maximum long to maximum short can occur within days, concentrating supply at precisely the moment other sellers are also emerging; a relatively modest move can initiate selling that amplifies into a substantially larger one. Margin debt creates a third channel: leveraged holders facing margin calls cannot wait for fundamental improvement, and the supply they generate is price-indifferent. The combined effect is a non-linear response to adverse fundamentals. A 5-7% fundamental-driven decline can cascade into a 15-20% technical drawdown before the positional overhang clears. Recovery requires not just fundamental stabilization but a full clearing of technical positions – CTAs rebuilding long exposure, margin debt being reduced etc. – a process that typically extends over weeks to months, well beyond what fundamental analysis alone would project.

The mega-IPO pipeline introduces a technical amplifier that operates through a distinct channel that is poorly captured by conventional risk models. Large index inclusions generate predictable, time-compressed, price-insensitive flows: passive vehicles must rebalance to the new index weights on a defined schedule, selling existing holdings in proportion to their current weight irrespective of valuation or momentum. This is mechanical selling driven by the arithmetic of capitalization weighting. For listings at the scale now under discussion, the rebalancing flow would represent some of the largest single-session institutional selling pressure seen in US markets outside of crisis conditions.

The interaction with existing technical positioning creates an adverse compounding effect. CTA momentum models hold existing large-cap technology names near maximum long. In such rebalancing-driven supply event in precisely those names (forced, scheduled, and price-indifferent), the ordinary self-reinforcing dynamics take hold: dealers sell into weakness, CTAs follow trend signals, margin calls cascade. The IPO supply event would not need to be the proximate cause of a broader market dislocation, it only needs to be large enough to move prices through the technical thresholds that convert a controlled rebalancing into a self-sustaining cascade. The lockup expiry cliff – typically 180 days post-listing, at which point insiders and pre-IPO institutional holders become sellers – adds a second, later-dated supply wave that extends the technical overhang well beyond the initial listing date. Markets are not pricing either wave.

In a downside scenario, all policy levers are simultaneously constrained.

The policy safety net is thinner than markets assume. A defining feature of equity behavior since 2010 has been the implicit assumption that policy will intervene when markets deteriorate sufficiently. This has been validated repeatedly: QE programs, fiscal stimulus, emergency rate cuts and geopolitically-motivated pivots have all been deployed in ways that supported equities. The result is that pricing reflects not just fundamental value but policy optionality. In 2026, markets appear to be pricing at least one of three possible interventions as available in any adverse path: a geopolitical deal (the TACO put), fiscal stimulus or central bank rate cuts. The implicit assumption is that whichever lever activates, the response will be sufficient to prevent a full cascade.

The underpriced risk is not the absence of any single put, it is the correlation of constraints across all three. The geopolitical channel has been partially activated: the ceasefire negotiations themselves are the response. But a full activation requires concessions exceeding prior pivots (e.g. verifiable nuclear rollback from Iran plus a reparations framework for Iran) and an administration that claims a partial settlement as a complete victory would exhaust this option without resolving the underlying risk. The fiscal channel has structural limits that Europe has already demonstrated. Substantial fiscal commitments to energy security, defense and industrial policy since 2022 provided some support, but they could not offset the structural competitiveness damage from sustained energy-price repricing, not for lack of fiscal support, but because energy-intensive sector damage is not curable by spending alone.

The central bank channel is the most powerful but the most constrained in this specific adverse scenario. Monetary easing at scale requires either dysfunction in sovereign bond markets or a banking-sector stress event – high-bar triggers. In an environment of persistent inflation, the direct consequence of sustained oil-driven energy costs, the central bank cannot cut pre-emptively without material credibility cost. The put is available, but its strike

is deep out of the money; the cost of waiting for conditions to deteriorate enough to activate it is borne by equity holders in the interim. By the time intervention arrives, the technical amplifier dynamic described in the previous section has already moved prices well past the level early action would have stabilized.

The four potential regimes going forward

Our six layers of analysis converge on two primary variables that govern the outcome space: whether the geopolitical situation resolves in the near term, and whether the real rate environment remains contained. The four combinations of these variables define materially distinct regimes for equity markets (see Figure 8).

Figure 8. Scenario matrix

	Rates falling / stable	Rates sticky / rising
Geopolitical resolution	Soft landing confirmed. Supply normalizes, inflation fades, central bank retains easing optionality. Policy support intact. Technical amplifiers operate constructively. Fundamental and valuation support both remain.	Stagflation lite. Earnings resilient but real rates compress multiples. Non-cyclical rate sensitivity exposed. Index rangebound with elevated sector dispersion. Policy put available.
Conflict persists	Geopolitical drag, monetary relief. Growth slows, consumption weakens, but central bank cut cycle is available as inflation pressures ease. Technical overhang clears as monetary pivot anchors. Near-term negative, recoverable over a medium horizon.	Adverse scenario. All layers deteriorate simultaneously. Earnings compress, multiples de-rate, technical amplifiers cascade, all three policy channels constrained. Structural drawdown without an available circuit breaker.

Source: Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Weed out the laggards: Sustainability as a credit filter

03 June 2026

Allianz Research

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Executive Summary



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- **Financial fundamentals remain the primary drivers of credit risk.** Across more than 7,400 companies and 280,000 firm-year observations, profitability (ROA), leverage and size collectively explain the vast majority of variation in credit risk. Sustainability is statistically significant but smaller in magnitude: financial fundamentals carry coefficients 3-7 times larger than the sustainability variable in our global sample. A firm's balance sheet, earnings power and scale determine whether it can service its debt - sustainability adds an incremental but real signal on top of that financial bedrock.
- **Sustainability acts as a credit-risk filter, but its signal concentrates at the bottom of the distribution.** Weaker sustainability is consistently associated with higher default risk. A one-decile improvement in sustainability (roughly +7.5 points on a 0-100 scale) corresponds to an approximately 0.25pp reduction in default probability - a 12-25% relative reduction for firms in the worst default-risk decile. The relationship is non-linear: moving from poor to average sustainability materially improves credit outcomes; moving from average to best-in-class delivers little additional benefit.
- **Environmental performance is the clearest predictor of default risk.** A 10-point improvement in the environmental score is associated with a 0.9-point gain in the Altman Z-score - enough to move borderline firms out of the distress zone. Governance, by contrast, is the key driver of broader credit quality, consistent with financial discipline and transparency underpinning financial strength.
- **European companies lead on sustainability scores and show the greatest sensitivity to ESG factors.** Regional baselines differ sharply: Europe leads (average combined score in the 50s), the US trails (40s, with environmental scores near 30) and Japan scores well on environment (near the high-40s) but weaker on social metrics. Environmental leadership carries the strongest credit premium in emerging markets, where best-in-class issuers benefit from a scarcity premium as standards tighten. In the US, holistic ESG profiles matter more than any single pillar; in Europe, the combined score links to both credit risk and credit quality.
- **The clearest ESG-credit links cluster in sectors exposed to energy transition, operational and reputational risk - notably communication services, consumer staples and energy.** Our sector analysis finds meaningful sustainability-credit relationships in seven of eleven sectors, with the largest effects in these three. Communication/software tends to concentrate financially strong firms that can fund sustainability investment and are rewarded for risk management that encompasses ESG. Consumer staples and energy include carbon-intensive sub-sectors - textiles to oil & gas - where pollution, labor issues, carbon pricing and stranded-asset dynamics can directly hit cash flows, asset values and funding access. The relationship is weaker in materials and IT, reinforcing that sustainability integration in credit requires a sector materiality lens, not a blanket score overlay.

- **These findings are robust across data providers and confirmed by our internal credit assessments.** Replicating the analysis with alternative sustainability scores yields virtually identical results. Using our proprietary internal credit grades, we find that sustainability is already implicitly embedded in analysts' assessments - the sustainability score and environmental and social pillars are all statistically significant predictors of internal grades. Analysts are capturing sustainability-related risks in practice, a more systematic approach could sharpen that signal further.
- **The practical implication is clear: screen out the weakest.** Banks should embed sustainability as a negative screen in credit pricing and due diligence. Credit investors should use it primarily as a downside protection tool. Corporate issuers face a real credit penalty for sitting below the sustainability threshold - and no proportionate reward for exceeding it. The imperative for all market participants is the same: clear the floor. Chasing the ceiling is a marginal bonus.



Unsustainable practices can generate financial stress

Sustainability has moved from a niche concern to a mainstream factor in extra-financial evaluation of corporates over the past decade. The underlying rationale is simple: sustainability issues directly or indirectly influence the fundamentals that eventually determine a company's ability to generate revenues, make a profit or just repay debt. If a company faces environmental fines or a factory is shut down by floods or wildfires, its revenues suffer and costs rise – impeding debt service. If poor labor practices lead to strikes or a consumer boycott, cash flows can dry up. And if weak governance leads to fraud or compliance breaches, a firm can quickly find itself facing legal penalties and loss of investor confidence. All these sustainability-related events hit the company's financial position, translating into traditional credit risk and increasing the probability of default.

Climate change and environmental management are now recognized as material credit factors. Physical risks (e.g. extreme weather, droughts, rising sea levels) can destroy assets and disrupt supply chains¹. For example, in 2019, wildfires in California saddled a major utility with staggering liabilities, contributing to its bankruptcy. Transition risks are also mounting: Governments imposing carbon taxes or banning high-emission activities can strand assets and force costly business model changes². Companies heavily reliant on fossil fuels or with poor pollution controls may face higher costs, asset write-downs and weaker financial ratios, all of which hurt credit quality. On the other hand, firms proactive in cutting emissions or investing in cleaner technology can reduce regulatory risks and operating costs over time, bolstering their credit outlook.

How a company manages its workforce, customers and communities also bears on creditworthiness. Neglecting safety and labor rights, for instance, can lead to workplace accidents or strikes that interrupt production. In the mining industry, fatal accidents not only incur direct costs but can prompt license suspensions. Similarly, poor product safety or customer data breaches can trigger lawsuits and erode a firm's reputation and revenues. By contrast, companies that invest in human capital (e.g. training employees, maintaining good labor relations) and uphold product quality tend to have more resilient operations and loyal customers. Although spending on social initiatives may add costs in the short term, that stability should translate into steadier cash flows and lower credit risk in the long run. Lastly, corporate governance has long been linked to credit outcomes. From Enron's collapse to more recent corporate scandals, history shows that failures were rooted in poor governance. Lax oversight, misaligned incentives or opaque disclosure can mask mounting problems until it is too late. When governance breaks down, firms are prone to accounting fraud, corruption or reckless risk-taking that can rapidly undermine their financial position. For creditors, strong governance is a first line of defense: it means competent management, prudent financial controls and transparency. In practice, governance issues can crystalize into credit events through regulatory penalties, adverse audit findings or sudden leadership crises. Any of these can spook investors and restrict a firm's access to funding.

¹ See our previous report [on adaptation](#)

² See our previous report [on stranded assets](#)



Sustainability matters for credit quality and credit risk

Over the last ten years, a growing body of research has examined whether strong sustainability actually translates into better credit outcomes. The evidence, especially from the most recent studies, generally finds that companies with higher ESG ratings tend to have lower credit risk. On the whole, firms that excel on ESG enjoy tangible credit benefits. They often face lower borrowing costs and fewer nasty surprises. For example, multiple studies of corporate bond markets have found that companies with high ESG scores pay narrower credit spreads than peers, even after controlling for their credit ratings. In other words, bond investors demand a smaller risk premium, viewing high-ESG issuers as safer bets. These firms also exhibit lower volatility and reduced tail risk of extreme losses. The upshot is that a strong ESG profile appears to enhance financial resilience, which lenders reward with cheaper debt capital (see appendix for more references on the sustainability-credit risk nexus).

To further investigate the sustainability-credit risk nexus, we conduct an extensive statistical analysis on a global dataset of companies. Our study covers firms worldwide over the period 1991-2025, using sustainability scores from Refinitiv (see appendix for more details). This dataset spans both developed and emerging markets, with a broad sector representation. We measured credit risk in two ways: (i) the Altman Z-score, a classic metric of default risk based on financial ratios, and (ii) a credit-quality index capturing broader aspects of financial health (primarily profitability, leverage and cash flow metrics). The Altman Z-score was available for a subset of about 2,100 firms (generally larger companies with complete

financial data), while the credit quality index covered a larger sample of 7,400 firms, giving us a comprehensive view of sustainability impacts on creditworthiness. We include firms from North America, Europe, Asia-Pacific and emerging economies. Sustainability scoring coverage has expanded significantly in the past decade; still, not all companies have sustainability ratings or all the inputs for the Altman Z-score. For example, among emerging market companies in our sample, only about 44% have sustainability scores available (as coverage tends to be thinner for smaller EM firms). By contrast, in the US sample almost 67% of firms have sustainability scores, reflecting better disclosure and data coverage in the US. Developed Europe falls in between (around 65% sustainability coverage). These proportions highlight that while sustainability data is now widespread, it is not yet universal. Nevertheless, our total sample comprises over 280,000 firm-year observations, providing a rich basis to detect patterns.

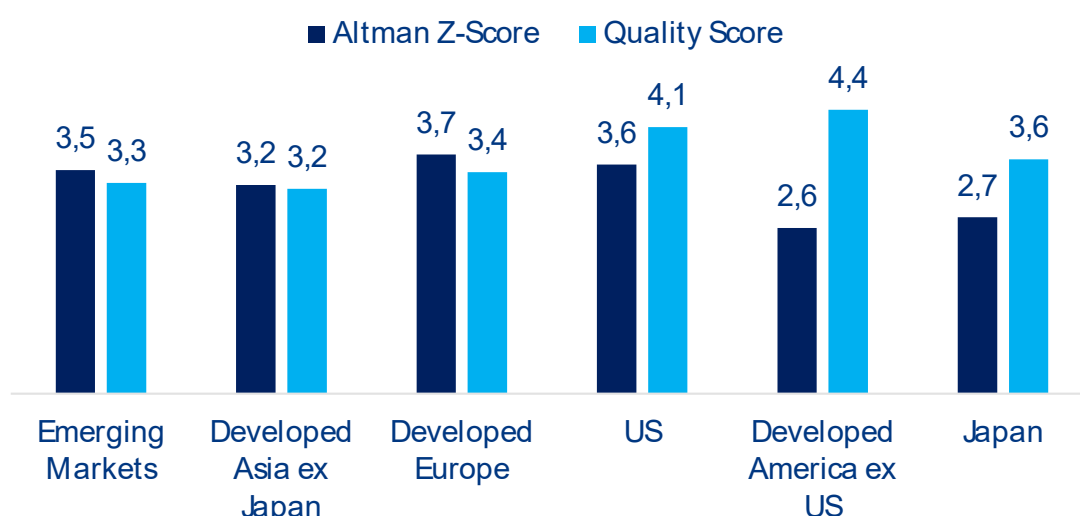
European companies boast the highest sustainability scores (the average combined score in Europe is about 50 out of 100), compared to the low 40s in emerging markets. US companies trail Europe: the average US sustainability score is around 40, with particularly low marks on environmental factors (the average US environmental pillar score is barely 30, reflecting that many American firms score poorly on carbon emissions and resource use relative to global peers). Japan, interestingly, scores well on environmental issues (average E score close to 48) but somewhat lower on social metrics. These regional differences suggest that European

businesses have led on sustainability, especially on social and environmental dimensions, while North American and some Asia-Pacific firms lag in environmental performance. Does strong sustainability in a region correspond with better credit profiles? The Altman Z-score in our sample averages around 3-4 for most regions (see Figure 1), with higher values indicating safer balance sheets³. Developed Europe had a slightly higher mean Z-score (3.7) than the US (3.6) or emerging markets (3.5). These averages put the median company safely out of distress, though the range is wide. We also construct a credit quality index for each firm-year to capture overall financial health. This composite index synthesizes profitability (return on assets) and cash-flow generation (operating cash flow to assets), interest coverage and leverage which are key inputs to credit quality. By design, higher values of this index indicate stronger credit fundamentals (high profits, robust cash flows, less debt). Not surprisingly, developed market firms tend to score better on this quality index on average, given their generally more mature, stable financials.

We find that a strong sustainability score is associated with lower credit risk. We employ panel regression analysis to quantify the relationship between sustainability performance and our credit risk measures⁴. In simple terms, when a company's sustainability score is higher, do we subsequently observe better credit outcomes? To answer this, we ran regressions of Altman Z-scores and the credit quality index on sustainability scores, controlling for other factors. Specifically, we include controls for firm size (larger firms often have more diversified risks)

and profitability (a higher return on assets naturally boosts credit metrics). We also account for fixed effects (essentially controlling for broad differences across sectors, years and regions) to isolate the impact of sustainability. Importantly, sustainability scores were lagged by one year in the models, meaning we related last year's sustainability performance to this year's credit risk indicators. This helps address causality (i.e. ensuring we are capturing sustainability influencing credit, not the other way around, and also to be consistent with the timing of availability of information, as today we know the sustainability score based on the past year's data). First of all, we need to underline that coefficients for fundamentals are significant and larger across all our regressions compared to sustainability measures. This means that financial metrics are the leading drivers of credit risk. However, we find that sustainability also matters to a significant extent: The coefficient on the Environmental Pillar Score implies that if a firm's environmental score rises by 10 points (say from 50 to 60), its Altman Z-score is expected to be about 0.9 higher on average (see Table 1). Such an effect is meaningful: given many firms' Z-scores hover in the 1.5-3.0 range, a nearly 1 point boost could move a firm from the borderline "grey zone" closer to safety. The overall ESG combined score (which integrates all pillars and accounts for controversies) had a positive coefficient of about 0.05. This suggests that, taken as a whole, strong sustainability performance is associated with better credit health (higher Altman Z-score), although among the individual components, the environmental factor is doing most of the work in explaining that link.

Figure 1: Distribution of credit risk and quality scores by regions



Sources: Refinitiv, Allianz Research

³ For context, a Z-score above 2.6 is considered "safe", while below 1.1 is considered "distressed"

⁴ See appendix for methodological details

Table 1: Distribution of credit risk and quality scores by regions

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals		
		Profitability	Size	Cashflow Strength
ESG Combined Score	0.05*	0.34***	0.28***	0.14***
E Pillar Score	0.09***	0.27***	0.36***	0.18***
S Pillar Score	-0.01	0.52***	0.42***	0.24***
G Pillar Score	0.02	0.49***	0.38***	0.22***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Sustainability is also linked to broad credit quality.

When looking at credit quality, like for credit risk, fundamentals are more important. Nevertheless, we also find here that the combined ESG score has a positive and statistically significant coefficient, indicating that firms with higher overall sustainability performance tend to exhibit better credit quality. In practical terms, a 10-point increase in the ESG score corresponds to about 0.16 higher on our credit quality index – a modest but non-trivial improvement in underlying financial strength (see Table 2). Interestingly, when we break it down into pillars, we see that the governance pillar shows a positive effect. This suggests that good governance practices (board structure, minority shareholder protection, transparency, etc.) are contributing to better credit fundamentals, which is intuitively consistent with the idea that well-governed firms have more disciplined financial management. At a global level, our findings imply that sustainability matters for credit risk and credit quality, but the relative importance of E, S, and G differs between the two. For default risk (Altman Z), environmental performance is the standout (i.e. companies with stronger environmental stewardship tend to be safer from a credit risk perspective). This resonates with the notion that environmental risks (climate, pollution, regulation) have become financially material and can drive default outcomes if unmanaged. In contrast, social and governance factors did not show up in the default risk regression, possibly because their effects on default are indirect or longer-term (or already reflected in

other control variables like profitability). For credit quality, we see that having a robust overall sustainability profile helps, and there is a hint that governance (and to an extent social) is playing a role.

E matters for EMs while overall sustainability is more important for US and Europe. Looking at different regions⁵, we find that environmental pillar impacts are strongest in emerging markets when it comes to credit quality. In this group, a higher environmental score was associated with a significantly better credit quality index. Emerging market firms that are environmental leaders tend to also have stronger credit fundamentals. One interpretation is that in developing economies, where environmental standards and enforcement are rapidly tightening, firms that proactively manage environmental risks gain a credit edge – perhaps by avoiding fines or attracting sustainability-linked capital. Additionally, emerging market companies with strong sustainability ratings might enjoy a reputational premium that improves investor confidence in their financial stability. Moreover, in emerging markets, the governance pillar had a positive relationship with credit quality, suggesting governance reforms are rewarded in those countries with better perceived creditworthiness. In the US, our models showed that overall sustainability performance had a mild positive relationship with credit quality, but none of the individual E, S, G pillars were statistically significant on their own. This might indicate that US credit investors take a holistic view

⁵ North America (US and other developed Americas), Europe, Asia-Pacific (developed Asia ex-Japan, and Japan) and Emerging Markets.

– i.e. a company has to be well-rounded on sustainability to get credit recognition, and no single pillar being enough. Europe showed a positive link with the overall sustainability performance for both credit risk and credit quality. Europe's long-standing focus on sustainability could mean investors and lenders have integrated these extra-financial criteria in their analysis.

Sustainability matters most for communication services, consumer staples and energy sector credit. Overall we find significant links between sustainability and credit risk/quality for seven sectors out of 11, underlining the broad relevance of extra-financial performance for credit worthiness. However, the influence of sustainability is the largest in communication services & software, consumer staples and energy (see Figure 2). For communication

services and software, our results most likely stem from the fact that the sector includes financially strong companies that spend and invest in sustainability. Meanwhile, both consumer staples and energy sectors include carbon-intensive segments – from textiles to oil & gas – for which sustainability-related risks represent substantial financial risk, too. For instance, pollution risks and labor related-issues could disrupt firms in textiles while carbon prices and stranded assets are clear hurdles for oil & gas companies. At the other end of the spectrum, although relevant, sustainability scores seem less important for credit metrics for firms in the materials and IT sectors.

Table 2: ESG and Credit Quality Index (global panel regression)

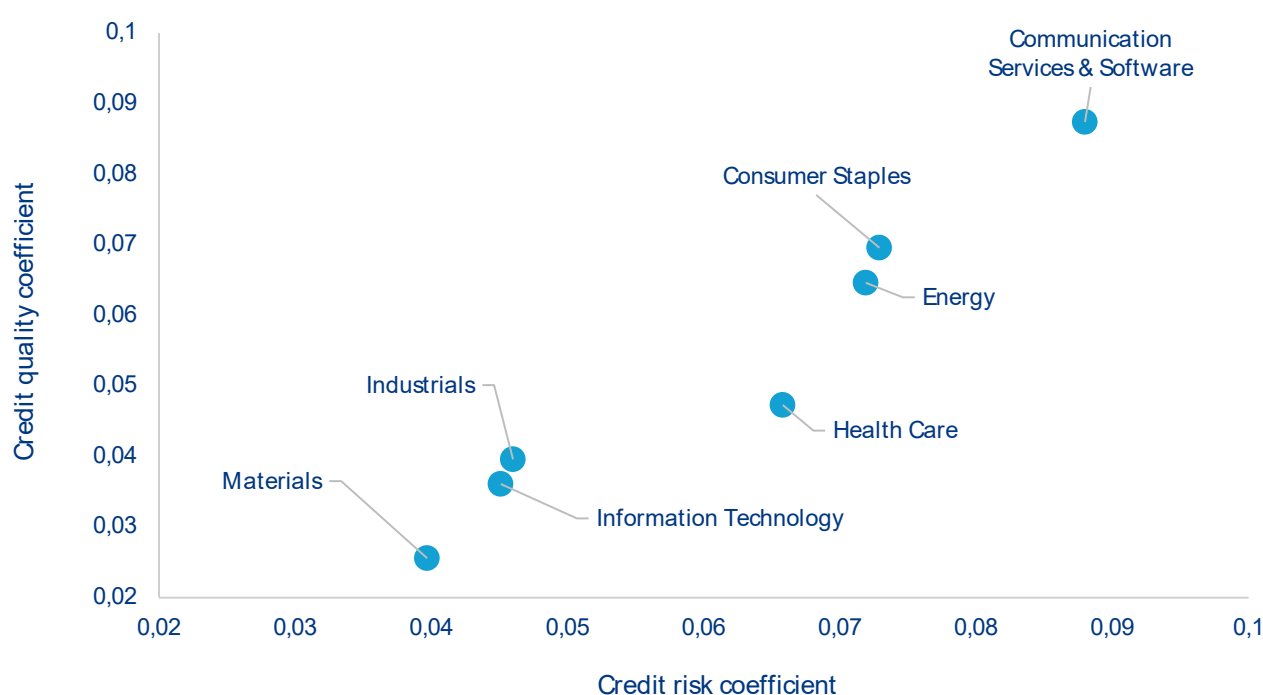
Sustainability measure vs credit quality	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals ⁱ
		Size
ESG Combined Score	0.02***	0.14***
E Pillar Score	0.01	0.14***
S Pillar Score	0.01	0.26***
G Pillar Score	0.01*	0.19***

*, ** and *** denote respectively 10%, 5% and 1% significance level

ⁱ We do not include profitability and cashflow here as they are used to compute credit quality

Sources: Refinitiv, Allianz Research

Figure 2: Sustainability score coefficients for credit risk and credit quality



Sources: Refinitiv & Allianz Research – Note: we only show significant coefficients.

Our findings hold when using MSCI sustainability data instead of Refinitiv, confirming that the sustainability-credit risk nexus is not an artefact of a single data provider.

A recurring concern in the sustainability-credit literature is that results may be sensitive to the choice of rating provider, given well-documented divergences across sustainability scoring methodologies (Berg, Kölbel & Rigobon, 2022). To address this, we replicate our core Altman Z-score regression using MSCI sustainability scores in place of Refinitiv. The MSCI-based sustainability coefficient is positive and highly significant ($\beta = 0.04$, $p < 0.01$ in Table 3), closely mirroring the Refinitiv result ($\beta = 0.05$ in Table 1). All control variables – profitability, size, leverage, cash-flow strength and interest coverage – retain the expected signs and significance. This consistency across two independent data providers strengthens our confidence that the underlying relationship between sustainability performance and credit risk is genuine and not driven by idiosyncratic scoring choices.

Using our proprietary internal credit grades, we confirm that sustainability carries credit-relevant information beyond what traditional financial analysis captures.

Beyond third-party credit-risk proxies such as the Altman Z-score, we test whether the sustainability-credit link holds against our own internal credit grading system, which

runs from 1 (strongest) to 10 (weakest) and incorporates qualitative analyst judgment alongside financial metrics. Table 4 reports the results using Refinitiv sustainability scores as explanatory variables. The sustainability score, environmental pillar and social pillar all exhibit negative and highly significant coefficients ($p < 0.01$), meaning that companies with stronger sustainability performance receive better internal credit grades. The governance pillar, however, is not statistically significant – likely because governance considerations are already explicitly embedded in our internal credit assessment process. While the economic magnitudes are modest – a move from the worst to the best sustainability score corresponds to roughly a quarter-notch improvement on the 1-to-10 scale – the statistical significance validates that sustainability captures credit-relevant information incremental to what traditional financial analysis already reflects. This supports using sustainability as a complementary signal – particularly as a negative screen, as we demonstrate below – rather than as a standalone driver of credit differentiation.

Table 3: Sustainability and Altman Z-score – MSCI data (global panel regression)

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals			
		Profitability	Size	Leverage	Cashflow Strength
MSCI Sustainability Score	0.04***	1.29***	0.10***	-1.06***	0.26***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Table 4: Refinitiv sustainability scores and internal credit grades

Sustainability measure vs internal credit grade	Beta coefficient	Significance
Sustainability Score	-0.04	***
Combined Sustainability Score	-0.03	***
Environmental Pillar Score	-0.03	***
Social Pillar Score	-0.04	***
Governance Pillar Score	-0.01	-

Negative coefficients indicate better (lower) credit grades. All regressions control for firm size.

*, ** and *** denote respectively 10%, 5% and 1% significance level. Sources: Refinitiv, Allianz Research

When we split firms into low, medium and high sustainability groups, the credit-risk relationship is concentrated in the bottom tercile – suggesting sustainability is most actionable as a filter to exclude the worst-scoring companies. To test whether the sustainability-credit link is linear or concentrated in specific segments, we estimate a spline regression that allows the slope to differ across sustainability terciles (using MSCI data). The results are striking (Table 5): the coefficient for the low sustainability group is 0.10 and highly significant at the 1% level, implying that among firms with the weakest sustainability profiles, each additional point of improvement is associated with a meaningful increase in the Altman Z-score. By contrast, the coefficients for the medium (0.01) and high (0.02) groups are small and statistically insignificant. In other words, the marginal credit benefit of better sustainability is strongest at the bottom of the distribution and tapers off once firms reach a moderate standard. This non-linearity has a clear practical implication: rather than rewarding top sustainability performers, the data supports using sustainability primarily as a negative screen – flagging or excluding firms in the lowest sustainability tercile, where poor practices are most strongly associated with elevated credit risk.

Translating our results into probability-of-default terms, a one-decile improvement in sustainability corresponds to a reduction in default probability of approximately 0.25pp. To make our findings directly actionable for

credit practitioners, we link the sustainability-to-Altman Z relationship to probability of default (PD) through a two-step estimation. First, from our MSCI regression, a one-decile improvement in sustainability score (approximately 7.5 points) is associated with a 0.29-point increase in the Altman Z-score. Second, using a logit regression of observed PD on winsorised Altman Z-scores ($\beta = -0.0096$, $p < 0.01$), we compute the implied change in default probability. The resulting chain implies that a one-decile sustainability improvement – approximately 7.5 points on the 0–100 sustainability scale, for example moving from a score of 12 to 20 for firms in the bottom decile – translates to approximately a 0.25pp reduction in PD. To put this in perspective, observed default probabilities in our sample range from near zero to around 5%. For a company in the worst decile of default risk, where PDs typically reach 1–2% or higher, a 0.25pp reduction represents a 12–25% relative improvement in creditworthiness – a meaningful shift that can alter risk classification. The effect compounds across large credit portfolios and is substantially larger for the environmental pillar using Refinitiv data: the stronger E-pillar coefficient ($\beta = 0.09$ per point, from Table 1) implies that a 10-point improvement in environmental performance corresponds to an even greater PD reduction – underscoring that environmental risk management is the single most credit-relevant dimension of sustainability.

Table 5: Sustainability tercile effects on Altman Z-score – Spline regression (MSCI data)

Sustainability group	Beta coefficient	Significance
Low (bottom tercile)	0.10	***
Medium (middle tercile)	0.01	-
High (top tercile)	0.02	-

Controls: Profitability, Size, Leverage, Cash-flow Strength, Interest Coverage – all significant at 1%.

Sources: MSCI, Allianz Research



Implications for financial institutions, investors and businesses

Commercial banks should embed the floor; pricing the premium is about conviction. The core finding for commercial banks is specific: Borrowers in the bottom sustainability tercile carry statistically higher default risk, and the 0.25pp PD reduction per sustainability decile is material in credit loss terms at portfolio scale. The practical response is to embed sustainability as a negative screen in credit due diligence: flagging and re-pricing exposure to low-sustainability borrowers, rather than rewarding high sustainability credentials with lower margins across the board. A borrower moving from moderate to strong sustainability does not generate a proportionate reduction in expected credit loss but a borrower in the bottom tercile generates elevated risk that traditional financial analysis may not fully surface. In practice, this integration means incorporating sustainability risk flags into internal ratings and pricing grids, particularly for borrowers in carbon-intensive sectors and those with governance deficiencies.

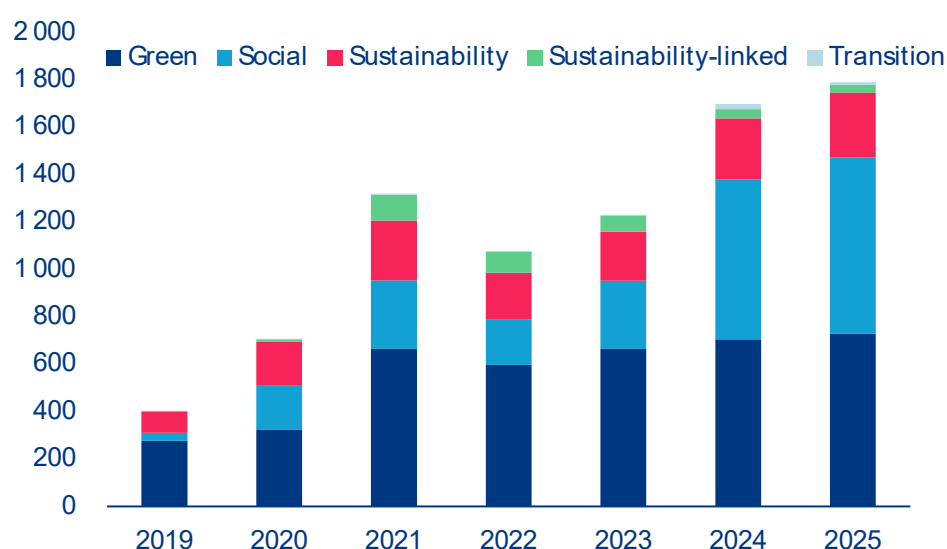
Broad-based credit investors should consider ESG filters as downside protection, not necessarily an alpha generator. For asset managers and credit investors, our findings support sustainability integration primarily as a downside risk mitigation tool. Indeed, the credit signal concentrates in identifying default-prone issuers, not in predicting outperformance by top-rated ones. A portfolio that systematically screens out bottom-tercile sustainability issuers captures the bulk of the credit risk

reduction; however, tilting toward the highest-rated sustainability issuers does not generate proportionate improvement in expected risk-adjusted returns. This reframes the analytical objective: the primary use case is loss avoidance, not return enhancement. This framing is consistent with fund-level evidence that sustainability-tilted credit portfolios have exhibited lower spread volatility and drawdowns during stress periods – precisely because the downside-protection effect is real and concentrated at the weakest issuers. Forward-looking sustainability indicators – transition strategies, green capex trajectories, emissions commitments – are increasingly relevant as transition risk begins to crystallize in credit spreads, particularly in the three sectors where our analysis shows the largest effects. Issuer engagement focused on sustainability risk management in communication services, consumer staples and energy represents a more targeted and defensible application of the credit signal than broad portfolio sustainability tilts. Investors who can distinguish genuine sustainability improvement at the bottom of the distribution from cosmetic score improvements elsewhere will have the most informative signal as the market matures.

Insurers are exposed to sustainability-related credit risk through both underwriting and investment activities and the sector concentration in our findings is directly relevant to where analytical effort should be focused. The credit signal from sustainability is largest in communication services, consumer staples and energy, and weakest in materials and IT. A uniform sustainability overlay applied to a corporate bond portfolio overstates the signal in low-materiality sectors and risks underweighting it in high-materiality ones – concentrating integration on the three high-signal sectors generates a materially better signal-to-noise ratio. The analytical focus for insurers is also shifting from static ESG scores toward forward-looking indicators to distinguish issuers that are credibly managing transition risk from those accumulating deferred exposure. During phases of spread decompression, when idiosyncratic credit risk dominates and return dispersion rises, sustainability-integrated analysis is most valuable precisely because the signal concentrates at the bottom of the distribution: it helps identify which issuers are deteriorating while others hold, rather than providing a uniform quality tilt. As climate change and shifting social expectations reshape sectoral risk profiles over the medium term, sector-level sustainability views should feed directly into strategic allocation decisions – the large dispersion our analysis shows across sectors in how strongly sustainability affects credit risk is itself a portfolio signal.

For corporations, improving sustainability can yield tangible benefits in terms of credit quality and funding costs. Our findings translate into a specific and financially grounded strategic priority for corporate issuers: the credit and financing benefits of sustainability concentrate at the threshold, and the penalty for falling below it is real. Companies with weak sustainability profiles (particularly weak environmental practices and poor governance) face measurably higher borrowing costs, tighter capital market access and elevated default risk. The strategic implication is to prioritize the most material sustainability risks rather than optimizing for the highest possible combined ESG score. Clear, measurable commitments to addressing material environmental and governance risks, transparent reporting and credible transition strategies are the inputs that matter for credit assessment. Companies that credibly clear the sustainability floor benefit from improved credit terms and a broader investor base; those that do not will face systematic pricing penalties as banks and investors formalize negative screen frameworks. The ESG investment that removes a credit penalty has a higher financial return than the ESG investment that earns marginal recognition at the top of a distribution where the signal is weak.

Figure 3: Labelled bond issuance volume (in USD bn)



Sources: Bloomberg. Includes corporate, government, supranational bonds, municipals and securitized products

Figure 4: 12M Normalized OAS

Source: Bloomberg

The expanding market for sustainability-labelled bonds provides investors with targeted exposure to environmental and social objectives, supported by use-of-proceeds and reporting frameworks that enhance transparency and allow for an assessment of issuers' sustainability trajectories. Issuance of GSSS bonds has risen markedly over the last six years, driven largely by issuers with comparatively strong sustainability profiles (see Figure 3). It remains dominated by green bonds, with activity concentrated in the financials and utilities sectors. While corporate green bonds historically traded at a premium ("greenium") due to scarcity amid strong demand, this premium has largely vanished at the aggregate index level as the market has scaled up, though sector-level dispersion persists (see Figure 4). Empirically, green bonds have exhibited lower spread volatility during market widening phases than the broader corporate bond universe, suggesting potentially more resilient risk-return characteristics, provided issuers' sustainability commitments are credible.

Appendix: Data and methodology details

Refinitiv Sustainability Scores – Methodology:

Refinitiv sustainability scores are constructed using a transparent, rules-based methodology that aggregates approximately 180–200 publicly available environmental, social, and governance indicators into standardized measures of corporate sustainability performance. Raw indicators, derived from company disclosures and publicly reported information, are mapped into defined sustainability categories and themes, normalized on an industry-relative basis, and combined using pre-determined, industry-specific weights to produce Environmental, Social, and Governance pillar scores on a 0–100 scale. In addition to the standalone ESG score, Refinitiv calculates an ESG Combined (ESGC) score, which incorporates an overlay for sustainability-related controversies by adjusting the ESG score based on the severity and frequency of documented incidents. Both ESG and ESGC scores are updated as new disclosures and controversy information become available.

Data sources and sample construction: We compiled firm-level data from Refinitiv covering the years 2011 through 2024. The ESG data comes from Refinitiv's ESG database, which provides an ESG Combined Score (a comprehensive rating including environmental, social, governance aspects and deducting for controversies) as well as separate pillar scores for Environmental, Social, and Governance (each scaled 0-100, with higher = better ESG performance). We also collected financial data from Refinitiv for the same firms, including balance sheet and income statement items required to calculate the Altman Z-score and other financial ratios. Our initial universe included over 130,000 companies globally, but many were missing ESG or financial data. We restricted the sample to firm-year observations with non-null values for the variables of interest. The final panel comprises about 280,000 observations on ~7,400 unique firms worldwide. The regional breakdown of firms (by headquarters) was: roughly 18% North America, 20% Developed Europe, 6% Japan, 7% other developed Asia-Pacific, 34% Emerging Markets, and 15% unclassified/unknown region (the latter are cases where region wasn't specified, often smaller frontier market firms). Data coverage varied, with developed markets generally having better ESG disclosure. For instance, 67% of U.S. firm-year observations had an ESG score available, versus ~44% in Emerging Markets, as noted earlier.

Altman Z-score calculation: We used the **Altman Z-score** as a proxy for credit risk or default risk of a firm. The Altman Z-score is a well-known composite financial indicator developed by Edward Altman, which combines five accounting ratios using fixed weights. We applied the version for publicly traded firms (Altman's Z-score for manufacturing, since most of our firms fall in non-financial sectors). The formula is:

$$Z = 1.2 \times \frac{\text{Working Capital}}{\text{Total Assets}} + 1.4 \times \frac{\text{Retained Earnings}}{\text{Total Assets}} + 3.3 \times \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \times \frac{\text{Market Value of Equity}}{\text{Total Liabilities}} + 1.0 \times \frac{\text{Sales}}{\text{Total Assets}}.$$

This formula produces a single score where higher values indicate a lower probability of bankruptcy. As mentioned, $Z > 2.6$ is usually considered "safe", $Z < 1.1$ is "distressed", and the 1.1–2.6 range is a grey zone. We winsorized extremely high or low Z-scores to avoid undue influence of outliers (e.g., a few observations had negative equity or abnormally large market capitalizations leading to very high Z).

In our dataset, not all firms had the market value data needed (especially if not publicly traded or if data was missing), which is why the Altman Z sample is smaller (~2,100 firms). Notably, financial sector firms (banks, insurance) are excluded from Altman Z analysis since the Z-score isn't designed for them and their balance sheets aren't comparable.

Credit quality composite: To measure credit quality more broadly (and to include more firms), we constructed a composite index reflecting a firm's earnings and cash flow capacity, which are crucial to creditworthiness. Specifically, we took three indicators: Return on Assets (ROA = Net Income / Total Assets), Leverage (Total Debt to Total Assets) and Operating Cash Flow to Total Assets (OCF/TA), which capture profitability and cash generation respectively. We standardized these within each year (to account for economic cycles) and took an average to form a composite score. This "credit quality index" is not a standard metric like Altman Z, but it serves as a general gauge of a firm's fundamental strength to service debt – high values mean the company is profitable and generating cash relative to its asset base (implying it likely can cover interest and reinvest, hence higher credit quality). We included this second measure because credit risk is multifaceted; a firm might have a low Z-score due to high market leverage but still strong cash flows to pay debts, or vice versa. The composite helps capture dimensions the Z-score might miss and greatly expands coverage to smaller firms that don't have market value data.

Regression specifications: We estimated panel regressions of the form:

$$\text{CreditMetric}_{i,t} = \alpha + \beta \times \text{ESGMetric}_{i,t-1} + \gamma \mathbf{X}_{i,t-1} + \text{FE} + \varepsilon_{i,t},$$

where $\text{CreditMetric}_{i,t}$ is either Altman Z or the credit quality index for firm i in year t . $\text{ESGMetric}_{i,t-1}$ is the ESG variable of interest (we ran separate regressions for the Combined ESG Score, ESG Score, Environmental pillar, Social pillar, and Governance pillar) lagged one year. $\mathbf{X}_{i,t-1}$ represents control variables, which included firm size (log of total assets) and profitability (ROA) – both lagged one year as well. “FE” denotes fixed effects; for the Altman Z models, we included sector-by-year fixed effects (SYFE), meaning we account for any shocks or trends affecting a whole sector in a given year (like an oil price spike affecting all oil & gas firms’ Z-scores in 2020, for example). In the credit quality models, we included both sector-year and region-year fixed effects (since that sample was much larger, we could control for regional macro conditions as well). These fixed effects absorb broad macroeconomic or industry effects, isolating the impact of ESG at the firm level. We did not include firm fixed effects because our ESG scores don’t exhibit enough within-firm time variation for many firms (many have short time series), and we aimed to capture between-firm differences as well – effectively assessing if higher-ESG firms have better credit metrics than lower-ESG firms, controlling for size and ROA.

We used robust standard errors clustered at the firm level to account for heteroskedasticity and serial correlation. The statistical significance was assessed at the 10%, 5%, and 1% levels as customary. The regression tables in the main text (Table 1 and 2) summarize the coefficient estimates for the ESG variables and their significance. The R-squared (within) for the Altman Z regressions were around 0.05 (i.e., 5% of the variation in Z within sectors/years was explained by ESG and controls – not surprising as many firm-specific factors drive Z). For the credit quality regressions, the within R-squared was higher (~0.69) because profitability (ROA) is itself part of the dependent variable and explains a lot of its variation, and fixed effects soak up a lot of variance as well.

Supplementary tables: Below we provide a summary comparison of regional results mentioned in the text for the credit quality regressions (coefficients of ESG variables by region), to give a fuller picture:

Table A1. ESG coefficient estimates by region – Credit Quality Index regressions (signs and significance of ESG Combined Score and Pillar effects)

Beta Coefficients of ESG vs Credit Risk	ESG Combined Score	E Pillar Score	S Pillar Score	G Pillar Score
US	positive & significant	positive & significant	positive	positive & significant
Developed Europe	positive & significant	positive & significant	positive & significant	positive & significant
Japan	positive & significant	positive & significant	positive	positive & significant
Developed America ex US	positive & significant	positive & significant	positive	positive
Developed Asia ex Japan	positive & significant	negative	negative	negative
Emerging Markets	negative	positive	positive	positive

Sources: Refinitiv & Allianz Research

References:

Date	Publication Source	Authors	Title	Main Findings
2011	Journal of Banking & Finance	Goss & Roberts	The impact of corporate social responsibility on the cost of bank loans	CSR concerns linked to higher loan spreads and tighter terms; lenders price ESG-related risk, especially for weaker borrowers.
2013	Journal of Business Ethics	Attig et al.	Corporate Social Responsibility and Credit Ratings	Higher CSR performance associated with better credit ratings, robust to endogeneity concerns.
2014	The Financial Review	Oikonomou et al.	The Effects of Corporate Social Performance on the Cost of Corporate Debt and Credit Ratings	Strong social performance linked to lower bond yield spreads and higher assessed credit quality.
2017	Principles for Responsible Investment (PRI)	PRI	Why ESG factors matter in credit risk analysis	ESG affects default risk, downgrade risk, and recovery values rather than just bond pricing.
2020	Banque de France / ACPR	Billio et al.	Inside the ESG ratings: (Dis)agreement and performance	Large ESG rating disagreement implies ESG–credit results depend strongly on provider choice.
2022	Review of Finance	Berg, Kölbel & Rigobon	Aggregate Confusion: The Divergence of ESG Ratings	Low correlation across ESG ratings; measurement noise complicates credit-risk inference.
2022	Journal of Risk Finance	Barth, Hübel & Scholz	ESG and corporate credit spreads	Higher ESG scores associated with lower CDS spreads; ESG shocks are priced in credit markets.
2022	NY Fed Staff Report	Seltzer, Starks & Zhu	Climate Regulatory Risk and Corporate Bonds	Climate regulatory risk causally worsens credit ratings and increases yield spreads.
2022	World Bank	Gratcheva et al.	Credit Worthy: ESG Factors and Sovereign Credit Ratings	Clarifies how ESG factors relate to sovereign creditworthiness; warns against mechanical ESG–rating mapping.
2023	International Review of Economics and Finance	Lian et al.	How does ESG performance affect bond credit spreads?	Better ESG linked to lower bond spreads via reduced financial risk and better transparency.
2025	International Review of Economics and Finance	Chen et al.	Impact of ESG factors on credit ratings	Environmental factors positively associated with European banks' credit ratings.

A photograph showing a group of diverse hands stacked on top of each other, resting on a tree trunk. The hands are of various skin tones and some have accessories like a black wristband and a silver ring. The background is a lush green forest. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

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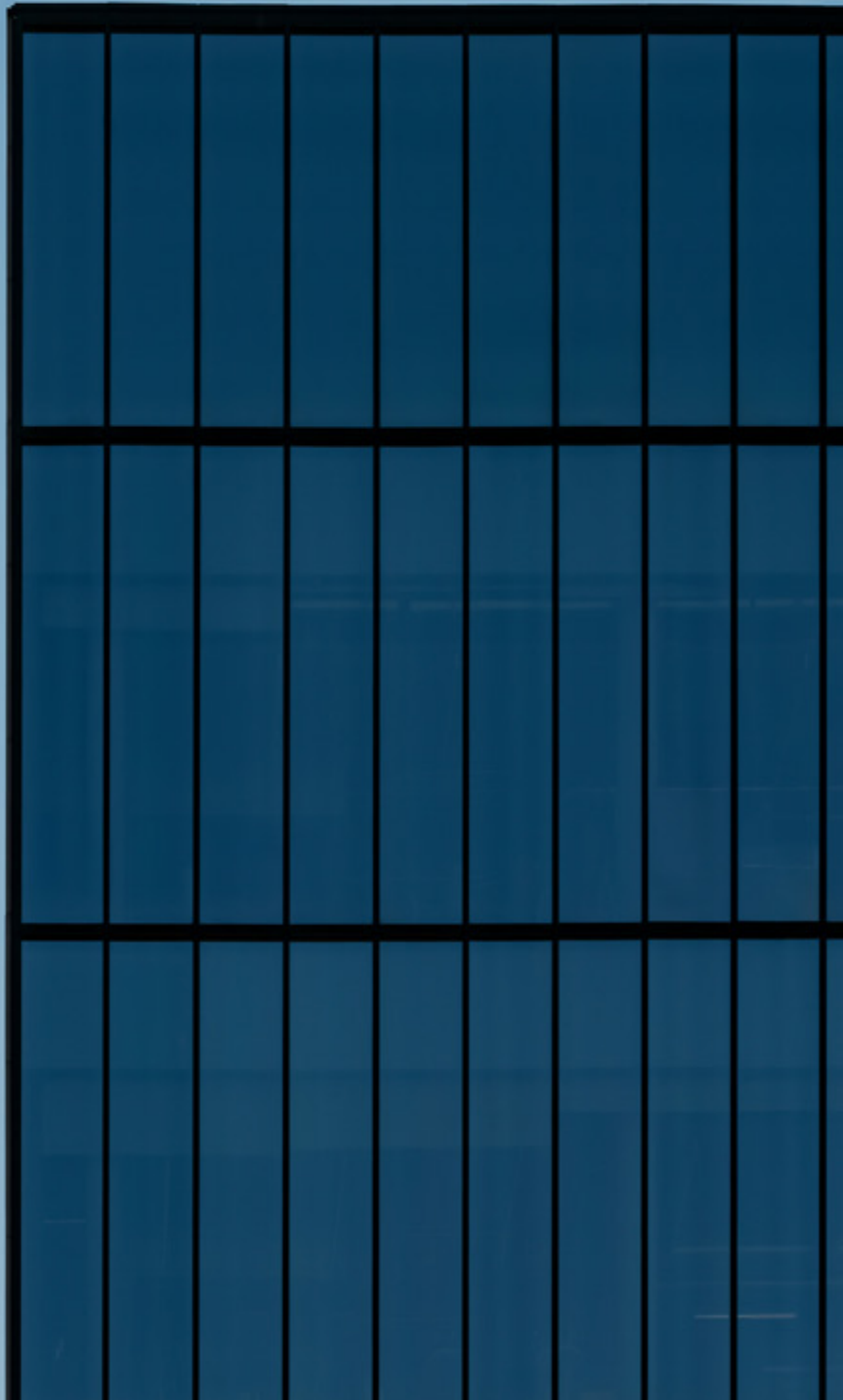
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Allianz Global Insurance Report 2026: The future of insurance in a fragmenting world

28 May 2026



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Executive Summary



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- The insurance industry is cooling from exceptional growth, not slowing into weakness.** Global premiums rose by +7.1% to EUR6.9trn in 2025, adding EUR456bn to the global premium pool. Although growth moderated from the exceptional +9.4% recorded in 2024, it remained comfortably above the ten-year CAGR of +5.6%, confirming that the industry's growth drivers remain firmly intact. Life insurance remained the largest segment (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn).
- Asia remains the structural center of growth even as advanced markets outperform cyclically.** Western Europe grew nearly twice as fast as its 2015–2025 average, while emerging market growth softened as China slowed to +7.4%, more than 3pps below its long-term average. Excluding China and Japan, however, Asia remained the industry's fastest-growing major region, with premium growth accelerating to +10.9% versus a long-term average of +4.9%.
- Much of the industry's recent expansion reflects higher prices rather than materially broader coverage.** Global insurance penetration rose only modestly to 7.2% of GDP in 2025 and remains below levels seen a decade ago. The disconnect is most visible in P&C, where penetration has stagnated around 2.5% of GDP despite strong premium growth. Trends across business lines are diverging: health-insurance penetration rose from 1.4% of GDP in 2015 to 1.8% in 2025, while life insurance at 3.0% remains well below the levels above 4% seen two decades ago.
- The P&C boom is normalizing while life and health remain structurally strong.** Global P&C growth slowed to +3.8% in 2025 as pricing cycles matured and claims inflation stabilized. North America remained dominant, accounting for 52% of global P&C premiums, while Asia continued to exhibit the world's largest protection gap, with penetration of just 1.3% compared with 4.3% in North America. Life insurance grew by a still robust +6.9%, though the US annuity boom is fading. Health insurance remained the industry's strongest growth story, expanding by +12.3%, the fastest pace since 2014, driven by rising medical costs and growing demand for private healthcare protection.
- Despite Asia's rise, global insurance remains overwhelmingly dominated by the US.** North America increased its share of global premiums from 42.5% to 46.4% over the past decade, meaning that almost every second euro written globally now originates from the region. China has emerged as a clear number two, increasing its market share from 7.5% to 10.9%. Yet, at EUR746bn in premiums, it remains less than a quarter of the size of the North America market at EUR3,191bn. Western Europe, by contrast, will continue to lose relative weight across most business lines.

- Geopolitics and fragmentation are becoming central forces shaping the insurance industry.** The Iran war is acting as a major external supply shock, disrupting energy markets, trade flows and supply chains. In our central scenario, global GDP growth is expected to slow to +2.6% in 2026, while Eurozone growth will fall to just +0.8%. If the conflict is not resolved during the summer, we expect additional upward pressure on inflation and a materially worse global growth outlook. More broadly, geopolitical fragmentation is creating a structurally more complex operating environment, challenging assumptions around global integration, capital mobility and cross-border risk diversification. Insurers will need to adapt by building more regionally resilient operating models, integrating geopolitical analysis more directly into underwriting and capital allocation and developing products tailored to emerging risks such as cyber escalation. While fragmentation raises costs and operational complexity, it also increases demand for protection and resilience, reinforcing the strategic relevance of insurance in a more uncertain global environment.
- Protection remains a structurally growing need in an increasingly uncertain and ageing world.** The global insurance market is expected to expand by +5.3% annually over the next decade, driven by rising demand for protection, demographic change, higher interest rates and growing health and retirement needs. Health insurance is projected to remain the fastest-growing segment at +6.7% p.a., followed by life insurance at +4.9% and P&C at +4.7%. Asia, particularly China and India, will remain the main growth engine, adding 5pps to their global market share, while North America is expected to retain its dominant global market position at roughly 46%. Europe, by contrast, is likely to continue ceding global market share over the coming decade (-4pps).
- Asia will generate most of the industry's future growth, with India emerging as the standout opportunity.** The global premium pool is expected to expand by EUR5,260bn by 2036, with more than half of additional premiums originating in Asia. China will remain the region's dominant market, but India is expected to emerge as the fastest-growing major insurance market. Despite already ranking among the world's ten largest, India remains significantly underinsured, with penetration of just 3.8% of GDP and per capita spending of roughly EUR85. Rising incomes, demographic change and recent reforms are expected to drive annual premium growth of around +10.7% over the next decade.
- Climate change is turning insurance affordability into a structural challenge for the industry.** Insured NatCat losses are rising by +5–7% annually in real terms, while higher premiums are increasingly colliding with weakening household purchasing power. The result is widening protection gaps, pushing lower-income households out of coverage. As affordability deteriorates, more climate losses are shifting onto already strained public balance sheets. Preserving long-term insurability will require moving beyond repricing toward stronger adaptation incentives, resilience-focused underwriting, targeted affordability mechanisms and deeper public-private risk-sharing frameworks, supported by large-scale investment in climate adaptation and resilient infrastructure.



Looking back: From boom to normalization

The global insurance industry returned to a more sustainable growth trajectory in 2025, with total gross written premiums (GWP) rising by +7.1% to EUR6.9trn. The expansion added EUR456bn to the global premium pool over the course of 2025. The life segment remained the largest (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn). However, growth decelerated from the exceptional +9.4% recorded in 2024, returning the industry toward a more sustainable trajectory after two years of super-charged growth. Crucially, even at this more moderate pace, the 2025 rate remained comfortably above the ten-year CAGR of +5.6%, confirming that the structural growth drivers that have fueled the global insurance market expansion since the mid-2010s remain in place. Over the full decade from 2015 to 2025, total GWP expanded from EUR3.97trn to EUR6.87trn – a cumulative addition of more than EUR2.9trn, driven by rising insurance penetration, expanding middle classes in emerging economies and the continued broadening of coverage across all three major segments.

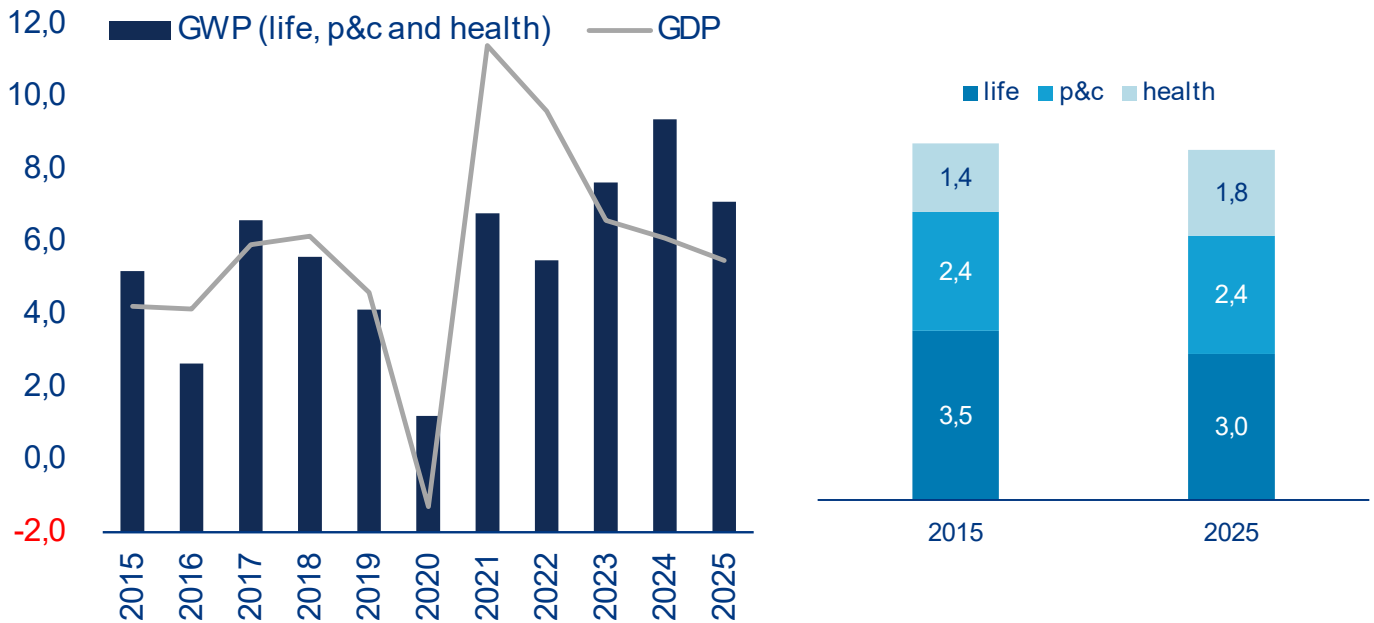
Much of the industry's recent premium growth reflects higher prices rather than a comparable deepening of insurance protection. With consumer price inflation running at approximately 2–3% across major markets in 2025, part of the headline growth merely reflects inflationary pass-through rather than additional coverage written. Insurance penetration, measured

as premiums relative to GDP, increased only modestly to 7.2% in 2025 from 7.0% in 2024 and remains below the 7.3% recorded a decade earlier. This divergence between nominal premium growth and real insurance deepening has become a defining feature of the post-2022 insurance cycle. While the industry has expanded substantially in monetary terms, the share of economic activity devoted to insurance has yet to recover to its long-run level. The disconnect is particularly visible in P&C, where higher claims costs have driven pricing increases without materially expanding insurance penetration.

The trend in global insurance penetration is increasingly uneven across business lines, with P&C broadly stable, health steadily gaining relevance and life insurance still recovering from the ultra-low-rate era. P&C penetration has remained broadly stable at around 2.5% of GDP, while health insurance has steadily gained importance, rising from 1.4% of GDP in 2015 to 1.8% in 2025 as demand for private healthcare protection increased. Life insurance, by contrast, continues to recover only gradually from the prolonged period of ultra-low interest rates. Although penetration has edged back up since 2022 to 3.0% of GDP in 2025, it is still below the 3.5% a decade earlier as the segment remains well below historical norms (Figure 1). The longer-term decline is even more striking: two decades ago, global life insurance penetration exceeded 4% of GDP.

Figure 1: Premium growth remains ahead of GDP

Global gross written premiums* and nominal GDP growth* (y/y, in %) and global gross written premiums* as % of GDP by segments



*The conversion into EUR is based on 2025 exchange rates.

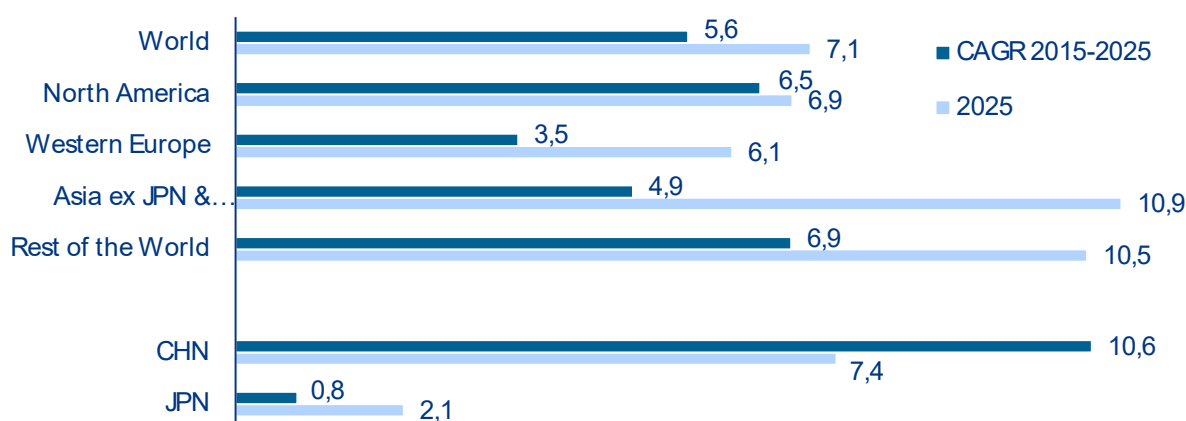
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

Growth remains strong globally, but the center of momentum is shifting as advanced markets outperform cyclically while Asia retains its structural growth lead. While all major regions grew faster than their respective long-term averages in 2025, advanced markets were the main drivers of this acceleration, with Western Europe (+6.1%) expanding nearly twice as fast as its 2015-2025 average. Emerging market growth, by contrast, softened relative to the past decade, largely reflecting slower expansion in China, which accounts for almost 60% of the emerging market aggregate. Chinese premium growth slowed to +7.4% in 2025, more than 3pps below its long-term average. While this is

to a large extent a function of past excellence rather than a weak current performance, nominal growth has also been held back by annual inflation registering just above 0%. Meanwhile Asia¹ (when excluding China and Japan) continues to significantly outperform Europe and North America, with premium growth reaching +10.9% in 2025 compared with a long-term average of +4.9%, reinforcing its position as the industry's key structural growth market.

¹ Throughout the report, Asia always refers to the region excluding China and Japan. If both countries are included, we use the term wider Asia.

Figure 2: 2025 growth outpaces long-term average
Gross written premium growth*, 2025 and CAGR 2015-2025 by region in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global P&C market is entering a phase of normalization after several years of exceptional pricing-driven growth. The segment grew by +3.8% in 2025 to reach EUR2,320bn globally, a notably softer performance than both the overall market and the segment's own ten-year CAGR of +5.6%, as well as last year's growth of +8.5%. The moderation reflects a maturing pricing cycle. After several years of sharp rate increases across motor, property and specialty lines, pricing has started to normalize in some regions, while claims costs, although still elevated, are no longer accelerating at the pace seen in 2022 and 2023. North America² remains the undisputed center of global P&C activity, accounting for 52.0% of world P&C premiums, with EUR1,205bn written in 2025. However, regional growth slowed to +2.2% from +9.7% in the previous year, reflecting both a more mature rate cycle and a -6.3% decline in US accident & health premiums³ that weighed on overall market growth (US: +2.2%); excluding A&H, however, the US P&C market is estimated to have grown by 4.1%⁴(Figure 3).

Western Europe's P&C market posted solid growth in 2025, although the exceptional post-inflation growth surge is now beginning to normalize. Premiums grew by +5.3% to EUR529bn, a solid expansion but notably below the exceptionally strong growth rates of +8.5% in 2023 and +6.8% in 2024 as inflation pressures continued to ease. Performance across the region was increasingly uneven. Southern European markets remained particularly dynamic, with several recording

high single-digit growth above +8%, while Scandinavia, the UK, Belgium and Switzerland expanded more modestly at around +1% to +4%. Among the largest Eurozone markets, growth remained comparatively robust, led by Germany at +7.7%, France at +7.0%, Spain at +6.5% and Italy at +6.4%. Wider Asia P&C premiums grew by +4.1% to EUR391bn in 2025, broadly in line with the previous year. Yet despite a population more than six times larger than Western Europe's, the region's P&C market remains significantly smaller. The gap is equally visible in insurance penetration and spending levels. P&C penetration stands at just 1.3% in wider Asia and 1.1% in China, compared with 2.5% in Western Europe and 4.3% in North America. Per capita P&C spending reached only EUR98 across Asia (China: EUR135), compared with EUR1,232 in Western Europe and EUR3,111 in North America. These disparities underline the scale of structural underinsurance and the region's long-term catch up potential.

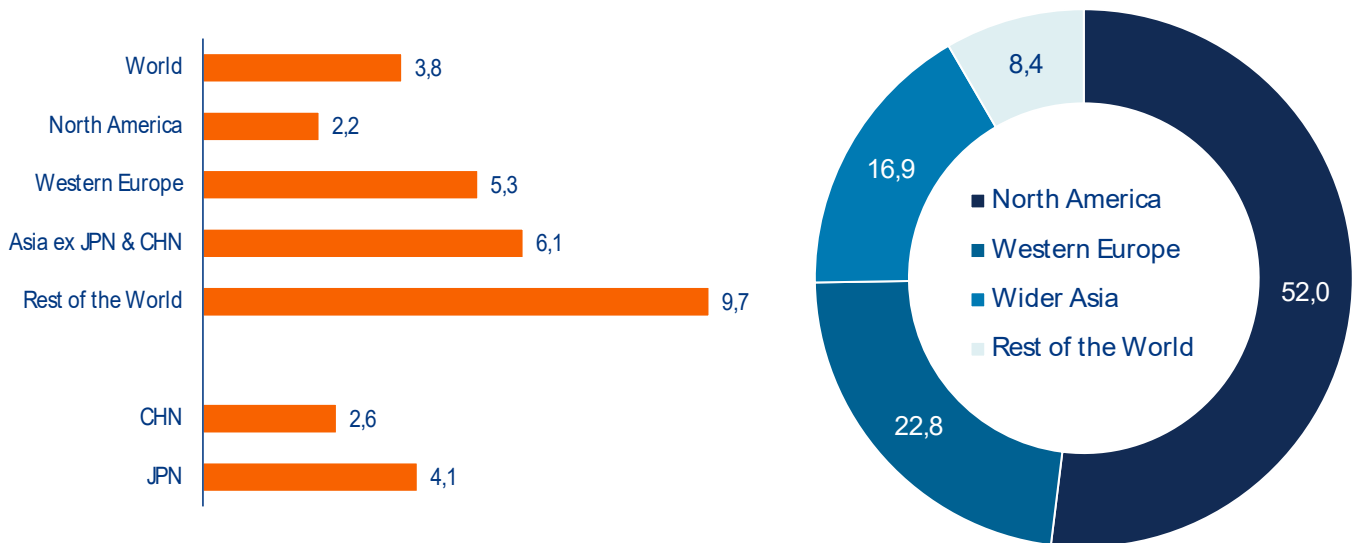
China is by far the largest market in the wider region, accounting for almost every second euro written. At +2.6% China registered one of the weakest performances in recent years outside the Covid-19 period and well below the country's average growth rate of +6.6% over the past decade. The slowdown largely reflects China's relatively muted nominal growth dynamics. All other insurance markets (rest of the world) recorded growth of +9.7% in the P&C segment, driven by strong (largely inflationary) increases of above +10% in Eastern Europe and Latin America.

² 96% of P&C premiums in North America are written in the US; the rest is attributable to Canada.

³ In this report, Accident & Health (A&H) premiums written by Life/A&H entities are included in the US P&C market figures, consistent with our general approach of classifying accident lines as part of the P&C market.

⁴ This estimate is based on growth in net premiums written as direct written premium figures had not yet been published at the time of writing.

Figure 3: Europe and Wider Asia power ahead
P&C gross written premium growth* and market share by region, 2025 in %



*The conversion into EUR is based on 2025 exchange rates.

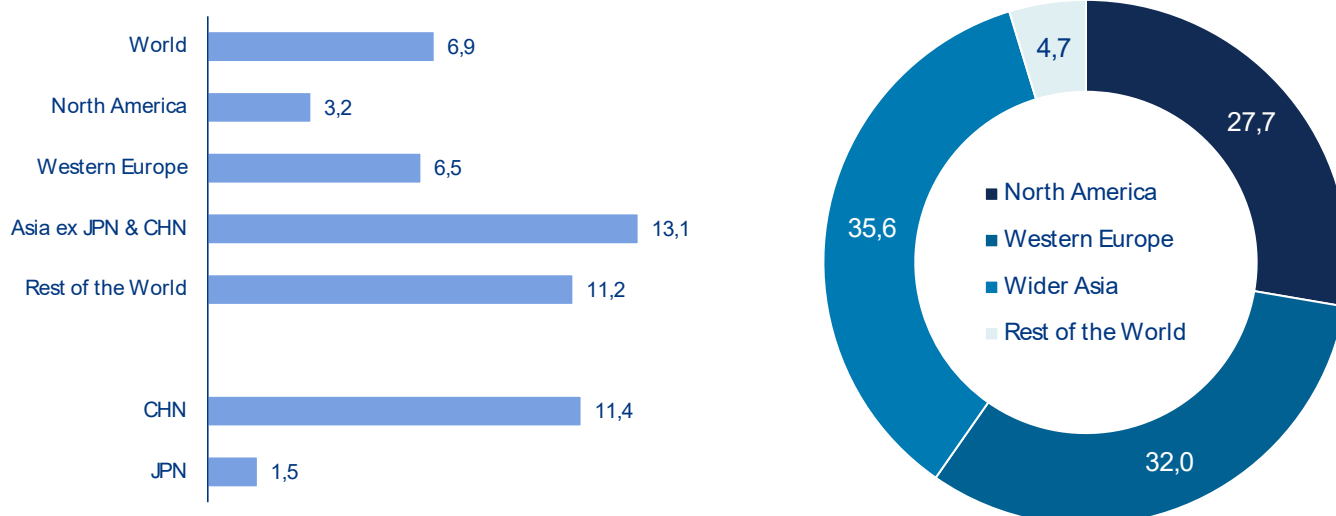
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The life insurance boom is cooling in North America, while Asia is reasserting itself as the industry's main growth engine. Global life insurance premiums grew by +6.9% in 2025, a clear moderation from the exceptionally strong +11.3% recorded in 2024 but still a robust performance by historical standards. The slowdown was driven primarily by North America, where the annuity boom triggered by households locking in higher interest rates has begun to lose momentum. Growth leadership has now shifted back toward Asia (Figure 4).

Higher interest rates continue to support life insurance globally, but regional momentum is increasingly diverging. In Western Europe, life premium growth moderated to +6.5% in 2025 from +9.9% in the previous year, although higher rates continued to support savings products. Performance remained uneven across markets. France, the Netherlands, Finland and Portugal posted another year of double-digit growth, while Switzerland continued to lag. Germany returned to positive territory with growth of +5.3%. Wider Asia recorded healthy +9.9% growth in 2025, with the majority of markets posting strong double-digit gains.

As a result, Asia as a whole remains the largest life insurance market, well ahead of North America and Europe (Figure 4), reflecting both demographic pressures and weaker public pension systems. This is also reflected in insurance penetration, which stands at 3.3% in Asia, driven by the popularity of life products in advanced Asian markets such as South Korea, Taiwan and Singapore. But China at 2.5% and India at 2.8% already approach North American levels of 2.8%, although still below Western Europe's 4.3%. These relatively high Asian penetration rates reflect greater reliance on private retirement savings in the absence of comprehensive public pension systems. By contrast, North America's comparatively low penetration partly reflects the widespread use of occupational retirement schemes such as 401(k) plans.

Figure 4: Asia pulls further ahead
Life gross written premium growth* and market share by region, 2025 in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

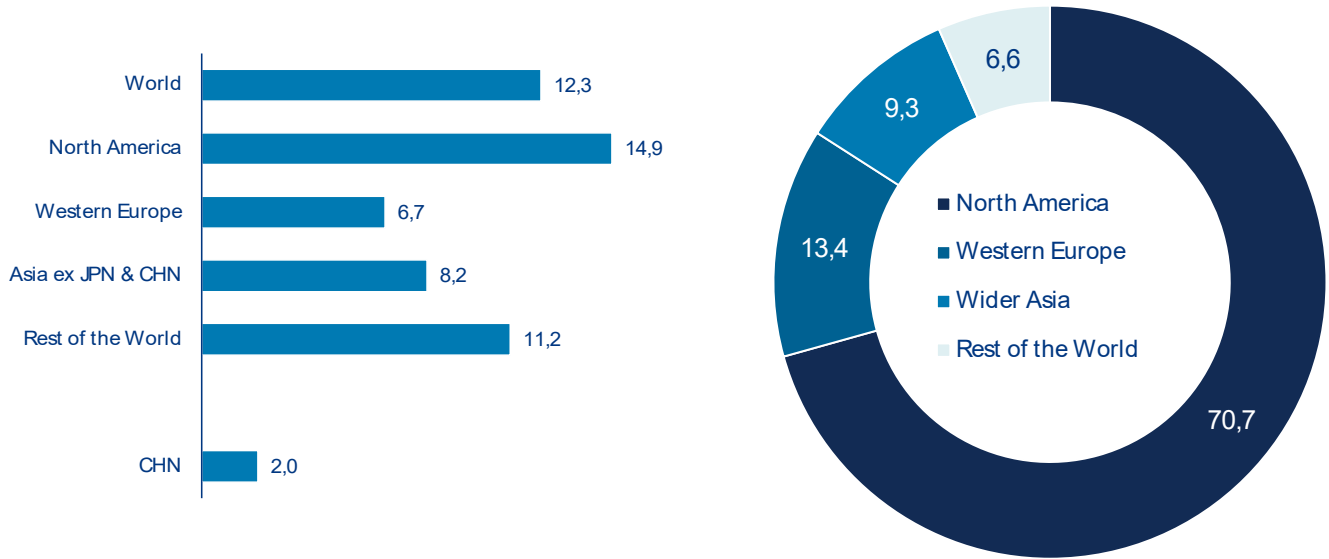
Health insurance⁵ remains the industry's fastest-growing segment, driven by rising medical costs and increasing demand for private protection. The US continues to dominate the global health market, accounting for more than 70% of worldwide health premiums (Figure 5). In many other markets, private health insurance is still a niche segment, albeit a very dynamic one: global premium growth averaged +8.5% p.a. over the last decade with particularly strong rates in recent years as Covid-19 boosted demand for additional health protection. In 2025, global health premiums increased by +12.3%, the strongest expansion since 2014, driven largely by North America where medical inflation pushed growth to +14.9%.

Despite a recent normalization in parts of Asia, the region's low health-insurance penetration continues to point to substantial long-term growth potential. Health-insurance growth in wider Asia slowed to +3.4% in 2025, reflecting a sharp deceleration in China, where growth dropped to just +2%. However, many markets

including Indonesia, Singapore, Hong Kong, Thailand and India continued to record double-digit growth, with several accelerating further in 2025. This strong momentum partially reflects the still low penetration in the region, which is below 1% in all markets except Taiwan. Even more than in life insurance, demand is mainly driven by the status of the social security system, i.e. the coverage and quality of the public healthcare system. This explains why even in some developed markets, such as the Scandinavian markets, private health insurance is almost non-existent.

⁵ In general, data visibility of health-insurance premiums is still rather low. Often, health premiums are included in other segments as health coverage is seen as part of other products and not separately reported; in Japan, for instance, health is treated as "third sector products" within life.

Figure 5: No slowdown in sight
Health gross written premium growth* and market share by region, 2025 in %



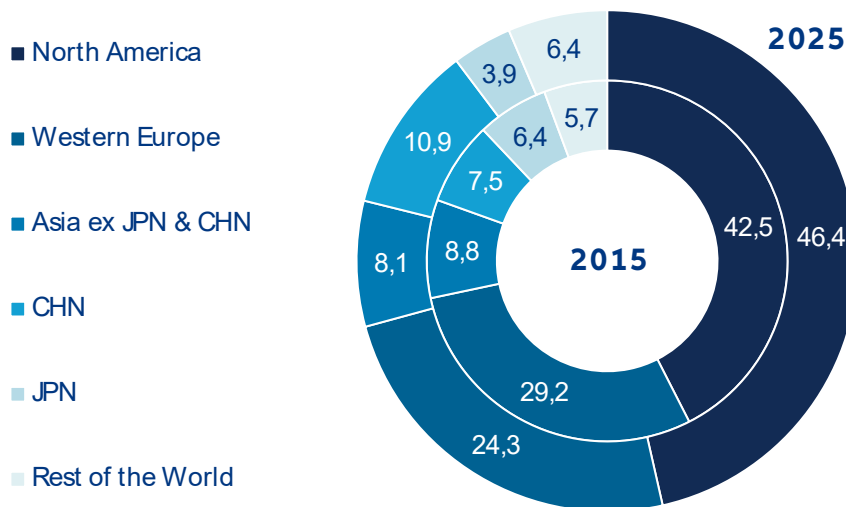
*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global insurance industry remains overwhelmingly dominated by the US despite the rise of China and broader Asia. Unlike many other industries where traditional markets are losing importance to new emerging markets, the global insurance industry is still dominated by the US. In fact, over the past decade, the North American insurance market has increased its share of global premiums from 42.5% to 46.4%, meaning that almost every second euro written globally now originates from the region. On the other hand, other „old“ markets such as Western Europe (-4.9pps) and

Japan (-2.5pps) behaved more or less as expected, losing market share mainly to China, which boosted its global share from 7.5% to 10.9%. However, the gap with the North America market is still huge: while the Chinese market is already the clear number two in terms of premium income (EUR746bn in 2025), it is dwarfed by the North America market, which generates more than four times as much premium income (EUR3,191bn). Given this huge discrepancy, it does not look like the US dominance in global insurance will end any time soon (Figure 6).

Figure 6: US dominance remains unchanged
Total gross written premiums*, 2015 and 2025 by region in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

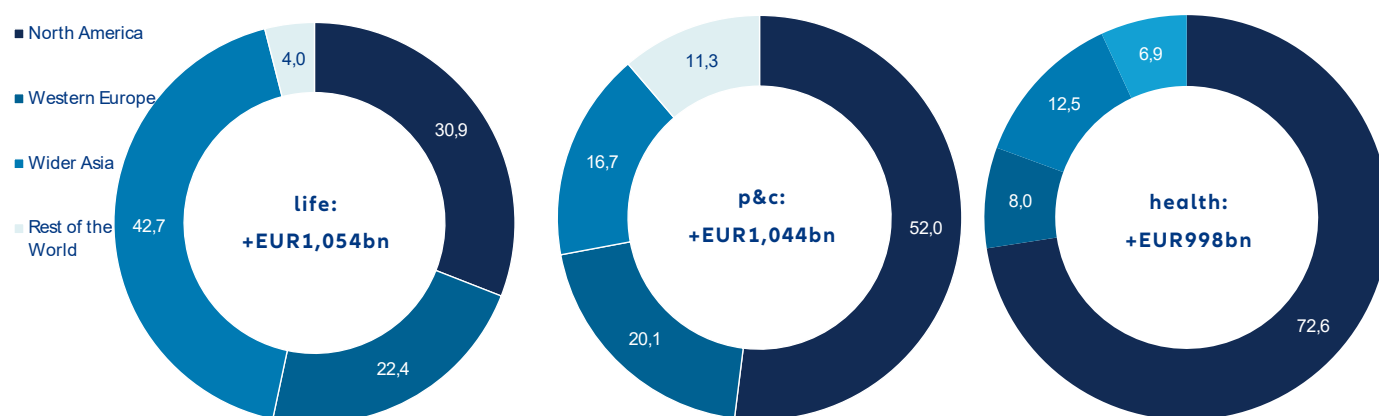
The past decade has reshaped the global insurance market unevenly across regions and business lines.

North America accounted for 52% of additional global P&C premiums and 71% of additional health premiums over the past ten years, underlining the region's overwhelming importance in both segments. In life insurance, however, the picture is more balanced. Despite the recent US annuity boom, only around one-third of additional life premiums originated in North America, while wider Asia captured by far the largest share of global growth. Western Europe, by contrast, continues to lose relative weight across all major business lines, falling behind not only the US market but increasingly also wider Asia.

Another notable shift is the growing convergence in absolute premium growth across segments.

Although life insurance remains substantially larger in size at EUR2,861bn compared with EUR2,320bn in P&C and EUR1,688bn in health, the absolute increase in premium volumes has recently been broadly similar. While life insurance growth was constrained for years by ultra-low interest rates, health insurance has benefited strongly from the post-Covid increase in risk awareness and demand for private health protection (Figure 7).

Figure 7: Western Europe continues to lose ground
Share of absolute premium growth* by region, 2015 - 2025 in %



*The conversion into EUR is based on 2024 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.



The changing risk landscape

Economic and capital market outlook

The global economy has entered a period of heightened uncertainty as the Iran war acts as a significant external supply shock, disrupting energy markets, raising transport costs and fragmenting supply chains. The result is renewed inflationary pressure alongside weaker growth at a time when the global outlook was already fragile. The scale and persistence of the economic impact remain highly uncertain and will depend critically on how the conflict evolves.

In our central scenario, in which a US-Iran agreement struck during the summer months leads to a normalization of oil and gas trade before the end of the year, we forecast global GDP growth of 2.6% in 2026, revised down by 0.5pp from pre-conflict expectations. The impact across major economies is uneven. In the US, growth is expected to remain

relatively more robust at +2.0% as substantial investment in artificial-intelligence infrastructure and stronger private consumption driven by tax cuts largely offset the energy shock. China would also prove comparatively resilient at +4.7%, supported by a more diversified energy mix and solid external demand. Europe, however, would bear the heaviest burden. Eurozone growth would nearly halve to +0.7% as the region's greater dependence on energy imports adds to longstanding structural weaknesses that continue to constrain economic momentum.

Inflation is set to rise on both sides of the Atlantic in 2026, interrupting the gradual normalization seen over the past year. In the US, the energy shock will push inflation back toward levels last seen during the post-pandemic surge. CPI inflation is forecast to rise to 3.6% in 2026, peaking at around 4.0% around mid-year

before gradually easing as energy prices stabilize and weaker activity dampens underlying price pressures. The Fed is expected to raise rates by 25bps in 2026, largely looking through the energy-driven spike as the risk of inflation expectations becoming de-anchored appears contained. After all, unlike in 2022, labor markets are softer and monetary conditions are no longer highly accommodative, limiting the scope for broad-based second-round effects. In the Eurozone, inflation dynamics would be similar in direction but more persistent in profile. Headline CPI is forecast at 3.1% for 2026 as the pass-through from higher wholesale gas prices into household utility bills keeps inflation elevated for longer than in the US. This leaves the ECB in an uncomfortable position: inflation is rising above target just as growth weakens sharply. We forecast the ECB to hike rates by 50bps in 2026, with a focus on keeping inflation expectations in check.

Government bond markets are undergoing a sharp repricing as the energy shock pushes inflation expectations higher while simultaneously weakening the growth outlook. Long-term sovereign yields are rising as investors demand greater compensation for inflation risk and increasingly question the sustainability of large fiscal deficits in a weaker growth environment. On balance, we expect 10-year Treasury yields to reach 4.6% by year-end, with persistent fiscal deficits of around 7% of GDP adding further upward pressure on US borrowing costs. In Europe, 10-year Bund yields are forecast at 3.0%, as weaker growth and safe-haven demand from investors diversifying away from US assets partially offset rising inflation expectations. In the periphery, however, Italian and French borrowing costs

are increasing more sharply than Germany's as markets grow more concerned about the capacity of heavily indebted sovereigns to absorb energy-related fiscal pressures without compromising fiscal consolidation commitments – a dynamic likely to intensify as France approaches its 2027 election cycle.

Risks to the outlook remain clearly skewed to the downside. A prolonged closure of the Strait of Hormuz into fall would produce a materially more adverse outcome. In this downside scenario, the Eurozone economy would fall into technical recession, with annual growth in 2026 slowing to just +0.2%, while US growth would weaken sharply to +1.5% as equity market corrections feed through to consumer confidence and spending. Inflation would rise to 4.1% in the Eurozone and 4.2% in the US, forcing central banks into a more aggressive tightening response despite deteriorating growth conditions – a policy dilemma with few good options. Beyond the direct energy channel, a prolonged conflict would amplify second-round effects through weaker consumer confidence, rising insolvencies, tighter financial conditions and growing fiscal strain across energy-importing economies. At a time of weakening international policy coordination and increasingly constrained public finances, policymakers may have less capacity to cushion additional shocks. In such an environment, non-linear dynamics could emerge quickly, turning an initial energy shock into a broader and more persistent global downturn.

How geopolitical fragmentation impacts insurance

The insurance industry was built for a world of open capital flows, stable rules and predictable cross-border commerce, and that world is receding. A series of geopolitical shocks, from the US-China trade conflict and the Covid-19 pandemic to the Russia-Ukraine war and the US-Iran conflict, has accelerated the fragmentation of trade, finance and supply chains, pushing the global economy toward a more fractured geoeconomic order. For insurers, this is not simply a challenging macroeconomic backdrop but a structural shift in the foundations of the global insurance business model.

The question is no longer whether fragmentation will reshape global insurance, but how far and how fast. The assumptions that supported global operating models for decades are weakening simultaneously: capital is becoming less mobile, regulatory regimes are diverging and borders are increasingly functioning as strategic barriers rather than manageable operational frictions. For insurers, this directly challenges the foundations of global risk pooling. Insurance depends on the ability to diversify exposures across geographies and move capital and reinsurance capacity freely across borders. Fragmentation weakens both. As cross-border activity becomes more constrained, volatility in claims, currencies and investment returns increases, while regional pools struggle to replicate the efficiency and resilience of global ones.

Regional pools are structurally less efficient because fragmentation increases correlation across exposures while reducing the fungibility of capital and reinsurance. Political shocks increasingly affect currencies, asset prices, trade flows and claims simultaneously, weakening diversification benefits precisely when they are most needed. At the same time, sanctions risk, capital controls and regulatory divergence increase the likelihood that capital becomes trapped within jurisdictions during periods of stress, forcing insurers to hold more capital against the same underlying risks. This undermines one of the key advantages of global insurance groups: efficient cross-border capital allocation and diversification. As trapped capital and duplicated solvency buffers rise, returns on

capital are likely to come under structural pressure. The challenge is therefore no longer preserving the hyper-globalized insurance model of the past, but maintaining sufficient diversification and risk-sharing capacity in a more regionalized world.

Regulatory fragmentation is reinforcing these pressures. The decades of supervisory convergence that produced Solvency II, IAIS standards and equivalence regimes are giving way to more modular and bloc-specific systems. Cross-border reinsurance recoveries, sanctions compliance and contract enforceability can no longer be treated as stable assumptions. Meanwhile, the weakening of multilateral cooperation is amplifying the very risks insurers must absorb. Climate change, cyber threats, pandemics and financial instability all require international coordination to contain; as geopolitical rivalry undermines that cooperation, these risks become larger, more interconnected and harder to price.

The investment consequences are equally significant. Insurers collectively manage trillions in long-duration assets and fragmentation permanently raises the cost and complexity of deploying that capital globally. When the movement of capital increasingly requires political approval rather than responding to market logic, global integration loses efficiency by design. Currency and convertibility risks rise alongside political tensions, sovereign risk becomes harder to hedge and friendshoring creates exposures that are often less liquid and operationally more complex.

The effects are unevenly distributed across business lines. Political-risk insurance is experiencing a structural increase in demand as multinational companies move from treating geopolitical exposure as episodic to embedding it permanently within risk-management frameworks. In trade credit, tariffs are squeezing margins and raising insolvency risk, particularly among smaller companies exposed to disrupted supply chains. Marine and aviation insurers must navigate rerouted shipping lanes and persistent geopolitical volatility; the closure of major maritime chokepoints has shown how quickly coverage availability and pricing can deteriorate when political

risk crystallizes into physical disruption. State-sponsored cyber threats are escalating, with attribution challenges straining the boundaries of standard policy language. In D&O, rising regulatory investigations and extra-territorial enforcement actions are increasing claims exposure for internationally active firms. Retail insurance faces a more indirect but persistent transmission through slower growth, higher inflation and weaker household purchasing power.

At the same time, fragmentation is also creating new insurable risk categories that reward analytical capability and underwriting specialization. The push for energy independence is increasing demand for specialized engineering insurance, while industrial policy initiatives including the CHIPS Act, European sovereignty investment programs and rising defense spending are generating substantial new property and infrastructure exposures. Geopolitics is on balance a net headwind for the sector's long-term growth prospects, but rising investment in defense, energy security, cyber resilience and strategic infrastructure is creating offsetting opportunities for insurance, risk transfer and long-duration capital deployment.

The fact that the insurance sector has remained broadly stable through successive geopolitical shocks so far is no guarantee of future success. Capital positions have remained healthy, combined ratios have held and capacity has not materially decreased. This is testament to the resilience of the insurance industry. However, each shock raises the structural baseline, which is not fully reversed during recovery periods. This means that the industry has to reprice for a genuinely higher level of structural risk. This makes strategically adapting to a fragmenting world a matter of current urgency.

The industry's strategic response must operate on several levels simultaneously.

- **First, operating models must become less centralized and more regionally resilient.** In a more fragmented world, resilience and adaptability matter more than pure scale efficiency. Capital must be increasingly pre-positioned within jurisdictions rather than assumed freely transferable, and entities must

be capable of operating with greater autonomy during periods of geopolitical disruption. Scenario planning must also replace narrow point forecasts. Tail risks such as severe bloc bifurcation or financial decoupling are no longer remote enough to ignore in strategic planning, even if the most likely trajectory remains one of escalating but contained protectionism.

- **Second, portfolio and underwriting decisions must incorporate explicit geopolitical assessment as a systematic input rather than a qualitative overlay.**

Insurers should classify markets along a clear spectrum from core to optional to potential exit candidates, combining underwriting analysis with regular geopolitical, sanctions and affordability assessments rather than making decisions reactively under pressure. Stress testing must extend beyond financial variables to include sanctions regimes, reinsurance enforceability and diversification failure under fragmentation scenarios. The industry cannot assume that mechanisms designed for a globalized world will continue to function under sustained geopolitical strain.

- **Third, product strategy must evolve to address the new exposures fragmentation is creating rather than relying primarily on exclusions and silent coverage.** Recent conflicts have exposed significant gaps in affirmative cover for war risk, political violence, trade disruption, cyber escalation and supply-chain interruption. Insurers that develop explicit and well-priced solutions for these exposures will benefit from a structural increase in corporate demand. Those that rely primarily on exclusions risk marginalization from some of the fastest-growing segments of the commercial market.

Fragmentation will reshape how the global insurance industry operates. A more regionalized world implies higher capital requirements, weaker diversification benefits and greater operational complexity. Yet the same forces that reduce efficiency are also increasing demand for protection, resilience and specialized risk transfer. Insurance may become structurally more complex and capital-intensive, but also strategically more important.



Looking ahead: Insurance remains a growth industry

Insurance will remain a structural growth industry over the next decade, although growth will become more regionally uneven. We expect the global insurance market to expand by +5.3% annually over the next ten years, broadly in line with last year's long-term outlook. Beneath this stable headline picture, however, regional dynamics are shifting meaningfully.

North America is likely to remain the industry's dominant profit and growth pool. Compared to last year, our outlook for the region has become more optimistic across all three business lines, particularly in P&C. Three forces are driving this shift: the expected AI investment boom and associated productivity gains, continued increases in household spending on protection – especially in health insurance – and elevated interest rates supporting life and annuity products.

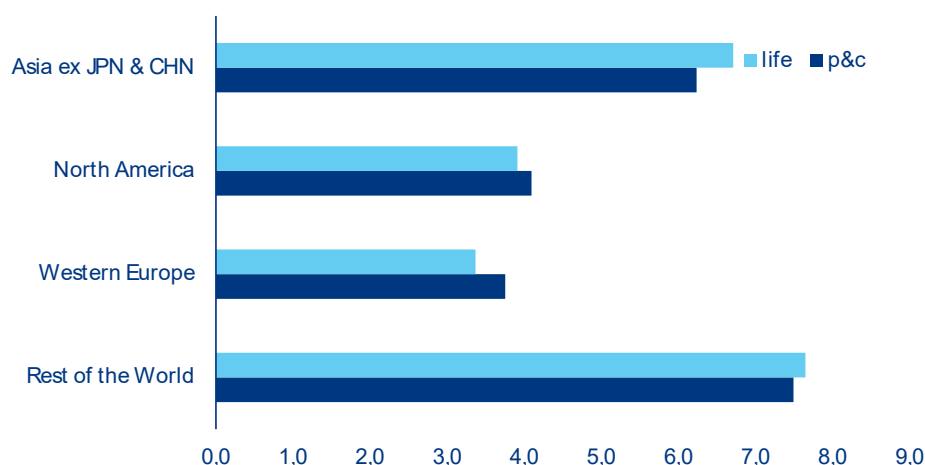
Europe faces a structurally weaker outlook. Our view on Western Europe has become more cautious compared with last year. The region is likely to continue struggling with the fallout from higher energy costs triggered by the escalation in the Middle East (see Chapter 3 for more details), slower than expected progress on strategic sovereignty and both weaker AI investment and slower AI adoption than the US. As a result, we expect European P&C growth to lag the US over the next decade at +3.8% versus +4.1%. Although

rising natural-catastrophe losses should continue to support premium growth, we now expect only a limited increase in penetration rates, rising by just +0.1pp to 2.6% by 2036.

Asia will remain the industry's principal long-term growth engine despite rising geopolitical headwinds. We expect P&C growth in Asia to slow modestly relative to previous forecasts as trade tensions and supply-chain restructuring weigh on economic momentum. Even so, projected annual growth of +6.2% in wider Asia and +6.3% in China remains comfortably above expected growth rates in Europe and North America. Overall, we expect global P&C premiums to expand by +4.7% annually through 2036 as demand for protection continues to rise worldwide.

Life and health insurance will remain the industry's key structural growth segments. We expect life insurance to grow at a CAGR of +4.9% over the next decade, benefiting from elevated interest rates and rising retirement needs. Asia and China should remain the principal growth engines as demographic ageing accelerates and pension systems continue to deepen. Health insurance is expected to remain the industry's fastest growing segment at +6.7% annually, supported by rising healthcare costs and still substantial unmet demand, particularly across Asia (Figure 8).

Figure 8: Solid growth
Gross written premium* growth, CAGR 2026 -2036 by region and segment in %



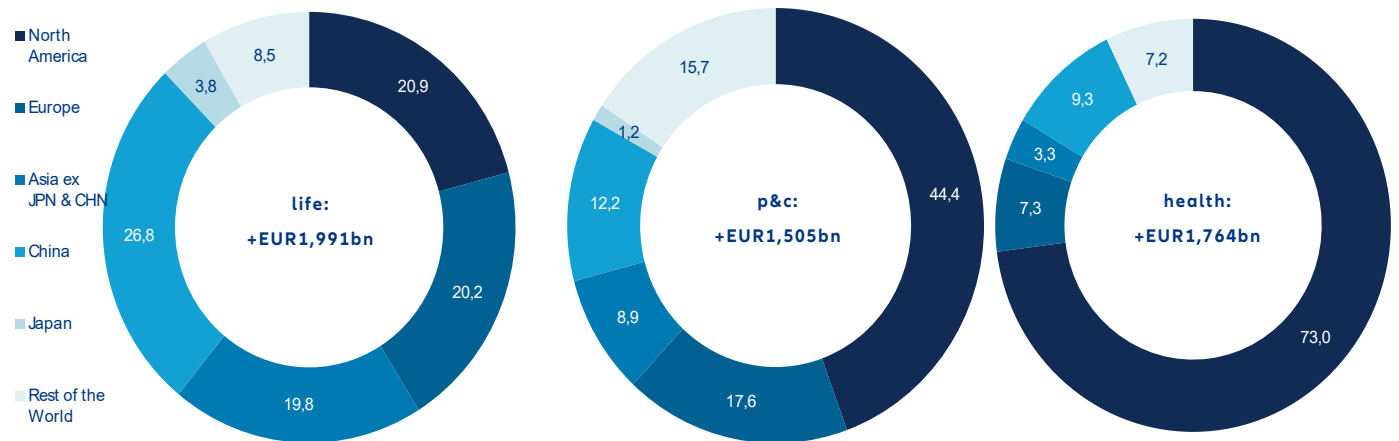
*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research..

Asia will generate most of the industry's additional premium growth. The global premium pool is expected to expand by EUR5,260bn by 2036 (Figure 9), with life insurance contributing the largest share at EUR1,991bn. More than half of all additional premiums will originate in Asia including China, exceeding the combined contribution of North America and Europe. While China will remain the region's dominant market in absolute terms with growth of +7.6% annually, India is expected to emerge as the fastest-growing major market at +10.5% annually, positioning it to challenge Japan as Asia's second-largest life insurance market by 2036.

Despite Asia's rise, the US insurance market will remain unrivaled in scale. Although North America's share of incremental P&C growth is expected to decline relative to the previous decade, the US market will continue to dominate globally. By 2036, US P&C premiums are projected to exceed EUR1,800bn, almost five times larger than China's market and more than twice the size of Europe's. In health insurance, US dominance is even more pronounced, with more than two-thirds of expected global premium growth projected to originate from the US market.

Figure 9: Life remains on top
Share of additional gross written premiums* by 2036, by region in %

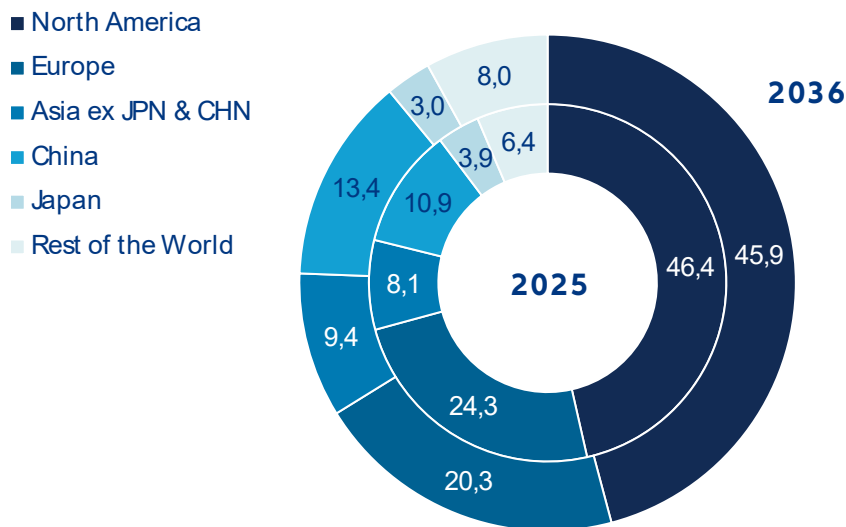


*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global insurance map will continue shifting eastward, albeit gradually. North America is expected to retain a global market share of roughly 46% through 2036, surrendering only marginal ground over the next decade (-0.5pp). India and China, by contrast, are expected to continue gaining relevance, together adding almost 4pps of global market share. Western Europe will

continue to lose relative weight. A glimmer of hope for the Old Continent: while it lost 5.3pps of market share in the last decade, it may lose „only“ 4pps in the next decade (Figure 10).

Figure 10: North America, a slightly smaller giant
Total gross written premiums*, 2025 and 2036 by region in %



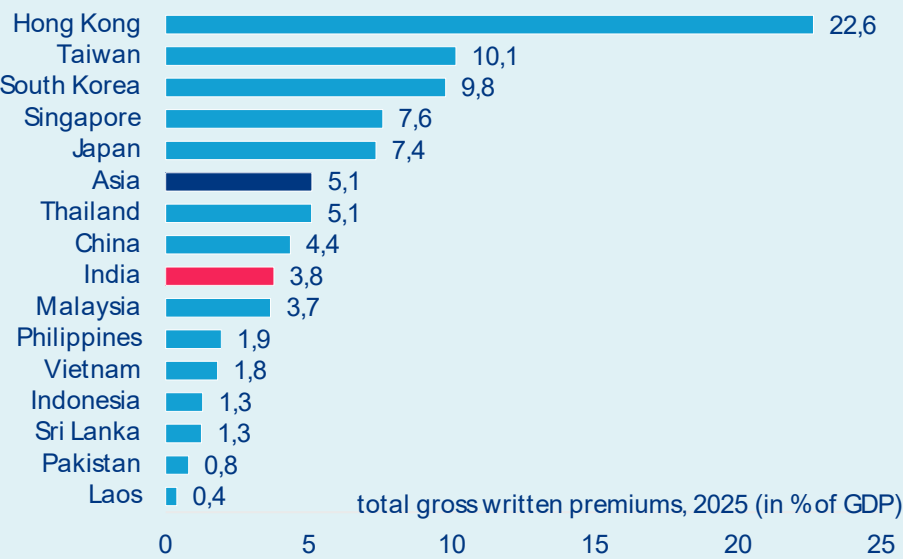
*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

Indian insurance market deep dive – turning scale into depth

India ranks among the world’s ten largest insurance markets. But by almost every measure of depth, it remains significantly underdeveloped. Despite double-digit growth rates and total gross written premium income of EUR124bn in 2025, insurance penetration and density still indicate substantial pent-up demand. In 2025, total gross written premium income corresponded

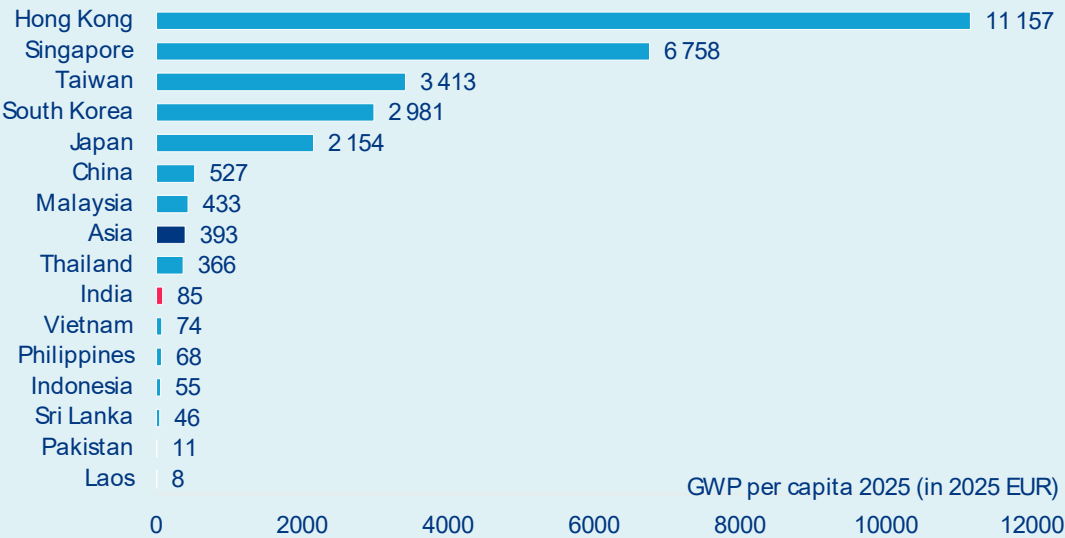
to 3.8% of GDP compared to 4.4% in China or 5.1% in Thailand. And while per-capita spending for insurance amounted to the equivalent of around EUR527 in China and EUR366 in Thailand, it was merely EUR85 in India (Figure 11 and 12).

Figure 11: Insurance penetration, selected countries (gross written premiums in percent of GDP)



Source: National financial supervisory authorities, insurance associations and statistical offices..

Figure 12: Insurance density, selected countries (gross written premiums per capita, in 2025 EUR)



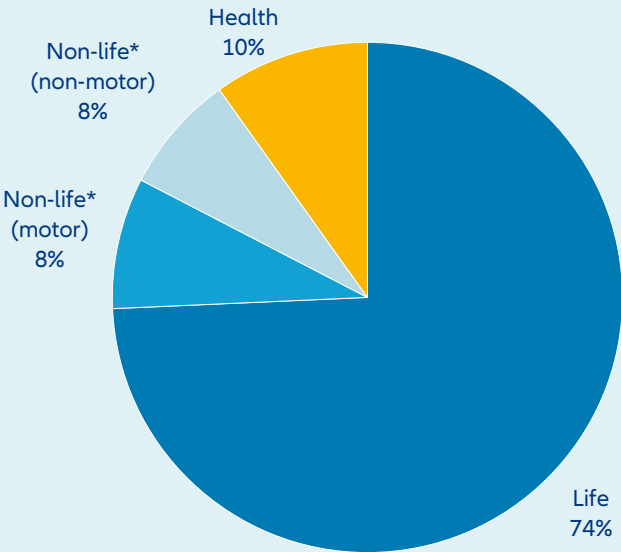
Source: National financial supervisory authorities, insurance associations and statistical offices.

Life insurance dominates the market, but the non-life and health segments reveal where the structural gaps and the growth potential are most acute. India’s insurance market is dominated by the life insurance segment, which accounted for 74% of total gross written premium income in 2024. Non-life insurance premiums, including personal accident and travel insurance, accounted for 16% and health insurance accounted for 10% of total gross written premiums (see Figure 13).

Mandatory motor coverage is the backbone of non-life insurance yet, half of all vehicles on Indian roads remain uninsured, illustrating the gap between legal requirements and market reality. As in most countries, motor insurance was the most important business line in the non-life segment (excluding health), with a share of 52% in fiscal year 2024–2025. Third-party liability business made up more than half of the motor segment

as such coverage is mandatory for all vehicles driven on public roads. Nevertheless, it is estimated that over 50% of vehicles remain uninsured⁶. Given the importance of the agricultural sector for the Indian economy, crop insurance accounted for 16% of general premium income. However, there were significant differences in the composition of premium income across individual states, largely reflecting their respective economic structures (see Figure 14).

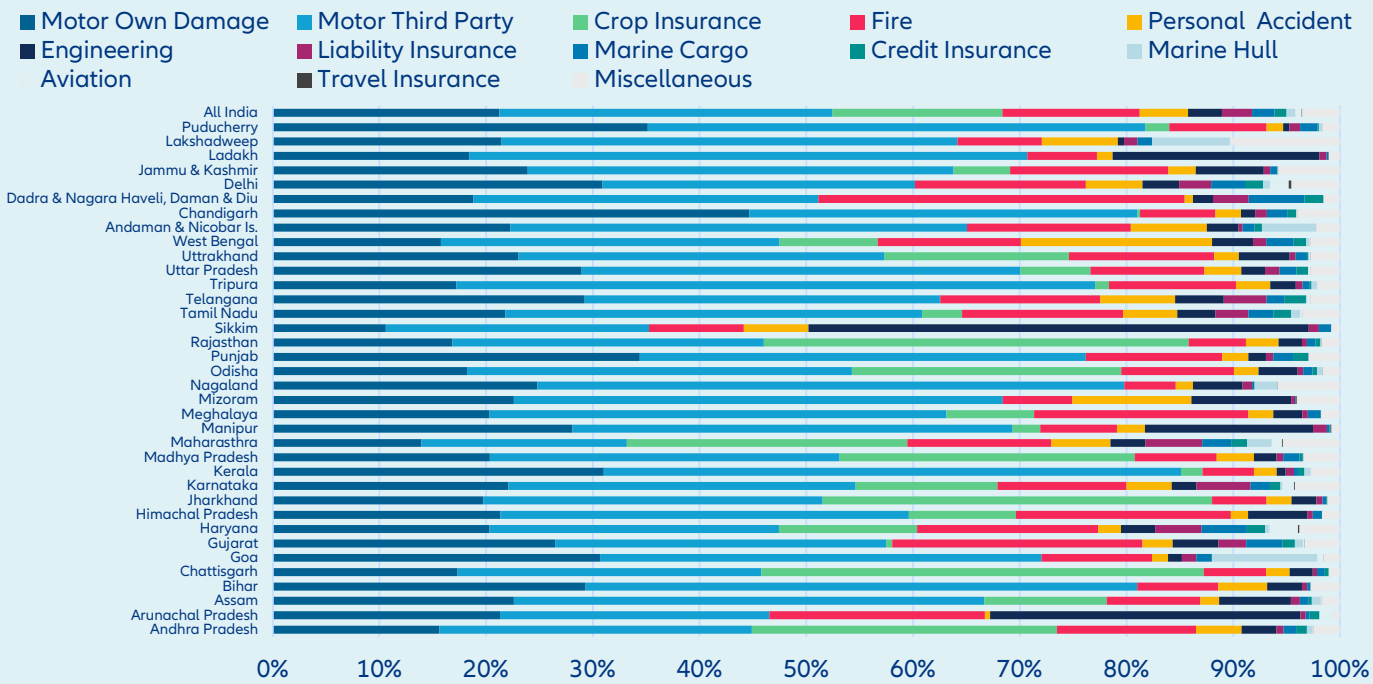
Figure 13: Gross written premiums by business line (in percent)



*including personal accident and travel insurance premiums
Source: IRDAI, Allianz Research.

⁶ See Axco (2025): Insurance market reports (non-life). India (as of March 2026),

Figure 14: Non-life insurance premium by line of business and state (in percent)

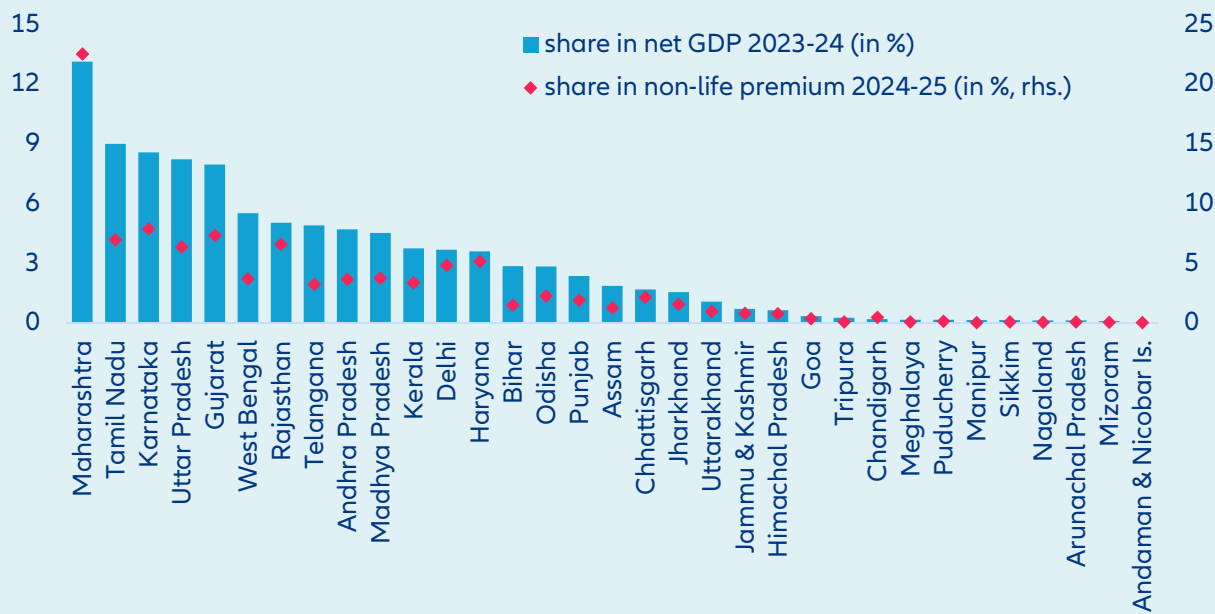


Source: IRDAI

Premium income is as unevenly distributed as economic activity itself, and product design that ignores this geography will systematically miss the largest underserved populations. Five states – Gujarat, Karnataka, Maharashtra, Rajasthan and Uttar Pradesh – accounted for 48% of general premium income, with Maharashtra alone accounting for 20% (see Figure 15). In health insurance, Karnataka, Maharashtra and Tamil Nadu accounted for 49.2% of premium income in

2023–2024, with 22.5% generated in Maharashtra. Despite signs of economic convergence, marked regional differences persist. In 2023–2024, per-capita net state domestic product ranged between EUR570 in Bihar, representing around a third of the national average of EUR1,790, to EUR5,568 in Sikkim. These differences need to be taken into account when it comes to the design and distribution of insurance products.

Figure 15: Non-life premium income by state



Source: IRDAI, Reserve Bank of India

The most powerful long-term growth driver for India's insurance market, alongside rising household wealth, is the still incomplete and uneven reach of the country's social protection system. Market expansion is therefore expected to be driven primarily by pent-up demand and the growing need for supplementary private risk protection, as public social security remains at a relatively early stage of development. According to the latest available ILO data, only 64.3% of India's population was covered by at least one branch of the social security system at all. Effective pension coverage is even more limited: the ILO estimates that only 45.8% of the working-age population was covered by the pension system, compared with close to universal coverage in most industrialized countries. Without targeted government initiatives such as Atal Pension Yojana and the e-Shram portal aimed at informal workers, as well as the supplementary state-level pension schemes, this share would be significantly lower. This is particularly relevant given that only around 12% of India's workforce is employed in the formal sector and therefore benefits from standard pension coverage. Against the backdrop of rising life expectancy, these structural gaps point to a growing need for supplementary, capital-funded private pension solutions to secure retirement income. A similar dynamic is evident in healthcare. Despite the successful rollout of Pradhan Mantri Jan Arogya Yojana (PM-JAY), out-of-pocket spending still accounts for 43.9% of total health expenditure - high by international standards. This underscores both the need for further reforms to broaden public healthcare coverage and the resulting demand for supplementary private health insurance. In this context, medical inflation in India, which is among the highest in Asia, poses a significant financial challenge for households and health insurance providers alike.

Regulatory reform and an ambitious universal coverage target are creating both the policy tailwind and the competitive catalyst for the next growth phase. Recent insurance market reforms, notably the opening up to foreign insurance participation, will increase the efficiency of the insurance sector and together with the IRDAI's „Insurance for All by 2047“ program are expected to spur double-digit insurance market growth of an estimated average 10.5% in life insurance, 10.2% in non-life and 12.5% in health insurance per annum until 2036.

India's insurance sector will expand not just in scale but in structural sophistication, shifting from a peripheral to a systemically important pillar of the financial system. Rising incomes and a growing middle class will lift demand for protection and savings products, driving higher penetration and density. Demand is expected to shift from basic, often mandatory covers such as motor and simple life towards more complex products including health, pensions and wealth-linked insurance. Insurers will increasingly act as institutional investors, supporting capital market development and long-term financing, particularly for infrastructure. Higher incomes and improving financial literacy would also help reduce underinsurance and encourage more proactive risk management by households and firms. Together with demographic trends, notably ageing, this would further boost demand for private health and retirement solutions. India's still-developing social security system implies a greater reliance on private provision compared to other markets such as China. Overall, sustained growth is likely to transform the sector into a more mature and systemically important pillar of the financial system.



The next frontier:

Climate risk, affordability and the future of Insurability

Affordability is emerging as a key structural challenge facing the insurance industry. In some markets and lines of business, the cost of adequate coverage is beginning to outpace households' capacity to pay, driven by claims severity, inflation and broader economic pressures. Where this gap becomes persistent, it translates into structural underinsurance, concentrating uninsured risk where financial resilience is weakest. Climate risk is where this dynamic is already most visible and most advanced.

Global insured NatCat losses are rising by 5–7% annually in real terms, underlining that climate risk is now a structural underwriting issue rather than a cyclical volatility event. Climate change is an undisputed driver, but observed loss trends are also shaped by socioeconomic factors, including exposure growth in hazard-prone areas and inflation rather than by climate effects alone⁷. Rising premiums are increasingly colliding with falling household purchasing power. Climate shocks alongside other socioeconomic pressures such as demographic change, wars and geopolitical instability are driving both insurance costs and greater financial stress. This dynamic is already

visible in household insurance markets: In the US, homeowners' insurance premiums rose by +38% between 2019 and 2024, roughly twice the rate of inflation. The same climate shocks driving up premiums also erode the capacity to pay: droughts reduce crop yields and push up food prices, while heat stress lowers labor productivity and wages. Global food prices could be around 13% above baseline by 2050 in a +2°C warming scenario, with Asia-Pacific facing increases of up to 27%, while the most exposed European economies could see cumulative GDP losses of 5–7% over five years⁹. Households are therefore being squeezed from both directions: rising insurance costs and weakening income resilience.

Lower-income households face a self-reinforcing vulnerability trap as higher premiums and weaker incomes push them out of insurance coverage. The poorest and most climate-exposed households are hit hardest, creating a cycle in which rising risks deepen both poverty and underinsurance. Evidence from developing countries shows that the poorest 40% experience climate-driven income losses 70% larger than the average¹⁰, while a 1°C increase in temperature raises poverty rates by 0.9 to 2.3pps¹¹. Similar patterns are emerging in developed

⁷ Swiss RE (Sigma 1/2025)

⁸ The primary driver to date has been exposure growth (urbanization, asset accumulation and rising construction costs) which accounts for over 80% of the long-term increase in weather-related insured losses. Climate change effects have so far been a smaller but growing contributor, and Swiss Re cautions that for certain perils and regions, such as wildfire in North America and severe convective storms in Europe, hazard intensification is already accelerating losses well beyond what exposure alone can explain.

⁹ Feeding a warming world: Securing food and economic stability in a changing climate | Allianz

¹⁰ Climate change through a poverty lens | Nature Climate Change

¹¹ World Bank Document

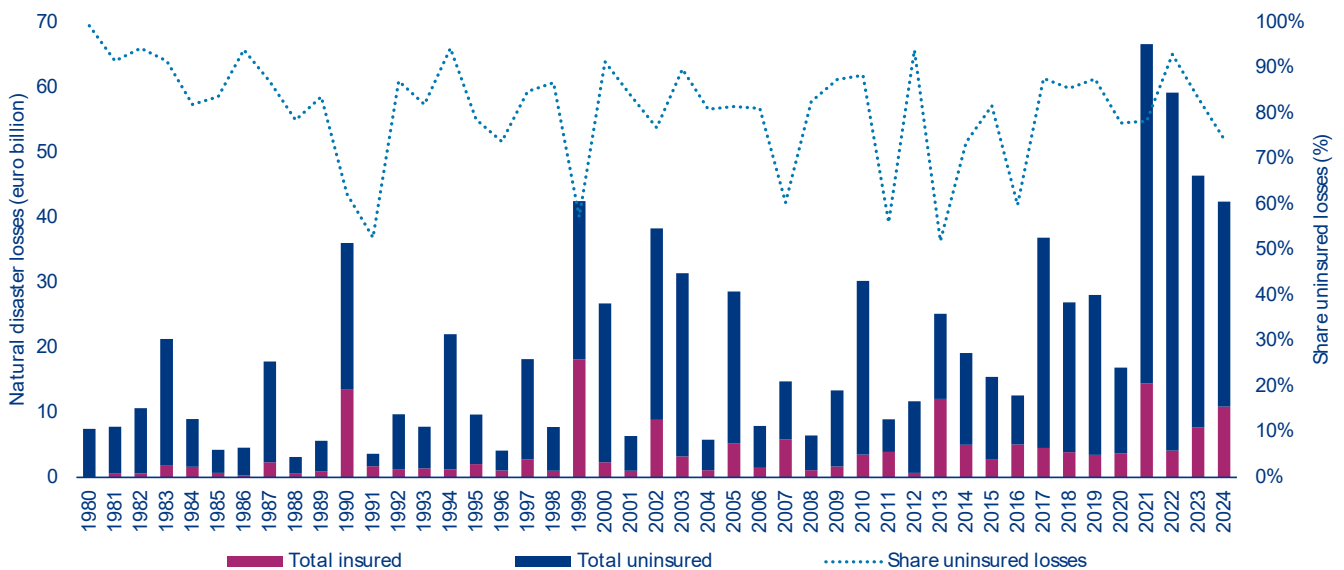
economies, where climate change also drives chronic rather than temporary poverty, making exclusion from insurance more persistent. As premiums rise, disaster insurance risks becoming a product accessible only for wealthier households. Evidence from Florida shows that without public investment in risk reduction and affordability mechanisms, both traditional and parametric insurance products become inaccessible for low- and middle-income households¹².

As private insurance affordability declines, climate losses increasingly shift onto public balance sheets that are themselves being weakened by climate shocks.

In the EU, uninsured direct disaster losses averaged EUR59bn annually between 2021 and 2023 (Figure 16), highlighting how a large share of climate losses continues to sit outside formal insurance coverage and is ultimately absorbed by households, firms and governments¹³. This gap is significantly demand-driven: EIOPA research highlights¹⁴ that low risk awareness, expectations of government

compensation and lack of product clarity mean many households and businesses simply do not purchase available NatCat coverage, a challenge distinct from the supply-side retreat seen in parts of the US market. This is also visible in the rapid expansion of state-backed insurance schemes. Heat stress alone could worsen European fiscal balances by around 0.5% of GDP annually. France faces additional fiscal pressure of 2.2% of GDP, while Italy and Spain face 1.9% and 1.7% respectively. The governments most exposed to climate risk are often those with the least fiscal space to subsidize insurance or invest in adaptation, creating a dangerous feedback loop of rising losses, weaker coverage and declining fiscal resilience. In the US, California's FAIR Plan increased insured exposure from USD220bn in September 2022 to USD750bn by early 2026¹⁵, and following the Los Angeles wildfires required a USD1bn emergency levy on insurers to maintain solvency. Florida's Citizens scheme peaked at USD618bn of insured exposure in 2023, showing how quickly public contingent liabilities can grow when private insurers retreat¹⁶.

Figure 16: Development of NatCat losses in Europe and share of uninsured losses



Source: CATDAT

¹² Insurance and climate risks: Policy lessons from three bounding scenarios | PNAS

¹³ Climate and nature risks fuel multi-billion insurance protection gaps | WWF

¹⁴ Climate insurance protection gaps: a demand-side challenge - European Insurance and Occupational Pensions Authority

¹⁵ California FAIR Plan Association (March 2026)

¹⁶ Citizens Property Insurance Corporation (Policies in Force)

Europe is increasingly moving toward mandatory and state-backed NatCat systems, but these solutions must avoid creating long-term moral hazard. Public-private NatCat systems are becoming a structural feature to adapt to extreme weather events. France already operates a mandatory state-backed NatCat reinsurance scheme funded by a surcharge on all property insurance policies, and after repeated climate-related losses that charge was nearly doubled last year to 20%. Since 2025, all businesses in Italy have been required to carry NatCat insurance, with the government covering up to half of the exposure through its state reinsurer. Germany is about to introduce a mandatory scheme, and the German Insurance Association has proposed a new private-sector pool for high-risk residential properties backed by a government stop-loss mechanism for catastrophe losses above EUR30bn. At the EU level, supervisors are exploring a Europe-wide risk-sharing system combining a premium-financed pool with a public loan-funded mechanism. These schemes can preserve insurability, but if public support becomes too strong, pricing signals could weaken and incentives for resilience investment, relocation and land-use discipline could be reduced. Equally, even well-designed public-private schemes face structural limits if investment in climate-change adaptation, from enforced flood defenses and resilient infrastructure to smart agriculture and water management systems, does not scale fast enough to reduce costs of the rising loss trends. Without combined strategies, the fiscal and insurance capacity underpinning the public-private NatCat schemes will be progressively eroded.

The deeper issue is that climate risk is exposing the limits of pricing alone as a long-term solution to maintaining insurability. Insurance pricing continues to rely primarily on claims experience, probabilistic NatCat models and increasingly granular hazard maps that capture location-specific risk differentials. These tools remain broadly effective for pricing incremental climate risks and preserving underwriting discipline. However, annual repricing becomes increasingly difficult when physical risks and claims severity rise faster than households' ability to absorb higher

premiums. For insurers, the challenge is no longer simply to reprice risk, but to preserve insurability in economies where affordability is structurally weakening.

Preserving long-term insurability requires stronger adaptation incentives, efficiency gains, targeted affordability measures and public-private solutions. This requires moving beyond pure risk transfer toward stronger risk reduction and resilience incentives. Insurance will increasingly need to incentivize adaptation through pricing, deductibles and prevention management, while efficiency gains become essential to offset structurally rising claims severity driven by technological complexity, specialized repair services and legal costs. Automation, AI and digital claims management already show significant potential to reduce operational costs without compromising coverage quality. Critically, however, affordability cannot be preserved through artificially suppressed premiums. Underpricing weakens incentives for risk reduction and ultimately accelerates rather than closes protection gaps. Where lower-income households are being priced out of coverage, targeted support mechanisms such as means-tested vouchers or public reinsurance are likely to prove more effective and transparent than below-risk pricing. In areas where physical risks have shifted permanently, managed relocation may ultimately be more sustainable than maintaining coverage at unsustainably low prices. More broadly, protection gaps cannot be solved by private insurance alone. Public-private partnerships, sovereign backstops and targeted affordability mechanisms will become increasingly necessary where risks become privately uninsurable, but must always be combined with credible adaptation investment at scale. Without enforced building codes, flood defenses and resilient infrastructure, public-private schemes absorb rising losses without addressing their source, progressively eroding the fiscal and insurance capacity that underpins them.

Appendix

Appendix A

Insurance markets

2025²⁾ KPIs

	Total premium income (in EUR bn) ¹⁾			Penetration (as % of GDP)			Density (per capita, in EUR) ¹⁾		
	p&c	life	health	p&c	life	health	p&c	life	health
Argentina	11,8	1,5	0,03	2,4	0,3	0,01	258	33	0,6
Australia	40,5	14,0	18,1	2,5	0,9	1,1	1 500	517	670,6
Austria	15,6	5,2	3,4	3,0	1,0	0,7	1 710	568	376,8
Bahrain	<i>0,4</i>	<i>0,1</i>	<i>0,2</i>	<i>1,1</i>	<i>0,2</i>	<i>0,5</i>	<i>260</i>	<i>41</i>	<i>134,6</i>
Belgium	<i>13,3</i>	<i>17,8</i>	<i>2,5</i>	<i>2,1</i>	<i>2,8</i>	<i>0,4</i>	<i>1 127</i>	<i>1 516</i>	<i>214,7</i>
Brazil	24,0	10,5	53,6	1,2	0,5	2,7	113	49	251,6
Bulgaria	<i>2,2</i>	<i>0,3</i>	<i>0,2</i>	<i>3,7</i>	<i>0,5</i>	<i>0,3</i>	<i>326</i>	<i>48</i>	<i>28,6</i>
Canada	<i>49,9</i>	<i>66,6</i>	<i>25,4</i>	<i>2,5</i>	<i>3,3</i>	<i>1,3</i>	<i>1 244</i>	<i>1 659</i>	<i>633,0</i>
Chile	4,9	9,1	1,1	1,5	2,9	0,4	247	460	57,6
China	190,9	433,2	121,5	1,1	2,5	0,7	135	306	85,8
Colombia	5,8	6,8	1,1	1,4	1,6	0,3	109	127	21,0
Croatia	1,6	0,3	0,2	1,7	0,4	0,2	411	87	39,4
Czech Republic	7,3	2,1	0,6	2,1	0,6	0,2	684	196	60,2
Denmark	<i>12,8</i>	<i>36,2</i>	<i>0,5</i>	<i>3,1</i>	<i>8,8</i>	<i>0,1</i>	<i>2 134</i>	<i>6 034</i>	<i>88,7</i>
Egypt	<i>0,9</i>	<i>1,1</i>	<i>0,3</i>	<i>0,3</i>	<i>0,3</i>	<i>0,1</i>	<i>8</i>	<i>9</i>	<i>2,6</i>
Finland	4,8	6,0	0,8	1,7	2,2	0,3	850	1 075	141,3
France	<i>98,2</i>	<i>192,1</i>	<i>48,4</i>	<i>3,3</i>	<i>6,4</i>	<i>1,6</i>	<i>1 474</i>	<i>2 882</i>	<i>726,7</i>
Germany	103,6	95,0	54,1	2,3	2,1	1,2	1 232	1 130	643,9
Greece	2,7	2,8	0,6	1,1	1,1	0,2	267	283	55,6
Hong Kong	<i>3,3</i>	<i>77,0</i>	<i>2,3</i>	<i>0,9</i>	<i>21,1</i>	<i>0,6</i>	<i>441</i>	<i>10 411</i>	<i>305,0</i>
Hungary	3,0	2,1	0,1	1,3	0,9	0,1	312	214	13,4
India	19,7	<i>92,1</i>	12,4	0,6	<i>2,8</i>	0,4	13	<i>63</i>	8,5
Indonesia	6,2	8,2	1,4	0,5	0,7	0,1	22	29	5,0
Ireland	<i>4,6</i>	<i>24,1</i>	<i>3,9</i>	<i>0,7</i>	<i>3,8</i>	<i>0,6</i>	<i>874</i>	<i>4 539</i>	<i>741,0</i>
Italy	<i>48,0</i>	<i>122,6</i>	5,0	<i>2,1</i>	<i>5,4</i>	0,2	<i>812</i>	<i>2 073</i>	84,0
Japan	58,5	206,7	n/a	1,6	5,7	n/a	475	1 679	n/a
Kazakhstan	1,2	1,8	0,1	0,4	0,7	0,1	55	86	6,6
Kenya	<i>0,9</i>	<i>1,4</i>	<i>0,6</i>	<i>0,8</i>	<i>1,2</i>	<i>0,5</i>	<i>16</i>	<i>25</i>	<i>10,9</i>
Laos	<i>0,1</i>	<i>0,01</i>	n/a	<i>0,3</i>	<i>0,1</i>	n/a	<i>6</i>	<i>1</i>	n/a
Malaysia	<i>4,7</i>	10,6	<i>0,3</i>	<i>1,1</i>	2,5	<i>0,1</i>	<i>130</i>	296	<i>7,4</i>
Mexico	18,1	22,0	8,7	1,1	1,3	0,5	137	167	65,6
Morocco	2,8	2,7	0,5	1,7	1,7	0,3	72	72	12,6
Netherlands	<i>16,7</i>	<i>15,7</i>	66,5	<i>1,4</i>	<i>1,3</i>	5,6	912	855	3 624,3
Nigeria	<i>0,9</i>	<i>0,4</i>	n/a	<i>0,4</i>	<i>0,2</i>	n/a	<i>4</i>	<i>2</i>	n/a
New Zealand	<i>5,5</i>	<i>1,6</i>	n/a	<i>2,5</i>	<i>0,7</i>	n/a	<i>1 039</i>	<i>303</i>	n/a
Norway	10,7	15,2	0,5	2,3	3,3	0,1	1 910	2 708	87,5
Pakistan	<i>1,5</i>	<i>1,2</i>	n/a	<i>0,4</i>	<i>0,4</i>	n/a	<i>6</i>	<i>5</i>	n/a
Peru	<i>2,1</i>	3,5	<i>0,4</i>	<i>0,7</i>	1,2	<i>0,1</i>	<i>61</i>	103	<i>12,5</i>
Philippines	2,1	5,6	<i>0,2</i>	0,5	1,4	<i>0,1</i>	18	48	<i>1,8</i>
Poland	<i>14,5</i>	<i>3,6</i>	1,7	<i>1,6</i>	<i>0,4</i>	0,2	<i>381</i>	<i>95</i>	43,8
Portugal	6,2	8,1	1,8	2,0	2,7	0,6	600	783	171,1
Romania	3,5	0,8	<i>0,2</i>	0,9	0,2	<i>0,1</i>	183	44	<i>11,7</i>
Saudi Arabia	<i>5,8</i>	<i>2,8</i>	<i>10,7</i>	<i>0,5</i>	<i>0,3</i>	<i>1,0</i>	<i>169</i>	<i>81</i>	<i>308,4</i>
Singapore	3,7	33,5	2,5	0,7	6,4	0,5	625	5 705	428,3
Slovakia	1,4	0,6	0,2	1,0	0,5	0,1	253	115	36,1
South Africa	11,8	40,2	n/a	3,0	10,2	n/a	183	622	n/a
South Korea	79,8	74,2	n/a	5,1	4,7	n/a	1 545	1 437	n/a
Spain	36,5	35,9	13,5	2,2	2,1	0,8	763	750	280,9
Sri Lanka	0,4	0,6	0,1	0,4	0,7	0,1	16	27	2,7
Sweden	10,8	39,4	n/a	1,8	6,5	n/a	1 017	3 697	n/a
Switzerland	22,1	24,3	14,4	2,4	2,6	1,5	2 462	2 710	1 609,0
Taiwan	9,8	56,1	13,0	1,3	7,2	1,7	424	2 428	560,3
Thailand	8,0	14,0	4,2	1,6	2,7	0,8	112	196	58,2
Türkiye	16,5	3,5	4,2	1,3	0,3	0,3	188	40	47,9
United Arab Emirates	7,3	1,9	8,2	1,5	0,4	1,7	642	170	726,3
United Kingdom	<i>122,0</i>	<i>275,3</i>	<i>10,4</i>	<i>3,5</i>	<i>7,9</i>	<i>0,3</i>	<i>1 754</i>	<i>3 958</i>	<i>150,1</i>
United States	<i>1155,5</i>	725,6	1167,6	<i>4,4</i>	2,8	4,5	<i>3 327</i>	2 089	3 362,1
Vietnam	<i>2,7</i>	<i>4,9</i>	n/a	<i>0,6</i>	<i>1,2</i>	n/a	<i>26</i>	<i>48</i>	n/a

¹⁾ 2025 exchange rates.

²⁾ Estimates are shown in italics.

Appendix B**Insurance markets****Long-term development**

	CAGR 2015-2025 (in %)			CAGR 2026-2036 (in %)			Total premium income 2036 (in EUR bn) ¹⁾		
	p&c	life	health	p&c	life	health	p&c	life	health
Argentina	61,0	54,4	61,0	17,5	18,8	22,8	69,8	10,1	0,3
Australia	6,5	-8,1	4,3	4,7	3,1	4,2	67,1	19,5	28,5
Austria	3,8	-2,3	5,6	3,1	1,2	5,0	21,8	5,9	5,9
Bahrain	1,2	-6,3	6,7	3,4	3,7	5,6	0,6	0,1	0,4
Belgium	3,2	0,9	5,2	3,0	4,0	6,1	18,4	27,4	4,8
Brazil	7,1	11,6	9,2	5,5	11,4	5,8	43,0	34,5	99,9
Bulgaria	10,5	7,2	16,9	7,2	7,4	15,4	4,7	0,7	0,9
Canada	4,1	5,3	4,8	3,7	4,1	4,7	74,8	104,0	42,1
Chile	8,2	9,3	11,1	6,4	7,3	6,3	9,7	19,9	2,2
China	6,6	11,3	18,2	6,3	7,6	8,1	374,3	966,7	285,6
Colombia	9,1	12,4	14,6	6,7	8,0	11,5	11,9	15,8	3,7
Croatia	7,0	-0,4	13,7	4,4	2,6	12,2	2,5	0,4	0,5
Czech Republic	6,6	-2,7	8,9	4,7	2,4	8,6	12,0	2,7	1,6
Denmark	2,8	6,2	7,2	3,0	5,6	8,1	17,8	66,0	1,3
Egypt	20,4	20,2	30,5	11,4	9,8	20,9	3,0	3,0	2,5
Finland	1,8	0,2	9,0	3,3	4,5	9,8	6,8	9,8	2,2
France	4,5	3,7	2,8	4,1	3,5	3,2	153,5	280,8	68,7
Germany	4,6	0,5	3,7	4,3	2,5	3,9	165,3	124,3	82,7
Greece	2,3	4,5	14,4	5,4	3,6	6,8	4,7	4,2	1,1
Hong Kong	2,3	8,2	6,8	4,1	5,2	5,9	5,1	135,1	4,2
Hungary	10,7	5,1	16,1	6,8	4,5	12,6	6,2	3,3	0,5
India	10,9	10,4	18,3	10,2	10,5	12,5	57,3	277,0	45,2
Indonesia	8,0	3,7	13,1	7,3	7,9	12,9	13,3	18,8	5,4
Ireland	5,1	8,0	4,9	3,5	3,4	4,9	6,8	34,8	6,7
Italy	2,9	0,5	8,3	3,6	2,8	7,6	70,6	167,0	11,1
Japan	1,9	0,5	n/a	2,5	2,9	n/a	77,2	282,9	n/a
Kazakhstan	12,0	32,8	12,7	7,1	11,8	12,1	2,4	6,1	0,5
Kenya	5,5	12,9	12,7	7,8	12,1	12,5	2,1	5,0	2,3
Laos	11,3	35,0	n/a	10,4	15,0	n/a	0,2	0,0	n/a
Malaysia	3,3	6,1	2,1	5,6	5,2	6,7	8,5	18,6	0,5
Mexico	9,2	9,8	12,2	8,2	8,5	9,7	43,0	53,8	24,0
Morocco	5,8	10,8	5,2	5,9	7,1	6,8	5,2	5,8	1,0
Netherlands	2,5	0,0	4,4	3,2	2,7	4,3	23,8	21,1	105,7
Nigeria	21,6	21,7	n/a	15,7	20,0	n/a	4,7	3,3	n/a
New Zealand	7,1	3,9	n/a	3,6	3,6	n/a	8,0	2,3	n/a
Norway	7,1	6,2	12,8	5,0	4,4	8,7	18,3	24,4	1,2
Pakistan	27,7	11,5	n/a	13,8	13,8	n/a	6,4	5,2	n/a
Peru	5,0	10,7	8,3	5,2	8,0	5,2	3,7	8,3	0,8
Philippines	7,6	8,7	n/a	8,5	10,0	n/a	5,1	16,0	n/a
Poland	6,9	-3,8	24,1	4,2	3,6	8,7	22,9	5,3	4,2
Portugal	6,1	-2,2	11,2	4,0	3,9	8,1	9,7	12,5	4,2
Romania	9,6	11,1	27,8	3,2	3,3	21,6	4,9	1,2	1,9
Saudi Arabia	5,8	26,8	10,5	6,2	11,7	8,2	11,3	9,4	25,3
Singapore	4,2	7,5	23,9	5,6	5,5	8,4	6,7	60,2	6,1
Slovakia	3,6	-5,4	n/a	4,0	3,5	10,5	2,1	0,9	0,6
South Africa	7,0	5,0	n/a	5,6	5,6	n/a	21,6	73,0	n/a
South Korea	5,4	0,8	n/a	4,9	4,4	n/a	135,6	119,0	n/a
Spain	4,1	3,4	5,8	3,4	3,6	6,0	52,9	52,8	25,6
Sri Lanka	8,8	15,9	10,6	10,6	12,8	8,3	1,1	2,4	0,2
Sweden	4,8	5,4	n/a	3,6	4,1	n/a	16,1	61,2	n/a
Switzerland	2,0	-3,3	3,1	2,9	1,6	2,4	30,4	28,9	18,8
Taiwan	5,9	-1,3	4,1	4,8	4,3	4,8	16,4	89,2	21,6
Thailand	2,5	0,5	9,7	4,5	3,6	8,6	12,9	20,8	10,4
Türkiye	40,5	43,8	48,1	11,8	14,4	15,3	56,3	15,6	20,0
United Arab Emirates	7,7	-0,4	11,2	4,7	5,3	6,3	12,0	3,4	16,1
United Kingdom	7,8	4,5	4,8	3,4	3,4	3,3	176,1	397,2	14,9
United States	5,7	4,9	9,0	4,1	3,9	6,9	1 799,6	1 103,9	2 437,4
Vietnam	10,6	16,4	n/a	9,5	7,0	n/a	7,3	10,2	n/a

¹⁾ 2025 exchange rates.



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Too hot to grow

The economic costs of extreme heat

28 May 2026

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Appendix

Executive Summary



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- **Extreme heat is emerging as a structural economic risk, with Europe highly exposed.**

Heat stress events have multiplied sevenfold since the 1980s while the average death toll per event has risen fivefold. That share partly reflects measurement: vital registration and excess-mortality surveillance are far more developed in Europe than in much of Africa and South Asia, where heat deaths go largely uncounted. But genuine structural vulnerability is also at work – ageing populations, dense urban building stock designed to retain warmth and severely underdeveloped cooling infrastructure, with air-conditioning penetration averaging 19% across Europe against roughly 90% in the US.

- **The economic transmission of heat stress is non-linear, with a critical threshold around 30°C beyond which productivity losses intensify sharply.**

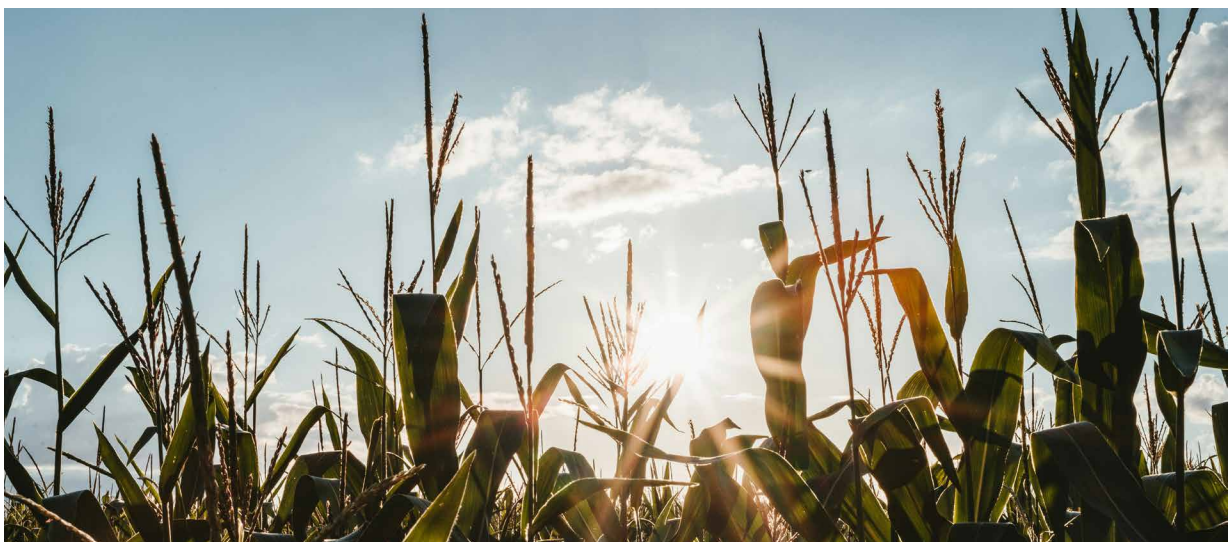
Below this level, warming reduces heating costs and is associated with modest productivity gains. Above this level, the relationship reverses and both channels worsen with each additional degree. The dominant effect operates through labor: output per hour declines by approximately USD1.3 (constant PPP, ~3% of mean hourly output in our 2014-2024 sample) for every degree across the 30-35°C range. Wage adjustments follow productivity with a lag, so the short-run cost falls disproportionately on firm profitability before gradually transmitting to household income and consumption. A second, smaller channel runs through energy: consumption rises by around 1.2% per degree, raising firms' input costs at exactly the temperatures where labor productivity is falling.

- **To gauge the macroeconomic stakes, we construct a stress scenario in which the five hottest years observed in each country between 2014 and 2024 are replayed in ascending order over 2026–2030 – the fifth-hottest year in 2026, the fourth in 2027 and so on, culminating in the country's hottest year on record in 2030.** Under this trajectory, cumulative implied GDP losses (2026 – 2030) could reach 5–7% for the most exposed economies: USD240bn for France, USD354bn for Japan, USD147bn for Italy, USD131bn for Germany and USD120bn for Spain. More consequentially for long-run growth, in such a scenario the decline in fixed capital formation systematically exceeds consumption losses, reaching 8% on average across affected countries: As heat compresses expected returns on capital, investment falls, reducing future productive capacity in a self-reinforcing drag. Moreover, stagflationary dynamics should be expected, with rising prices alongside rising unemployment, placing monetary authorities in a binding trade-off that is especially acute in the Eurozone, where a single policy rate must serve economies with sharply diverging climate exposures.

- **The fiscal consequences fall most heavily on the economies least able to absorb them.** The loss of economic output due to heat reduces tax revenues: Estimated annual losses would reach 1.8% in France, 1.3% in Italy and Spain and 0.7% in Germany – partly because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. Simultaneously inflation-indexed transfers, healthcare costs and emergency infrastructure repair raises public

expenditure. Fiscal balances deteriorate by around 0.5% of GDP annually on average. Italy and Spain risk breaching the Maastricht deficit ceiling (again) once heat-related pressures are incorporated. France, already carrying a projected deficit of –4.9% of GDP, faces additional heat-related pressure of 2.2%.

- **Insured losses remain a small fraction of total damages, reflecting a structural mismatch between what heat destroys and what conventional insurance was designed to cover.** In 2022, total climatological losses in Europe reached EUR46bn while the insured share rose only marginally. Most heat damage accumulates through excess mortality, lost working hours, healthcare-system pressure and infrastructure stress – channels that indemnity contracts are not designed to handle. This makes extreme heat harder to insure than other climate risks because losses are widespread and are often indirect – like lower productivity or health impacts – making them difficult to measure and price. Closing the protection gap is therefore as much a product-design challenge as a capacity one, and the insurance toolkit is already evolving in response: parametric instruments that pay out on objective temperature or duration thresholds, public–private risk-sharing arrangements for systemic exposures and dedicated public backstops where private capacity cannot reasonably reach.
- **Heat-adaptation policy in Europe is built mainly to compensate for losses rather than prevent them.** Following the multi-actor frame of IPCC, closing the gap in Europe requires coordinated action on four fronts: labor regulation, buildings, public finance and households. A workable occupational regime needs binding temperature thresholds, automatic work restrictions when those thresholds are crossed, paid compensation for lost hours and coverage that reaches fixed-term, seasonal and platform workers. No major European economy has all four, and the gap is concentrated on the last: protections were designed around standard contracts and leave the workers most exposed to heat largely outside the regime. Prevention itself is also underused, with shifted hours, partial mechanization and indoor cooling still rare. For buildings, four pieces have to fit together: overheating standards in new construction, mandatory passive cooling in renovation, cooling access for vulnerable households as a social entitlement and grid-adequacy planning that accounts for coincident summer cooling demand and the thermal derating of generators. The revised EU Energy Performance of Buildings Directive delivers the first; the gap is the other three, where indoor temperatures, mortality and peak electricity demand are actually held down. On fiscal architecture, every major European economy has a national adaptation strategy but almost none has translated it into a multi-year budget envelope, so the response defaults to ad-hoc emergency packages – and each episode quietly consumes the fiscal space that ex-ante adaptation would have used to lower the next. The missing layer is households. EU households hold almost EUR40trn in financial assets, including a very large stock of deposits, while much of Europe’s housing stock remains poorly adapted to hotter summers. Mobilizing even a small, well-targeted share – through incentives for retrofit, passive cooling and affordable parametric cover – could close part of the gap that public budgets alone cannot reach. But this is not a private-finance solution: the households most exposed are often not those with the greatest liquid savings, so public guarantees, subsidies and distributional safeguards are what turn household wealth into resilience rather than into inequality.



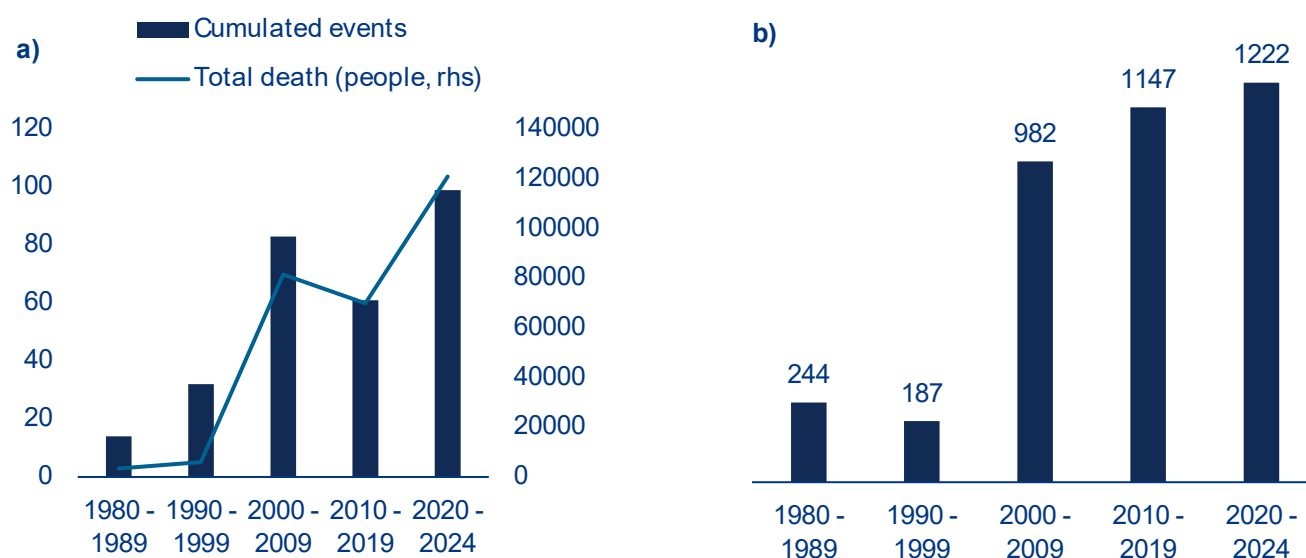
The economic transmission of heat stress

From one decade to another, the data are recording more frequent and more lethal heat stress events.

Between 1980 and 1989, only 14 heat-wave events¹ were catalogued globally; the figure more than doubled in the 1990s, jumped to 83 in the 2000s and has already reached 99 in the truncated 2020–2024 period, meaning five years have nearly matched what entire decades produced two cycles earlier (Figure 1a). Mortality has followed an even steeper trajectory: from around 3,400

deaths in the 1980s to more than 121,000 in the first half of the 2020s alone. The most analytically significant metric is the average death toll per event, which has risen fivefold from 244 in the 1980s to 1,222 in 2020–2024. This decoupling between event frequency and per-event lethality indicates that the intensity and duration of individual episodes are increasing faster than societies can adapt (Figure 1b).

Figure 1: Increasing exposure to heat stress: a) Evolution of number of events and total death (1980 – 2024); b) Average number of death per heat stress event (1980 – 2024)



Sources: EM-DAT, Allianz Research

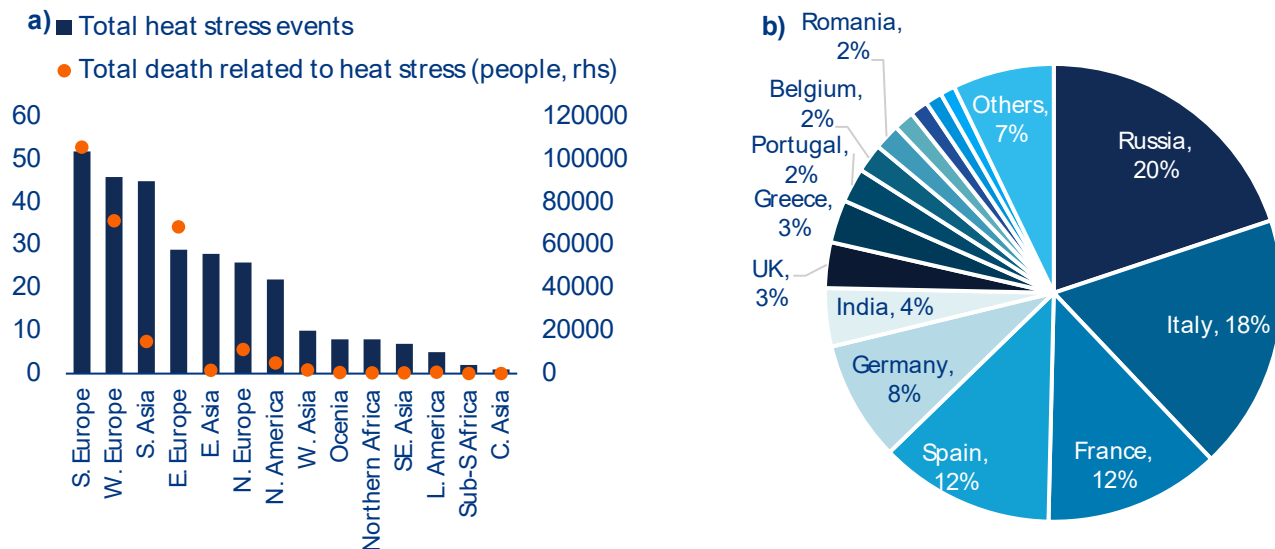
¹ EM-DAT defines a heat wave as a period of abnormally hot and/or humid weather lasting typically two or more days, with location-specific thresholds. Only events with ≥ 10 deaths, ≥ 100 people affected, a declared emergency, or an international assistance call are catalogued.

The cross-sectional distribution of heat stress events and mortality adds a second layer of insight, showing that exposure and impacts do not fully align geographically (Figure 2a). At the regional level, Southern and Western Europe concentrate the highest number of recorded events (52 and 46 respectively), closely followed by South Asia (45). Yet, this apparent convergence in frequency masks a profound divergence in outcomes. Southern Europe alone accounts for more than 105,000 deaths, while Western and Eastern Europe contribute a further 71,000 and 68,500 respectively. By contrast, South Asia, despite a comparable number of events, records fewer than 15,000 deaths over the same period. This asymmetry becomes even more pronounced when examined at the country level. A limited set of European countries dominates the global mortality profile: Russia (20%), Italy (18%), France (12%), Spain (12%) and Germany (8%) together account for 68% of total deaths. The concentration is too strong to be explained by hazard occurrence alone. Recent valuations suggest that heat-related mortality in European cities already entails welfare losses of EUR192–314 per adult per year, on par with air pollution, with avoided mortality alone worth more than

EUR150bn annually.² This is only one channel of impact. On the productivity side, heatwaves cost 0.3–0.5% of European GDP in anomalously hot years such as 2003, 2010, 2015 and 2018, and more than 1% in the most exposed southern regions. Losses are projected to rise fivefold by 2060 in the absence of further mitigation or adaptation.³

The pattern in Figure 2b needs to be read carefully: EM-DAT counts reflect reporting practices as much as biophysical exposure. In hot regions such as the Gulf, India and interior Australia, populations have adapted over time through acclimatization, building design and work practices, so many high-temperature episodes do not register as disaster events, even though heat continues to kill outdoor workers, the poor and the elderly. Europe sits at the opposite end, combining ageing populations with impaired thermoregulation, dense historic urban form, uneven air-conditioning access and stronger vital registration. The European mortality concentration therefore reflects exposure meeting structural vulnerability and more complete death recording, not exceptional hazard frequency.

Figure 2: Global distribution of heat stress events: a) Number of events and total death across regions (1980 – 2024); b) Heat stress related death by country (1980 – 2024)



Sources: EM-DAT, Allianz Research

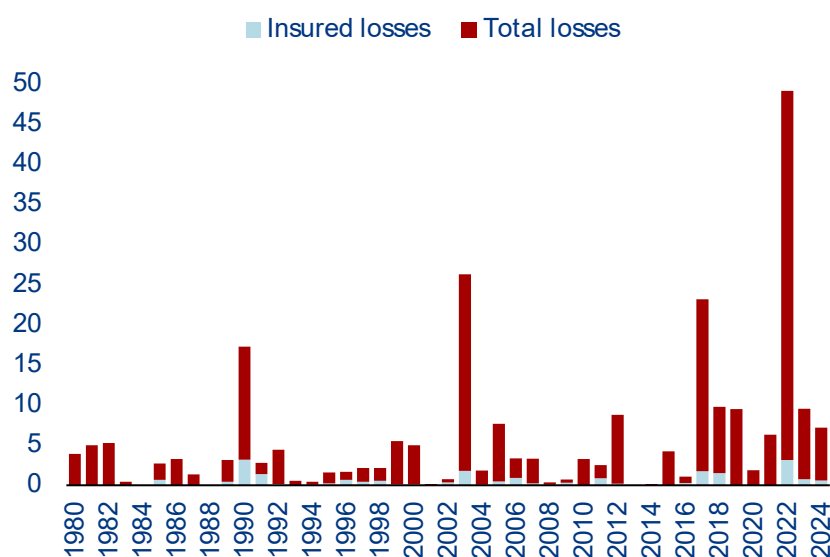
² [Economic valuation of temperature-related mortality attributed to urban heat islands in European cities | Nature Communications](#)

³ [Current and projected regional economic impacts of heatwaves in Europe | Nature Communications](#)

Insured and uninsured losses tell a parallel story. Total losses from climatological hazards in Europe have risen across 1980–2024, with sharp peaks in 2003 and 2022, while the insured share has stayed limited (Figure 3). In 2022, total losses reached around EUR46bn, of which insurance covered only a small fraction. The gap is not simply a matter of market lag; it also reflects what heat actually damages. Compared with storms or floods, extreme heat less often shows up as direct damage to insured assets. Most of its cost appears through excess mortality, lost working hours, healthcare-system pressure and the disruption that follows when grids, railways and water systems are pushed past their design limits. The more conventionally insurable parts, such as direct property damage, heat-amplified wildfire losses, some crop and equipment failures etc., make up only part of the total. EEA data for 1980–2022⁴ show that only around 10% of economic losses from heatwaves, droughts, wildfires and other climatological events in Europe were insured, against more than a third for storms, hail and wind. These figures cover direct asset damage only and exclude mortality, healthcare costs and lost productivity.

Heat is also hard to insure through standard indemnity contracts. Heatwaves can be defined meteorologically, but turning a temperature episode into a compensable loss is diffuse, lagged and hard to attribute. Losses tend to be highly correlated across large regional portfolios, which limits diversification. And the largest impacts – deaths, morbidity, lost output, public-service stress – are not the kind of damage private indemnity contracts were written for. Heat therefore sits at the boundary of conventional insurability: some asset-based losses can be transferred, but the largest welfare and productivity costs are hard to transfer at scale. Closing the gap will require new instruments, such as parametric cover, public–private risk-sharing and dedicated public backstops, without which much of the cost will keep falling on households, firms and public budgets.

Figure 3: Climatological (including heat waves) insured and uninsured losses in Europe (1980 – 2024)



Sources: EEA, Allianz Research

⁴ Economic losses from weather- and climate-related extremes in Europe | Indicators | European Environment Agency (EEA)

Heat translates into measurable macroeconomic cost through two channels: additional energy demand and lower labor productivity.⁵ Both relationships are sharply nonlinear, with a critical threshold around 30°C beyond which damages intensify. The remainder of this section estimates the magnitude of each channel using panel regressions covering 35–49 countries over three decades, and quantifies what those estimates imply when each country experiences its recent peak-heat conditions relative to a 1991–2010 climatological baseline. The output of these two channels feeds directly into the global macroeconomic scenario presented in Section 2.

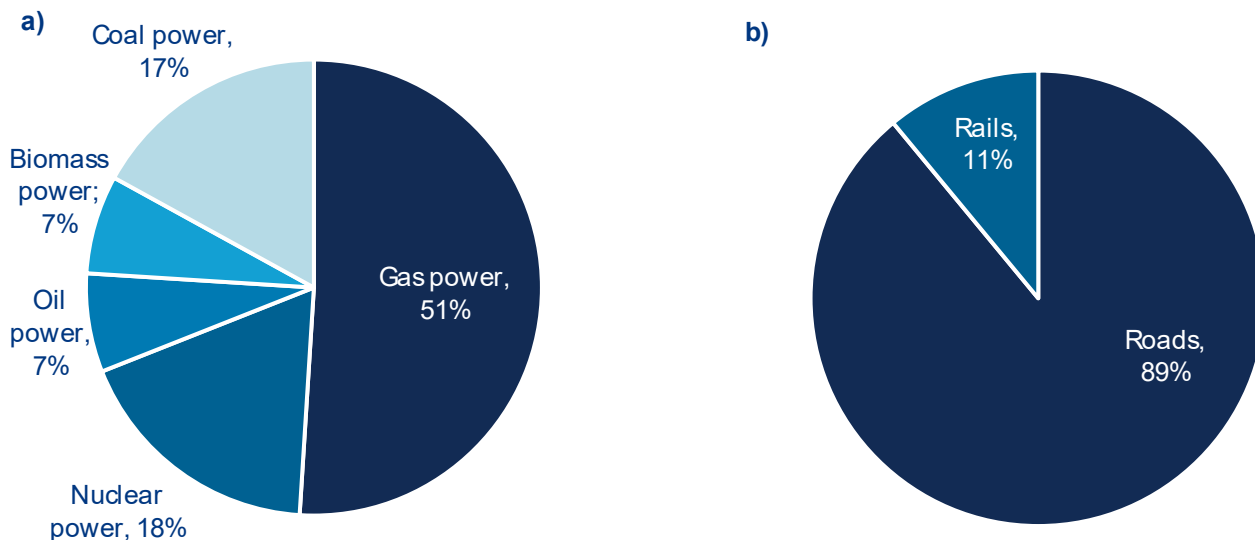
Energy demand

The relationship between temperature and energy demand is inherently non-linear, and the structure of Europe’s power mix amplifies the risk on both demand and supply sides. At moderate temperatures, milder conditions reduce space-heating needs and consumption falls; above a threshold, cooling dominates and consumption rises sharply, producing a convex U-shaped response (Auffhammer and Mansur, 2014)⁶. The supply side magnifies the demand-side pressure. A large share of European generation capacity exposed to heat stress relies on thermoelectric technologies – gas (51%), nuclear (18%) and coal (17%) – all of which

depend on water availability and cooling efficiency. Elevated air and river temperatures directly constrain these systems, forcing output reductions precisely when cooling-driven electricity demand surges (Figure 4a).

Recent episodes illustrate the systemic nature of the resulting grid stress, and analogous pressures affect transport infrastructure. During the August 2020 California heatwave, electricity demand peaked near 47,000 MW while generation capacity was simultaneously constrained, producing the first rolling blackouts in nearly two decades. During the 2019 French heatwave, nuclear output was reduced due to cooling constraints, tightening supply and triggering sharp electricity-price spikes. Heat also degrades transmission networks, lowering the carrying capacity of generators, transformers and power lines, while even non-thermal renewables, in particular solar PV, experience efficiency losses at high temperatures. Transport infrastructure faces analogous physical pressures: road networks account for nearly 89% of heat-exposed transport assets and rail for 11% (Figure 4b), with asphalt softening and rutting under sustained heat and rail tracks prone to thermal expansion, illustrated by the UK in July 2022, when temperatures first exceeded 40°C and rail operators imposed strict speed limits or suspended services entirely.

Figure 4: Exposure of critical European infrastructure to heat stress : a) Energy infrastructure; b) Transport infrastructure



Sources: Forzieri et al. (2018)⁷, Allianz Research

⁵ What to watch | July 01, 2025 | Allianz

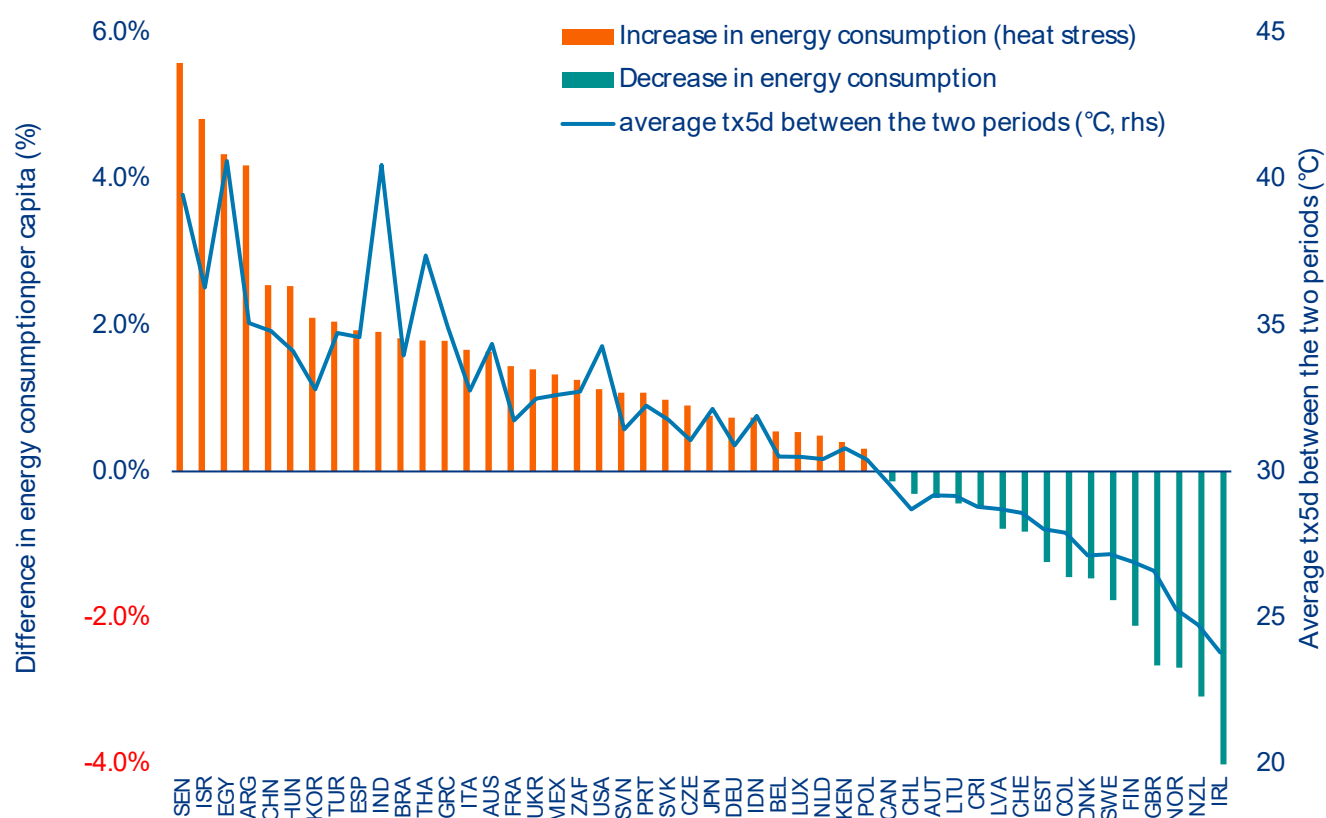
⁶ Measuring climatic impacts on energy consumption: A review of the empirical literature - ScienceDirect

⁷ Escalating impacts of climate extremes on critical infrastructures in Europe - ScienceDirect

Our empirical results confirm the non-linear structure and identify a threshold close to 30°C (Appendix 1). Using the Tx5d indicator, which is the mean daily maximum temperature over the hottest five consecutive days of the year, on an unbalanced panel of 49 countries over 1991–2023, the estimated marginal effect of heat stress on per capita energy consumption is sign-switching and temperature-dependent. At 25°C, a one-unit increase in Tx5d is associated with an approximately 0.6% decline in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase implies a 1.2% rise. The twofold differential illustrates how disproportionately energy-system stress intensifies once thermal thresholds are crossed.

The country-level counterfactual reveals a sharply asymmetric pattern, with heat already operating as a structural driver of energy demand across much of the G20 (Figure 5). Comparing per capita energy consumption during each country's most intense post-2014 heat episode (maximum Tx5d) with its 1991–2010 baseline average, countries already operating at high temperature levels (Senegal, Egypt, China, South Korea, Spain and Brazil) register increases above 4–5%, pushed further into the convex segment of the response. Almost all G20 economies in the tropics see rising energy demand across sectors, while only Europe and Oceania post aggregate reductions, driven by lower residential heating needs in cooler latitudes. The cooling gains are structurally fragile and likely to erode as temperatures continue to rise.

Figure 5: Per capita energy demand under peak heat stress (maximum Tx5d after 2014) vs. the 1991–2010 average baseline



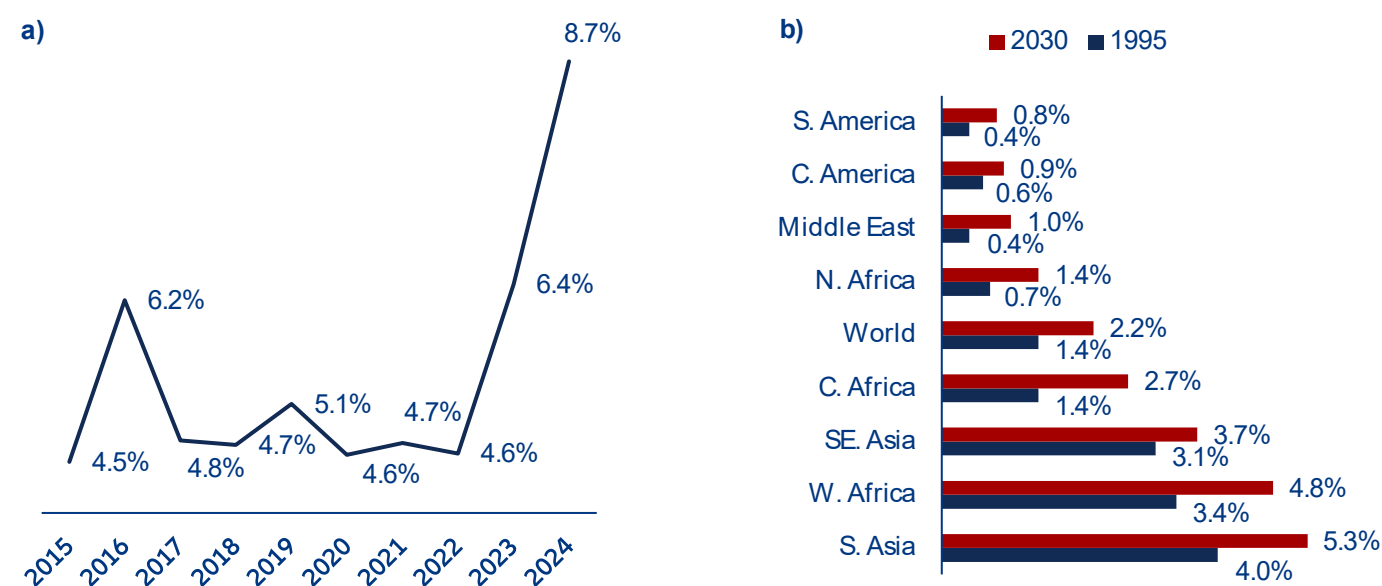
Sources: China National Customs, Allianz Research

Labor productivity and wages

Heat stress reduces labor output through physiological, cognitive and sleep-related channels that extend well beyond outdoor work (Figure 6). At the worker level, sustained exposure above the thermoneutral zone induces cardiovascular strain, dehydration and cognitive impairment, with field studies consistently documenting productivity losses across sectors (Ebi et al., 2021)⁸. The share of global working hours lost to heat stress is projected to rise from 1.4% in 1995 to 2.2% by 2030, with much higher impacts in already-warm regions – 5.3% in South Asia, 4.8% in West Africa – and a near doubling even in traditionally less exposed regions

(Figure 6b). Indoor workers are not insulated: night-time temperatures, which are rising faster than daytime averages in many regions, increased total sleep time lost by around 6% over 2020–2024 relative to the 1986–2005 baseline, reaching a record 8.7% in 2024, with up to 12 additional hours of sleep loss per person per year in the most affected locations (Figure 6a). Degraded sleep duration and quality has measurable next-day consequences for cognition, cardiovascular health and decision-making, transmitting heat impacts to indoor as well as outdoor labor.

Figure 6: Heat stress and productivity : a) Loss of sleeping time for the period 2015 – 2024, compared to a baseline 1986 – 2005 (%); b) Decline in working hours (%)



Sources: The Lancet Countdown (2025), ILO (2019), Allianz Research

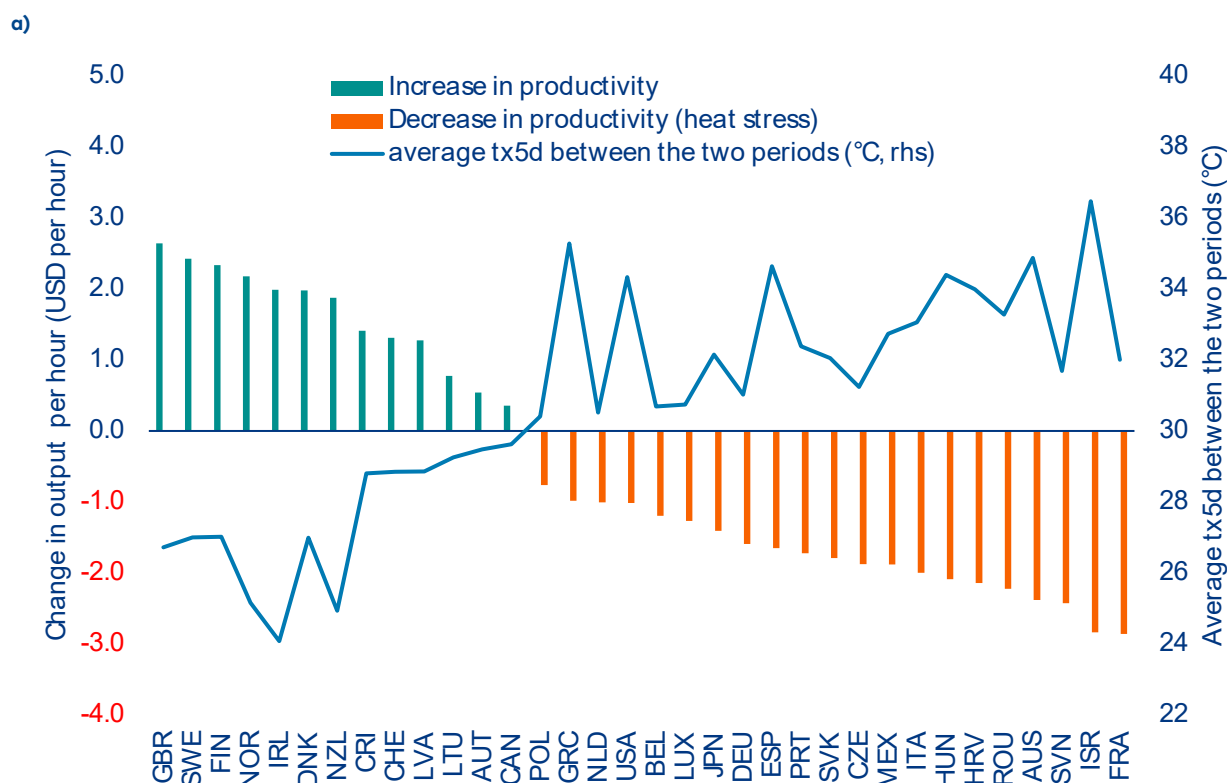
Our regressions on output per hour reveal a clear threshold effect at 30°C, consistent with the empirical consensus (Appendix 2). Estimated on a panel of 35 countries over 1997–2023, the coefficient on Tx5d below 30°C is positive and marginally significant (+0.59), consistent with mild productivity gains at moderate temperatures in temperate economies. Above 30°C, the interaction coefficient turns negative and statistically significant (–1.30), implying that each additional degree of heat beyond the threshold reduces output per hour by

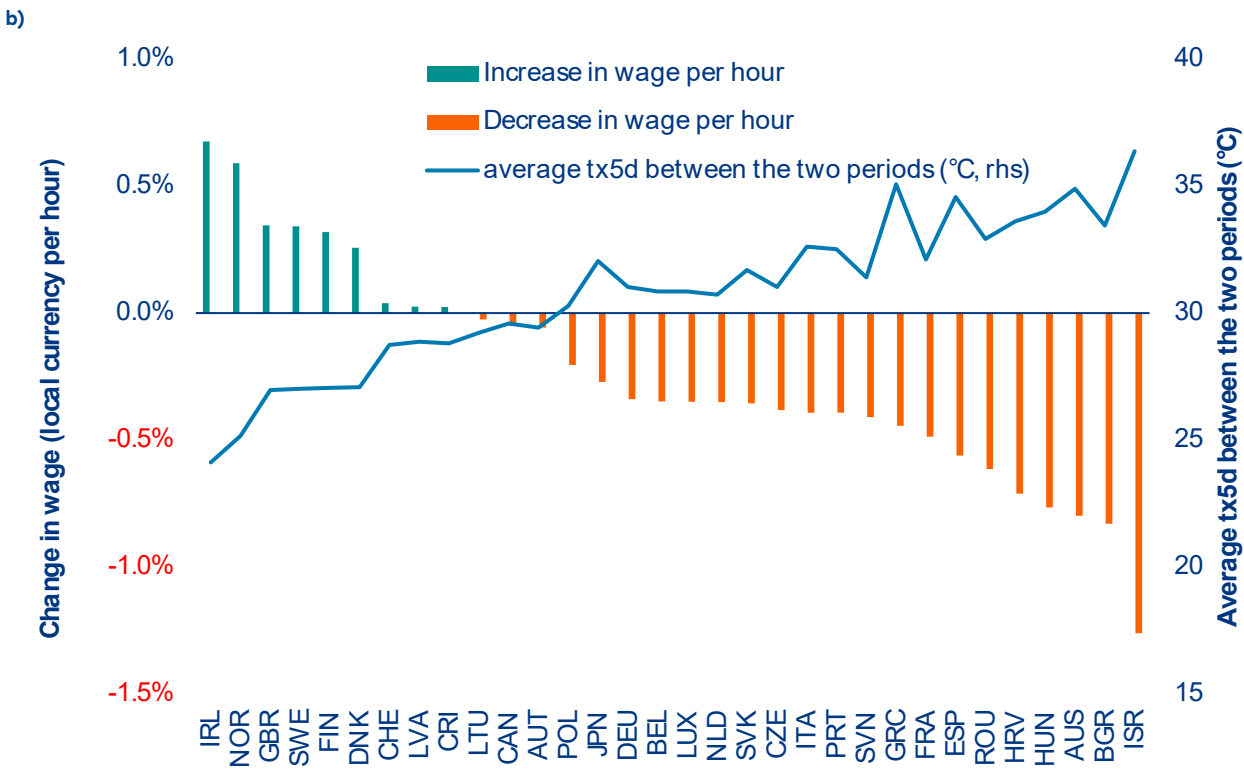
approximately USD1.30 (constant PPP, representing 3% of average output per hour in our sample between 2014 and 2024). Applied to the country-level counterfactual, where each country experiencing its maximum Tx5d observed over 2014–2024, against its 1991–2010 average, France, Slovenia, Australia and Italy emerge as the most affected, with output per hour declining by up to USD 1–3 in the most exposed cases. The UK, Finland, Ireland, New Zealand and Canada, all below the 30°C threshold, exhibit modest productivity gains (Figure 7a).

The wage response materializes with a lag, reflecting institutional rigidities in labor markets (Appendix 3). Using a one-year lag specification on a panel of 33 countries over 1995–2023, the relationship follows an inverted U-shape: below 30°C, higher heat stress in year t is associated with a modest acceleration of wage growth in year $t+1$ (approximately +0.2% at 25°C); above 30°C, the relationship reverses (approximately –0.2% at 35°C). The lag is consistent with the combination of collective-bargaining frictions, nominal wage rigidities and minimum-wage floors, all of which delay the transmission of productivity losses into labor compensation. The counterfactual maps the same threshold logic onto observed peak heat: Israel, Bulgaria, Australia, Hungary, Romania and Spain register wage declines of approximately 0.7–1.3%, while Ireland, Norway, the UK, Sweden and Finland post wage gains of 0.3–0.7% (Figure 7b).

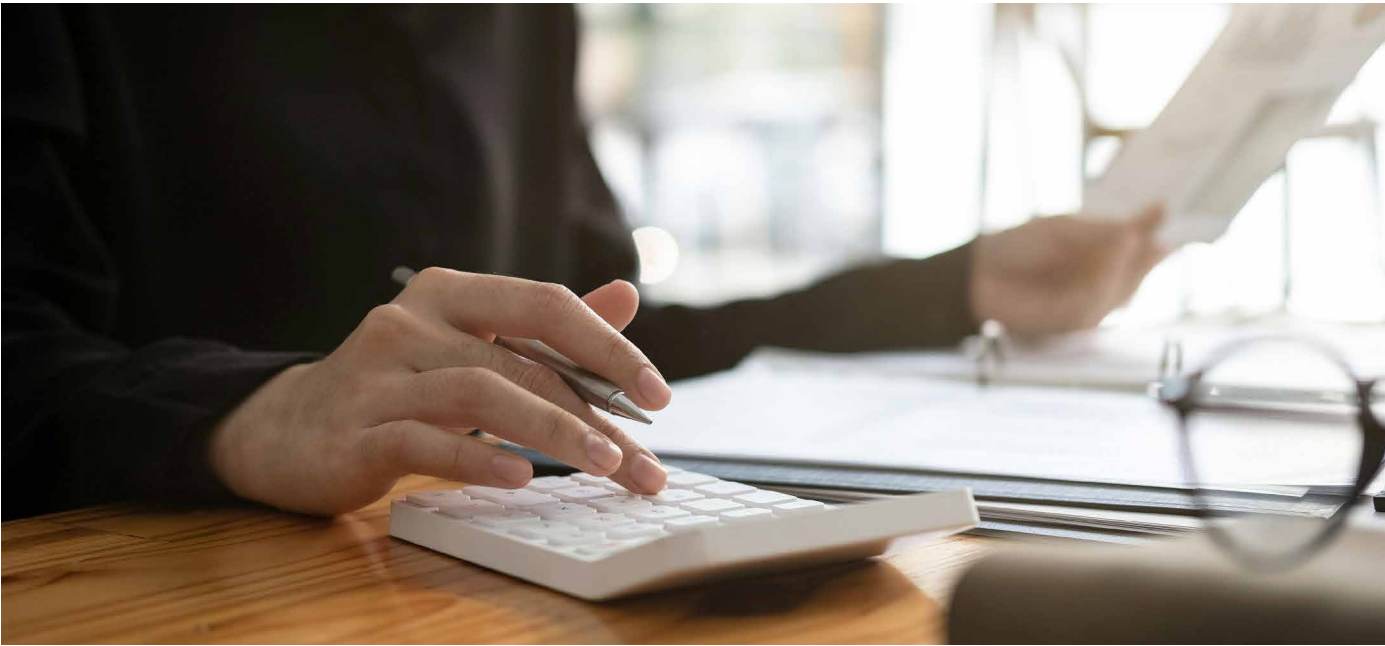
The distributional implication is structurally important and motivates the macroeconomic scenario that follows. In the short run, the cost of heat stress falls disproportionately on firm profitability as wage rigidities prevent compensation from adjusting in line with productivity. Over time, however, that wedge closes: real wages decline, household purchasing power erodes and consumption weakens, precisely in the economies already most exposed to rising temperatures. The two channels estimated in this section, additional energy demand and lower labor productivity, transmitted partially and with a lag into wages, are the inputs to the global macroeconomic simulation in the next section, which quantifies the general-equilibrium consequences in terms of GDP, prices, employment and fiscal balances.

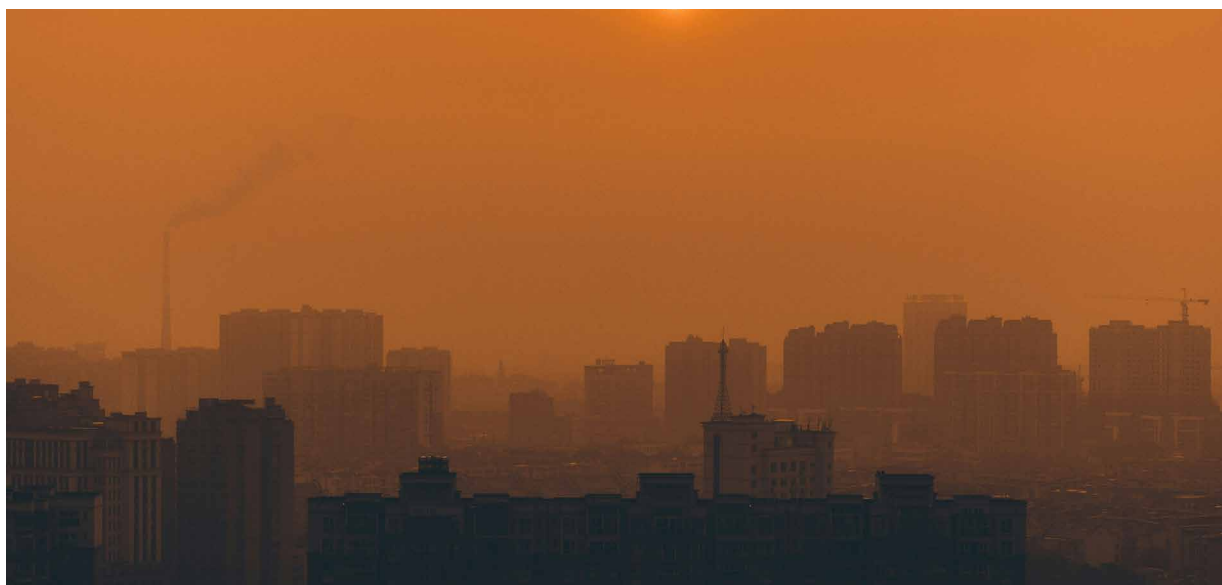
Figure 7: The change in labor productivity under peak heat stress (maximum Tx5d post-2014) relative to the 1991–2010 baseline, measured as (a) output per hour in USD and (b) wage per hour (%).





Source: Allianz Research





The macroeconomic implications of heat stress

To quantify the macroeconomic cost of these effects, we translate the empirical estimates into a structural scenario using the Oxford Economics Global Economic Model. The counterfactual exercises described above provide country-specific measures of the productivity and energy demand impact of heat stress, but they remain partial-equilibrium estimates, as they capture the direct effect of temperature on each variable in isolation, without accounting for the general-equilibrium feedbacks that propagate through the broader economy. As the recent climate-macroeconomics literature has increasingly recognized, reduced-form empirical estimates of climate damages, while essential for identification, cannot capture the full macroeconomic cost because they omit the general-equilibrium feedbacks, through trade, prices, investment and fiscal balances, that amplify the initial shock.⁹

To capture these interdependencies, we impose three simultaneous shocks on the model, each calibrated directly from our regression results. The first is a shock to potential output¹⁰, reflecting the estimated decline in output per hour worked under extreme heat conditions. Entering the productivity loss through potential output ensures that the shock operates on the

supply side of the economy, reducing the economy's productive capacity. The second shock targets wages and salaries per employee¹¹, capturing the lagged and partial transmission of productivity losses to labor compensation that our analysis identifies. The exogenous wage shock therefore captures the residual labor market adjustment that our regressions document beyond the model's endogenous response. The third shock targets energy demand¹², reflecting the additional per capita energy consumption induced by heat stress. This channel translates higher cooling needs into increased household and firm expenditure, higher energy imports for net-importing economies and upward pressure on consumer prices.

To assess the macroeconomic cost of heat stress, we construct a counterfactual scenario and compare it against a baseline projection. The baseline simulates economic outcomes over the period 2026–2030 under climatological normal conditions, defined as the average Tx5d per country observed over the 1991–2010 reference period. This baseline represents the economic trajectory each country would follow if heat stress remained at its historical norm. Against this baseline, we impose a stress scenario in which each country experiences,

⁹ [Climate Change through the Lens of Macroeconomic Modeling | NBER](#)

¹⁰ YHAT in Oxford Economics

¹¹ PEWFP in Oxford economics

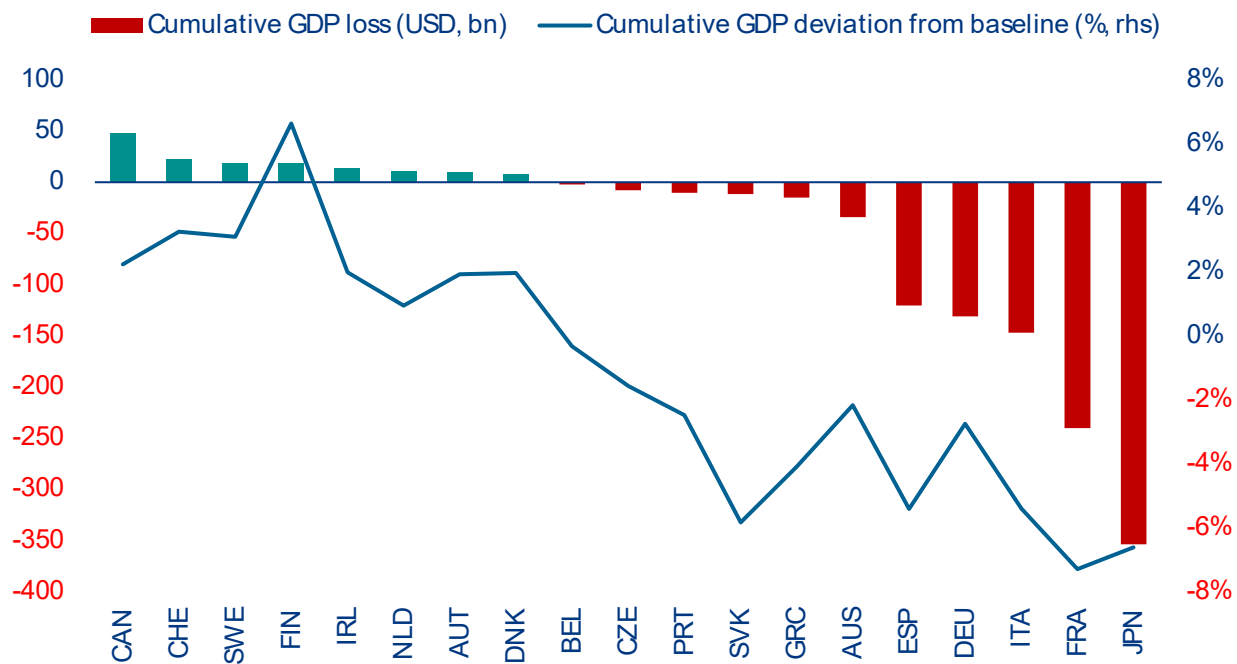
¹² DENERGY in Oxford economics

consecutively from 2026 to 2030, five years with the most important heat waves recorded during the 2014–2024 period, ranked by Tx5d. This is not a climate scenario based analysis but a historically grounded exercise: it asks what the macroeconomic consequences would be if the extreme heat conditions already observed in the recent past were to recur over a sustained five-year window.

The results reveal a clear divide across economies (Figure 8). Countries where average Tx5d remains below the 30°C threshold under the stress scenario – Canada, Switzerland, Sweden and Ireland – experience cumulative GDP gains of 1% to 3% relative to the historical climate baseline, leading to absolute gains

ranging from USD10bn to USD47bn.¹³ These gains reflect the net balance of reduced heating costs and marginally improved working conditions, benefits that are structurally fragile and likely to erode as temperatures continue to rise. Beyond this group, the transition into net losses is sharp. France, Japan, Slovakia, Spain and Italy register the highest cumulative GDP losses of 5% to 7% below baseline, reflecting the direct productivity drag that our shocks impose. The largest absolute losses, however, are concentrated in the biggest economies in the sample: Spain (USD120bn), Germany (USD131bn), Italy (USD147bn), France (USD240bn) and Japan (USD354bn).

Figure 8: Cumulative GDP loss from sustained extreme heat stress (2026–2030) relative to historical climate baseline



Sources: Oxford economics, Allianz Research

¹³ Finland is an outlier, with a potential cumulative GDP gain of 7% under sustained extreme heat stress

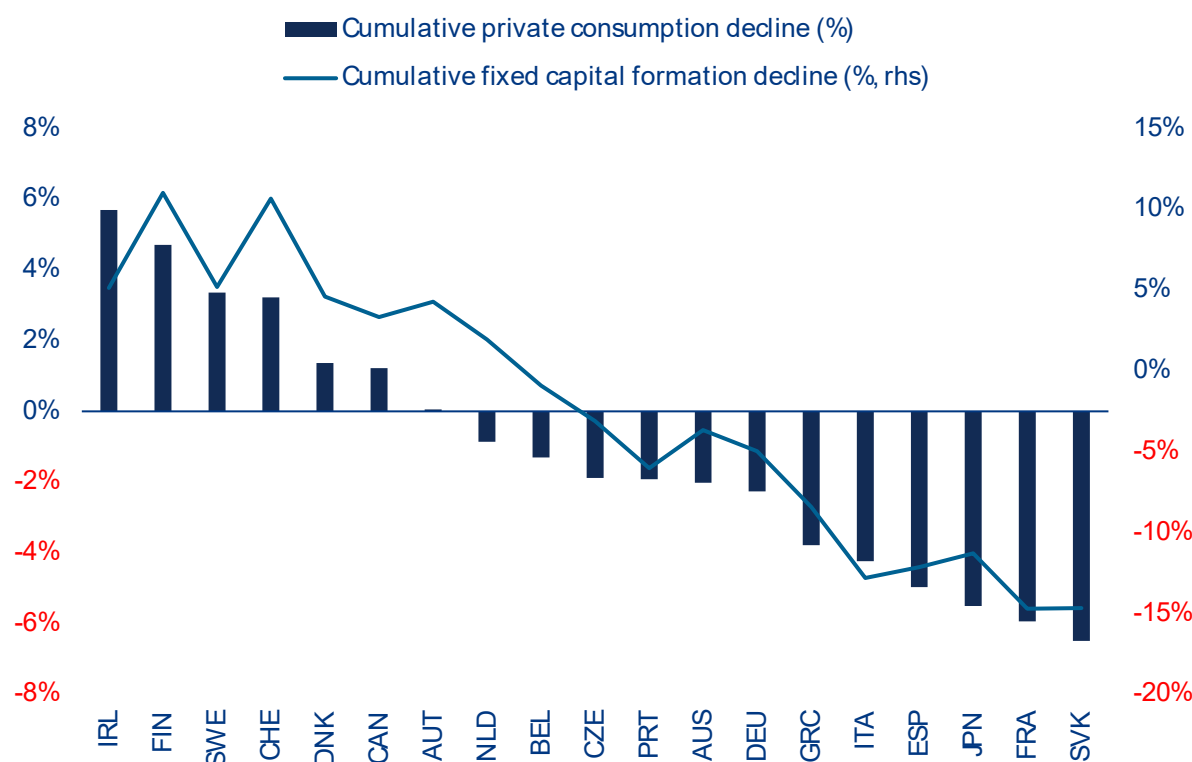
Decomposing the GDP impact into its demand-side components reveals that the macroeconomic cost of heat stress operates through two reinforcing channels: weaker household consumption and lower investment leading to a long-run decline in fixed capital formation (Figure 9).

In countries where moderate warming produces net economic benefits – Ireland, Finland, Sweden, Switzerland and Denmark – both private consumption and fixed capital formation rise relative to baseline, with investment gains generally exceeding consumption gains. The pattern inverts sharply for heat-exposed economies, and here the most striking feature of the results is the systematic amplification through investment. In virtually every country where the heat stress scenario produces economic losses, the decline in fixed capital formation exceeds the decline in private consumption, often by a wide margin. Portugal sees a consumption decline of 1.9% but an investment decline of 6.0%. Italy registers consumption losses of 4.2% against an investment collapse of 12.8%. France and the Slovak Republic, the two most affected economies, experience investment declines of 14.7% alongside consumption losses of 5.9% and 6.5% respectively. This wedge between the consumption and investment

response reflects a powerful amplification mechanism: as heat stress reduces potential output and compresses firm margins, the expected return on new capital falls, discouraging investment. Lower investment in turn reduces future productive capacity, reinforcing the initial productivity shock and creating a self-reinforcing drag on the economy's growth trajectory. This asymmetry between the investment and consumption response is consistent with Casey et al. (2022)¹⁴, who show that standard climate-economy models systematically understate investment damages because climate change disproportionately affects capital-intensive sectors exposed to outdoor heat.

The consumption response, while smaller in percentage terms, is nonetheless consequential. It captures the combined effect of lower real wages, which our wage shock imposes directly, and higher energy expenditure, which erodes household purchasing power. In the most affected economies, cumulative consumption declines of 4% to 6.5% compared to the baseline, representing a material reduction in living standards over a five-year horizon, with implications for domestic demand, retail activity and ultimately tax revenues.

Figure 9: Impact of sustained extreme heat stress on private consumption and fixed capital formation, 2026–2030 (cumulative deviation from baseline)



Sources: Oxford economics, Allianz Research

¹⁴ [Understanding climate damages: Consumption versus investment - ScienceDirect](#)

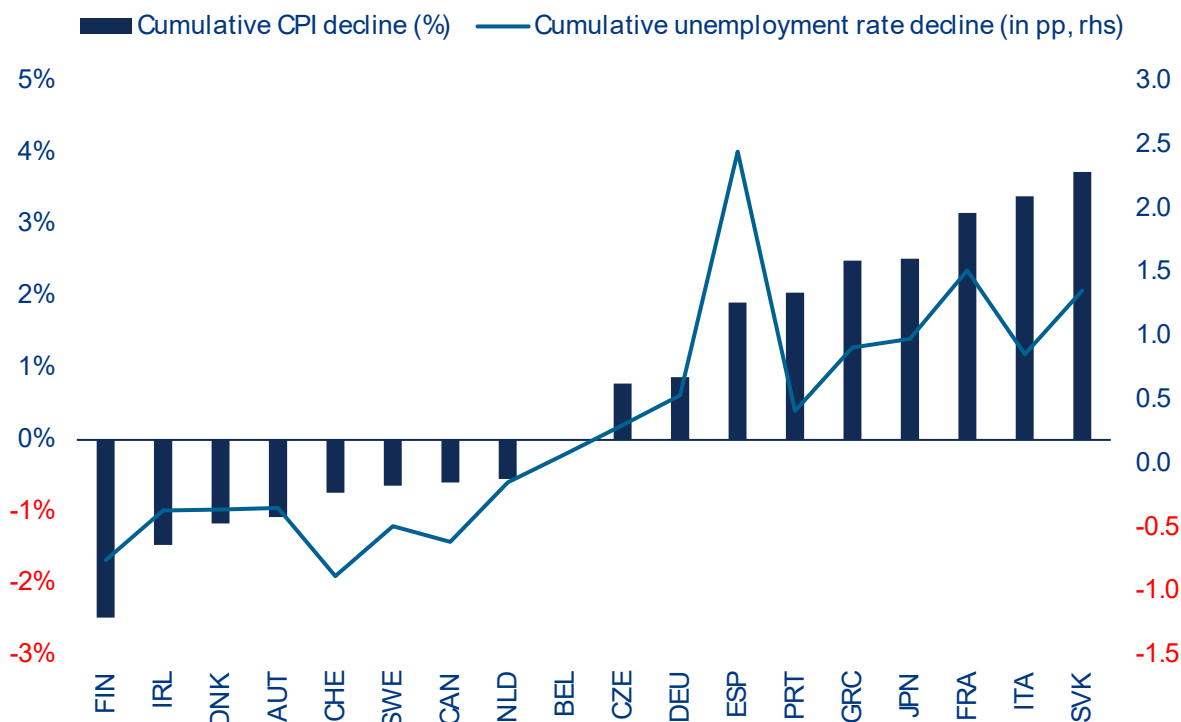
The joint behavior of consumer prices (CPI) and the unemployment rate under the stress scenario reveals the stagflationary nature of the heat stress shock, a finding with direct implications for monetary and fiscal policy. In cooler economies – Finland, Ireland, Denmark, Austria and Switzerland – both indicators improve relative to the baseline: prices fall by 0.5% to 2.5% cumulatively, while unemployment declines by 0.14 to 0.88pp. In these countries, the modest productivity gains associated with below-threshold heat stress expand supply, easing price pressures and strengthening labor demand simultaneously.

The picture reverses as heat exposure rises. For hotter countries in our samples – the Czech republic, Germany, Spain, Portugal, Greece, Japan, France, Italy and Slovakia – both prices and unemployment move in the same adverse direction, which is a defining characteristic of a negative supply-side disturbance. Spain registers a cumulative CPI increase of 1.9% alongside a 2.45pp rise in unemployment. France, Italy and Slovakia experience price increases of 3.2% to 3.7% combined with unemployment increases of 0.86pp to 1.51pp. This co-movement of rising inflation and rising unemployment is the hallmark of stagflation, and it stands in sharp contrast to the pattern produced by a standard demand

shock, where the two variables move in opposite directions along the Phillips curve. In the New Keynesian framework, Blanchard and Galí (2007)¹⁵ show that such a trade-off arises when real imperfections, such as wage rigidities, prevent the economy from adjusting smoothly to supply-side shocks. The heat stress shock in our simulation operates through precisely this type of supply-side channel: reduced potential output pushes up unit costs and prices, while simultaneously contracting wages and salaries.

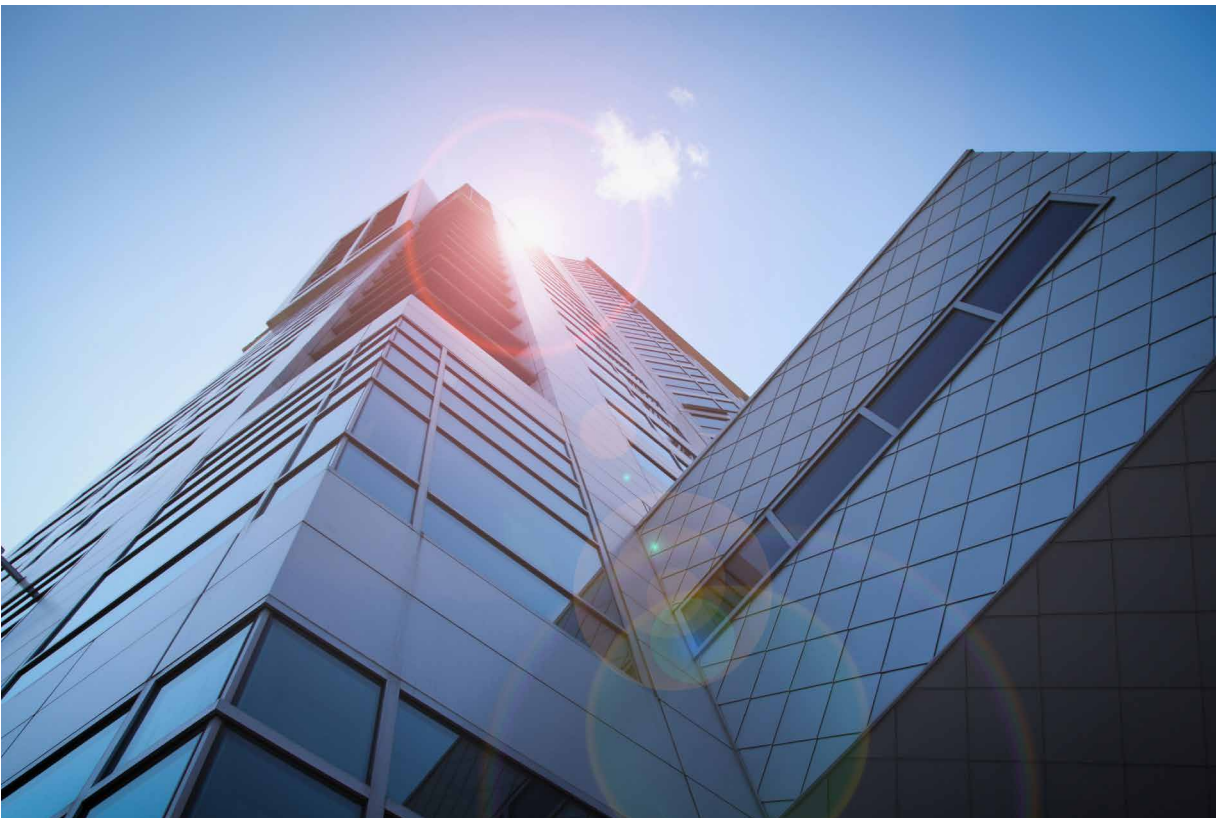
The result of such heat stress scenario is a policy environment in which conventional stabilization tools face a binding tradeoff: tightening monetary policy to contain inflation would deepen the employment losses, while loosening policy to support the labor market would accommodate further price increases. This dilemma is especially acute in the Eurozone context, where a single monetary policy must simultaneously address economies experiencing positive supply effects from moderate warming, such as Finland and Austria, and economies facing stagflationary pressures from extreme heat, such as Germany, France, Italy and Spain, a form of climate-induced macroeconomic divergence that existing policy frameworks are not designed to manage.

Figure 10: Consumer prices and labor market response to sustained extreme heat stress, 2026–2030 (cumulative deviation from baseline)



Sources: World Bank, Microsoft, Allianz Research

¹⁵ Real Wage Rigidities and the New Keynesian Model - BLANCHARD - 2007 - Journal of Money, Credit and Banking - Wiley Online Library

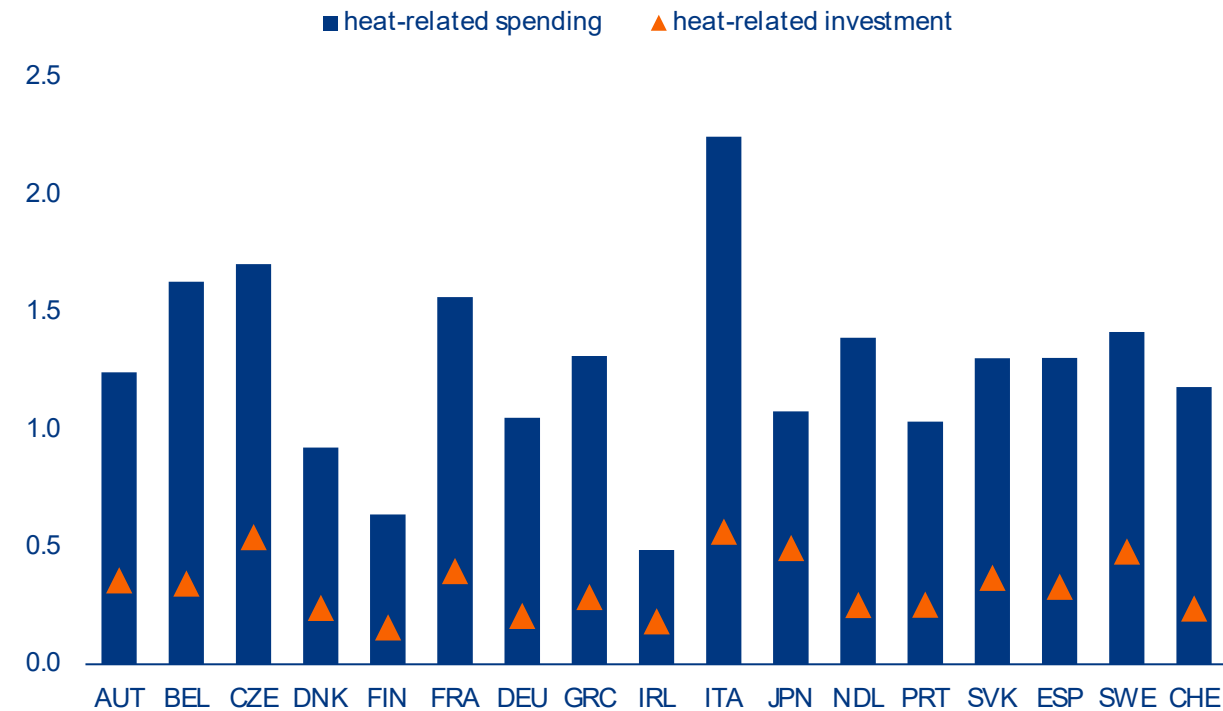


The fiscal burden

Even though it already receives about 50% of climate-related investments, extreme heat puts a strain on public finances by cutting revenues and raising adaption costs. Heat reduces GDP and productivity, which lowers tax collection, while also disrupting supply chains and energy systems, which further weakens revenues. Estimated annual revenue losses are -1.8% in France, -1.3% in Italy and Spain, -1.0% in Greece and -0.7% in Germany – higher than estimated GDP losses because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. At the same time, public spending rises. Governments face higher inflation-linked costs (e.g. wages, pensions and social benefits) as well as heat-specific expenditures (e.g. healthcare, energy subsidies, disaster response, infrastructure repair and

social protection). Demand for emergency investments in cooling infrastructure also increases. In Europe, heat-related spending accounts for around 50% of climate-related public expenditure (2.1% of GDP), but this figure varies widely: 40% in Austria, 51% in Germany, 58% in France, 63% in Spain and 65% in Italy. Flooding accounts for about 35%, with the remainder going to wildfires, drought and other areas like biodiversity. As a percentage of GDP, heat-related spending ranges from 2.2% in Italy to 1.6% in France, 1.3% in Spain, 1.1% in Germany and 0.5% in Ireland (Figure 11). Adaptation to heatwaves accounts for a significant proportion of more than a quarter of heat-related budgets, primarily for cooling and urban greening measures. However, overall investment in heat resilience is still lacking to provide for a heat proof society.¹⁶

¹⁶ From invisible to investible: An investment taxonomy for climate adaptation | Allianz

Figure 11: Heat-related spending and investment, in % of GDP 2023

Sources: OECD, Allianz Research

On average, the fiscal balances of European countries could deteriorate by around 0.5% of GDP annually due to heat stress, creating a structural tension between the fiscal damage caused by heat and the investment required to adapt to it. A rapidly growing literature has established that climatic risks impose economically significant costs on public finances¹⁷, operating through both the erosion of tax revenues as climate damages reduce economic output and the increase in public expenditure as governments face rising costs for existing commitments and new demands for adaptation. Our results confirm both channels operating simultaneously. On the revenue side, the productivity and wage shocks documented in our simulations translate directly into lower tax receipts: economies producing less and paying lower real wages generate less income tax, social security contributions and value-added tax revenue. On the expenditure side, the stagflationary character of the heat shock through rising prices combined with contracting output, inflates the cost of government programs indexed to prices or wages. The share of public spending subject to such

indexation varies considerably across our sample, from approximately 20% in Ireland to 58% in Finland, creating heterogeneous fiscal exposure to the same inflationary impulse.

We identify three groups of countries based on the interaction between heat-related fiscal pressure and existing budgetary constraints. The first group comprises countries where this tension is most acute: heat-related fiscal pressure is significant and public finances are already strained exceeding the Maastricht deficit criterion of –3%. France is the most exposed case. Heat stress could worsen an already high expected fiscal deficit of –4.9% (average 2025-2030) by an additional 2.2% of GDP. At this level, fiscal space for discretionary adaptation spending is severely compromised. Germany, with an estimated deficit of –3.6% and additional heat-related pressure of 0.9%, faces a tighter version of the same constraint. Italy's fiscal deficit of –3.1% could worsen by an additional 1.9% of GDP while Belgium, where 0.2% of additional pressure meet a deficit of –4.0%, completes this group

¹⁷ [Climate Change Impacts on Public Finances Around the World | Annual Reviews](#)

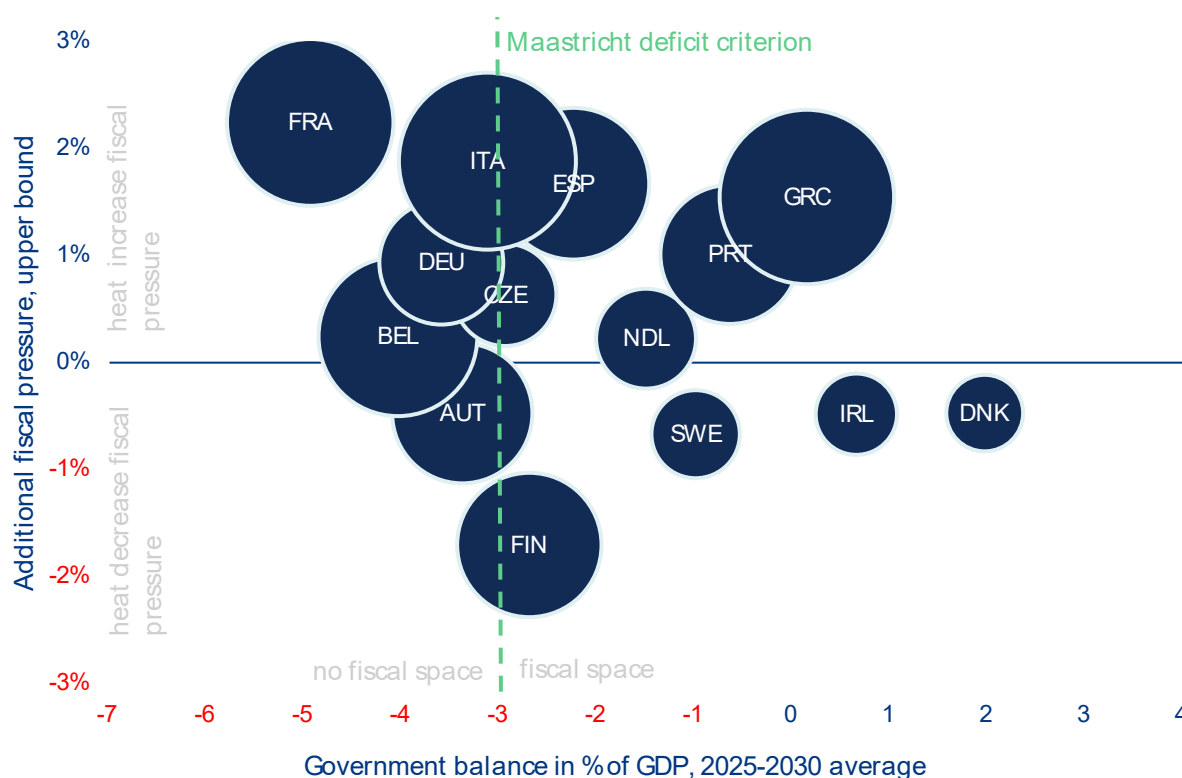
in the upper quadrant (Figure 12). For these economies, the fiscal damage from heat is actively eroding the budgetary room from which the adaptation response must be financed. Considering in addition the projected debt burden of a country also makes clear that the heat-related fiscal pressure on countries like France (121% of GDP on average between 2025 and 2030, bubble size), Italy (136% of GDP) or Belgium (109% of GDP) is more worrying than for countries like Germany with a projected debt burden of about 66%.

The second group includes countries where the fiscal deficit still provides some room but where heat-related pressures from lower growth, higher inflation and adaptation investment needs are large enough to threaten the Maastricht threshold. Italy and Spain, with estimated deficits of -2.8% and -2.4% respectively, could clearly breach the -3% criterion once heat-related pressures of 1.9% and 1.7% of GDP are factored in. The Czech Republic faces a similar dynamic, with 0.6% of heat pressure meeting a deficit hovering just above

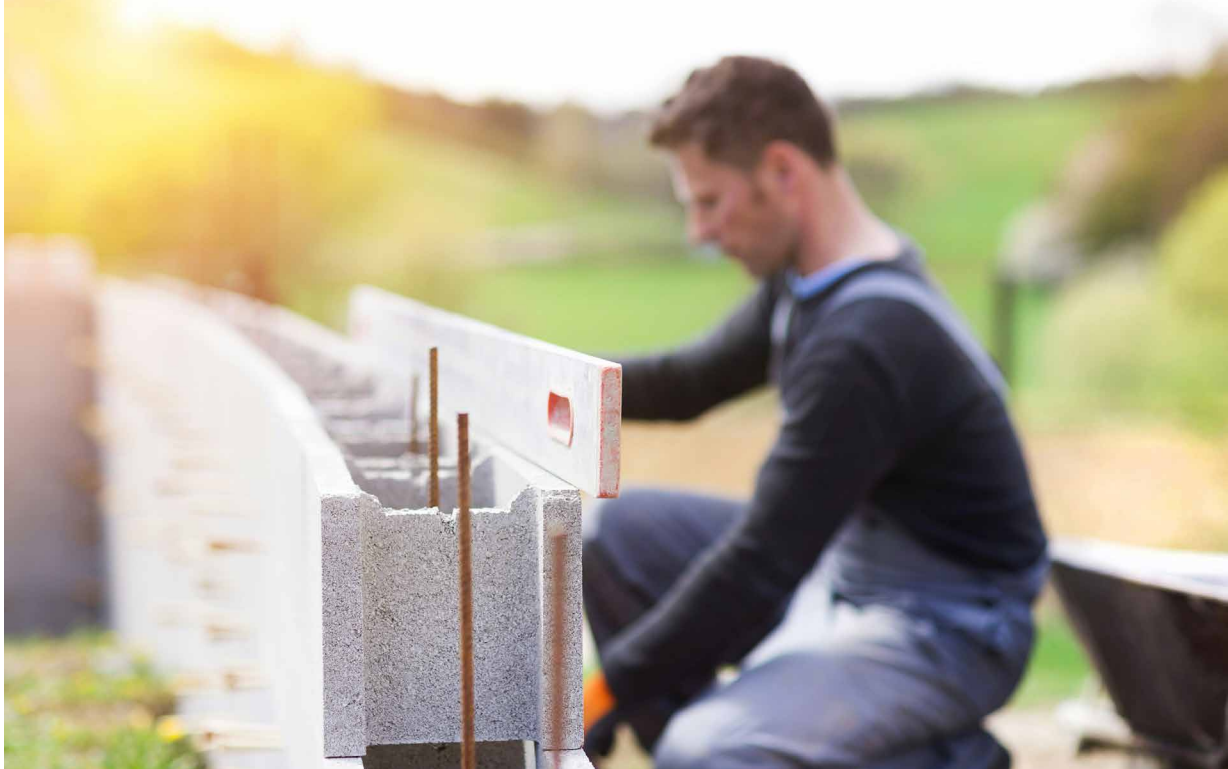
-3%. Other countries in this group, such as Greece (1.5% of heat pressure), Portugal (1.0%) and the Netherlands (0.2%), have more fiscal room and are thus less immediately constrained, but the trajectory might point toward increasing pressure.

The third group, which contains cooler regions such as the Nordic countries, may experience an improvement in their fiscal positions as moderate warming boosts economic activity through reduced heating costs and marginally improved working conditions. For these countries, the fiscal space created by a temporarily favorable climate position should be used for preventive adaptation investment, including updating building codes, establishing heat-health action plans, introducing workplace heat protocols, before the temperature distribution shifts further. Preventive action taken now, when fiscal conditions permit, yields substantially higher returns than remedial action taken later under fiscal constraint.

Figure 12: Average fiscal pressure due to heat in % of GDP (annually between 2026 and 2030) and government balance in % of GDP (2025-2030 average)



Sources: OECD, COFAS, Oxford economics, Allianz Research. Note: Bubble size indicates the projected debt burden by country 2025-2030 on average. These figures take into account the average change in inflation due to heat between 2026 and 2030, the share of indexed spending by country and heat-specific expenditure as a share of total climate-related public expenditure



From exposure to resilience: the three-pillar adaptation plan

Pillar 1. Protecting workers and productivity

The labor-productivity exposure shows that above 30°C, each additional degree of Tx5d reduces output per worker-hour by approximately USD1.30. Globally, without a tailored adaptation strategy, working hours lost to heat stress are projected to rise from 1.4% in 1995 to 2.2% by 2030 (ILO 2019). Four regulatory mechanisms shape how much of this will be mitigated: quantitative temperature thresholds, automatic work-restriction or suspension triggers, compensation for hours lost and contractual extension to fixed-term, seasonal, self-employed and platform workers.

No major European economy currently pulls all four, and the open cells cluster on the dimensions most relevant to outdoor and non-standard workers (Table 1). Spain comes closest, combining binding indoor thresholds (Royal Decree 486/1997) with up to four days of paid adverse-weather leave (Royal Decree-Law 8/2024), though the indoor focus and standard-contract design leave outdoor and non-standard workers partially exposed. France introduced procedural employer obligations from July 2025 but stopped short of quantitative thresholds or compensation. Germany's ASR A3.5 sets graduated thresholds at 26°C, 30°C and 35°C with a recognized compliance route under workplace-safety law, but does not by itself trigger automatic work restrictions. Finland regulates exposure time above 28°C through binding work-rest limits. The UK and Ireland rely on general workplace-safety duties.

Table 1: Coverage matrix — occupational heat-stress regulation across selected economies, scored on four mechanism dimensions.

Country / Economy	Quantitative threshold	Automatic work restriction / suspension trigger	Compensation for lost time	Extension to non-standard workers
Spain	● indoor	● adverse-weather trigger	● RDL 8/2024	● standard contract bias
France	○	● procedural only	○	○
Germany	● ASR A3.5 graduated	○ compliance route only	○	○
Finland	● 28°C / 33°C	● work-rest limits	○	○
Italy	○	● protocols	● wage support for heat stoppages	○
UK & Ireland	○	○	○	○
Slovenia	● indoor reported	○	○	○
Australia	○ no national heat-specific mechanism	○	○	○

Legend: ● fully covered, ● partial or conditional, ○ not covered

Source: Allianz Research

Read in this format, the gap is not that most countries are silent, it is that no country closes all four cells, and the open cells concentrate where exposure is highest.

Of the four economies most affected in the output-per-worker-hour counterfactual (France, Italy, Slovenia and Australia) none appears to combine comprehensive binding heat thresholds, automatic work-restriction rules and broad compensation coverage. France has recently strengthened procedural obligations; Italy provides some wage-support channels during extreme heat, recognized for temperatures above 35°C and, in some cases, below it depending on perceived temperature and work conditions; Australia relies mainly on general work-health-and-safety duties rather than a national

heat threshold and Slovenia's coverage should be treated separately depending on whether the analysis focuses on indoor thresholds or outdoor exposure. Several EU countries have no explicit heat-specific workplace provision. The most consistent open cell across all groups is contractual: compensation provisions, where they exist, are built around standard employment relationships; they may cover fixed-term and seasonal employees in principle, but access is weaker in practice, while self-employed and platform workers are often excluded or only indirectly covered. The cost of adaptation therefore falls disproportionately on the workers least able to bear it.

Pillar 2. Cooling buildings and stabilizing the grid

The exposure is two-sided and must be addressed simultaneously: poorly adapted buildings on the demand side, where indoor temperatures rise and cooling demand and heat mortality follow, and thermoelectric generation on the supply side, which derates under heat stress precisely when cooling demand peaks. Four regulatory levers shape how much of this is mitigated: overheating thresholds in building codes, mandatory passive cooling in renovation, cooling access as social policy and summer adequacy planning that models coincident cooling demand and supply derating. No major European economy currently pulls all four, and the gaps cluster on renovation, social-policy access and adequacy planning (Table 2). The

revised EPBD (Directive 2024/1275), with a transposition deadline of 29 May 2026, brings overheating and passive cooling into the energy-performance framework but is silent on cooling access and grid resilience. At the national level, France's RE2020 is the most advanced overheating regime, integrating a degree-hour indicator calibrated on a 2003-heatwave reference; England's Approved Document O and the Netherlands' TOjuli indicator play comparable new-build roles. Spain and Germany sit a tier below – Spain capping cooling energy demand without a dedicated overheating definition and Germany addressing summer thermal protection through DIN 4108-2 rather than a dedicated regime.

Table 2: Coverage matrix — building heat resilience and summer peak-demand management across selected economies, scored on four mechanism dimensions.

Country / Economy	Quantitative overheating threshold (new build)	Mandatory passive cooling in renovation	Cooling access as social policy	Heat-adjusted summer adequacy & distribution planning
EU EPBD 2024/1275 (from May 2026)	● EPC reflection, passive cooling in min. standards	● depends on transposition	○ out of scope	○ out of scope
France	● RE2020 DH indicator	● new-build focus	○	● RTE summer adequacy assessment
Germany	● DIN 4108-2 (indirect)	○	○	○ winter-oriented
Spain	● cooling-energy cap (CTE)	● partial	○	○ no documented mechanism
United Kingdom (England)	● Approved Doc O	○ new-build only	○	○ winter-peak focus
Italy	○ no dedicated national overheating indicator	○	○	○
Netherlands	● TOjuli indicator	● partial	○	○
Legend: ● fully covered, ● partial or conditional, ○ not covered				

Source: Allianz Research

Read in this format, the gap is twofold: cooling access remains far less institutionalized than winter heating support, and even the strongest national codes apply mainly to new build rather than to the existing stock where most exposure sits. The forthcoming EU floor in May 2026 will raise the minimum on overheating and passive cooling but does not address cooling access or grid resilience, and its bite depends entirely on transposition. The most consistent open cell is cooling access: no major European economy has developed a systematic summer counterpart to the winter heating-support architecture, and the gap falls on the groups that drive the European mortality concentration – older people, low-income households and tenants in the least-insulated dwellings. A workable national plan therefore rests on closing the four open cells together: binding overheating thresholds extended from new build to deep renovation; mandatory passive cooling triggered by a transparent renovation threshold; cooling access as a coordinated package – passive retrofits and efficient-appliance subsidies first, targeted bill support and medically prioritized access second, paired with a regulated right-to-install subject to structural and heritage constraints and heat-adjusted summer adequacy planning that explicitly models coincident cooling demand and supply derating, paired with residential demand response.

Pillar 3. Income protection and adaptation finance

Pillar 3 addresses the income-transmission channel of heat risk: when productivity losses become wage losses, the burden falls unevenly across workers, households and public budgets. The wage compression documented in previous sections lands first on outdoor workers, fixed-term, seasonal and platform workers and lower-income households. Existing stabilizers, such as unemployment insurance, short-time work, indexed transfers and energy-poverty support, absorb part of the shock, but with a structural twist: the same indexation and triggering that protect household purchasing power also raise public expenditure or compress tax receipts, lifting the fiscal cost of each heat episode.

This is the fiscal-stabilizer paradox at the heart of Pillar 3. The more effectively a welfare state cushions households from heat-driven income and price shocks, the more it shifts the climate burden onto public budgets. If that burden is met through ex-post compensation rather

than ex-ante adaptation, it consumes the fiscal space needed to lower future exposure. The paradox has two legs: a household-income leg, where existing social protection determines who absorbs the shock, and a fiscal-architecture leg, where public finance determines whether the resulting cost goes into prevention or compensation.

Only Spain (Royal Decree-Law 8/2024 paid adverse-weather leave, including heatwave situations) and partially Italy (CIG/CIGS wage supplementation activated for extreme heat) have operational heat-specific income-protection mechanisms. The remaining major economies – France, Germany, the UK, the Netherlands and most other member states – have general schemes that could in principle absorb heat-driven income loss but require ad-hoc activation each time, with no automatic heat trigger in legislation. Parametric risk-transfer for residual income volatility remains essentially absent at the national level in Europe; the European Commission's Climate Resilience Dialogue (2024) classifies it as an evolving tool requiring further analysis.¹⁸ National adaptation strategies are now in place across all major European economies but remain strategic documents rather than multi-year budget envelopes. At EU level, several frameworks enable national action without amounting to a European income-protection or adaptation-budget regime.

To reduce the gap, income protection, residual risk-transfer, budget planning and investment appraisal have to be redesigned as one fiscal architecture, working on both sides of the problem. Existing social-protection instruments should carry explicit heat triggers conditional on workplace adaptation by employers. Parametric instruments calibrated to occupational heat thresholds can complement social insurance for the residual income volatility it does not absorb, with prioritized access for outdoor and non-standard workers and a public-backstop component for affordability. Multi-year adaptation budget envelopes, set out within medium-term expenditure frameworks, should replace reliance on ex-post emergency packages, paired with climate-risk screening across public investment management and coordinated oversight bringing together the health, housing, environment, energy, transport and labor ministries.

¹⁸ Climate Resilience Dialogue - Final Report - Zurich Climate Resilience Alliance

Appendix

A1 – The impact of heat stress on energy demand

To empirically assess the relationship between heat stress and energy demand, we construct a panel regression framework that captures how rising heat stress translates into higher energy consumption – primarily through the increased use of cooling technologies such as air conditioning and refrigeration. As climate change drives more frequent and intense heat extremes, this mechanism is expected to become an increasingly important driver of energy demand, with significant implications for grid stability, energy security and the pace of the low-carbon transition.

Our analysis draws on two data sources. Energy consumption data are sourced from the IEA's World Energy Balances 2025, expressed in per capita megajoules (MJ) to allow for cross-country comparability. To capture heat stress, we use the Tx5d indicator from the ERA5 reanalysis dataset, defined as the mean daily maximum temperature over the hottest five consecutive days of a given year. This metric is particularly well-suited to our analysis, as it reflects the kind of sustained heat episodes most likely to trigger a behavioral and technological response in energy use, rather than isolated daily peaks.

The empirical model is estimated on an unbalanced panel covering 49 countries over the period 1991–2023. It provides both broad geographic coverage and sufficient temporal depth to identify meaningful climate-energy relationships. We employ a two-way fixed effects specification, controlling for both country-level heterogeneity and common time trends, with robust standard errors clustered at the regional level to account for potential spatial correlation in residuals. The regression model reads

$$\ln(\text{energy}) = \beta_1 \text{tx5d} + \beta_2 \text{tx5d}^2 + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 1}$$

The relationship between heat stress and energy consumption is nonlinear, following a convex U-shaped curve (Table A1), confirming previous literature results.¹⁹ At moderate temperature levels, energy demand declines as milder conditions reduce the need for space heating and other thermal energy services. This downward trend reverses once temperatures exceed approximately 30°C, a commonly used threshold for heat stress²⁰, at which point cooling demand begins to dominate, driven primarily by the widespread use of air conditioning.

This non-linearity is reflected in the regression results, which show that the marginal effect of heat stress on energy consumption is both temperature-dependent and sign-switching. At 25°C, the effect is negative: a one-unit increase in the Tx5d indicator is associated with an approximately 0.6% decrease in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase in heat stress is associated with a 1.2% rise in energy consumption. This twofold difference in magnitude illustrates how the energy implications of heat stress intensify disproportionately at higher temperature levels. A dynamic with important consequences for energy planning and infrastructure resilience in a warming climate.

¹⁹ [Electricity Consumption and Temperature: Evidence from Satellite Data, WP/21/22, February 2021](#)

²⁰ [Keeping cool in a hotter world is using more energy, making efficiency more important than ever – Analysis - IEA](#)

Table A1: Regression results: The impact of heat stress on energy consumption

Variable	Coefficient	p-value
Intercept	10.3390	0.000
tx5d	-0.0514	0.005
tx5d ²	0.0009	0.008

Source: Allianz Research

A2 – The impact of heat stress on productivity

The adverse effects of heat stress on labor productivity are well established in the academic literature. Growing evidence from both micro and macro studies shows that exposure to extreme temperatures affects health, labor supply, and labor productivity in the short run.²¹ At the physiological level, heat stress affects workers through physiological and behavioral responses, in turn reducing labor supply, productivity and overall labor capacity, with the relationship largely non-linear, declining sharply beyond peak temperature thresholds.²² Despite considerable heterogeneity across work sectors and study designs, nearly all field studies consistently find a reduction in productivity due to occupational heat exposure.²³

Building on this evidence, we adopt the same empirical framework used for our energy consumption analysis to derive an econometric estimate of heat stress impacts on productivity. Heat stress is measured using the Tx5d index, as defined previously. For productivity, we draw on two complementary indicators from the OECD database²⁴, output (gross value added) per hour worked, expressed in USD, and wages per hour worked, which together capture both the economic output and the compensation dimension of labor performance. The models reads as follow

$$GVA_{\text{per hour}} = \text{tx5d} + \text{tx5d}_{\geq 30} + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 2}$$

$$\Delta \ln(\text{wage}_{\text{per hour}}) = \text{tx5d}_{(t-1)} + \text{tx5d}_{(t-1)}^2 + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 3}$$

Equations 2 and 3 depart from the baseline specification of Equation 1 in two important ways, each motivated by the distinct channels through which heat stress affects labor productivity. Equation 2 models gross value added per hour as a function of both the Tx5d index and an interaction term, Tx5d when heat stress exceeds 30°C, to account for the non-linear intensification of heat impacts at higher temperature levels. Equation 3, which models the log change in wages per hour, follows the same structure as the energy consumption model (Equation 1), but introduces a one-year lag specification for both the linear and quadratic Tx5d terms. This lagged structure reflects the hypothesis that wage adjustments to heat stress occur with a delay, as labor market responses, such as renegotiated contracts or sectoral reallocation, tend to materialize more slowly than changes in physical output.²⁵

²¹ Reflections—Temperature Stress and the Direct Impact of Climate Change: A Review of an Emerging Literature | Review of Environmental Economics and Policy: Vol 10, No 2

²² Heat stress and the labour force | Nature Reviews Earth & Environment

²³ Frontiers | Occupational heat stress, heat-related effects and the related social and economic loss: a scoping literature review

²⁴ OECD Data Explorer • Productivity database

²⁵ Economic impact of labor productivity losses induced by heat stress: an agent-based macroeconomic approach | Climatic Change | Springer Nature Link

The output per hour regression (Eq. 2) is estimated on a panel of 35 countries over the period 1997–2023 (Table A2).

The results reveal a clear threshold effect consistent with the non-linear relationship hypothesized in our framework. Below 30°C, the coefficient on Tx5d is positive and marginally significant at the 10% level (0.59), suggesting that moderate heat is associated with a slight increase in output per hour – potentially reflecting seasonal productivity gains in temperate economies. Above 30°C, however, the picture changes markedly: the coefficient on the interaction term is negative and statistically significant at the 5% level (–1.30), indicating that each additional degree beyond this threshold is associated with a reduction in output per hour of approximately 1.30 USD. These findings underscore the non-linearity of heat’s impact on labor productivity, where the results point to 30°C as a critical inflection point beyond which heat stress begins to impose measurable economic costs on labor output.

Table A2: Regression results: The impact of heat stress on output per hour

Variable	Coefficient	p-value
Intercept	13.7401	0.142
tx5d	0.5900	0.072
tx5d _{≥30}	-1.3039	0.042

Source: Allianz Research

The wage regression (Eq. 3) is estimated on a panel of 33 countries over the period 1995–2023 (Table A3). The results are broadly consistent with those obtained for output per hour, with a turning point again emerging around 30°C. The use of a one-year lag structure reflects the hypothesis – supported by the data – that labor compensation adjusts to heat stress with a delay, as wage-setting mechanisms respond more slowly to productivity shocks than output itself.

The relationship between heat stress and wage growth follows an inverted U-shape. In the range below 30°C, higher heat stress in year t is associated with a modest acceleration in wage growth in year $t+1$, consistent with the productivity gains observed at moderate temperatures in the output per hour specification. Above 30°C, this relationship reverses: additional heat stress dampens wage growth the following year, reflecting the transmission of heat-induced productivity losses into labor compensation over time.

The marginal effect of heat stress on wage growth is both temperature-dependent and sign-switching. At 25°C, the effect is positive, with a one-unit increase in Tx5d associated with an approximate 0.2% increase in wage growth. At 35°C, the relationship reverses, with the same marginal increase in heat stress associated with a 0.2% decline in wage growth. While the magnitude of the effect may appear modest, its consistency with the output per hour results and its statistical significance lend confidence to the finding that heat stress above 30°C exerts a meaningful drag on labor compensation, with implications for household income, consumer demand and the broader macroeconomic costs of climate change.

Table A3: Regression results: The impact of heat stress on per hour wage growth

Variable	Coefficient	p-value
Intercept	-0.0756	0.364
tx5d _(t-1)	0.0120	0.027
tx5d _(t-1) ²	-0.0002	0.013

Source: Allianz Research

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text "Our team" is overlaid in the center, with "Our" in white and "team" in orange.

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